QUASINORMAL MODES ON KERR SPACETIMES

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Abstract. We introduce a rigorous framework for defining quasinormal modes on stationary, asymptotically flat spacetimes as isolated eigenvalues of the infinitesimal generator of time translations. We consider time functions corresponding to a foliation of asymptotically hyperboloidal hypersurfaces and restrict to suitable Hilbert spaces of functions. These functions have finite Sobolev regularity in bounded regions, but need to be Gevrey-regular at null infinity. This framework is developed in the context of sub-extremal Kerr spacetimes, but also gives uniform-in-Λ resolvent estimates on Kerr–de Sitter spacetimes with a small cosmological constant Λ. As a corollary, we also construct the meromorphic continuation (in a sector of the complex plane) of the cut-off resolvent in Kerr that is associated to the standard Boyer–Lindquist time function. The framework introduced in this paper bridges different notions of quasinormal modes found in the literature. As further applications of our methods, we prove stability of quasinormal frequencies in a sector of the complex plane, with respect to suitably small perturbations and establish convergence properties for Kerr–de Sitter quasinormal frequencies when the cosmological constant approaches zero.

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1. Introduction

The quantitative behaviour of gravitational radiation emitted at late times, as black hole solutions to the Einstein equations settle down to a final, stationary state, is intimately tied to the geometry of the stationary spacetime that emerges as the final state. In the limit as time $t \to \infty$, the dynamics are expected to be governed by solutions to wave equations, which decay inverse-polynomially in time, with powers that, in many cases, only depend on the asymptotic flatness of the spacetime and therefore are not characteristic to the particular stationary spacetime under consideration; see [Pri72, AAG18b, Hin22, AAG23a, AAG23b, LO24] and references therein.[1]

In late, *but finite*, time intervals, however, an important role is instead expected to be played by *quasinormal modes*. These are oscillating and exponentially damped solutions to wave equations that govern the spacetime dynamics, with frequencies that form a discrete set and are in fact characteristic to the particular geometry of the stationary spacetime in question. They form a dispersive analogue to the purely oscillating, vibrational *normal modes* of a vibrating string or drumhead.

In contrast with the decay rates of inverse-polynomial tails, the complex quasinormal frequencies, which determine the temporal behaviour of quasinormal modes, may serve as characteristic signatures of the geometry of the stationary spacetime that can be read off from gravitational radiation emitted to future null infinity. This type of inverse problem is known as "black hole spectroscopy". For this reason, quasinormal modes are of particular interest to gravitational wave experiments, where they have been used to estimate the masses and angular momenta of Kerr black hole spacetimes that are expected to arise asymptotically as the final states of various observed black hole mergers; see for example [Abb21].

While they have played a prominent role in the theoretical and experimental physics literature on black hole dynamics, the status of quasinormal modes on asymptotically flat spacetimes as well-defined mathematical objects is far less developed.

In this paper, we present a mathematical construction of quasinormal modes in the context of the geometric wave equation on fixed sub-extremal Kerr black hole exteriors [Ker63]:

$$\Box_{g_{M,a}}\phi = 0, \tag{1.1}$$

with M the mass of the Kerr spacetime and a its specific angular momentum. In fact, the results in the present paper apply uniformly (in Λ) to Kerr(–de Sitter) spacetimes with cosmological constant $\Lambda \geq 0$. Indeed, we consider also the more general, conformally covariant Klein-Gordon equation:

$$\Box_{g_{M,a,\Lambda}}\phi = \frac{2}{3}\Lambda\phi, \tag{1.2}$$

with $g_{M,a,\Lambda}$ a Kerr–de Sitter metric with $\Lambda \geq 0$, $M^2\Lambda \ll 1$, such that the event horizon is non-degenerate[2]. We summarize the main results below in two theorems.

The first theorem demonstrates that the spectrum of $\mathcal{A}$, the infinitesimal generator of time translations associated to (1.2) with $\Lambda \geq 0$, consists of isolated eigenvalues, when $\mathcal{A}$ is restricted to appropriate Hilbert space domains, and it moreover includes the poles of meromorphic continuations

[1] Extremal black holes form a notable exception. In the case of extremal Kerr black holes, the decay rates can differ significantly from the sub-extremal case, due to the dominant role played by the degenerate event horizon. In this case, the inverse-polynomial decay rates of gravitational radiation could actually be used as a signature of the extremality of the stationary spacetime [Gaj23]. See also [AAG18a, BKS23, AKS25] for more work on observational signatures of extremality. We refer §5 for a further discussion on quasinormal modes on extremal Kerr.

[2] This means that the surface gravity associated to the event horizon is strictly positive (but the surface gravity of the cosmological horizon goes to zero as $\Lambda \downarrow 0$).

of cut-off resolvent operators as a subset. This therefore motivates a new definition of quasinormal frequencies as eigenvalues of restrictions of $\mathcal{A}$, extending a definition of quasinormal frequencies as *scattering resonances*, i.e. poles of meromorphic continuations of cut-off resolvent operators.

Theorem 1.1 (Rough version)**.** *Let $\mathcal{A}$ be the infinitesimal generator of time translations corresponding to (1.2) on a Kerr(–de Sitter) metric with a non-degenerate event horizon, with initial data on an asymptotically hyperboloidal, horizon-intersecting hypersurface. Then:*

(i) *There exists a family of subsets $\Omega_\alpha \subset \mathbb{C}$ and a family of restrictions $\mathcal{A}_\alpha$ of $\mathcal{A}$ to suitable initial data Hilbert spaces $\mathbf{H}_\alpha$, such that $\mathcal{A}_\alpha$ has a pure point spectrum of isolated eigenvalues in Ω_α ("**regularity quasinormal frequencies**") with corresponding finite-dimensional eigenspaces (spanned by "**regularity quasinormal modes**"). The regularity quasinormal frequencies are independent of the precise choice of initial data hypersurface and of the index α.*

(ii) *The union of all Ω_α satisfies*

$$\Omega := \bigcup_\alpha \Omega_\alpha = \{|\arg z| < \varphi_0\}\,,$$

with $\varphi_0 = \pi - \arcsin\left(\sqrt{\frac{1}{\gamma_{GR}}}\right) \approx 0.712\pi$, with $\gamma_{GR} := \frac{1}{2}(1+\sqrt{5})$ the golden ratio and the corresponding union of all eigenvalues, denoted $\mathcal{Q}_{\mathrm{reg}}$, has no accumulation points in Ω.

(iii) *The standard cut-off resolvent associated to (1.1) with respect to the Boyer–Lindquist time function t can be meromorphically continued to Ω and its poles are contained in $i\mathcal{Q}_{\mathrm{reg}}$.*

In the second theorem, we establish the stability of the quasinormal spectrum under suitably small perturbations and the existence of convergent sequences of quasinormal frequencies in the limit $\Lambda \downarrow 0$.

Theorem 1.2 (Rough version)**.** *The following additional perturbative properties hold for regularity quasinormal frequencies associated to (1.2) with $\Lambda \geq 0$:*

(i) *When $\Lambda = 0$, the quasinormal frequencies in $\mathcal{Q}_{\mathrm{reg}}$ are stable under perturbations of the wave operator that are suitably small with respect to the norms on $\mathbf{H}_\alpha$.*

(ii) *Let $\{\Lambda_n\}$ be a sequence of cosmological constants with $\Lambda_n \downarrow 0$ as $n \to \infty$. For each quasinormal frequency $s_* \in \mathcal{Q}_{\mathrm{reg}}$ with respect to $\Lambda = 0$, there exists a sequence $\{s_n\}$ of quasinormal frequencies in $\mathcal{Q}_{\mathrm{reg}}$ with respect to $\Lambda = \Lambda_n$, with limit s_*. Conversely, for each sequence $\{s_n\}$ of quasinormal frequencies in $\mathcal{Q}_{\mathrm{reg}}$ with respect to $\Lambda = \Lambda_n$ and with limit $s_* \in \Omega$, s_* must be a quasinormal frequency in $\mathcal{Q}_{\mathrm{reg}}$ with respect to $\Lambda = 0$.*

Precise versions of the above statements (including definitions of the Hilbert spaces $\mathbf{H}_\alpha$) are given in §10, following the introduction of the necessary notation and geometric and functional analytic concepts in §§6–9. The main ideas and techniques in the proof of Theorem 1.1 are already outlined in §4.

By Theorem 1.1(iii), the set $i\mathcal{Q}_{\mathrm{reg}}$ includes the set of scattering resonances in an appropriate sector of the complex plane, which we will take to be the poles of meromorphically continued cut-off resolvent operators that are defined with respect to "standard" time functions with asymptotically flat level sets that intersect the bifurcation sphere; see also §3.3.[3] In this sense, regularity quasinormal frequencies are more general objects than scattering resonances. In contrast with scattering resonances, regularity quasinormal frequencies are natural objects to consider in the setting of initial data whose support includes the event horizon of the Kerr black hole, or which extend to future null infinity in a conformal sense.

[3]The set $i\mathcal{Q}_{\mathrm{reg}}$ also includes poles of meromorphically continued cut-off resolvents with respect to time functions that intersect the future event horizon to the future of the bifurcation sphere.

The definition of regularity quasinormal modes in the present paper does *not* rely on a decomposition of the corresponding solution to the wave equation in terms of oblate spheroidal harmonics via Carter separation [Car68]. Indeed, for frequencies $\omega \in i\mathcal{Q}_{\text{reg}} \subset \mathbb{C} \setminus \mathbb{R}$, such a decomposition is *not guaranteed*; see Remark 10.2 for more details.

The tools developed in the present paper can be applied to more general wave equations on Kerr, like the Teukolsky equations, and to wave equations on more general stationary, asymptotically flat spacetimes; see §5 for a further discussion.

In the final stages of writing this paper, we learned about upcoming related, but independent, work by Stucker [Stu24] on scattering resonances for (1.1) that complement the results in the present paper. See section 3.7 for a comparison of the results.

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3. Related works on quasinormal modes

In this section, we give a overview of previous literature on quasinormal modes and provide some context to the results in Theorems 1.1 and 1.2.

3.1. **Early numerical work.** The first numerical observation of the dominance of characteristic oscillations (independent of the precise initial data under consideration) in the time evolution of linear wave equations on asymptotically flat black holes was made by Vishveshwara [Vis70]. It was observed that at late, but finite time intervals, angular momentum $\ell = 2$ solutions to the Regge–Wheeler wave equations on fixed Schwarzschild background spacetimes are dominated by a damped oscillation at a single-frequency, depending only on the Schwarzschild mass M.

Further numerical investigations for higher angular momenta ℓ were carried out by Press [Pre71], who coined the term "quasinormal mode" to describe these characteristic oscillations; "normal mode" because of the occurrence of characteristic oscillations, analogous to the normal mode solutions of wave equations on compact spatial domains and "quasi" because, in contrast with normal modes, these oscillations seemed to be damped and do not fully determine the global temporal behaviour of solutions arising from generic initial data.

An identification of quasinormal modes as a particular class of solutions to the wave equation with exact time dependence $e^{-i\omega t}$, where $\omega \in \mathbb{C}$, was made in a numerical analysis by Chandrasekhar–Detweiler [CD75]; see also related earlier work by Zerilli [Zer70]. Solutions to (1.1) on Schwarzschild of the form $\phi(t, r_*, \theta, \varphi) = e^{-i\omega t}\frac{u(r_*)}{r}Y_{\ell m}(\theta, \varphi)$, with $Y_{\ell m}$ spherical harmonics, satisfy the following radial ODE:

$$u'' + (\omega^2 - V_\ell)u = 0, \tag{3.1}$$

where

$$V_\ell(r_*) \sim \begin{cases} e^{-\frac{r_*}{4M}} & r_* \to -\infty, \\ \frac{\ell(\ell+1)}{r_*^2} & r_* \to +\infty. \end{cases}$$

By standard asymptotic ODE analysis, it can be seen that $u \sim e^{\pm i\omega r_*}$ as $r_* \to \pm\infty$. In [CD75], quasinormal modes were characterized as solutions to (1.1) with a radial part u that satisfies (3.1), such that $u \sim e^{-i\omega r_*}$ as $r_* \to -\infty$ ("ingoing at the horizon") and $u \sim e^{+i\omega r_*}$ as $r_* \to \infty$ ("outgoing at infinity").“Quasinormal frequencies" would then correspond to the values of ω which allow for non-trivial solutions u satisfying these boundary conditions to exist.

When $\operatorname{Im}\omega < 0$, this characterization amounts to demanding exponential growth in $|r_*|$ as $r_* \to \pm\infty$. However, as the relation $u \sim e^{\pm i\omega r_*}$ is merely an asymptotic one, this characterization

is ill-defined:[4] it is unclear how to filter out solutions with exponential decay in $|r_*|$, as they are always dominated by exponential growth. This ambiguity was already observed in [CD75].

3.2. **Quasinormal modes and converging power series.** In the numerical scheme of [CD75], ingoing/outgoing boundary conditions are implemented by requiring u to take the form of convergent power series towards $r_* = -\infty$ and $r_* = \infty$:

$$u(r_*) = \begin{cases} e^{i\omega r_*} \sum_{j=0}^{\infty} a_j r^{-j} & r \text{ large} \\ e^{-i\omega r_*} \sum_{j=0}^{\infty} b_j (r-2M)^j & r - 2M \text{ small} \end{cases} \tag{3.2}$$

and then estimating ω for which these power series (up to some large number of terms) match at an intermediate value of r, for example at $r = 3M$.

Consistent with this approach is a characterization of quasinormal modes due to Leaver [Lea85], who defined quasinormal modes in Schwarzschild as solutions to (1.1) of the form $\phi(t, r_*, \theta, \varphi) = e^{-i\omega t} \frac{u(r_*)}{r} Y_{\ell m}(\theta, \varphi)$ with a single power series expression for u converging on $[2M, \infty)$:

$$u(r) = \left(1 - \frac{2M}{r}\right)^{-i2M\omega} r^{i2M\omega} e^{i\omega(r-2M)} \sum_{j=0}^{\infty} c_j \left(1 - \frac{2M}{r}\right)^j. \tag{3.3}$$

The requirement that u can indeed be written as the above power series imposes restrictions on the class of solutions u to (3.1) and implements therefore a notion of "ingoing at the horizon" and "outgoing at infinity. Indeed, by rewriting (3.3) in terms of the tortoise radial coordinate r_*, it is straightforward to show that (3.2) follows from (3.3).

Note that the convergence of the power series at $r = 2M$ is consistent with the fact that $r = 2M$ is a regular singular point of the ODE (3.1), so it admits solutions that are analytic at $r = 2M$. The boundary condition "ingoing at the horizon" can be interpreted as the requirement that $e^{+i\omega r_*} u$ be *analytic* in r at $r = 2M$. In fact, for solutions u with the asymptotic behaviour $e^{+i\omega r_*}$, the expression $e^{+i\omega r_*} u$ has only finite regularity in r at the event horizon (with the order of regularity depending on $\operatorname{Im}\omega$), so the ingoing boundary condition can already be implemented by restricting to u with *smooth* $e^{i\omega r_*} u$ (or C^k, with k sufficiently large, depending on $\operatorname{Im}\omega$).

The situation at $r = \infty$ is more complicated, because it corresponds to an *irregular* singular point of the ODE (3.1). In this case, the power series appearing in the expression for general u is only asymptotic and the rescaled $e^{-i\omega r_*} u$ will *always* be smooth in x at $x = 0$, with $x = \frac{1}{r}$. Demanding the convergence of (3.3) therefore implies that the following power series around $x = \frac{1}{2M}$ converges in $[0, \frac{1}{2M}]$:

$$(1 - 2Mx)^{i4M\omega} \left(e^{-i\omega r_*} u\right)(x) = \sum_{j=0}^{\infty} c_j (-2M)^j \left(x - \frac{1}{2M}\right)^j.$$

This can be thought of as a possible characterization of the "outgoing at infinity" boundary condition. Note however that this is a strictly weaker condition than analyticity in $[0, \frac{1}{2M}]$, since that would also require convergence for (suitably small) $x < 0$ and not just for $x \geq 0$. In contrast with the boundary condition at the horizon, it is not clear *a priori* if there exist *any* solutions with the desired convergence property at $x = 0$.

The ingoing and outgoing boundary conditions of [CD75, Lea85] have a cleaner interpretation when expressed in terms of a time function τ whose level sets are appropriate asymptotically hyperboloidal or null hypersurfaces intersecting the future event horizon and future null infinity. In that case, a quasinormal mode in the sense of Leaver would correspond to a solution to (1.1) on

[4] Note that if V_ℓ were compactly supported in r_*, then the behaviour $e^{\pm i\omega r_*}$ would be exact, so there would be no ambiguity in selecting the desired boundary conditions.

Schwarzschild of the form $\phi(\tau, x = \frac{1}{r}, \theta, \varphi) = e^{-i\omega\tau} x \hat{\psi}_{\ell m}(x) Y_{\ell m}(\theta, \varphi)$, such that $\hat{\psi}_{\ell m}$ is smooth (or analytic) at the horizon $x = \frac{1}{2M}$ and such that $\hat{\psi}_{\ell m}$ can be expanded as follows for $x \geq 0$:

$$\hat{\psi}_{\ell m}(x) = p(x; M, \omega) \sum_{j=0}^{\infty} c_j (-2M)^j \left(x - \frac{1}{2M} \right)^j , \tag{3.4}$$

with p analytic at $x = 0$ and smooth at $x = \frac{1}{2M}$, depending on the precise choice of asymptotically hyperboloidal or null foliation.

The ingoing and outgoing boundary conditions can therefore be thought of as *regularity conditions* at the future event horizon and future null infinity imposed on solutions to (1.1). See also [JMAS21] for recent numerical methods exploiting the advantages of the asymptotically hyperboloidal setting.

Using the characterization (3.3), Leaver developed a numerical scheme for determining quasinormal frequencies in [Lea85] (i.e. values of ω for which (3.3) holds) based around a continued fraction method introduced by Jaffé [Jaf34] and generalized by Baber–Hassé [BH35] in the context of the Schrödinger equation for hydrogen-molecule-like ions. This method was also applied to the Kerr setting, where quasinormal modes were taken to be solutions of the form

$$\phi(t, r_*, \theta, \varphi) = e^{-i\omega t} \frac{u(r_*)}{r} S_{\ell m}(\theta; a\omega) e^{im\varphi}, \tag{3.5}$$

with $S_{\ell m}(\theta; a\omega)e^{im\varphi}$ oblate spheroidal harmonics and u satisfying an equation analogous to (3.3), but with a dependence on the angular eigenvalues associated to $S_{\ell m}(\theta; a\omega)e^{im\varphi}$; see Remark 10.2 for more details regarding subtleties connected with the assumption (3.5).

For a comprehensive overview of additional numerical results on quasinormal modes, we refer the reader to the review papers [KS99, BCS09].

3.3. **Scattering resonances.** An alternative way to characterize quasinormal modes is motivated by the study of scattering resonances. When $\operatorname{Im}\omega > 0$, we can umambiguously fix ingoing boundary conditions at the horizon and outgoing boundary conditions at null infinity, since in this case, this amounts to selecting solutions that *decay* exponentially in $|r_*|$, rather than those that grow exponentially, as would be the case for $\operatorname{Im}\omega < 0$.

Let $\operatorname{Im}\omega > 0$ and let $u_+(r_*)$ and $u_\infty(r_*)$ denote solutions to (3.1) such that $u_+(r_*) \sim e^{-i\omega r_*}$ as $r_* \to -\infty$ and $u_\infty(r_*) \sim e^{+i\omega r_*}$ for $r_* \to \infty$. Let W denote the Wronskian corresponding to the solutions u_+ and u_∞, which depends only on ω.

By an energy conservation argument, it follows that W does not have any zeroes when $\operatorname{Im}\omega > 0$ and the resolvent operator;

$$R_\ell(\omega) = [\partial_{r_*}^2 + (\omega^2 - V_\ell)]^{-1} : L^2(\mathbb{R}) \to L^2(\mathbb{R})$$

can be defined via Green's formula:

$$R_\ell(\omega)(f) = W^{-1} \left(u_\infty(r_*) \int_{-\infty}^{r_*} u_+(y) f(y)\, dy + u_+(r_*) \int_{r_*}^{\infty} u_\infty(y) f(y)\, dy \right). \tag{3.6}$$

If we assume that we can make sense of the analytic continuation of W to $\mathbb{C} \setminus i(-\infty, 0]$, the meromorphic continuation of the *cut-off resolvents* $\chi R_\ell(\omega)\chi : L^2(\mathbb{R}) \to L^2(\mathbb{R})$ to $\operatorname{Im}\omega \leq 0$, where χ are arbitrary smooth compactly supported function, is well-defined. The poles of this meromorphic continuation correspond to zeroes of the analytic continuation of W and are known as *scattering resonances*.

In light of the above, Detweiler suggested an alternative characterization of "quasinormal frequencies" in [Det77], by identifying them with scattering resonances, where quasinormal modes are the corresponding resonant states. In contrast with Leaver's characterization, which is based around (3.3), the relevance of scattering resonances to the temporal behaviour of solutions to (1.1)

is clearer. Indeed, by denoting $s = -i\omega$, one can express $\phi_{\ell m}(t,r)$ via the *Bromwich integral*: let $s_0 \in (0,\infty)$, then

$$(\chi \cdot \phi_{\ell m})(t,r) = \frac{1}{2\pi i} \lim_{S\to\infty} \int_{s_0-iS}^{s_0+iS} [(\chi R_\ell(is)\chi)(f_{\ell m})](r)\, ds, \tag{3.7}$$

where $f_{\ell m}$ is determined by initial data for (1.1) at $t = 0$: $f_{\ell m} = -s\phi_{\ell m}|_{t=0} - \partial_t\phi_{\ell m}|_{t=0}$.

The above contour of integration in $\{\operatorname{Re} s > 0\}$ (i.e. $\operatorname{Im}\omega > 0$) can be deformed to obtain a contour intersecting the region $\{\operatorname{Re} s < 0\}$ (i.e. $\operatorname{Im}\omega < 0$) that encloses potential poles of $\chi R_\ell(\omega)\chi$. By the residue theorem, the poles will therefore contribute to $\chi\phi_{\ell m}(t,r)$ as terms with time dependence $e^{-i\omega t}$, multiplied by a solution to (3.1) (the resonant state).

Note that from their definitions, it is not immediately clear whether scattering resonances actually correspond to the same values of $\omega \in \mathbb{C}$ as those that give solutions to (3.1) satisfying Leaver's power series expansion (though there is ample numerical evidence in favour of such a correspondence). See §3.5 for a further discussion on this matter.

In [BMB93], Bachelot–Motet-Bachelot proved that the cut-off resolvents $\chi R_\ell(\omega)\chi$ corresponding to the resolvent in (3.6) can indeed be meromorphically continued to $\mathbb{C}\backslash i(-\infty, 0]$ for all fixed angular momenta ℓ, so the notion of quasinormal frequencies as scattering resonances is mathematically well-defined. Furthermore, the results in [BMB93] imply that for fixed ℓ, there are no accumulation points away from $i(-\infty, 0]$, so the set of quasinormal frequencies in any sector of the complex plane that excludes $i(-\infty, 0]$ is a discrete subset, provided one restricts to fixed ℓ.

The proof of the meromorphic continuation of the resolvent in [BMB93] uses the method of *complex scaling*, developed in [AC71, BC71, Sim73], which relies on the analyticity of V in r_*.

Scattering resonances have a long history in mathematical physics and quantum mechanics. A general theory for defining and studying scattering resonances for the wave equation on Minkowski with compactly supported potentials was introduced by Lax–Phillips [LP68]. We refer the interested reader to [DZ19] for an introduction to scattering resonances as well as compendium of relevant references.

3.4. **Regularity quasinormal modes in asymptotically de Sitter and anti de Sitter spacetimes.** A drawback of the Leaver characterization of quasinormal modes is that it does not offer a clear connection to the temporal evolution of solutions to (1.1) arising from prescribed initial data. Furthermore, it is not immediate how to place this method on a solid mathematical footing and derive properties about the quasinormal spectrum like its discreteness or the asymptotic behaviour of quasinormal frequencies with large frequencies $|\operatorname{Re}\omega| \gg M^{-1}$.

The characterization of quasinormal modes as resonant states of cut-off resolvents, on the other hand, is more amenable to a mathematical study. It also provides a clear link with the time evolution of solutions to (1.1), but only for *compactly supported initial data* that are moreover supported away from the event horizon. Furthermore, due to appearance of the cut-off function χ on the left-hand side of (3.7), it is not clear how resonant states contribute to the time evolution along future null infinity, the region that is most relevant for modelling gravitational wave observations.

Regularity quasinormal modes provide a way to bridge these two different characterizations of quasinormal modes and resolve the above mentioned issues. They were first introduced in [Vas13] in the context of asymptotically hyperbolic manifolds and on a subfamily of Kerr–de Sitter spacetimes. A related characterization was provided by the second author in [War15] in the context of asymptotically anti de Sitter spacetimes.

The definition of regularity quasinormal modes in asymptotically (anti) de Sitter spacetimes can be motivated by the following two observations:

- With respect to a time function τ whose level sets intersect the future event horizon and the future cosmological horizon (in the asymptotically de Sitter case) away from the bifurcation sphere(s), the ingoing (at the event horizon) and outgoing (at the cosmological horizon)

boundary conditions can be implemented by requiring the analogues of the functions $\widehat{\psi}_{\ell m}$ from (3.4) to be elements of the Sobolev space of functions H^k, where k is determined by the subset of the complex plane under consideration and is related to the surface gravities κ_+ and κ_c of the event horizon and cosmological horizon, respectively.

- It is possible to directly construct a resolvent operator $R(s) : H^k \to H^{k+1}$ with respect to the time function τ for $\operatorname{Re} s > -\frac{\min\{\kappa_+,\kappa_c\}}{2}(1+2k)$. This is different from constructing the resolvent operator in $\{\operatorname{Re} s > 0\}$ and then meromorphically continuing it after composition with cut-off functions to a sector in $\{\operatorname{Re} s < 0\}$.

  In fact, the poles of the resolvent operator $R(s) : H^k \to H^{k+1}$ correspond to the eigenvalues of the infinitesimal generator of times translations $\mathcal{A}$, with $S(\tau') = e^{s\tau'\mathcal{A}}$ the map that sends initial data $(\phi, \partial_\tau\phi)$ at $\tau = 0$ to $(\phi, \partial_\tau\phi)$ at $\tau = \tau'$.

Regularity quasinormal modes in this setting can therefore be *defined* as the eigenfunctions of the infinitesimal generator of time translations $\mathcal{A}$, restricted to $H^{k+1} \times H^k$ with $k \in \mathbb{N}_0$ suitably large. The corresponding eigenvalues s are the regularity quasinormal frequencies.

It can easily be shown that the set of regularity quasinormal frequencies *includes* the set

$$\{-i\omega \,|\, \omega \text{ scattering resonance}\};$$

see for example [War15]. This inclusion can, however, be *strict*. This, for example, is the case in the model problem studied in §6 of [War15] and for the wave equation on the de Sitter spacetime [HX21, Joy22].

Let us also mention the earlier work of Sá Barreto–Zworski [BZ97], who constructed and studied the properties of scattering resonances on Schwarzschild–de Sitter at large values of $|\operatorname{Re}\omega|$ and Bony–Häfner [BH08], who derived resonance expansions on Schwarzschild–de Sitter. See also the recent work on the distribution of Schwarzschild(–de Sitter) scattering resonances in [HZ24]. The results of [BZ97, BH08] were generalized to the slowly-rotating Kerr–de Sitter setting by Dyatlov [Dya11, Dya12].

The construction of regularity quasinormal modes from [Vas13] has recently also been extended to the full sub-extremal Kerr–de Sitter family of spacetimes in [PV24]. In the asymptotically anti de Sitter setting, it was shown by Gannot [Gan14] that scattering resonances on Schwarzschild-anti de Sitter approach the real axis as $|\operatorname{Re}\omega| \to \infty$, which is intimately connected to the presence of stable trapping of null geodesics and the sharpness of energy decay estimates with slow, logarithmic decay in Kerr-anti de Sitter; see [HS14].

Recently, the existence of slowly-damped ("zero-damped") modes in a neighbourhood of $0 \in \mathbb{C}$ was proved in [HX22a] in the setting of Schwarzschild–de Sitter with $M^2\Lambda \ll 1$, and similarly in near-extremal Reissner–Nordström–de Sitter spacetimes (where the difference between the event and inner horizon radii is very small) in [Joy22]. See also [Hin24] for analogous results in sub-extremal Kerr–de Sitter spacetimes. These constructions make use of *quasinormal co-modes* or *dual resonant* states; see also Theorem 10.1(iv) for a characterization of such objects in the asymptotically flat setting.

3.5. **Regularity quasinormal modes in asymptotically flat spacetimes.** In the asymptotically flat setting, a definition of regularity quasinormal modes is more complicated, due to the fact that the outgoing boundary condition at future null infinity cannot be implemented by considering Sobolev Hilbert spaces. Indeed, the convergence of the power series in (3.4) cannot even be guaranteed by restricting to the space of smooth functions.

This difficulty was first overcome in the setting of the geometric wave equation on extremal Reissner–Nordström spacetimes by the authors in [GW21] and in a one-dimensional toy model setting in [GW20]. The key insight in these papers is the use of L^2-based spaces of functions which are $(2,\sigma)$-Gevrey-regular at infinity, with $\sigma > 0$ depending on the region of the complex plane under consideration (i.e. Gevrey-regular in $x = \frac{1}{r}$ at $x = 0$). A function $f : [0,1] \to \mathbb{R}$ can be said

to be (α,σ)-Gevrey-regular if there exists a constant $C_\sigma > 0$, such that for all $k \in \mathbb{N}_0$:

$$\|f^{(k)}\|_{L^2([0,1])} \leq C_\sigma \sigma^{-k} k!^\alpha.$$

If the above holds for all $\sigma \in \mathbb{R}$, the function f is said to be α-Gevrey-regular.

The space of (α,σ)-Gevrey-regular functions with $\alpha > 1$ therefore may be thought of as sitting between the space of analytic functions ($\alpha = 1$) and the space of smooth functions in terms of regularity.

In [GW21, GW20] it was moreover shown that the set of regularity quasinormal frequencies includes the set $\{-i\omega \text{ , } \omega \text{ scattering resonance}\}$ in a sector of the complex plane excluding the half-line $(-\infty, 0]$.

Furthermore, [GW20] demonstrated that the convergence of the power series in (3.4) implies Gevrey regularity of $\widehat{\psi}_{\ell m}$ and therefore the Leaver quasinormal frequencies are also included when considering regularity quasinormal frequencies.

As a *corollary* of the methods developed [GW21, GW20], it was moreover shown that, in the one-dimensional toy model setting of [GW20], the construction of the meromorphic continuation of the standard cut-off resolvent applies even when the potential under consideration is $(2,\sigma)$-Gevrey-regular, rather than analytic. A straightforward implementation of the complex scaling method fails in this case, but it was observed in [GZ21] that $(2,\sigma)$-Gevrey-regular potentials can be split into an analytic part and an exponentially decaying part, the latter which can be treated as a small perturbation when studying cut-off resolvents. With this splitting, classical implementations of the complex scaling method therefore also apply to this $(2,\sigma)$-Gevrey-regular toy model, at least in the context of cut-off resolvents.

In comparison with the results in [GW21, GW20], Theorem 1.1 of the present paper identifies L^2-based spaces of functions that are applicable in a much more general setting of asymptotically flat spacetimes, including the sub-extremal Kerr spacetimes, where a suitable notion of Gevrey-like regularity must also be imposed at infinity.

Besides providing a definition of quasinormal modes that is more general and widely applicable than the definition in terms of scattering resonances, Theorem 1.1(iii) provides the first construction of the meromorphic continuation of the cut-off resolvent on sub-extremal Kerr spacetimes with respect to the standard Boyer–Lindquist time function. Theorem 1.1 moreover provides an extension of the fixed-ℓ result in Schwarzschild [BMB93] in a sector of the complex plane, showing that scattering resonances do not feature any accumulation points in the limit $\ell \to \infty$ (in the above mentioned sector). See also [Stu24] and the discussion at the end of §1.

As Theorem 1.1 and Theorem 1.2(ii) demonstrate, in contrast with methods involving cut offs and complex scaling, regularity quasinormal frequencies (and therefore also scattering resonances) in Kerr and Kerr–de Sitter with suitably small $M^2\Lambda$ can be studied simultaneously and uniformly in Λ with the methods of the present paper.

3.6. **Stability of the quasinormal mode spectrum.** A natural question in the context of the eigenvalue problem of $\mathcal{A}$ is the stability of the point spectrum under suitably small perturbations of the corresponding wave operator. Since $\mathcal{A}$ is not self-adjoint, one cannot simply appeal to classical stability results in spectral theory.

Stability properties of quasinormal spectra were investigated numerically by Nollert and Nollert–Price in [Nol96, NP99] in the context of (toy models for) perturbed wave operators of the form $\Box_{g_{M,0}} + \delta V$, with V a discontinuous potential function. More recently, the quasinormal spectrum (in)stability question was revisited for (1.1) on Schwarzschild in [DGM20, JMAS21, CDM+22] for different choices of V.

Results pertaining to "instability" in the above works are in fact all consistent with the *stability* statement in Theorem 1.2(i), because **in all of the above cases, the potentials under consideration do <u>not</u> correspond to small perturbations for $0 < \delta \ll 1$, as measured by the**

norms on $\mathbf{H}_\alpha$. Roughly, speaking $\mathbf{H}_\alpha$ consists of functions with finite regularity in a bounded r region $r \leq R$, but with $(2,\sigma)$-Gevrey regularity in $x = \frac{1}{r}$ as $r \to \infty$ with σ determined by the index α. In particular, the rescaled potential function r^2V therefore needs to be small and more regular than smooth in x near $x = 0$ for Theorem 1.2(i) to apply, but away from $x = 0$, far less regularity is required.

Let us emphasize however, that it remains open how to even *define* quasinormal modes (either in the sense of Leaver, scattering resonances or regularity quasinormal modes), let alone study their stability properties mathematically, when r^2V is not at least $(2,\sigma)$-Gevrey at $x = 0$ for some $\sigma > 0$.

Recently, the question of (in)stability and the characterization of "smallness' have also been addressed in the context of the conformally covariant wave operator in de Sitter by the second author [War24].

3.7. **Comparison with [Stu24].** In [Stu24], meromorphic continuations are constructed for cut-off resolvents corresponding to time functions with future-horizon intersecting, asymptotically flat level sets by combining the constructions introduced by Vasy in [Vas13] in a spacetime region of bounded radius, together with a general complex scaling argument in the remaining region, analogous to the complex scaling argument introduced in [SZ91].

An advantage of a complex scaling-based approach is that it allows for a construction of scattering resonances in the whole region $\mathbb{C}\setminus i(-\infty, 0]$, indeed the cut-off resolvent can be meromorphically continued beyond this to a portion of the logarithmic Riemann surface. This contrasts with Theorem 1.1(iii), which only provides such a construction in a sector of the complex plane.

One disadvantage of such an approach is that, in contrast with the methods introduced in the present paper, it relies fundamentally on the conjugation of the resolvent operator with a cut-off function at large radius. From studying such a cut-off resolvent it is possible to extract information about the behaviour in a bounded r-region of a solution arising from initial data of compact support. The restriction to a bounded region is necessary to observe wave decay due to dispersion, since the time slices all end at i^0 and so energy is conserved near infinity.

In the standard theoretical framework for gravitational wave measurements, ideal observers are located on future null infinity $\mathcal{I}^+$, and the physically relevant observable is the radiation field at $\mathcal{I}^+$: in our scalar model this is given by $r\phi|_{\mathcal{I}^+}$. Since our method does not involve introducing a large-r cutoff, and our time slices extend to $\mathcal{I}^+$, our results extend uniformly to null infinity.

The method of complex scaling moreover does not directly allow for a uniform analysis of Kerr–de Sitter spacetimes with $\Lambda > 0$ (where instead the methods of [Vas13, War15, PV24] apply) as $\Lambda \downarrow 0$, whereas the methods introduced in the present paper enable a uniform analysis of resolvent operators for $\Lambda \geq 0$, with $M^2\Lambda \ll 1$. This enables the study of the convergence properties of Kerr–de Sitter quasinormal frequencies as $\Lambda \downarrow 0$ in Theorem 1.2(ii).

The spectral convergence results we obtain in fact suggest that our results are necessarily limited to a smaller sector than achieved with complex scaling. The limit $\Lambda \downarrow 0$ involves a cosmological horizon becoming extremal. There is strong evidence that such a limit is always accompanied by a sequence of QNFs approaching 0 approximately along the negative real axis (in our preferred orientation) [Joy22, HX22b, Hin25]. This appears to represent an obstacle to spectral convergence on sets that do not avoid some sector around the negative real axis.

4. Main ideas and techniques

In this section, we give an outline of the proofs of Theorems 1.1 and 1.2 and highlight the main new ideas and tools that are introduced in the present paper.

Infinitesimal generators and Laplace transformed operators. Let ϕ be a solution to the wave equation (1.2) and let $S(\tau)$ denote the following time translation map:

$$S(\tau') : (\varrho\cdot\phi, \varrho\cdot\partial_\tau\phi)|_{\tau=0} \mapsto (\varrho\cdot\phi, \varrho\cdot\partial_\tau\phi)|_{\tau=\tau'},$$

where ϱ is an appropriate radial coordinate (see the later paragraphs of this section for a discussion on the choice of radial coordinate).

By standard local-in-time energy estimates, $S(\tau)$ is well-defined for all $\tau \in [0, \infty)$ with initial data that are smooth in a conformal sense, i.e. with respect to the differentiable structure $(x = \frac{1}{\varrho}, \vartheta, \varphi_*)$, with (ϑ, φ_*) appropriate angular coordinates. The corresponding *infinitesimal generator of time translations* is defined as the time derivative $\mathcal{A} := \frac{d}{d\tau}|_{\tau=0} S(\tau)$ and, formally, takes the form:

$$\mathcal{A}\left((\varrho \cdot \phi, \varrho \cdot \partial_\tau \phi)|_{\tau=0}\right) = (\varrho \cdot \partial_\tau \phi, \varrho \cdot \partial_\tau^2 \phi)|_{\tau=0}.$$

The time function τ is chosen so that its level sets Σ_τ are spacelike hypersurfaces that intersect $\mathcal{H}^+$ strictly to the future of the bifurcation sphere and: 1) are asymptotically hyperboloidal in the $\Lambda = 0$ case; or 2) intersect the cosmological horizon to the future of the bifurcation sphere in the $\Lambda > 0$ case.

The level sets Σ_τ capture the radiation of energy through $\mathcal{H}^+$ and future null infinity $\mathcal{I}^+$ or the cosmological horizon $\mathcal{C}^+$, in contrast with the standard Boyer–Linquist time function t, whose level sets Σ'_t are spacelike hypersurfaces that intersect the bifurcation sphere and approach spacelike infinity i^0. See Figure 1 for an illustration in the $\Lambda = 0$ case.

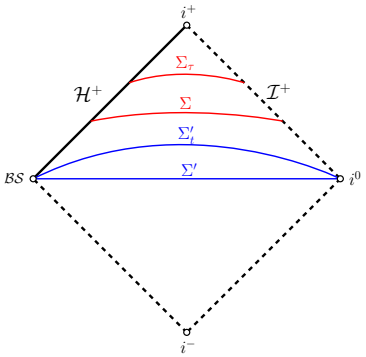


FIGURE 1. A Penrose diagrammatic representation of the Kerr black hole exterior (including the future event horizon $\mathcal{H}^+$), with $\Sigma' = \{t = 0\}$ and Σ'_t the t-level sets, and with $\Sigma = \{\tau = 0\}$ and Σ_τ the τ-level sets. The point $\mathcal{BS}$ represents the bifurcation sphere and i^+, i^0, i^- represent future timelike/spacelike/past timelike infinity.

In §9, we show that $\mathcal{A}$ can be related to the operators $L_{s,\Lambda}$, which are defined via the Laplace transform of the rescaled wave operator $\varrho^3 \circ \left(\Box_{g_{M,a,\Lambda}} - \frac{2}{3}\Lambda\right) \circ \varrho^{-1}$, i.e.

$$L_{s,\Lambda}\hat{\psi} = e^{-s\tau}\varrho^3 \left(\Box_{g_{M,a,\Lambda}} - \frac{2}{3}\Lambda\right)(e^{s\tau}\varrho^{-1}\hat{\psi}),$$

with $\hat{\psi}$ a function on Σ. We will show that:

$$\mathcal{A} - s\mathbf{1} = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & P - s\mathbf{1} \end{pmatrix} \begin{pmatrix} \frac{1}{b} L_{s,\Lambda} & 0 \\ 0 & \mathbf{1} \end{pmatrix} \begin{pmatrix} \mathbf{1} & 0 \\ -s\mathbf{1} & \mathbf{1} \end{pmatrix},$$

with b a bounded function that is moreover bounded away from 0 and with P an appropriate first order differential operator.

Formally, the corresponding resolvent operator $(\mathcal{A} - s\mathbf{1})^{-1}$ then takes the form:

$$(\mathcal{A} - s\mathbf{1})^{-1} = \begin{pmatrix} \mathbf{1} & 0 \\ s\mathbf{1} & \mathbf{1} \end{pmatrix} \begin{pmatrix} L_{s,\Lambda}^{-1} \circ b\mathbf{1} & 0 \\ 0 & \mathbf{1} \end{pmatrix} \begin{pmatrix} -(P - s\mathbf{1}) & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}.$$

Hence, we can determine the spectrum $\mathrm{Spect}(\mathcal{A})$ of $\mathcal{A}$ from a suitable characterization of the invertibility properties of the operator $L_{s,\Lambda}$. In particular, eigenvectors of $\mathcal{A}$ are elements of the form $(\hat{\psi}_s, s\cdot\hat{\psi}_s)$ with $\hat{\psi}_s \in \ker L_{s,\Lambda}$. Theorem 1.1 can therefore be thought of as a *corollary* of the following theorem:

Theorem 4.1 (Rough version)**.** *Let $\Lambda \geq 0$ and let $L_{s,\Lambda}$ with $s \in \mathbb{C}$ be the Laplace transform of the operator $\varrho^3 \circ (\Box_{g_{M,a,\Lambda}} - \frac{2}{3}\Lambda) \circ \varrho^{-1}$ with respect to the time function τ associated to an asymptotically hyperboloidal and future-horizon-intersecting foliation of hypersurfaces. Let $N_+ \in \mathbb{N}_0$, $\sigma \in \mathbb{R}$ and $R_\infty > r_+$. Denote by $H^{N_+}_{\sigma,R_\infty}$ and $H^{N_+,1}_{\sigma,R_\infty,2}$ a corresponding family of Hilbert spaces of functions on Σ (see §8 precise definitions). Then, for suitably large $M^{-1}R_\infty$:*

(i) *For $M^2\Lambda$ suitably small, there exists a family of subsets $\Omega_{\sigma,N_+} \subset \mathbb{C}$ with*

$$\Omega := \bigcup_{\sigma\in\mathbb{R}, N_+\in\mathbb{N}_0} \Omega_{\sigma,N_+} = \{|\arg z| < \varphi_0\},$$

with $\varphi_0 = \pi - \arcsin\left(\sqrt{\frac{2}{1+\sqrt{5}}}\right) \approx 0.712\pi$, such that $L_{s,\Lambda}$ is defined as follows: $L_{s,\Lambda} : D^{N_+}_{\sigma,R_\infty}(L_s) \to H^{N_+}_{\sigma,R_\infty}$, with $H^{N_+,1}_{\sigma,R_\infty,2} \subseteq D^{N_+}_{\sigma,R_\infty}(L_s) \subseteq H^{N_+}_{\sigma,R_\infty}$ and $D^{N_+}_{\sigma,R_\infty}(L_s)$ a suitable closed subspace, admits an inverse $L^{-1}_{s,\Lambda}$ for s in the complement of a discrete subset of Ω_{σ,N_+}, such that $s \mapsto L^{-1}_{s,\Lambda} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+,1}_{\sigma,R_\infty,2}$ is meromorphic in Ω_{σ,N_+}. The set of poles of $s \mapsto L^{-1}_{s,\Lambda}$ does not depend on the precise choice of time function τ and indices σ and N_+.

(ii) *The union $\mathcal{Q}_{\mathrm{reg}}$ over $\sigma \in \mathbb{R}$ and $N_+ \in \mathbb{N}_0$ of all poles of $s \mapsto L^{-1}_{s,\Lambda}$ has no accumulation points in Ω.*

(iii) *The cut-off resolvent of (1.2) corresponding to the Boyer–Lindquist time function t can be meromorphically continued to Ω and its poles are contained in $i\mathcal{Q}_{\mathrm{reg}}$ provided $M^2\Lambda$ is suitably small.*

Similarly, Theorem 1.2 can be thought of as a corollary of the following theorem:

Theorem 4.2 (Rough version)**.**

(i) *The poles of $L^{-1}_{s,0}$ are stable under small perturbations $L_s + \delta\widetilde{Q}$, with $\widetilde{Q}$ a bounded operator with respect to appropriate Hilbert spaces and $\delta > 0$ a suitably small constant.*

(ii) *Let $\{\Lambda_n\}$ be a sequence of cosmological constants with $\Lambda_n \downarrow 0$ as $n \to \infty$. For each pole s_* of $s \mapsto L^{-1}_{s,0}$, there exists a sequence $\{s_n\}$ of poles of $s \mapsto L^{-1}_{s,\Lambda_n}$ with limit s_*. Conversely, for each sequence $\{s_n\}$ of poles of $s \mapsto L^{-1}_{s,\Lambda_n}$ with limit $s_* \in \Omega$, s_* is a pole of $s \mapsto L^{-1}_{s,0}$.*

We will now sketch the proofs of Theorems 4.1 and 4.2. References to detailed versions of the steps below can be found in §11.

Sketch of the proof of Theorem 4.1(i). The proof of (i) constitutes the main part of Theorem 4.1. The goal is to show that the following operator is meromorphic in s:

$$L^{-1}_{s,\Lambda} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+,1}_{\sigma,R_\infty,2} \subset H^{N_+}_{\sigma,R_\infty},$$

with $H^{N_+,1}_{\sigma,R_\infty,2}$ a Hilbert space consisting of functions in $H^{N_+}_{\sigma,R_\infty}$ whose derivatives up to first order are also contained in $H^{N_+,1}_{\sigma,R_\infty,2}$ and whose second-order derivatives, restricted outside a suitable bounded set, with possibly additional degenerate factors, are also contained in $H^{N_+}_{\sigma,R_\infty}$; see §8 for a precise definition. Key to the proof is that $H^{N_+,1}_{\sigma,R_\infty,2}$ is *compactly* embedded in $H^{N_+}_{\sigma,R_\infty}$.

In the discussion below, we will first restrict to the $\Lambda = 0$ setting. We denote $L_s = L_{s,0}$. As we will see a simplified version of $L_{s,\Lambda}$ with $\Lambda > 0$ will play an important role, even when investigating L_s.

We first show that we can decompose L_s as follows:

$$L_s = \mathfrak{L}_{s,\lambda} + \epsilon_0 \cdot B_s + K_{s,\lambda}, \tag{4.1}$$

with $\epsilon_0 > 0$ arbitrarily small and with $\lambda = \mathfrak{l}(\mathfrak{l}+1)$, where $\mathfrak{l} \in \mathbb{N}_0$ is a large parameter that arises from an *angular frequency shift* applied to low angular frequencies. Furthermore:

$$\begin{aligned}
\mathfrak{L}_{s,\lambda}^{-1} &: H^{N_+}_{\sigma,R_\infty} \to H^{N_+ +1}_{\sigma,R_\infty,2} \subset\subset H^{N_+}_{\sigma,R_\infty} \quad \text{is bounded for all } s \in \Omega_{\sigma,N_+}, \text{ for suitably large } \mathfrak{l} \in \mathbb{N}_0,\\
B_s &: H^{N_+ +1}_{\sigma,R_\infty,2} \to H^{N_+}_{\sigma,R_\infty} \quad \text{is bounded for all } s \in \Omega_{\sigma,N_+},\\
K_{s,\lambda} &: H^{N_+ +1}_{\sigma,R_\infty,2} \to H^{N_+ +1}_{\sigma,R_\infty,2} \subset\subset H^{N_+}_{\sigma,R_\infty} \quad \text{is bounded for all } s \in \Omega_{\sigma,N_+} \text{ and } \mathfrak{l} \in \mathbb{N}_0.
\end{aligned}$$

For $\epsilon_0 > 0$ suitably small, it follows that $\|\epsilon_0 B_s \circ \mathfrak{L}_{s,\lambda}^{-1}\| < 1$, so

$$(\mathfrak{L}_{s,\lambda} + \epsilon_0 \cdot B_s)^{-1} = \mathfrak{L}_{s,\lambda}^{-1}(\mathbf{1} + \epsilon_0 B_s \mathfrak{L}_{s,\lambda}^{-1})^{-1}$$

is a well-defined and bounded linear operator from $H^{N_+}_{\sigma,R_\infty}$ to $H^{N_+ +1}_{\sigma,R_\infty,2}$. Meromorphicity of $s \mapsto L_s^{-1}$ then follows from a direct application of the analytic Fredholm theorem, using that $K_{s,\lambda} \circ \mathfrak{L}_{s,\lambda}^{-1}$ is a compact operator from $H^{N_+}_{\sigma,R_\infty}$ to $H^{N_+}_{\sigma,R_\infty}$. At this step, we also use that $(\mathfrak{L}_{s,\lambda} + K_{s,\lambda})^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ is well-defined and bounded for $\operatorname{Re} s > s_0$ if s_0 is sufficiently large to conclude that $s \mapsto L_s^{-1}$ is holomorphic when $\operatorname{Re} s > s_0$.

Coordinate choices. In order to guarantee the splitting of L_s in (4.1), we do not work in ingoing Kerr coordinates $(v, r, \theta, \varphi_*)$; see §6 for more details on these coordinates. Instead, we modify r and θ in the region $r \geq R_\infty$, with $R_\infty \gg M$ to obtain a new radial function ϱ and a new angular function ϑ. This change is motivated by the fact that in the case $M = 0$, the Kerr spacetime in Boyer–Lindquist coordinates (t, r, θ, φ) is isometric to the Minkowski spacetime in non-standard coordinates, where the surfaces of constant t and r are ellipsoids with eccentricity parameter a^2. The functions ϱ and ϑ are chosen so that in the $M = 0$ case the surfaces of constant t and ϱ are round spheres. In other words, we choose coordinates so that as our radial coordinate goes to infinity, the Kerr metric approaches the Minkowski metric in spherical coordinates in a suitably fast way.[5]

In practice, it is often more convenient to work with the inverted coordinate $x = \frac{1}{\varrho}$, which has a finite range on the spacetime.

In the $\Lambda > 0$ case, we consider a related change of coordinates. In the $M = 0$ case, these correspond to standard coordinates on a static patch of de Sitter. We refer to §18 for the precise coordinate transformation.

Inverting $\mathfrak{L}_{s,\lambda}$. By the discussion above, property (i) follows from invertibility of $\mathfrak{L}_{s,\lambda}$ for $s \in \Omega_{\sigma,N_+}$. Rather than establishing this directly, we first *regularize* the operator at infinity by considering

$$\mathfrak{L}_{s,\lambda,\kappa} := \kappa \partial_x(\chi_{\varrho \geq R_\infty} x \partial_x(\cdot)) + \mathfrak{L}_{s,\lambda},$$

with $\kappa > 0$ and $\chi_{\varrho \geq R_\infty}$ a smooth cut-off function that is equal to 1 in $\varrho \geq R_\infty$. The inclusion of the κ term can be interpreted in two ways:

(1) (A geometric perspective) Consider the Kerr-de Sitter metric $g_{M,a,\Lambda}$, which features an additional parameter $\Lambda > 0$, such that $g_{M,a} = g_{M,a,0}$, and consider the conformally invariant

[5]The coordinate transformation $(r, \theta) \mapsto (\varrho, \vartheta)$ in the $M = 0$ setting also plays an important role in the strategy for constructing double null coordinates on Kerr spacetime by Pretorius–Israel in [PI98]. In [PI98], however, the choice of (ρ, ϑ) is M-dependent, so that the spheres of constant t and ρ form the intersection of the null hypersurfaces of a double null foliation. In the present paper, it suffices to take (ρ, ϑ) to agree with the $M = 0$ expressions for suitably large values of r.

wave operator $\Box_{g_{M,a}} - \frac{2\Lambda}{3}\mathbf{1}$. The level set $\{x = 0\}$ then corresponds to the future cosmological horizon $\mathcal{C}^+$. The consideration of $\Lambda > 0$ leads to the following perturbation of L_s:

$$2\kappa_c \partial_x(\chi_{\varrho \geq R_\infty} x \partial_x(\cdot)) + \dots$$

with $\kappa_c > 0$ the surface gravity of $\mathcal{C}^+$ and with $\dots$ denoting terms that can be treated as small perturbations. Hence, the κ-term in $\mathfrak{L}_{s,\lambda,\kappa}$ *effectively* adds a non-zero surface gravity to future null infinity $\mathcal{I}^+$ (interpreted as a null hypersurface in a conformal extension of the Kerr spacetime). See §18 for details.

(2) (An ODE perspective) In the region $\varrho \geq R_\infty$, $\mathfrak{L}_{s,\lambda,\kappa}$ takes the following form when restricted to *spherical harmonic modes* $\widehat{\psi}_\ell$, which are projections of $\widehat{\psi}$ to the spherical harmonics $Y_{\ell m}$ with angular frequency ℓ:

$$(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell = \partial_x(x(\kappa + x)\partial_x\widehat{\psi}_\ell) + 2s\partial_x\widehat{\psi}_\ell - \mathfrak{l}(\mathfrak{l}+1)\widehat{\psi}_\ell,$$

where $\mathfrak{l} = \ell + l$ for $\ell \leq l$ and $\mathfrak{l} = \ell$ for $\ell > l$. The ODE $(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell = f_\ell$ has a regular singular point at $x = 0$ if $\operatorname{Re} s < 0$ and $\kappa > 0$, but the singular point becomes irregular as $\kappa = 0$. Hence, κ has a regularizing effect on the ODE.

In contrast, $\varrho = r_+$ (the future event horizon $\mathcal{H}^+$) is always a regular singular point, since the surface gravity of the event horizon κ_+ is always positive on sub-extremal Kerr spacetimes.

The advantage of considering $\mathfrak{L}_{s,\lambda,\kappa}$ is that it is possible to obtain estimates for $\mathfrak{L}^{-1}_{s,\lambda,\kappa}$ involving function spaces with only a finite number of derivatives. Using the existence of $\mathfrak{L}^{-1}_{s,\lambda,\kappa}$ as an operator on appropriate Sobolev spaces, restricted to finite-m azimuthal modes, as derived in [War15], it is possible to show that for $\operatorname{Re} s > -\frac{\kappa}{2}(1 + 2N_\kappa)$,

$$\mathfrak{L}^{-1}_{s,\lambda,\kappa} : H^{N_\kappa}(\widehat{\Sigma}) \to H^{N_\kappa+1}(\widehat{\Sigma}),$$

is well-defined and bounded, where $N_\kappa > 0$ is suitably large depending on κ. In particular, $\|\mathfrak{L}^{-1}_{s,\lambda,\kappa}\|_{H^{N_\kappa}(\widehat{\Sigma}) \to H^{N_\kappa+1}(\widehat{\Sigma})}$ is finite and depends on κ.

The core of part (i) is to show that for $s \in \Omega_{\sigma,N_+}$, we can in fact estimate uniformly in κ:

$$\|\mathfrak{L}^{-1}_{s,\lambda,\kappa} f\|_{H^{N_+ +1}_{\sigma,R_\infty,2}} \leq C_{\lambda,\sigma,R_\infty}\|f\|_{H^{N_+}_{\sigma,R_\infty}}. \tag{4.2}$$

Then $\mathfrak{L}^{-1}_{s,\lambda}$ can be constructed as the following limit:

$$\mathfrak{L}^{-1}_{s,\lambda} := \lim_{\kappa \downarrow 0} \mathfrak{L}^{-1}_{s,\lambda,\kappa},$$

with respect to the operator norm.

Most of the analysis in the paper concerns the uniform-in-κ estimate (4.2).

Nature of the relevant norms. The $H^{N_+}_{\sigma,R_\infty}$ norm can roughly be decomposed into four parts, which each have a different character: for $R_\infty \gg M$, with $\nabla\!\!\!/_{\mathbb{S}^2}$ denoting the covariant derivative on the unit round sphere $\mathbb{S}^2$ with respect to the angular coordinates (ϑ, φ_*) and $\Delta\!\!\!/_{\mathbb{S}^2}$ the corresponding angular Laplacian we have:

(a) **Sobolev norms in bounded regions**: $\|\cdot\|_{H^{N_+}(\Sigma \cap \{\varrho \leq 3R_\infty\})} + \|\nabla\!\!\!/_{\mathbb{S}^2}(\cdot)\|_{H^{N_+}(\Sigma \cap \{\varrho \leq \frac{R_\infty}{3}\})}$,

(b) Sobolev norms in far-away, but bounded regions that are **microlocal in the angular directions**: $\|(\mathbf{1} - \Delta\!\!\!/_{\mathbb{S}^2})^{-\frac{w_p}{4}}(\cdot)\|_{H^{N_+}(\Sigma \cap \{3R_\infty \leq \varrho \leq 4R_\infty\})}$, with w_p a smooth cut-off function such that $w_p(\varrho) = 0$ for $\varrho \leq 3R_\infty$ and $w_p(\rho) = p$ for $\varrho \geq 4R_\infty$, where $p > 0$ is a suitably large constant.

(c) **1-Gevrey-type boundary norms** at $\varrho = 2R_\infty$ with infinitely many derivatives:

$$\|\cdot\|^2_{B_{R_\infty}} = \sum_{\ell,n\in\mathbb{N}_0} \frac{c^n}{\max\{n+1,\ell+1\}^{2n}} \|(\varrho^2\partial_\varrho)^n(\cdot)|_{\varrho=2R_\infty}\|^2_{L^2(\mathbb{S}^2)},$$

for some constant $c > 0$.

(d) **L^2-based 2-Gevrey-type norms**: for lower radial derivatives

$$\sum_{\ell\in\mathbb{N}_0} \sum_{n=0}^{\max\{\ell-1,0\}} \frac{(4\sigma)^{2n}(\ell-(n+1))!^2}{(\ell+n+1)!^2} \int_{2R_\infty}^\infty \|(\varrho^2\partial_\varrho)^n(\cdot)\|^2_{L^2(\mathbb{S}^2)}(x)\, \varrho^2 d\varrho$$

and for higher radial derivatives

$$\sum_{\ell\in\mathbb{N}_0} (\ell+1)^3 \sum_{n=\ell}^\infty \frac{\sigma^{2n}}{n!^2(n+1)!^2} \int_{2R_\infty}^\infty \|(\varrho^2\partial_\varrho)^n(\cdot)\|^2_{L^2(\mathbb{S}^2)}\, \varrho^2 d\varrho.$$

Note that (c) and (d) involve so-called "Gevrey-type norms" in $0 \le x \le x_0$, with $x_0 = \frac{1}{2R_\infty}$. For a function $f : [0, x_0] \times \mathbb{S}^2 \to \mathbb{C}$, we define the L^2-based (α, b)-Gevrey norm as follows:

$$\sqrt{\sum_{n\in\mathbb{N}_0} \sum_{n_1+n_2=n} \frac{b^{2n}}{n!^{2\alpha}} \int_0^{x_0} \|\nabla\!\!\!/^{n_1}_{\mathbb{S}^2} \partial_x^{n_2} f\|^2_{L^2(\mathbb{S}^2)}\, dx}$$

$$\sim \sqrt{\sum_{\ell\in\mathbb{N}_0} \sum_{n\in\mathbb{N}_0} \sum_{m\in\mathbb{N}_0} \frac{(e\cdot b)^{2(n+m)}(\ell(\ell+1))^m}{(n+m)^{2\alpha(n+m+1)}} \int_0^{x_0} \|\partial_x^n f_\ell\|^2_{L^2(\mathbb{S}^2)}\, dx},$$

where the equivalence of norms follows from the orthogonality of the spherical harmonic modes f_ℓ of f and from an application of Stirling's approximation formula: $n! \sim n^{n+1}e^{-n}$.

Similarly, we consider L^∞-based (α, b)-Gevrey norms:

$$\sqrt{\sum_{n\in\mathbb{N}_0} \sum_{n_1+n_2=n} \frac{b^{2n}}{n!^{2\alpha}} \|\nabla\!\!\!/^{n_1}_{\mathbb{S}^2} \partial_x^{n_2} f\|^2_{L^\infty([0,x_0]\times\mathbb{S}^2)}}$$

$$\sim \sqrt{\sum_{\ell\in\mathbb{N}_0} \sum_{n\in\mathbb{N}_0} \sum_{m\in\mathbb{N}_0} \frac{(e\cdot b)^{2(n+m)}(\ell(\ell+1))^m}{(n+m)^{2\alpha(n+m+1)}} \|\partial_x^n f_\ell\|^2_{L^\infty([0,x_0]\times\mathbb{S}^2)}}$$

Note that the space of $(1, b)$-Gevrey norms corresponds to the space of analytic functions with analyticity radius depending on the exponent b.

It is straightforward to show that the boundary norm in (iii) can be bounded above by L^∞-based $(1, b)$-Gevrey norms restricted to an arbitrarily small neighbourhood of $x = x_0$ for a suitable exponent b.

In (d), the $n \ge \ell$ part of the norm can immediately be bounded above by an L^2-based $(2, b)$-Gevrey norm with suitable exponent b. The same holds for the $n \le \ell - 1$ part of the norm, but this is less straightforward. Suppose $n \le (1-\epsilon)(\ell-1)$, for some $0 < \epsilon < 1$. Without loss of generality, we can take $\ell \ge 1$. Then, by Stirling's approximation formula:

$$\frac{(\ell-(n+1))!^2}{(\ell+(n+1))!^2} \lesssim e^{4(n+1)}\epsilon^{2(\ell-(n+1)+1)} \frac{\ell^{2(\ell-(n+1)+1)}}{(2\ell+1)^{2(\ell+(n+1)+1)}} \le e^{4(n+1)}\ell^{-4(n+1)} \le e^{4(n+1)}(n+1)^{-4(n+1)}.$$

Hence, the $n \le (1-\epsilon)(\ell-1)$ part of the form can also be bounded by an L^2-based $(2, b)$-Gevrey norm with suitable exponent b. Finally, consider the part where $(1-\epsilon)(\ell-1) \le n \le \ell-1$, where we

may assume without loss of generality that $n \geq 1$. Then we apply Stirling's approximation formula again to estimate:

$$\frac{(\ell-(n+1))!^2}{(\ell+(n+1))!^2} \lesssim \frac{1}{(2\ell)!^2} \lesssim e^{-2\ell}(2\ell)^{-4\ell-2} \leq e^{-2n}2^{-4n-2}n^{-4n-2}.$$

Deriving (4.2). The different norms in (a)–(d) arise from different types of fixed-frequency analogues of energy estimates and elliptic estimates in different ranges of the radial coordinate ϱ, which are indicated in Figure 2.

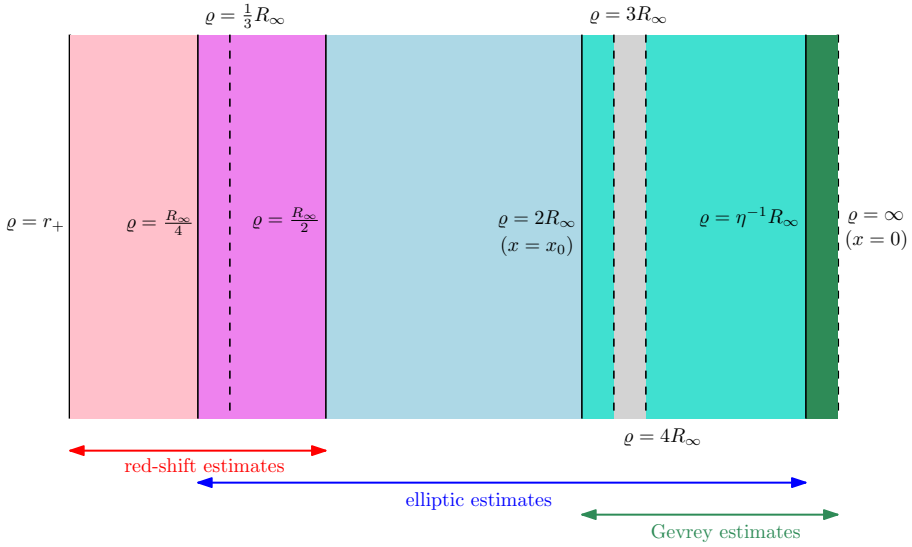


FIGURE 2. The spacetime regions labelled by the radial function ϱ where the different types of estimates are applied.

These are:

- **Red-shift estimates**: We consider the integral of:

$$\mathrm{Re}(\varrho^\delta \overline{\partial_\varrho(\varrho^{-1}\widehat{\psi})} \cdot \mathfrak{L}_{s,\kappa}\widehat{\psi})$$

  for $\delta > 0$ suitably small and integrate by parts. These estimates rely crucially on the positivity of κ_+, the surface gravity of the Kerr event horizon, and they may be thought of as zero-frequency analogues of the physical space red-shift estimates of [DR09]. In contrast with their physical space analogues, it is important that these estimates are valid for large values of ϱ and not just in a neighbourhood of $\varrho = r_+$, with r_+ the radius of the event horizon. The *ergoregion* of the black hole, where the vector field ∂_τ fails to be timelike, is entirely included in the region where we apply red-shift estimates. With these estimates, we gain control of <u>one</u> extra derivative of $\widehat{\psi}$ compared to $\mathfrak{L}_{s,\kappa}\widehat{\psi}$.
- **Elliptic estimates**: These estimates concern a far-away but bounded region, excluding the ergoregion, where ∂_τ is timelike and the operator $\mathfrak{L}_{s,\lambda,\kappa}$ is therefore elliptic. As a result, we can consider elliptic estimates obtained by splitting

$$\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi} = \text{ second order derivatives} + \mathfrak{R}_{s,\lambda}$$

  and integrating the weighted square

$$x^{1-\delta}(\mathfrak{l}+1)^{-w_p}|\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi} - \mathfrak{R}_{s,\lambda}\widehat{\psi}|^2.$$

  For suitably large λ, we can arrive at an elliptic estimate by integrating by parts, with boundary terms on the right-hand side. With these estimates, we gain control of <u>two</u> extra

derivatives of $\widehat{\psi}$ compared to $\mathfrak{L}_{s,\kappa}\widehat{\psi}$. The red-shift and elliptic estimates combined provide control of the norms in (a) and (b).

- **Higher-order gevrey estimates**: These estimates take place in the region $\varrho \geq 2R_\infty$ where $\mathfrak{L}_{s,\lambda,\kappa}$ takes the simple form:

$$(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell + \mathfrak{l}(\mathfrak{l}+1)\widehat{\psi}_\ell = \partial_x(x(\kappa+x)\partial_x\widehat{\psi}_\ell) + 2s\partial_x\widehat{\psi}_\ell.$$

Ignoring the κ term, this is simply the Laplace transform of $\varrho^3\Box(\varrho^{-1}\cdot)$ with $x = \frac{1}{\varrho}$ and $\Box$ the standard wave operator with respect to the Minkowski spacetime and standard spherical coordinates $(\varrho, \vartheta, \varphi_*)$, where τ-level sets are outgoing null hypersurfaces.

The above equation can easily be commuted with $\partial_x = -\varrho^2\partial_\varrho$ and we integrate:

$$\left|[\mathfrak{l}(\mathfrak{l}+1) - n(n+1)]\partial_x^n\widehat{\psi} + \partial_x^n(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell\right|^2 \tag{4.3}$$

to obtain weighted L^2-estimates for derivatives (up to second order) of $\widehat{\phi}$. In order for these type of estimates to close, we sum them up with the following weights:

$$\sum_{n=\mathfrak{l}}^{N_\infty} \frac{|2s|^2\tilde{\sigma}^{2n}}{n!^2(n+1)!^2},$$

with $\tilde{\sigma}^2 \in (0,\infty)$ restricted to an appropriate range.

Here it is important to note that for $n = \mathfrak{l}$, the following equation holds:

$$\partial_x(x(\kappa+x)\partial_x^{\mathfrak{l}+1}\widehat{\psi}_\ell) + 2\left[s + (n+1)\left(x + \frac{\kappa}{2}\right)\right]\partial_x^{\mathfrak{l}+1}\widehat{\psi}_\ell = (\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell.$$

When $\kappa = 0$, the physical space analogue of the above equation is:

$$\partial_x(x^2\partial_x^{\mathfrak{l}+1}\psi_\ell) + 2\left[\partial_\tau + (n+1)x\right]\partial_x^{\mathfrak{l}+1}\psi_\ell = 0 \tag{4.4}$$

with $\psi = \varrho\phi$. Evaluating the above expression at $x = 0$ gives $\partial_\tau\partial_x^{\mathfrak{l}+1}\psi_\ell|_{\mathcal{I}^+} = 0$, which is a conservation law that was first pointed out by Newman and Penrose [NP68]. *The existence of this conservation law in Minkowski is directly related to the fact that the Gevrey estimates close when summing over* $n \geq \mathfrak{l}$. With these estimates, we gain control of <u>two</u> extra derivatives of $\widehat{\psi}$ compared to $\mathfrak{L}_{s,\kappa}\widehat{\psi}$. These estimates are the origin of the restriction to the sector $\{\arg z < \varphi_0\}$.

- **Lower-order gevrey estimates**: It remains to control an appropriately weighted sum terms with $n < \mathfrak{l}$. By summing elliptic-type estimates in the region $\varrho \geq 2R_\infty$, we obtain control of lower-order derivatives in terms of the higher-order derivatives (which we already control independently!) and boundary terms at $\varrho = 2R_\infty$. To control the boundary terms, we simply use the equation (4.3) to express an appropriately weighted sum of higher-order derivatives $\partial_\varrho^{k+2}\widehat{\psi}$ in terms of $\widehat{\psi}$ and $\partial_\varrho\widehat{\psi}$. In this step, we obtain, in particular, control over the boundary norm in (c) above and we can use this to control the remaining lower-order (in n) Gevrey norms in (d). With these estimates, we gain control of <u>two</u> extra derivatives of $\widehat{\psi}$ compared to $\mathfrak{L}_{s,\kappa}\widehat{\psi}$.

The estimates above that take place in different ranges of ϱ are all *coupled*, apart from the higher-order Gevrey estimates, which can be closed independently of all the other estimates. To enable a coupling of the estimates, we need to take λ suitably large compared to the other parameters and we need to choose the value p in the weight function w in the elliptic estimates appropriately.

Now consider the more general operator $L_{s,\Lambda}$, with $\Lambda \geq 0$. Let $\kappa_c \geq 0$ be the surface gravity associated to the cosmological horizon $\mathcal{C}^+$. Note that $\kappa_c \lesssim \sqrt{\Lambda}$. Then we show that we can decompose $L_{s,\Lambda}$ as follows:

$$L_{s,\Lambda} = \mathfrak{L}_{s,\lambda,2\kappa_c} + \epsilon_0 \cdot B_{s,\Lambda} + K_{s,\lambda,\Lambda} + \kappa_c^2 B_{s,\Lambda}^{\mathcal{C}^+}.$$

The operator $\mathfrak{L}_{s,\lambda,2\kappa_c}$ closely resembles the κ-regularized operator $\mathfrak{L}_{s,\lambda,\kappa}$, with $\kappa = 2\kappa_c$, which plays a key role in the proof of Theorem 4.1(i) for $\Lambda = 0$, with only minor differences in the region $r \leq \frac{R_\infty}{2}$ that do not affect the relevant estimates. Boundedness of the operator $\kappa_c B^{\mathcal{C}^+}_{s,\Lambda} \circ \mathfrak{L}^{-1}_{s,\lambda,2\kappa_c} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ follows by applying the following κ_c-dependent generalization of (4.2):

$$\|\mathfrak{L}^{-1}_{s,\lambda,\kappa_c} f\|_{H^{N_++1}_{\sigma,R_\infty,2}} + \kappa_c \|x\partial_x^2 \mathfrak{L}^{-1}_{s,\lambda,2\kappa_c}(f)\|_{G_{\sigma,R_\infty}} + \kappa_c \|\not{\Delta}_{\mathbb{S}^2} \mathfrak{L}^{-1}_{s,\lambda,2\kappa_c}(f)\|_{G_{\sigma,R_\infty}} \leq C_{\lambda,\sigma,R_\infty} \|f\|_{H^{N_+}_{\sigma,R_\infty}} \tag{4.5}$$

and exploiting the control over additional second-order derivatives on the left-hand side.

Hence, for $M\kappa_c \ll 1$, $\kappa_c^2 B^{\mathcal{C}^+}_{s,\Lambda} \circ \mathfrak{L}^{-1}_{s,\lambda,2\kappa_c}$ can be interpreted as a small bounded operator, just like $\epsilon_0 \cdot B_{s,\Lambda} \circ \mathfrak{L}^{-1}_{s,\lambda,2\kappa_c}$. The operator $K_{s,\lambda,\Lambda} \circ \mathfrak{L}^{-1}_{s,\lambda,2\kappa_c}$, on the other hand, is shown to be compact, as in the $\Lambda = 0$ case.

Sketch of the proof of Theorem 4.1(ii). We would like to take the union of all s for which $\ker L_s \neq \{0\}$ with L_s restricted to a subspace of $H^{N_+}_{\sigma,R_\infty}$. A priori, it is not clear how the non-triviality of the kernel of L_s depends on the parameters σ, N_+ and R_∞. We show that the kernel of restrictions of L_s to $H^{N_+}_{\sigma,R_\infty}$ is *independent* of the choice of parameters. The underlying reason is that we can use the estimates for $\mathfrak{L}^{-1}_{s,\lambda,\kappa}$ to further improve the regularity of elements of $\ker L_s$. In particular, we conclude in this step that elements of the $\ker L_s$ are smooth on Σ.

Here, we moreover use that any element of $\ker L_s$ can be approximated by elements of the kernel of the regularized operator $L_{s,\kappa} = L_s + \partial_x(x(\kappa + x)\chi_{\varrho \geq R_\infty} \partial_x \widehat{\psi})$ with $\kappa > 0$, by an application of Hurwitz's theorem of complex analysis. This is very closely related to the quasinormal frequency convergence property in Theorem 4.2(ii).

Sketch of the proof of Theorem 4.1(iii). Note that of the norms in (a)–(d) in the sketch of the proof of (i), only (a) is relevant when restricting to functions supported in $\{\varrho \leq \frac{1}{2}R_\infty\}$ and R_∞ can be chosen arbitrarily large. In particular, the following operator can easily shown to be well-defined for $s \in \Omega \setminus \mathscr{Q}_{\mathrm{reg}}$:

$$L_s^{-1} : C_c^\infty(\Sigma) \to C^\infty(\Sigma).$$

Consider $\widehat{\phi} : \Sigma' \to \mathbb{C}$. Then we define:

$$A(\omega)\hat{\phi} = e^{i\omega t} \Box_{g_{M,a}}(e^{-i\omega t}\hat{\phi}).$$

The inverse $R(\omega) = A(\omega)^{-1}$ on appropriate function spaces is called the *resolvent operator* associated the the wave operator $\Box_{g_{M,a}}$ and the Boyer–Lindquist time function t.

Let Q_s be the map that takes the trivial extension of a function on Σ' to the rest of the spacetime and restricts it to the hypersurface $\Sigma = \{\tau = 0\}$. Then we can relate $A(\omega)$ to L_s in the following way:

$$A(\omega) = Q_s^{-1} \circ (\varrho^{-3} L_s(\varrho \cdot)) \circ Q_s$$

with $s = -i\omega$, so the cut-off resolvent, defined as follows: $\chi R(\omega)\chi$ for arbitrary $\chi \in C_c^\infty(\Sigma')$, satisfies

$$\chi R(\omega)\chi = (\chi Q_s^{-1}) \circ (\varrho^{-1} L_s(\varrho^3 \cdot))^{-1} \circ (Q_s \chi).$$

Since χQ_s^{-1} and $Q_s\chi$ define a bounded linear operator on Sobolev spaces of arbitrary order, it follows that the right-hand side of the equation above is a well-defined map from $C^\infty(\Sigma)$ to $C^\infty(\Sigma)$ that is meromorphic in s for $s \in \Omega$ and hence the cut-off resolvent can be meromorphically continued to the subset $\Omega \subset \mathbb{C}$ with poles contained in $\mathscr{Q}_{\mathrm{reg}}$.

Sketch of the proof of Theorem 4.2. Stability of the quasinormal spectrum follows from the following two observations: 1) Invertibility of L_s implies invertibility of $L_s + \delta\widetilde{Q}$ for $\delta\|\widetilde{Q}L_s^{-1}\| < 1$. Hence, small perturbations of L_s are invertible for s contained in compact subsets of Ω_{σ,N_+} that exclude arbitrarily small disks around the poles of $s \mapsto L_s^{-1}$. The smaller the disks, the larger $\|\widetilde{Q}L_s^{-1}\|$ can be, so δ needs to be taken appropriately small. 2) $\ker L_s$ can be approximated by elements of $\ker(L_s+\delta\widetilde{Q})$ (by an application of Hurwitz's theorem), so arbitrarily small disks around the poles of $s \mapsto L_s^{-1}$ contain poles of $s \mapsto (L_s + \delta\widetilde{Q})^{-1}$ for suitably small $\delta > 0$.

Theorem 4.2(ii) follows from a similar application of Hurwitz's theorem, with κ_c taking the role of the smallness parameter δ and the factor that the estimates for $L_{s,\Lambda}^{-1}$ are uniform in Λ.

5. Extensions and further applications

We describe below future directions of research related to the results and methods derived in the present paper.

5.1. **General asymptotically flat spacetimes and wave equations.** The methods developed in the present paper can be applied to more general settings of wave operators on asymptotically flat spacetimes. They rely on the following key assumptions, which we will only sketch: Let $\Sigma_\tau \cong \mathbb{R}\times\mathbb{S}^2$ or $\Sigma_\tau \cong \mathbb{R}^3$ be isometric asymptotically hyperboloidal hypersurfaces that foliate a stationary, asymptotically flat spacetime. Let $\mathcal{L}_s$ denote the Laplace transform of a wave operator with all time derivatives replaced by multiplications with the frequency $s \in \mathbb{C}$. Let r be a radial-like coordinate.

(1) There exists coordinates on $\Sigma = \Sigma_0$ with respect to which the coefficients of $\mathcal{L}_s$ approach the coefficients of the Laplace transform of the wave operator on Minkowski, with respect to an asymptotically hyperboloidal time function. Furthermore, these coefficients are suitably regular (in a Gevrey sense, after suitable rescaling) at infinity.

(2) The zero-frequency operator $\mathcal{L}_0$ satisfies the following estimate: there exists $\beta \in \mathbb{R}$, such that for all $k \in \mathbb{N}_0$

$$\frac{1}{C_k}\|r^{-1+\delta}\widehat{\phi}\|^2_{H^{k+1}(\Sigma)} + (k-\beta)\|r^{-1+\delta}\partial_r^{k+1}\widehat{\phi}\|^2_{L^2(\Sigma)} \leq \|r^{-1+\delta}\mathcal{L}_0\widehat{\phi}\|^2_{H^k(\Sigma)} + \|r^{-1+\delta}\widehat{\phi}\|^2_{H^k(\Sigma)},$$

where $C_k > 0$ are uniform constants. The above estimate may be thought of as a "zero-frequency" resolvent estimate, incorporating moreover the presence of an "enhanced red shift effect" for higher radial derivatives due to the growing factor k on the left-hand side.

One could state the above assumptions precisely and incorporate them into the arguments of the present paper as a "black box" to extend the validity of the results to a much more general setting. We do not carry out such a procedure in the present paper.

Analogues of (2) also play an important role in the derivation of sharp decay estimates and late-time (inverse-polynomial) asymptotics for wave equations on asymptotically flat spacetimes; see for example [Tat13, AAG18b, Hin22, AAG23a, LO24].

In the case of sub-extremal Reissner–Nordström spacetimes (which include Schwarzschild spacetimes), (2) can be derived by exploiting the (degenerate) ellipticity of $\mathcal{L}_0$ and combining it with red-shift estimates in a neighbourhood of the event horizon. As already mentioned in §4, this argument does not work on sub-extremal Kerr, due to the presence of an ergoregion. Nevertheless, as shown in the present paper, the estimate in (2) does hold. Note also that the necessary convergence to Minkowski in (1) does not hold with respect to Boyer–Lindquist coordinates (r,θ,φ), but does hold with respect to modified coordinates $(\varrho,\vartheta,\varphi)$.

The Teukolsky wave equations (of arbitrary spin) on sub-extremal Kerr can be shown to satisfy (2). Property (1) can similarly be shown to hold by considering as the main variables appropriate ϱ-rescalings and $\varrho^2\partial_\varrho$-derivatives of the Teukolsky variables, which are closely related to the conserved Newman–Penrose charges [NP68] for the Teukolsky equations on Minkowski; see for example the expressions in [GK22][§3].

5.2. **Higher spacetime dimensions.** The methods in the present paper extend also to wave equations on stationary asymptotically flat spacetimes with spacetime dimensions $n+1$, where $n \geq 3$. Important here is that the wave equation on the $n+1$-dimensional Minkowski spacetime is equivalent to the following equation with respect to standard spherical coordinates: let $\psi = r^{\frac{n-1}{2}}\phi$, then:

$$0 = -\partial_t^2\psi + \partial_r^2\psi - \frac{1}{4r^2}(n-1)(n-3)\psi + r^{-2}\not{\Delta}_{\mathbb{S}^{n-1}}\psi.$$

Using that higher-dimensional spherical harmonics $Y_{\ell m}$ satisfy $\not{\Delta}_{\mathbb{S}^{n-1}}Y_{\ell m} = -\ell(\ell+n-2)Y_{\ell m}$, this results in the following equation for ψ_ℓ, the projection of ψ onto the spherical harmonics with angular frequency ℓ:

$$0 = -\partial_t^2\psi_\ell + \partial_r^2\psi_\ell - \frac{1}{4r^2}[4\ell(\ell+n-2) + (n-1)(n-3)]\psi_\ell. \tag{5.1}$$

When $n = 2k+3$, $k \in \mathbb{N}_0$, (5.1) is equivalent to:

$$0 = -\partial_t^2\psi_\ell + \partial_r^2\psi_\ell - \frac{1}{r^2}(\ell+k)(\ell+k+1)\psi_\ell.$$

each ψ_ℓ satisfies a conservation law after taking appropriately many outgoing null derivatives, analogous to (4.4). This is directly connected to the validity of the strong Huygens principle. The higher-order Gevrey estimates, which rely on this conservation law, therefore proceed analogously to the $n = 3$ case.[6]

5.3. **Extending the sector of the complex plane.** The construction of regularity quasinormal mode frequencies in the present paper applies only to the sector

$$\Omega = \{|\arg z| < \varphi_0 \approx 0.712\pi\} \subset \mathbb{C}.$$

In contrast, scattering resonances, $is \in \mathbb{C}$, are well-defined in the case of fixed spherical harmonics on Schwarzschild for $s \notin (-\infty, 0]$ by [BMB93]; see also the work [Stu24] for an analogous statement on sub-extremal Kerr.

The exclusion of a sector around the negative real axis in the present paper is, however, of a different nature than the exclusion that arises when considering cut-off resolvents as in [BMB93, Stu24]. Indeed, in contrast with the case of cut-off resolvents, an exclusion of a neighbourhood of the real axis is necessary when considering regularity quasinormal frequencies, even when considering the wave equation on the Minkowski spacetime where the cut-off resolvent is in fact holomorphic in the whole complex plane.[7]

We illustrate this as follows. The characterization of regularity quasinormal frequencies in the present paper can be used to obtain a uniform understanding of quasinormal frequencies when considering wave equations on families of spacetimes depending on $\Lambda \geq 0$. In particular, we obtain

[6]When $n = 2k+2$, $k \in \mathbb{N}_0$, the conservation law breaks down (and the strong Huygens principle does not hold). However, in that case, using that the relevant wave operator in the far-away region $\{\varrho \geq 2R_\infty\}$ is just the Minkowski operator, one can apply the method of descent, by considering as an auxiliary operator, the Minkowski wave operator in $n+2$ spacetime dimensions, restricted to functions that do not depend on the Cartesian coordinates x_{n+1}. Then estimates can be carried out in the far-away region for the rescaled functions $\tilde{r}^{\frac{n}{2}}\phi$, with $\tilde{r}^2 = r^2 + x_{n+1}^2$ by considering the Laplace transform of the $(n+2)$-dimensional wave operator, with respect to the time function $\tilde{\tau} = t - \tilde{r}$. This can be used to prove the existence of L_s^{-1} on appropriate Hilbert spaces, away from a subset of isolated $s \in \mathbb{C}$.

[7]Note that the condition $|\arg z| < \varphi_0$ purely arises at the level of estimates for the exact Minkowski wave operator in the far-away region $r \geq 2R_\infty$.

uniform convergence of regularity quasinormal frequencies in the limit $\Lambda \downarrow 0$ in the sector where regularity quasinormal frequencies are well-defined (Theorem 1.2(ii)). In the case of the wave equation on Minkowski, this implies that regularity quasinormal frequencies corresponding to the conformally invariant wave equation on the Sitter must converge to regularity quasinormal frequencies on Minkowski. Regularity quasinormal frequencies in de Sitter take the form $s_n = -n\sqrt{\frac{\Lambda}{3}}$; see [Joy22, HX22b, Hin25]. This implies, in particular, the absence of regularity quasinormal frequencies in Ω, away from $\mathbb{R}_{\leq 0}$. Furthermore, since the de Sitter quasinormal frequencies accumulate at every point on the negative real axis as $\Lambda \downarrow 0$, this also implies that the largest sector for which uniform convergence of quasinormal frequencies could possibly hold must necessarily exclude $\mathbb{R}_{\leq 0}$.

It remains an open problem whether the excluded sector around $\mathbb{R}_{\leq 0}$ can be made *arbitrarily* small in the setting of regularity quasinormal modes. Since the methods in the present paper are robust to the consideration of more general operators than the operators $L_{s,\Lambda}$ and L_s on Kerr–de Sitter and Kerr, it would be interesting to investigate the possible existence of operators with sequences of regularity quasinormal frequencies analogous to $\{s_n\}$ in de Sitter, but which accumulate somewhere away from $\mathbb{R}_{\leq 0}$, in $\mathbb{C} \setminus \Omega$, as $\Lambda \downarrow 0$, in order to obtain an upper bound on the angle appearing in the definition of Ω.

5.4. **Extremal black hole spacetimes.** In the case of extremal black hole spacetimes, (2) does not hold, due to the absence of the horizon red-shift effect. However, modifications exist with additional degeneracies at the event horizon; see [AAG20, GW21, Gaj23]. For this reason, regularity quasinormal modes can still be defined on extremal Reissner–Nordström. We expect that the methods of the present paper can also be extended to extremal Kerr, provided the near-horizon region is treated in a similar way to the far-away region in the sub-extremal case. An important difference with the sub-extremal setting is that the frequencies $\omega = m\boldsymbol{\omega}_+$, with $\boldsymbol{\omega}_+$ the angular velocity of the black hole and $m \in \mathbb{Z}$ the azimuthal number, play the role of $\omega = 0$ in the far-away region. The region of the complex plane for which regularity quasinormal frequencies can be defined with the methods of the present paper would therefore have to exclude in addition sectors around the lines $s = (-\infty, 0] + im\boldsymbol{\omega}_+$ for all $m \in \mathbb{Z}$.

5.5. **Zero-damped frequencies and high frequencies.** With the results in the present paper and the methods developed in [HX22a, Joy22, Hin24], the properties of zero-damped quasinormal frequencies, which are expected to accumulate at $\omega = m\boldsymbol{\omega}_+$ with $m \neq 0$ in the extremal limit of sub-extremal Kerr can in principle be studied.

The behaviour of regularity quasinormal frequencies in sub-extremal Kerr for bounded $|\operatorname{Re} s|$ (or bounded $|\operatorname{Im} \omega|$) and very large $|\operatorname{Im} s|$ (or very large $|\operatorname{Re} \omega|$), which is closely related to the structure of the trapping of null geodesics in the spacetime, would also be an interesting topic for further research in a uniform-in-Λ setting. See [BZ97, Pra99, Dya12, HZ24] for related results in de asymptotically de Sitter and Schwarzschild (fixed spherical harmonic, scattering resonance) setting.

6. Preliminaries: geometry

We start by defining the main geometric objects related to Kerr spacetimes. We will briefly discuss relevant geometric aspects of Kerr–de Sitter spacetimes in §18.

6.1. **Coordinates and vector fields.** Let $(\mathcal{M}_{M,a}, g_{M,a})$ denote the the black hole exterior of a Kerr spacetime, including the future event horizon, with mass M and angular momentum/mass a. With respect to *ingoing Kerr coordinates* $(v, r, \theta, \varphi_*)$, we can express

$$\mathcal{M}_{M,a} = \mathbb{R}_v \times [r_+, \infty)_r \times \mathbb{S}^2_{\theta,\varphi},$$

where

$$\begin{aligned} g_{M,a} = &- \rho^{-2}\left(\Delta - a^2\sin^2\theta\right) dv^2 + 2dvdr - 4Mar\rho^{-2}\sin^2\theta dvd\varphi_* - 2a\sin^2\theta drd\varphi_* + \rho^2 d\theta^2 \\ &+ \rho^{-2}((r^2+a^2)^2 - a^2\Delta\sin^2\theta)\sin^2\theta d\varphi_*^2, \end{aligned} \tag{6.1}$$

with

$$\begin{aligned} \Delta &= r^2 - 2Mr + a^2 = (r_+ - r)(r - r_-), \\ r_{+,-} &= M \pm \sqrt{M^2 - a^2}, \\ \rho^2 &= r^2 + a^2\cos^2\theta. \end{aligned}$$

We obtain the *Boyer–Lindquist coordinates* $(t, r, \theta, \varphi_*)$ via the following relations:

$$\begin{aligned} t &= v - r_*, \\ \varphi &= \varphi_* + \int_r^\infty \frac{a}{\Delta}\, dr' \mod 2\pi, \end{aligned}$$

where $r_* : (r_+, \infty) \to \mathbb{R}$ is a solution to

$$\frac{dr_*}{dr} = \frac{r^2 + a^2}{\Delta},$$

defined uniquely up to a constant. Note that the Boyer–Lindquist coordinates are only well defined on $\mathcal{M}_{M,a} \setminus \{r = r_+\}$.

It will be convenient to state moreover the inverse metric $g_{M,a}^{-1}$:

$$\begin{aligned} g_{M,a}^{-1} = &\, a^2\rho^{-2}\sin^2\theta\partial_v \otimes \partial_v + \rho^{-2}(r^2+a^2)[\partial_v \otimes \partial_r + \partial_r \otimes \partial_v] + \Delta\rho^{-2}\partial_r \otimes \partial_r \\ &+ a\rho^{-2}[(\partial_v + \partial_r) \otimes \partial_{\varphi_*} + \partial_{\varphi_*} \otimes (\partial_v + \partial_r)] + \rho^{-2}[\partial_\theta \otimes \partial_\theta + \sin^{-2}\theta\partial_{\varphi_*} \otimes \partial_{\varphi_*}]. \end{aligned} \tag{6.2}$$

The *surface gravity* κ_+ associated to the future event horizon $\mathcal{H}^+ = \{r = r_+\}$ takes the following expression:

$$\kappa_+ = \frac{r_+ - M}{2Mr_+}.$$

We introduce the following vector fields, defined with respect to $(v, r, \theta, \varphi_*)$ coordinates:

$$\begin{aligned} Y &= \partial_r, \\ T &= \partial_v, \\ \Phi &= \partial_{\varphi_*}. \end{aligned}$$

We will now introduce the coordinate chart $(\tilde{v}, \varrho, \vartheta, \varphi_*)$ which we motivate by first considering the $M = 0$ case where the Kerr spacetime is locally isometric to the Minkowski spacetime and the coordinates $(v, r, \theta, \varphi_*)$ correspond to a foliation of the spacetime by axisymmetric ellipsoids with eccentricity parameter a^2.

In the $M = 0$ case, the transformation from (v, r, θ) to $(\tilde{v}, \varrho, \vartheta)$ corresponds to passing from ellipsoids with a non-zero eccentricity parameter to round spheres. More precisely, (r, θ) have the following relation to Cartesian coordinates $(\tilde{x}, \tilde{y}, \tilde{z})$ on Minkowski:

$$\begin{aligned} \tilde{z} &= r\cos\theta, \\ \tilde{x}^2 + \tilde{y}^2 &= (r^2 + a^2)\sin^2\theta. \end{aligned}$$

Let $(\varrho, \vartheta, \varphi)$ denote standard spherical coordinates on Minkowski, then we also have that

$$\begin{aligned} \tilde{z} &= \varrho\cos\vartheta, \\ \tilde{x}^2 + \tilde{y}^2 &= \varrho^2\sin^2\vartheta. \end{aligned}$$

From the above, it is straightforward to obtain the following relations in Minkowski:

$$\varrho^2 = r^2 + a^2 \sin^2\theta, \tag{6.3}$$

$$\sin^2\vartheta = \frac{r^2+a^2}{r^2+a^2\sin^2\theta}\sin^2\theta, \tag{6.4}$$

$$r^2 = \frac{1}{2}(\varrho^2 - a^2)\left(1 + \sqrt{1 + \frac{4a^2\varrho^2}{(\varrho^2-a^2)^2}\cos^2\vartheta}\right), \tag{6.5}$$

$$\sin^2\theta = \frac{1}{2a^2}\left(\varrho^2 + a^2 - \sqrt{(\varrho^2+a^2)^2 - 4a^2\varrho^2\sin^2\vartheta}\right). \tag{6.6}$$

Consider now the $M \neq 0$ case. In this case, we define $\varrho : [r_+,\infty)\times(0,\pi) \to [r_+,\infty)$ and $\vartheta : [r_+,\infty)\times(0,\pi)\to(0,\pi)$ as smooth functions satisfying:

$$\varrho(r,\theta) = \sqrt{r^2 + \check{a}^2\sin^2\theta},$$
$$(\sin\vartheta)(r,\theta) = \sqrt{\frac{r^2+\check{a}^2}{r^2+\check{a}^2\sin^2\theta}}\sin\theta,$$

with $\check{a} : [r_+,\infty)\to\mathbb{R}$ a smooth function with $|\check{a}| \leq |a|$, such that $\check{a}(r) = a$ for $r \geq \frac{1}{\sqrt{2}}R_\infty$ and $\check{a}(r) = 0$ for $r \leq \frac{1}{2}R_\infty$. Then $r \leq \varrho(r,\theta) \leq r\sqrt{1+a^2r^{-2}} \leq \sqrt{2}r$ and, for $r \leq \frac{1}{2}R_\infty$ we have that $\varrho = r$. Hence, in particular, we have that $\check{a} = a$ in the region $\{\varrho \leq \frac{1}{2}R_\infty\}$ and $\check{a} = 0$ in the region $\{\varrho \geq R_\infty\}$.

Note that for $|a|R_\infty^{-1}$ suitably small, $\begin{pmatrix}\partial_r\varrho & \partial_r\vartheta,\\ \partial_\theta\varrho & \partial_\theta\vartheta\end{pmatrix}$ is clearly invertible everywhere.

In particular, for $r \geq \frac{1}{\sqrt{2}}R_\infty$:

$$\partial_r\vartheta = \frac{a^2}{r(r^2+a^2)}\sin^2\vartheta\cos^2\vartheta,$$
$$\partial_r\varrho = \frac{r}{\varrho}.$$

Furthermore, it follows that we can write for all $r \geq r_+$:

$$r(\varrho,\vartheta) = \varrho - f_r(\varrho,\vartheta),$$
$$\theta(\varrho,\vartheta) = \vartheta - f_\theta(\varrho,\vartheta),$$

where f_r and f_θ are smooth everywhere, analytic in $r \geq \frac{1}{\sqrt{2}}R_\infty$ with respect to the variable $\frac{1}{r}$ and with:

$$f_r(\varrho,\vartheta) = a^2 O_\infty\left(\frac{1}{\varrho}\right),$$
$$f_\theta(\varrho,\vartheta) = a^2 O_\infty\left(\frac{1}{\varrho^2}\right).$$

In particular,

$$\partial_r\varrho = \frac{1}{2\varrho}\partial_r(\varrho^2) = \varrho^{-1}(r + \check{a}\partial_r\check{a}\sin^2\theta) = 1 + a^2\sin^2\vartheta O_\infty(\varrho^{-2}), \tag{6.7}$$

$$\partial_r\vartheta = \frac{\partial_r(\sin^2\vartheta)}{2\sin\vartheta\cos\vartheta} = \frac{\partial_r(\check{a}^2(r^2+\check{a}^2\sin^2\theta)^{-1})\sin^2\theta\cos^2\theta}{2\sin\vartheta\cos\vartheta} = a^2 O_\infty(\varrho^{-3})\sin\vartheta\cos\vartheta \tag{6.8}$$

We define $\tilde{v} = t + r_*(\varrho)$ so that $\tilde{v} = v + r_*(\varrho) - r_*(r)$. Then $\tilde{v} = v$ in $r \leq \frac{R_\infty}{2}$.

6.2. **Asymptotically hyperboloidal foliations.** We will define a time function τ, such that the level sets of τ, denoted Σ_τ, are appropriate asymptotically hyperboloidal hypersurfaces. Let

$$\tau = \tilde{v} - \int_{r_+}^{\varrho} \mathbb{h}(\varrho')\, d\varrho' - v_0,$$

where $v_0 \in \mathbb{R}_{>0}$ and $\mathbb{h} : [r_+, \infty) \to \mathbb{R}$ is a smooth, non-negative function, such that

$$2 - \mathbb{h}(\varrho)(1 - 2M\varrho^{-1}) = O(\varrho^{-2}), \tag{6.9}$$

$$\rho^{-2}\left[-\mathbb{h}(2(r^2+a^2) - \mathbb{h}\Delta) + a^2\sin^2\theta\right] = g^{-1}_{M,a}(d\tau, d\tau) < 0. \tag{6.10}$$

By construction, $T(\tau) = 1$.

Let $x = \frac{1}{\varrho}$ and $h : [0, \frac{1}{r_+}]_x \to \mathbb{R}$, with

$$M^2 h(x) = 2x^{-2} - \mathbb{h}(x^{-1})(x^{-2} - 2Mx^{-1}).$$

By assumption on $\mathbb{h}$, $h(x) = O(1)$. **We will assume that h is analytic in an arbitrarily small neighbourhood of $x = 0$ and that $h(0) \neq 0$.** By $g^{-1}_{M,a}(d\tau, d\tau) < 0$, we then necessarily have that $h(0) > 0$.

Note that we can obtain $\tau = t$, by defining instead $\mathbb{h} : (r_+, \infty) \to \mathbb{R}$ by $\mathbb{h}(r) = \frac{r^2+a^2}{\Delta}$, which cannot be smoothly extended to $r = r_+$ and does not satisfy (6.9).

If $M = 0$, then we can simply take $\mathbb{h}(\varrho) = 2$ and Σ_τ are outgoing null hypersurfaces.

For the sake of later convenience, we moreover assume that $1 \leq \mathbb{h} \leq \frac{r^2+a^2}{\Delta}$ in the region $\varrho \leq R_\infty$, where $R_\infty \gg M$ will be defined later.

We can extend the manifolds-with-boundary $\Sigma_{\tau'} = \{\tau'\} \times (0, r_+^{-1}]_x \times \mathbb{S}^2_{\vartheta,\varphi_*}$ as follows:

$$\widehat{\Sigma}_{\tau'} = \{\tau'\} \times [0, r_+^{-1}]_x \times \mathbb{S}^2_{\vartheta,\varphi_*}.$$

We denote moreover $\mathcal{I}^+ = \mathbb{R}_\tau \times \{0\}_x \times \mathbb{S}^2_{\vartheta,\varphi_*}$ and also $\Sigma := \Sigma_0$ and $\widehat{\Sigma} := \widehat{\Sigma}_0$.

The extended manifold $[0,\infty)_\tau \times [0, r_+^{-1}]_x \times \mathbb{S}^2_{\vartheta,\varphi_*}$ can be equipped with an extension of the conformally rescaled metric $r^{-2} g_{M,a}$ and forms a conformal compactification of the domain of dependence $D^+(\Sigma)$. The subset $\mathcal{I}^+$ is called future null infinity. See Figure 1 for an illustration of the above geometric notions.

The tuple $(\tau, \varrho, \vartheta, \varphi_*)$ forms a well-defined coordinate chart on the spacetime. With respect to these coordinates, we can express:

$$\begin{aligned} T &= \partial_\tau \quad \text{and we define} \\ \Theta &:= \partial_\vartheta, \\ X &:= \partial_\varrho - \mathbb{h}\partial_\tau. \end{aligned}$$

Alternatively, we can consider the coordinates $(\tau, x, \vartheta, \varphi_*)$, to obtain the following expression for X:

$$-X = x^2\partial_x + \frac{2 - M^2x^2h(x)}{1 - 2Mx}\partial_\tau.$$

By applying (6.7) and (6.8), we can further express Y in terms of X and Θ:

$$Y = (\partial_r \varrho)X + (\partial_r \vartheta)\Theta = (1 + a^2\sin^2\vartheta O_\infty(\varrho^{-2}))X + a^2 O_\infty(\varrho^{-3})\sin\vartheta\cos\vartheta\Theta.$$

7. Preliminaries: differential operators

In this section, we introduce the main differential operators that are relevant for the analysis in the paper. We will first express the Kerr wave operator $\Box_{g_{M,a}}$ with respect to suitable spacetime coordinates, introduced in §6, and we will then write down the corresponding rescaled Laplace-transformed operator L_s, as well as $\mathfrak{L}_{s,\lambda,\kappa}$, which is a modification of L_s that plays a leading role

in the main estimates. In §18, we will briefly discuss analogous objects in the context of Kerr–de Sitter spacetimes.

7.1. **Wave operator.** We will first express the wave operator $\Box_{g_{M,a}}$ with respect to different choices of vector fields, adapted to the spacetime coordinates under consideration. With respect to the vector fields $(T, Y, \partial_\theta, \Phi)$, introduced in §6, we have that:

$$\Box_{g_{M,a}}\phi = \rho^{-2}Y(\Delta Y\phi) + 2a\rho^{-2}Y\Phi\phi + \rho^{-2}\not\Delta_{\mathbb{S}^2,\theta}\phi + 2\rho^{-2}(r^2+a^2)YT\phi + 2\rho^{-2}rT\phi \\ + a^2\sin^2\theta\rho^{-2}T^2\phi + 2a\rho^{-2}T\Phi\phi. \quad (7.1)$$

We will now express $\Box_{g_{M,a}}$ in terms of the vector fields (T, X, Θ, Φ), where we recall that in the region $\{\varrho \leq \frac{R_\infty}{2}\}$, $X = Y - \mathbb{h}T$ and $\Theta = \partial_\theta$. We also introduce the following spherical differential operators:

$$\not\Delta_{\mathbb{S}^2,\theta}(\cdot) = \frac{1}{\sin\theta}\partial_\theta(\sin\theta\partial_\theta(\cdot)) + \frac{1}{\sin^2\theta}\Phi^2,$$
$$\not\Delta_{\mathbb{S}^2,\vartheta}(\cdot) = \frac{1}{\sin\vartheta}\partial_\vartheta(\sin\vartheta\partial_\vartheta(\cdot)) + \frac{1}{\sin^2\vartheta}\Phi^2.$$

We can decompose $f \in L^2(\mathbb{S}^2)$ into azimuthal modes:

$$f = \sum_{m\in\mathbb{Z}} f_m,$$

with $\Phi f_m = imf_m$.

We can also decompose $f \in L^2(\mathbb{S}^2)$ into spherical harmonic modes:

$$f(\vartheta,\varphi_*) = \sum_{\ell\in\mathbb{N}_0} f_\ell(\vartheta,\varphi_*) = \sum_{m\in\mathbb{Z}}\sum_{\ell\geq|m|} f_{m\ell}Y_{\ell m}(\vartheta,\varphi_*),$$

with $\not\Delta_{\mathbb{S}^2,\vartheta}f_\ell = -\ell(\ell+1)f_\ell$.

We will reserve the notation $\not\Delta_{\mathbb{S}^2}$ for the Laplacian $\not\Delta_{\mathbb{S}^2,\vartheta}$ and $\not\nabla_{\mathbb{S}^2}$ for the covariant derivative $\not\nabla_{\mathbb{S}^2,\vartheta}$, both associated to spheres of constant τ and ϱ with respect to the unit round sphere metric.

First, we restrict to the region $\{\varrho \geq R_\infty\}$ and use the fact that in this region $\check{a}(r) = 0$. In the $M = 0$ case, the operator $\Box_{g_{M,a}}$ with respect to $(T, X, \partial_\vartheta, \Phi)$ corresponds simply to the wave operator on Minkowski in outgoing Eddington–Finkelstein coordinates $(\tilde{u}, \varrho, \vartheta, \varphi_*)$, with $T = \partial_{\tilde{u}}$, $X = \partial_\varrho$ and $\Phi = \partial_{\varphi_*}$.

We can therefore express in $\{\varrho \geq R_\infty\}$:

$$\rho^{-2}Y(r^2Y\phi) + 2a\rho^{-2}Y\Phi\phi + \rho^{-2}\not\Delta_{\mathbb{S}^2,\theta}\phi + 2\rho^{-2}(r^2+a^2)YT\phi + 2rT\phi + a^2\sin^2\theta\rho^{-2}T^2\phi + 2a\rho^{-2}T\Phi\phi \\ = \Box_{g_{0,a}}\phi = \varrho^{-2}X(\varrho^2X\phi) + \varrho^{-2}\not\Delta_{\mathbb{S}^2,\vartheta}\phi + 2\varrho^{-1}X(\varrho T\phi).$$

Hence, in $\{\varrho \geq R_\infty\}$, we have that

$$\Box_{g_{M,a}}\phi = \varrho^{-2}X(\varrho^2X\phi) + \varrho^{-2}\not\Delta_{\mathbb{S}^2,\vartheta}\phi + 2\varrho^{-1}X(\varrho T\phi) - 2M\varrho^{-2}Y(rY\phi). \quad (7.2)$$

We now turn to the full region $\{r \geq r_+\}$.

Lemma 7.1. *Define $\psi := \varrho\cdot\phi$. With respect to the coordinates $(\tau, x = \varrho^{-1}, \vartheta, \varphi_*)$, we can write:*

$$\varrho^3\Box_{g_{M,a}}\phi = \partial_x(x^2\partial_x\psi) + 2\partial_x\partial_\tau\psi + \not\Delta_{\mathbb{S}^2,\vartheta}\psi + Mx\cdot\mathcal{B}^\infty\psi + \mathcal{K}^\infty\psi, \quad (7.3)$$

with

$$\mathcal{B}^\infty f := b_{xx}\cdot x^2\partial_x^2 f + b_{x\tau}\cdot\partial_x\partial_\tau f + b_{\tau\tau}\partial_\tau^2 f + a^2(xb_{x\vartheta}\cdot x^2\partial_x + xb_{\tau\vartheta}\partial_\tau)\sin\vartheta\partial_\vartheta f \\ + a^4x^4\cdot b_{\vartheta\vartheta}(\sin\vartheta\partial_\vartheta)^2 f + b_x\cdot x\partial_x f + a^2x^2b_\vartheta\cdot\sin\vartheta\partial_\vartheta f + b_\tau\partial_\tau f + b\cdot f,$$
$$\mathcal{K}^\infty f := -2M^2h(0)\partial_\tau^2 f,$$

where the functions $b_{\star\star}, b_\star, b$, with $\star \in \{x, \tau, \vartheta\}$, are smooth and bounded functions of (ϱ, ϑ), by (7.2). We moreover have that $b_{\star\star}, b_\star, b$ are analytic in $[0, x_0]_x \times \mathbb{S}^2_{\theta,\varphi}$ and there exist constants $C, B_1, B_2 > 0$, such that

$$\frac{1}{(n+m)!}(\|\nabla\!\!\!/^m_{\mathbb{S}^2}\partial_x^n b\|_{L^\infty([0,x_0]\times\mathbb{S}^2)}+\|\nabla\!\!\!/^m_{\mathbb{S}^2}\partial_x^n b_\star\|_{L^\infty([0,x_0]\times\mathbb{S}^2)}+\|\nabla\!\!\!/^m_{\mathbb{S}^2}\partial_x^n b_{\star\star}\|_{L^\infty([0,x_0]\times\mathbb{S}^2)}) \leq CB_1^n(a^2x_0^2B_2)^m.$$

In $\{\varrho \leq \frac{R_\infty}{2}\}$ we can express:

$$(Mxb_{\tau\tau} - 2M^2h(0)) = -\mathbb{h}(2(r^2+a^2) - \Delta\mathbb{h}) + a^2\sin^2\theta, \tag{7.4}$$

which is strictly negative by (6.10).

Proof. Note that the form (7.3) and the smoothness property of $b_{\star\star}, b_\star, b$ hold trivially in $\{\varrho \leq R_\infty\}$. Using that in $\{\varrho \leq \frac{R_\infty}{2}\}$, we have that $\varrho = r$ and $\vartheta = \theta$, so $Y = X + \mathbb{h}T$ and the property (7.4) also follows immediately after expanding the Y derivatives in (7.1) and considering the factor appearing in front of $T^2\phi$.

It remains to verify the stated asymptotic properties of $b_{\star\star}, b_\star, b$ as $x \to 0$, for which we can restrict to the region $\{\varrho \geq R_\infty\}$. We first recall that:

$$-X = x^2\partial_x + \frac{2 - M^2x^2h(x)}{1-2Mx}\partial_\tau,$$

so

$$\begin{aligned}
&\varrho X(\varrho^2 X\phi) + \varrho\Delta\!\!\!/_{\mathbb{S}^2,\vartheta}\phi - 2\varrho^2 X(\varrho T\phi) = \varrho^2X^2(\varrho\phi) + 2\varrho^2XT(\varrho\phi) + \varrho\Delta\!\!\!/_{\mathbb{S}^2,\vartheta}\phi\\
&= \left(\partial_x + x^{-2}\frac{2-M^2x^2h(x)}{1-2Mx}\partial_\tau\right)\left(x^2\partial_x\psi + \frac{2-M^2x^2h(x)}{1-2Mx}\partial_\tau\psi\right) - 2\left(\partial_x + x^{-2}\frac{2-M^2x^2h(x)}{1-2Mx}\partial_\tau\right)\partial_\tau\psi\\
&\qquad + \Delta\!\!\!/_{\mathbb{S}^2,\vartheta}\psi\\
&= \partial_x(x^2\partial_x\psi) + 2\partial_x\partial_\tau\psi + \Delta\!\!\!/_{\mathbb{S}^2,\vartheta}\psi - 2M^2\frac{h(x)}{1-2Mx}\partial_x\partial_\tau\psi\\
&\qquad + \frac{4Mx^{-1} - M^2h}{(1-2Mx)^2}(2 - M^2x^2h(x))\partial_\tau^2\psi + \partial_x\left(\frac{2-M^2x^2h(x)}{1-2Mx}\right)\partial_\tau\psi.
\end{aligned} \tag{7.5}$$

Using that:

$$Y = (1 + a^2\sin^2\vartheta O_\infty(\varrho^{-2}))X + a^2O_\infty(\varrho^{-3})\sin\vartheta\cos\vartheta\partial_\vartheta,$$

we moreover obtain:

$$\begin{aligned}
&-2M\varrho Y(rY\phi)\\
&= 2Mx^{-1}\left[(1 + a^2\sin^2\vartheta O_\infty(x^2))\left(x^2\partial_x + \frac{2-M^2x^2h(x)}{1-2Mx}\partial_\tau\right) + a^2O_\infty(x^3)\sin\vartheta\cos\vartheta\partial_\vartheta\right](x\psi)\\
&-2Mx^{-1}r\left((1 + a^2\sin^2\vartheta O_\infty(x^2))\left(x^2\partial_x + \frac{2-M^2x^2h(x)}{1-2Mx}\partial_\tau\right) + a^2O_\infty(x^3)\sin\vartheta\cos\vartheta\partial_\vartheta\right)^2(x\psi).
\end{aligned} \tag{7.6}$$

Using that $r = x^{-1} + a^2O(x)$, terms involving $\partial_\tau^2\psi$ appearing on the very right-hand side combine to obtain:

$$-2Mx^{-1}\frac{2 + M^2O_\infty(x^2) + a^2O_\infty(x^2)}{(1-2Mx)^2}(2 - M^2x^2h(x))\partial_\tau^2\psi,$$

which will cancel to leading-order in x with the $\partial_\tau^2\psi$ term on the very right-hand side of (7.5), to obtain a total leading-order term:

$$-2M^2h(0)\partial_\tau^2\psi$$

when we consider, via (7.2),

$$\varrho^3\Box_{g_{M,a}}\phi = \varrho X(\varrho^2 X\phi) + \varrho\not{\Delta}_{\mathbb{S}^2,\vartheta}\phi - 2\varrho^2 X(\varrho T\phi) - 2M\varrho Y(rY\phi)$$

in the region $\{\varrho \geq R_\infty\}$.

Similarly, the leading-order term in the factor multiplying $\partial_\tau\psi$ in (7.6) is:

$$-4M\partial_\tau\psi,$$

which cancels with the leading-order contribution in the $\partial_\tau\psi$ term on the very right-hand side of (7.5).

For the remaining coefficients, the asymptotics in x follow straightforwardly.

The analyticity properties of $b, b_\star, b_{\star\star}$ in $\{r \geq R_\infty\}$ follow from the analyticity of $r(\varrho,\vartheta)$ by (6.5) and the $a^2x_0^2$ factor in the L^∞ estimate follows from the factor $\frac{4a^2\varrho^2}{(\varrho^2-a^2)^2}$ in front of the ϑ-dependent term in (6.5), which plays the role of a scale factor when taking ϑ-derivatives of r □

7.2. **Laplace-transformed operator.** We define the differential operator L_s to be the Laplace transform of the operator

$$\varrho^3 \circ \Box_{g_{M,a}} \circ \varrho^{-1}$$

with respect to the time function τ. More explicitly, for $\widehat{\psi} \in C^\infty(\widehat{\Sigma})$:

$$L_s\widehat{\psi} = e^{-s\tau}\varrho^3\Box_{g_{M,a}}(e^{s\tau}\varrho^{-1}\widehat{\psi}).$$

In view of (7.3), we can then write:

$$L_s = \partial_x(x^2\partial_x(\cdot)) + 2s\partial_x(\cdot) + \not{\Delta}_{\mathbb{S}^2,\vartheta} + Mx\cdot B_s^\infty + K_s^\infty, \tag{7.7}$$

with B_s^∞ and K_s^∞ the Laplace transforms of $\mathcal{B}^\infty$ and $\mathcal{K}^\infty$, defined as follows:

$$\begin{aligned} B_s^\infty f :=\ & b_{xx}\cdot x^2\partial_x^2 f + sb_{x\tau}\cdot\partial_x f + s^2 b_{\tau\tau} f + a^2(xb_{x\vartheta}\cdot x^2\partial_x + sxb_{\tau\vartheta})\sin\vartheta\partial_\vartheta f \\ & + a^4x^4\cdot b_{\vartheta\vartheta}(\sin\vartheta\partial_\vartheta)^2 f + b_x\cdot x\partial_x f + a^2x^2 b_\vartheta\cdot\sin\vartheta\partial_\vartheta f + sb_\tau f + b\cdot f, \\ K_s^\infty f :=\ & -2M^2h(0)s^2 f. \end{aligned}$$

7.3. **The Laplace-transformed operator as a perturbation.** The main estimates in this paper will not directly involve L_s. Instead we will treat L_s as a *perturbation* of a suitable operator that we will denote $\mathfrak{L}_{s,\lambda}$. Here, λ will be a large parameter that can be interpreted as an angular frequency shift. In §16, we will show that L_s can be split up in the following way:

$$L_s = \mathfrak{L}_{s,\lambda} + \epsilon_0\cdot B_s + K_{s,\lambda}, \tag{7.8}$$

with $B_{s,\lambda} : H^{N_+ +2}_{\sigma,R_\infty,2} \to H^{N_+}_{\sigma,R_\infty}$ and $K_{s,\lambda} : H^{N_+ +1}_{\sigma,R_\infty,2} \to H^{N_+ +1}_{\sigma,R_\infty,2}$ bounded operators, with $H^{N_+ +1}_{\sigma,R_\infty,2} \subset H^{N_+}_{\sigma,R_\infty}$ Hilbert spaces that will be introduced in §8.

We will also show that that the composition $K_{s,\lambda}\circ R$, with $R : H^{N_+}_{\sigma,R_\infty} \to H^{N_+ +1}_{\sigma,R_\infty,2}$ a suitable bounded operator, is a compact operator and that we can arrange ϵ_0 to be suitably small to obtain $\|\epsilon_0 B_s\circ R\| < 1$. This splitting of L_s will allow us to obtain *meromorphicity* of the map $s \mapsto L_s^{-1}$ from the invertibility of $\mathfrak{L}_{s,\lambda}$ for s restricted to suitable domains, via the analytic Fredholm theorem.

We will in fact introduce a family of operators $\mathfrak{L}_{s,\lambda,\kappa}$, where $\kappa \in [0,\infty)$, with $\mathfrak{L}_{s,\lambda} := \mathfrak{L}_{s,\lambda,0}$. The parameter $\kappa > 0$ plays a regularizing role, which can be interpreted as making "surface gravity of future-null infinity $\mathcal{I}^+$" positive. These operators can also be thought of as simplifications of the operators that are relevant on Kerr–de Sitter spacetimes.

Using that $\varrho = r$ and $\vartheta = \theta$ in $\{r \leq \frac{1}{2}R_\infty\}$ and applying (7.1), we also have that for $\varrho \leq \frac{1}{2}R_\infty$:

$$\begin{aligned} r^{-2}\rho^2 L_s = r\partial_r(\Delta\partial_r(r^{-1}\cdot)) + 2ar\partial_r\partial_{\varphi_*}(r^{-1}\cdot) + \not{\Delta}_{\mathbb{S}^2,\theta}(\cdot) - 2r[\mathbb{h}\Delta - (r^2+a^2)]s\partial_r(r^{-1}\cdot) + \left[2r - \frac{d}{dr}(\Delta\mathbb{h})\right]s(\cdot) \\ + [\mathbb{h}^2\Delta - 2\mathbb{h}(r^2+a^2) + a^2\sin^2\theta]s^2(\cdot) - 2a(\mathbb{h}-1)s\partial_{\varphi_*}(\cdot). \end{aligned}$$

Define

$$r^{-2}\rho^2 K_s^+ f := \left[2r - \frac{d}{dr}(\Delta \mathbb{h})\right] s(r^{-1}f) + [\mathbb{h}^2\Delta - 2\mathbb{h}(r^2+a^2) + a^2\sin^2\theta]s^2 f - 2a(\mathbb{h}-1)s\partial_{\varphi_*} f.$$

Then we can express in $\{\varrho \leq \frac{1}{2}R_\infty\}$:

$$L_s = r^2\rho^{-2}\left[r\partial_r(\Delta\partial_r(r^{-1}\cdot)) + 2ar\partial_r\partial_{\varphi_*}(r^{-1}\cdot) + \not{\Delta}_{\mathbb{S}^2}(\cdot) + (4Mr^2 - 2(\mathbb{h}-1)r\Delta)s\partial_r(r^{-1}\cdot)\right] + K_s^+.$$

Let $\lambda \in [0,\infty)$ and define

$$S_\lambda f = 2l\sum_{\ell\in\mathbb{N}_0} \ell f_\ell,$$

with $l \in \mathbb{R}_{\geq 0}$ the non-negative solution to $l(l+1) = \lambda$. Then, in $\{r \geq R_\infty\}$, we can write

$$(\not{\Delta}_{\mathbb{S}^2} - (\lambda\mathbf{1} + S_\lambda))f_\ell = -(\ell+l)(\ell+l+1)f_\ell.$$

The operator $-(\lambda\mathbf{1}+S_\lambda)$ therefore has the effect of shifting the angular frequency ℓ by l. Later, we will need to restrict to λ such that $l \in \mathbb{N}_0$ (and we need to take l suitably large).

Denote with $\pi_{\leq l}$ the projection of f to spherical harmonics with angular frequency $\leq l$. Then

$$(\not{\Delta}_{\mathbb{S}^2} - (\lambda\mathbf{1} + S_\lambda)\pi_{\leq l})f_\ell = -\mathfrak{l}(\mathfrak{l}+1)f_\ell,$$

with $\mathfrak{l} = \ell + l$ if $\ell \leq l$ and $\mathfrak{l} = \ell$ if $\ell > l$. We therefore have that $\mathfrak{l} \geq l$ for all $\ell \in \mathbb{N}_0$.

Let $\chi_{\varrho\geq R_\infty}$ be a smooth cut-off function that is 1 in $\varrho \geq R_\infty$ and 0 in $\varrho \leq \frac{1}{2}R_\infty$. Let $\chi_{\varrho\leq\frac{1}{2}R_\infty}$ be a smooth cut-off function that is 1 in $\varrho \leq \frac{1}{2}R_\infty$ and 0 in $\varrho \geq R_\infty$

We now define the main differential operator with respect to which we will derive energy estimates and elliptic estimates:

$$\begin{aligned}\mathfrak{L}_{s,\lambda,\kappa} := L_s + \kappa\partial_x(\chi_{\varrho\geq R_\infty}x\partial_x(\cdot)) - Mx\chi_{\varrho\geq R_\infty}B_s^\infty \\ - (\lambda\pi_{\leq l} + \chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l} + (1-\chi_{\varrho\leq\frac{1}{2}R_\infty})K_s^\infty + \chi_{\varrho\leq\frac{1}{2}R_\infty}K_s^+).\end{aligned} \tag{7.9}$$

In particular, L_s takes the form (7.8) for $\kappa = 0$, if we define:

$$\begin{aligned}B_s &:= Mx\epsilon_0^{-1}\chi_{\varrho\geq R_\infty}B_s^\infty, \\ K_{s,\lambda} &:= \lambda\pi_{\leq l} + \chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l} + K_s^\infty + \chi_{\varrho\leq\frac{1}{2}R_\infty}K_s^+.\end{aligned}$$

Note that $\mathfrak{L}_{s,\lambda,\kappa}$ takes on rather simple forms when $\varrho \in [r_+,\infty)\setminus(\frac{R_\infty}{2}, R_\infty)$:

$$\begin{aligned}r^{-2}\rho^2\mathfrak{L}_{s,\lambda,\kappa} &= r\partial_r(\Delta\partial_r(r^{-1}\cdot)) + 2ar\partial_r\partial_{\varphi_*}(r^{-1}\cdot) + \not{\Delta}_{\mathbb{S}^2}(\cdot) \\ &\quad + sr(4Mr - 2(1-\mathbb{h})\Delta)\partial_r(r^{-1}\cdot) - \lambda\pi_{\leq l} \quad \text{for} \quad r = \varrho \leq \frac{R_\infty}{2},\end{aligned} \tag{7.10}$$

$$(\mathfrak{L}_{s,\lambda,\kappa}f)_\ell = \partial_x(x(\kappa+x)\partial_x f_\ell) + 2s\partial_x f_\ell - \mathfrak{l}(\mathfrak{l}+1)f_\ell \quad \text{for} \quad \frac{1}{x} = \varrho \geq R_\infty. \tag{7.11}$$

In general, we can express:

$$\begin{aligned}(\mathfrak{L}_{s,\lambda,\kappa}f) = \partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x f) + 2s\partial_x f + \not{\Delta}_{\mathbb{S}^2}f - \lambda\pi_{\leq l}f - \chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l}f + (1-\chi_{\varrho\leq R_\infty})MxB_s^\infty \\ - \chi_{\varrho\leq\frac{1}{2}R_\infty}(K_s^+ - K_s^\infty) \quad \text{for all} \quad \varrho \geq r_+.\end{aligned} \tag{7.12}$$

Finally, we define the κ-regularized operator $L_{s,\kappa}$ as follows:

$$L_{s,\kappa} := L_s + \kappa\partial_x(x\chi_{\varrho\geq R_\infty})\partial_x f). \tag{7.13}$$

8. Preliminaries: Hilbert spaces

In this section, we will define all the Hilbert spaces that are relevant for the present paper.

Let $S \subseteq \Sigma$ be closed. We define $H^k(S)$ as the Hilbert space obtained from the completion of $C^\infty(S)$ with respect to the Sobolev inner product:

$$\langle f, g\rangle_{H^k(S)} = \sum_{k_1+k_1\leq k} \int_S \nabla\!\!\!/^{k_1}_{\mathbb{S}^2} \partial_\varrho^{k_2} f \cdot \nabla\!\!\!/^{k_1}_{\mathbb{S}^2} \partial_\varrho^{k_2} \overline{g}\, \varrho^2 d\sigma d\varrho.$$

Here, $d\sigma$ denotes the following standard volume form on the unit round sphere: $d\sigma := \sin\vartheta d\vartheta d\varphi_*$.

We denote moreover

$$\langle f, g\rangle_{H^k_{\nabla\!\!\!/}(S)} = \langle f, g\rangle_{H^k(S)} + \langle \nabla\!\!\!/_{\mathbb{S}^2} f, \nabla\!\!\!/_{\mathbb{S}^2} g\rangle_{H^k(S)},$$

and we let $H^k_{\nabla\!\!\!/}(S)$ denote the completion of $C^\infty(S)$ with respect to this inner product.

Consider $\widehat{\Sigma} \cong [0, r_+^{-1}]_x \times \mathbb{S}^2_{\vartheta,\varphi_*}$. Let $\sigma \in \mathbb{R}$ and $\delta > 0$. We introduce the following sequilinear forms on $C^\infty([0, x_0]_x \times \mathbb{S}^2_{\vartheta,\varphi_*})$ where $x_0 > 0$, $d_0 > 0$ and $\sigma \in \mathbb{R}$:

$$\begin{aligned}
\langle f, g\rangle_{B_{R_\infty}} &:= \sum_{\ell\in\mathbb{N}_0}\sum_{n=0}^{\infty} \frac{(d_0 x_0)^{2n}}{(\max\{\ell+1, n+1\})^{2n}} \langle \partial_x^n f_\ell, \partial_x^n g_\ell\rangle_{L^2(\mathbb{S}^2)}(x_0),\\
\langle f, g\rangle_{B_{R_\infty,2}} &:= \sum_{\ell\in\mathbb{N}_0}\sum_{n=0}^{\infty} \frac{(d_0 x_0)^{2n}}{(\max\{\ell+1, n+1\})^{2n}} \Big[(\ell+1)^4 \langle \partial_x^n f_\ell, \partial_x^n g_\ell\rangle_{L^2(\mathbb{S}^2)}(x_0)\\
&\quad + (\ell+1)^2 \langle \partial_x^{n+1} f_\ell, \partial_x^{n+1} g_\ell\rangle_{L^2(\mathbb{S}^2)}(x_0) + \langle \partial_x^{n+2} f_\ell, \partial_x^{n+2} g_\ell\rangle_{L^2(\mathbb{S}^2)}(x_0)\Big],\\
\langle f, g\rangle_{G_{\sigma,R_\infty}} &:= \sum_{\ell\in\mathbb{N}_0} \sum_{n=0}^{\max\{\ell-1,0\}} \frac{(4\sigma)^{2n}(\ell-(n+1))!^2}{(\ell+n+1)!^2} \int_0^{x_0} \langle \partial_x^n f_\ell, \partial_x^n g_\ell\rangle_{L^2(\mathbb{S}^2)}(x)\, dx\\
&\quad + (\ell+1)^3 \sum_{n=\ell}^{\infty} \frac{\sigma^{2n}}{n!^2(n+1)!^2} \int_0^{x_0} \langle \partial_x^n f_\ell, \partial_x^n g_\ell\rangle_{L^2(\mathbb{S}^2)}(x)\, dx\\
\langle f, g\rangle_{G_{\sigma,R_\infty,2}} &:= \sum_{\ell\in\mathbb{N}_0} \sum_{n=0}^{\max\{\ell-1,0\}} \frac{(4\sigma)^{2n}(\ell-(n+1))!^2}{(\ell+n+1)!^2}\\
&\quad \times \int_0^{x_0} (\ell-n)^2(\ell+n+1)^2 \langle \partial_x^n f_\ell, \partial_x^n g_\ell\rangle_{L^2(\mathbb{S}^2)}(x)\\
&\quad + (1+x^2(\ell+1)^2) \langle \partial_x^{n+1} f_\ell, \partial_x^{n+1} g_\ell\rangle_{L^2(\mathbb{S}^2)}(x) + x^4 \langle \partial_x^{n+2} f_\ell, \partial_x^{n+2} g_\ell\rangle_{L^2(\mathbb{S}^2)}(x)\, dx\\
&\quad + (\ell+1)^3 \sum_{n=\ell}^{\infty} \frac{\sigma^{2n}}{n!^2(n+1)!^2} \int_0^{x_0} (n+1)^4 \langle \partial_x^n f_\ell, \partial_x^n g_\ell\rangle_{L^2(\mathbb{S}^2)}(x)\\
&\quad + (1+(n+1)^2 x^2) \langle \partial_x^{n+1} f_\ell, \partial_x^{n+1} g_\ell\rangle_{L^2(\mathbb{S}^2)}(x) + x^4 \langle \partial_x^{n+2} f_\ell, \partial_x^{n+2} g_\ell\rangle_{L^2(\mathbb{S}^2)}(x)\, dx.
\end{aligned}$$

We denote moreover

$$\|f\|_{B_{R_\infty}} := \sqrt{\langle f, f\rangle_{B_{R_\infty}}} \quad \text{and} \quad \|f\|_{B_{R_\infty,2}} := \sqrt{\langle f, f\rangle_{B_{R_\infty,2}}}$$
$$\|f\|_{G_{\sigma,R_\infty}} := \sqrt{\langle f, f\rangle_{G_{\sigma,R_\infty}}} \quad \text{and} \quad \|f\|_{G_{\sigma,R_\infty,2}} := \sqrt{\langle f, f\rangle_{G_{\sigma,R_\infty,2}}}.$$

Let $w_p : [0, r_+^{-1}] \to \mathbb{R}$ be a smooth cut-off function that is defined as follows: let $p \in [0, \infty)$, then

$$w_p(x) = \begin{cases} p & x \leq \frac{1}{4R_\infty}, \\ 0 & x \geq \frac{1}{3R_\infty}. \end{cases} \tag{8.1}$$

Let $N_+ \in \mathbb{N}_0$ and $\sigma \in \mathbb{R}$. We fix $p = 4(N_+ + 1) + 2$. We define the following inner products on $C^\infty(\widehat{\Sigma})$: let $x_0 = \frac{1}{2R_\infty}$, then

$$\begin{aligned}
\langle f, g\rangle_{H^{N_+}_{\sigma,R_\infty}} &:= \langle f, g\rangle_{H^{N_+}(\Sigma\cap\{\varrho\leq 3R_\infty\})} + \langle \nabla\!\!\!/_{\mathbb{S}^2} f, \nabla\!\!\!/_{\mathbb{S}^2} g\rangle_{H^{N_+}(\Sigma\cap\{\varrho\leq \frac{1}{3}R_\infty\})} \\
&\quad + \langle (\mathbf{1} - \Delta\!\!\!/_{\mathbb{S}^2})^{-\frac{w_p}{4}} f, (\mathbf{1} - \Delta\!\!\!/_{\mathbb{S}^2})^{-\frac{w_p}{4}} g\rangle_{H^{N_+}(\Sigma\cap\{3R_\infty\leq\varrho\leq 4R_\infty\})} + \langle f, g\rangle_{B_{R_\infty}} + \langle f, g\rangle_{G_{\sigma,R_\infty}}, \\
\langle f, g\rangle_{H^{N_++1}_{\sigma,R_\infty,2}} &:= \sum_{k=0}^{1} \langle \nabla\!\!\!/^k_{\mathbb{S}^2} f, \nabla\!\!\!/^k_{\mathbb{S}^2} g\rangle_{H^{N_++1}(\Sigma\cap\{\varrho\leq\frac{1}{3}R_\infty\})} + \langle f, g\rangle_{H^{N_++2}(\Sigma\cap\{\frac{1}{3}R_\infty\leq\varrho\leq 3R_\infty\})} \\
&\quad + \langle (\mathbf{1} - \Delta\!\!\!/_{\mathbb{S}^2})^{-\frac{w_p}{4}} f, (\mathbf{1} - \Delta\!\!\!/_{\mathbb{S}^2})^{-\frac{w_p}{4}} g\rangle_{H^{N_++2}(\Sigma\cap\{3R_\infty\leq\varrho\leq 4R_\infty\})} + \langle f, g\rangle_{B_{R_\infty,2}} + \langle f, g\rangle_{G_{\sigma,R_\infty,2}}.
\end{aligned}$$

We define the Hilbert spaces $H^{N_+}_{\sigma,R_\infty}$ and $H^{N_++1}_{\sigma,R_\infty,2}$ as the completions of $C^\infty(\widehat{\Sigma})$ with respect to the inner products $\langle\cdot,\cdot\rangle_{H^{N_+}_{\sigma,R_\infty}}$ and $\langle\cdot,\cdot\rangle_{H^{N_++1}_{\sigma,R_\infty,2}}$, respectively, and we denote the corresponding norms by $\|\cdot\|_{H^{N_+}_{\sigma,R_\infty}}$ and $\|\cdot\|_{H^{N_++1}_{\sigma,R_\infty,2}}$, respectively.

We also introduce the subspace $H^{N_++1}_{\sigma,R_\infty,1} \subset H^{N_+}_{\sigma,R_\infty}$, which is defined as follows:

$$H^{N_++1}_{\sigma,R_\infty,1} := \{f \in H^{N_+}_{\sigma,R_\infty} \,|\, \partial_x f, \nabla\!\!\!/_{\mathbb{S}^2} f \in H^{N_+}_{\sigma,R_\infty}\},$$

and we denote $\|f\|^2_{H^{N_++1}_{\sigma,R_\infty,1}} := \sum_{1\leq k_1+k_2\leq 1} \|\nabla\!\!\!/^{k_1}_{\mathbb{S}^2}\partial_x^{k_2} f\|^2_{H^{N_+}_{\sigma,R_\infty}}$.

Let $P : H^1(\widehat{\Sigma}) \to L^2(\widehat{\Sigma})$ be the following linear operator:

$$Pf := \frac{1}{2M^2h(0) - b_{\tau\tau}} \left((2 + Mxb_{x\tau})\partial_x + Ma^2 \sin\vartheta x^2 b_{\tau\vartheta}\partial_\vartheta + Mxb_\tau \mathbf{1}\right),$$

with the coefficients $b_{\star\star}$ and $b_\star$ defined in Lemma 7.1.

The role of the operator P will be explained in Theorem 9.1. We define the following space:

$$\mathbf{H}^{N_+}_{\sigma,R_\infty} = H^{N_+}_{\sigma,R_\infty}(P) \oplus H^{N_+}_{\sigma,R_\infty},$$

where $H^{N_+}_{\sigma,R_\infty}(P) = \{f \in H^{N_+}_{\sigma,R_\infty} \,|\, Pf \in H^{N_+}_{\sigma,R_\infty}\}$ and we equip it with the inner product:

$$\langle (f, g), (\tilde{f}, \tilde{g})\rangle_{\mathbf{H}^{N_+}_{\sigma,R_\infty}} = \langle f, \tilde{f}\rangle_{H^{N_+}_{\sigma,R_\infty}} + \langle Pf, P\tilde{f}\rangle_{H^{N_+}_{\sigma,R_\infty}} + \langle g, \tilde{g}\rangle_{H^{N_+}_{\sigma,R_\infty}}$$

to obtain a Hilbert space of pairs of functions on $\widehat{\Sigma}$. We note the corresponding norm by $\|\cdot\|_{\mathbf{H}^{N_+}_{\sigma,R_\infty}}$.

9. Preliminaries: the infinitesimal generator of time translations

We define the time evolution operators $\mathcal{S}(\tau)$ associated to the wave equation $\Box_{g_{M,a}}\phi = 0$ and the foliation $\{\widehat{\Sigma}_\tau\}_{\tau\in\mathbb{R}}$ as follows: let $N \geq 0$ and $\tau \in [0, \infty)$, then

$$\begin{aligned}
\mathcal{S}(\tau) : H^{N+1}(\widehat{\Sigma}) \times H^N(\widehat{\Sigma}) &\to H^{N+1}(\widehat{\Sigma}) \times H^N(\widehat{\Sigma}), \\
(\Psi, \Psi') &\mapsto (\psi(\tau,\cdot), \partial_\tau\psi(\tau,\cdot)),
\end{aligned}$$

where $\psi = \varrho \cdot \phi$, with $\phi : \mathcal{R} \to \mathbb{C}$ the solution to $\Box_{g_{M,a}}\phi = 0$ corresponding to initial data $(\psi|_\Sigma, \partial_\tau\psi|_\Sigma) = (\Psi, \Psi')$. The map $\mathcal{S}(\tau)$ is well-defined, by local-in-time energy estimates on $\widehat{\Sigma}_\tau$; see for example [AAG23a][Proposition 3.4]. The results in this section also hold with very minimal modifications on Kerr–de Sitter spacetimes for the equation $(\Box_{g_{M,a,\Lambda}} - \frac{2}{3}\Lambda)\phi = 0$.

From standard, local-in-time energy estimates, it follows moreover that $\{\mathcal{S}(\tau)\}_{\tau\in\mathbb{R}}$ form a C_0 (strongly continuous) semigroup, i.e.

$$\begin{aligned}\mathcal{S}(0) &= \mathrm{id},\\ \mathcal{S}(\tau_1)\mathcal{S}(\tau_2) &= \mathcal{S}_{\tau_1+\tau_2} \quad \text{for all } \tau_1,\tau_2\in\mathbb{R},\\ \lim_{\tau\to 0}\|\mathcal{S}(\tau)(\Psi,\Psi')-(\Psi,\Psi')\|_{H^{N+1}(\widehat{\Sigma})\times H^N(\widehat{\Sigma})} &= 0 \quad \text{for all } (\Psi,\Psi')\in H^{N+1}(\widehat{\Sigma})\times H^N(\widehat{\Sigma}).\end{aligned}$$

We define the *infinitesimal generator* corresponding to the semigroup $\{\mathcal{S}(\tau)\}_{\tau\in\mathbb{R}}$ as follows:

$$\begin{aligned}\mathcal{A}&: C^\infty(\widehat{\Sigma})\times C^\infty(\widehat{\Sigma})\to H^{N_+ +1}(\widehat{\Sigma})\times H^{N_+}(\widehat{\Sigma}),\\ \mathcal{A}(\Psi,\Psi') &= \lim_{\tau\to 0}\frac{1}{\tau}\big(\mathcal{S}(\tau)(\Psi,\Psi')-(\Psi,\Psi')\big) \text{ for all } (\Psi,\Psi')\in C^\infty(\widehat{\Sigma})\times C^\infty(\widehat{\Sigma}),\end{aligned}$$

where the limit is taken with respect to the $H^{N_+ +1}(\widehat{\Sigma})\times H^{N_+}(\widehat{\Sigma})$ norm.

Note that for $(\Psi,\Psi')\in C^\infty(\widehat{\Sigma})\times C^\infty(\widehat{\Sigma})$, the function $\psi=\varrho\phi$ with $\phi: D^+(\Sigma)\to\mathbb{C}$ the solution to $\Box_{g_{M,a}}\phi=0$ corresponding to initial data $(\psi|_\Sigma,\partial_\tau\psi|_\Sigma)=(\Psi,\Psi')$ satisfies

$$\mathcal{A}(\Psi,\Psi')=(\partial_\tau\psi|_\Sigma,\partial_\tau^2\psi|_\Sigma)\in C^\infty(\widehat{\Sigma})\times C^\infty(\widehat{\Sigma})=:(C^\infty(\widehat{\Sigma}))^2.$$

In the lemma below, we will relate $\mathcal{A}$ to the operator L_s introduced in §7.2. For this, we introduce the following domain for L_s:

$$\mathrm{dom}\,(L_s):=\{f\in C^\infty(\widehat{\Sigma})\cap H^{N_+}_{\sigma,R_\infty}\,|\,L_sf\in H^{N_+}_{\sigma,R_\infty}\}.$$

We define the Hilbert space $D^{N_+}_{\sigma,R_\infty}(L_s)$ as the closure of $\mathrm{dom}\,(L_s)$ with respect to the graph norm $\|\cdot\|_{H^{N_+}_{\sigma,R_\infty}}+\|L_s(\cdot)\|_{H^{N_+}_{\sigma,R_\infty}}$. By construction, L_s can be extended as a bounded linear operator to obtain:

$$L_s: D^{N_+}_{\sigma,R_\infty}(L_s)\to H^{N_+}_{\sigma,R_\infty}.$$

Proposition 9.1. *The infinitesimal generator of time translations $\mathcal{A}$ (corresponding to Kerr spacetimes) satisfies the following properties:*

(i) *Let $s\in\mathbb{C}$. Then we can express:*

$$\mathcal{A}-s\mathbf{1}=\begin{pmatrix}0 & \mathbf{1}\\ \mathbf{1} & P-s\mathbf{1}\end{pmatrix}\begin{pmatrix}\frac{1}{2M^2h(0)-Mxb_{\tau\tau}}L_s & 0\\ 0 & \mathbf{1}\end{pmatrix}\begin{pmatrix}\mathbf{1} & 0\\ -s\mathbf{1} & \mathbf{1}\end{pmatrix}. \tag{9.1}$$

(ii) *Assume that $D^{N_+}_{\sigma,R_\infty}(L_s)\subseteq H^{N_+ +1}_{\sigma,R_\infty,2}$ and that the inverse of L_s, $L_s^{-1}: H^{N_+}_{\sigma,R_\infty}\to D^{N_+}_{\sigma,R_\infty}(L_s)\subseteq H^{N_+ +1}_{\sigma,R_\infty,2}$, is a bounded linear operator with respect to the operator norm $\|\cdot\|_{H^{N_+}_{\sigma,R_\infty}\to H^{N_+ +1}_{\sigma,R_\infty,2}}$. Then the operator*

$$\begin{aligned}\mathcal{R}(s)&:\mathbf{H}^{N_+}_{\sigma,R_\infty}\to\mathbf{H}^{N_+}_{\sigma,R_\infty},\\ \mathcal{R}(s)&=\begin{pmatrix}\mathbf{1} & 0\\ s\mathbf{1} & \mathbf{1}\end{pmatrix}\begin{pmatrix}L_s^{-1}\circ(2M^2h(0)-Mxb_{\tau\tau})\mathbf{1} & 0\\ 0 & \mathbf{1}\end{pmatrix}\begin{pmatrix}-(P-s\mathbf{1}) & \mathbf{1}\\ \mathbf{1} & 0\end{pmatrix}\end{aligned}$$

is a bounded linear operator on $\mathbf{H}^{N_+}_{\sigma,R_\infty}$, which satisfies $(\mathcal{A}-s\mathbf{1})\mathcal{R}(s)(f,g)=(f,g)$ for all $(f,g)\in\mathbf{H}^{N_+}_{\sigma,R_\infty}\cap(C^\infty(\widehat{\Sigma}))^2$ and $\mathcal{R}(s)(\mathcal{A}-s\mathbf{1})(\Psi,\Psi')=(\Psi,\Psi')$ for all $(\Psi,\Psi')\in\mathrm{ran}\,\mathcal{R}(s)\cap(C^\infty(\widehat{\Sigma}))^2$.

(iii) *Assume that there exists a $s_0\in\mathbb{C}$, such that $H^{N_+}_{\sigma,R_\infty}(L_{s_0})\subseteq H^{N_+ +1}_{\sigma,R_\infty,2}$ and $L_{s_0}^{-1}: H^{N_+}_{\sigma,R_\infty}\to D^{N_+}_{\sigma,R_\infty}(L_{s_0})\subseteq H^{N_+ +1}_{\sigma,R_\infty,2}$ is a bounded linear operator with respect to the operator norm*

$$\|\cdot\|_{H^{N_+}_{\sigma,R_\infty}\to H^{N_+ +1}_{\sigma,R_\infty,2}}.$$

Then the operator $\mathcal{A}$ can be extended as a closed linear operator on the following domain:

$$\mathcal{A} : \mathbf{H}^{N_+}_{\sigma,R_\infty} \supseteq \operatorname{ran}\mathcal{R}(s_0) \to \mathbf{H}^{N_+}_{\sigma,R_\infty}$$

and

$$D^{N_+}_{\sigma,R_\infty}(L_s) \oplus H^{N_+}_{\sigma,R_\infty}(P) = \operatorname{ran}\mathcal{R}(s) = \operatorname{ran}\mathcal{R}(s_0)$$

for all $s \in \mathbb{C}$ such that L_s^{-1} is well-defined.

The above results generalize to Kerr–de Sitter spacetimes with suitably small $M^2\Lambda$ by replacing the factor $\frac{1}{2M^2h(0)-Mxb_{\tau\tau}}$ with an analogous factor, depending on the choice of future-horizon-intersecting, spacelike foliation.

Proof. We will restrict to Kerr in the proof. The Kerr–de Sitter case follows with minimal modifications. We compare the expression for $\varrho^3\Box_{g_{M,a}}\phi$ in (7.3) to the expression for L_s in (7.7) in order to obtain:

$$\begin{aligned}\varrho^3\Box_{g_{M,a}}\phi &= L_s\psi + 2\partial_x\partial_\tau\psi + Mxb_{x\tau}\cdot\partial_x\partial_\tau\psi + (Mxb_{\tau\tau} - 2M^2h(0))\partial_\tau^2\psi + Ma^2x^2\sin\vartheta b_{\tau\vartheta}\partial_\tau\partial_\vartheta\psi + Mxb_\tau\partial_\tau\psi\\ &\quad - 2s\partial_x\psi - sMxb_{x\tau}\cdot\partial_x\psi - s^2(Mxb_{\tau\tau} - 2M^2h(0))\psi - Ma^2sx^2\sin\vartheta b_{\tau\vartheta}\partial_\vartheta\psi - Msxb_\tau\psi\\ &= L_s\psi + (Mxb_{\tau\tau} - 2M^2h(0))\partial_\tau^2\psi + (2M^2h(0) - Mxb_{\tau\tau})P\partial_\tau\psi - s(2M^2h(0) - Mxb_{\tau\tau})P\psi + s^2(2M^2h(0) - Mxb_{\tau\tau})\psi.\end{aligned}$$

Using that $\Box_{g_{M,a}}\phi = 0$, we can therefore express:

$$\partial_\tau^2\psi = (2M^2h(0) - Mxb_{\tau\tau})^{-1}L_s\psi + P\partial_\tau\psi - sP\psi + s^2\mathbf{1}$$

and therefore, $\mathcal{A}$ takes the following form

$$\mathcal{A}(\Psi,\Psi') = \begin{pmatrix}\Psi'\\ (2M^2h(0) - Mxb_{\tau\tau})^{-1}L_s\Psi + (s^2\mathbf{1} - sP)\Psi + P\Psi'\end{pmatrix}.$$

The expression for $\mathcal{A} - s\mathbf{1}$ in (i) can then be easily verified.

We consider now (ii). We have that

$$\mathcal{R}(s)(f,g) = \begin{pmatrix}-L_s^{-1}\left((2M^2h(0) - Mxb_{\tau\tau})((P - s\mathbf{1})f - g)\right),\\ -sL_s^{-1}\left((2M^2h(0) - Mxb_{\tau\tau})((P - s\mathbf{1})f - g)\right) + f\end{pmatrix}. \tag{9.2}$$

We can use that $L_s^{-1} : H^{N_+}_{\sigma,R_\infty} \to D^{N_+}_{\sigma,R_\infty}(L_s) \subseteq H^{N_++1}_{\sigma,R_\infty,2}$ is a bounded linear operator and $f, Pf, g \in H^{N_+}_{\sigma,R_\infty}$ for $(f,g) \in \mathbf{H}^{N_+}_{\sigma,R_\infty}$ to ensure that $\mathcal{R}(s)$ is in fact well-defined and bounded on the domain $\mathbf{H}^{N_+}_{\sigma,R_\infty}$ and $\mathcal{R}(s_0)$.

Note moreover that $(f,g) \in \mathbf{H}^{N_+}_{\sigma,R_\infty} \cap (C^\infty(\widehat{\Sigma}))^2$ implies that $\mathcal{R}(f,g) \in (C^\infty(\widehat{\Sigma}))^2$. The property $(\mathcal{A} - s\mathbf{1})\mathcal{R}(s) = \mathcal{R}(s)(\mathcal{A} - s\mathbf{1})$ can then easily be verified by using the block diagonal form of $\mathcal{A} - s\mathbf{1}$ from (i) to formally invert the operator.

We will now prove (iii). First observe that we can extend the domain of $\mathcal{A}$ to $\operatorname{ran}\mathcal{R}(s_0)$ as follows. Let $(\Psi,\Psi') \in \operatorname{ran}\mathcal{R}(s_0)$. Then there exists an element $(f,g) \in \mathbf{H}^{N_+}_{\sigma,R_\infty}$ such that $(\Psi,\Psi') = \mathcal{R}(s_0)(f,g)$. We then define:

$$(\mathcal{A} - s\mathbf{1})(\Psi,\Psi') := (f,g)$$

By (ii), this extends the operator $\mathcal{A}$ defined on $\operatorname{ran}\mathcal{R}(s_0) \cap (C^\infty(\widehat{\Sigma}))^2$ to $\operatorname{ran}\mathcal{R}(s_0)$. Closedness of $\mathcal{A} - s\mathbf{1}$ (and hence $\mathcal{A}$) then follows immediately from the boundedness of its inverse $\mathcal{R}(s_0)$.

We will now prove the remaining statement in (iii). Let $(\Psi,\Psi') \in \operatorname{ran}\mathcal{R}(s)$. Then there exist $(f,g) \in \mathbf{H}^{N_+}_{\sigma,R_\infty}$ satisfying (9.2). From (i), we have that formally

$$(\mathcal{A} - s_0)(\Psi,\Psi') = (\mathcal{A} - s)(\Psi,\Psi') + (s - s_0)\begin{pmatrix}\Psi\\ \Psi'\end{pmatrix} = \begin{pmatrix}f + (s - s_0)\Psi\\ g + (s - s_0)\Psi'\end{pmatrix}.$$

Motivated by the above, using that $(\Psi,\Psi') = \mathcal{R}(s)(f,g)$, we define

$$\tilde{f} := f + (s - s_0)\Psi,$$

$$\tilde{g} := g + (s - s_0)\Psi'.$$

It then follows that $(\tilde{f}, \tilde{g}) \in \mathbf{H}^{N_+}_{\sigma,R_\infty}$. Furthermore, $\mathcal{R}(s_0)(\tilde{f}, \tilde{g}) = (\Psi, \Psi')$, so $(\Psi, \Psi') \in \operatorname{ran} \mathcal{R}(s_0)$. We conclude that $\operatorname{ran} \mathcal{R}(s) \subseteq \operatorname{ran} \mathcal{R}(s_0)$. Reversing the roles of s_0 and s in the above argument allows us to conclude that $\operatorname{ran} \mathcal{R}(s) = \operatorname{ran} \mathcal{R}(s_0)$. □

The operator $\mathcal{R}(s)$ from Proposition 9.1 satisfies the defining properties of the *resolvent operator* corresponding to $\mathcal{A} : \operatorname{ran} \mathcal{R}(s_0) \to \mathbf{H}^{N_+}_{\sigma,R_\infty}$, where $L_{s_0}^{-1}$ is well-defined. We can denote

$$(\mathcal{A} - s\mathbf{1})^{-1} := \mathcal{R}(s).$$

Corollary 9.2. *Assume that for $s \in O$, with $O \subset \mathbb{C}$ a connected domain, either $L_s^{-1} : H^{N_+}_{\sigma,R_\infty} \to D^{N_+}_{\sigma,R_\infty}(L_{s_0}) \subseteq H^{N_++1}_{\sigma,R_\infty,2}$ is bounded with respect to the operator norm $\|\cdot\|_{H^{N_+}_{\sigma,R_\infty} \to H^{N_++1}_{\sigma,R_\infty,2}}$, or $\ker L_s \neq \{0\}$. Assume also that there exists a $s_0 \in O$ such that $\ker L_{s_0} = \{0\}$.*

Consider $\mathcal{A} : \operatorname{ran} \mathcal{R}(s_0) \to \mathbf{H}^{N_+}_{\sigma,R_\infty}$. Then the spectrum $\operatorname{Spect}(\mathcal{A})$ satisfies:

$$\operatorname{Spect}(\mathcal{A}) \cap O = \operatorname{Spect}_{\text{point}}(\mathcal{A}) \cap O = \{s \in O \mid \ker L_s \neq \{0\}\}.$$

Furthermore, $(\Psi, \Psi') \in \operatorname{ran} \mathcal{R}(s_0)$ is an eigenvector of $\mathcal{A}$ if and only if $(\Psi, \Psi') = (\widehat{\psi}_s, s\widehat{\psi}_s)$, with $\widehat{\psi}_s \in \ker L_s$.

Proof. In view of the form of $\mathcal{R}(s)$ from (ii) of Proposition 9.1, boundedness of $\mathcal{R}(s)$ is equivalent to boundedness of L_s^{-1}. Hence, $s \in \operatorname{Spect}(\mathcal{A}) \cap O$ if and only if L_s^{-1} is not bounded. By assumption, the latter implies that $\ker L_s \neq \{0\}$. Let $\widehat{\psi}_s \in \ker L_s$. Then

$$\mathcal{A}(\widehat{\psi}_s, s\widehat{\psi}_s) = s \cdot (\widehat{\psi}_s, s\widehat{\psi}_s) \in \mathbf{H}^{N_+}_{\sigma,R_\infty}$$

and $(f, g) = (\mathcal{A} - s_0\mathbf{1})(\widehat{\psi}_s, s\widehat{\psi}_s) = (s - s_0)(\widehat{\psi}_s, s\widehat{\psi}_s) \in \mathbf{H}^{N_+}_{\sigma,R_\infty}$, so $\mathcal{R}(s_0)(f, g) = \begin{pmatrix} \widehat{\psi}_s \\ s\widehat{\psi}_s \end{pmatrix}$ and therefore $(\widehat{\psi}, s\widehat{\psi}_s) \in \operatorname{ran} \mathcal{R}(s_0)$, so $s \in \operatorname{Spect}_{\text{point}}(\mathcal{A}) \cap O$, with $\mathcal{A}$ defined on the domain $\operatorname{ran} \mathcal{R}(s_0)$. □

The *Hermitian adjoint* of $L_s : D^{N_+}_{\sigma,R_\infty}(L_s) \supset \operatorname{dom}(L_s) \to H^{N_+}_{\sigma,R_\infty}$, with respect to the $H^{N_+}_{\sigma,R_\infty}$ inner product is defined as follows: let

$$D^{N_+}_{\sigma,R_\infty}(L_s^*) := \left\{ f \in H^{N_+}_{\sigma,R_\infty}, \langle f, L_s(\cdot)\rangle_{H^{N_+}_{\sigma,R_\infty}} : D^{N_+}_{\sigma,R_\infty}(L_s) \supset \operatorname{dom}(L_s) \to \mathbb{C} \text{ is bounded} \right\}.$$

By the Riesz representation theorem, we have that for all $f \in D^{N_+}_{\sigma,R_\infty}(L_s^*)$, there exists a unique $g \in D^{N_+}_{\sigma,R_\infty}(L_s)$, such that $\langle g, \widehat{\psi}\rangle_{H^{N_+}_{\sigma,R_\infty}} = \langle f, L_s\widehat{\psi}\rangle_{H^{N_+}_{\sigma,R_\infty}}$ for all $\widehat{\psi} \in \operatorname{dom}(L_s)$. Then we define the Hermitian adjoint of L_s by $L_s^* : D^{N_+}_{\sigma,R_\infty}(L_s^*) \to H^{N_+}_{\sigma,R_\infty}$, with $L_s^* f = g$, with f, g as above.

Similarly, we define the Hermitian adjoint of $\mathcal{A}$, with respect to the $\mathbf{H}^{N_+}_{\sigma,R_\infty}$ inner product, as follows: let $s_0 \in \mathbb{C}$ be such that $L_{s_0}^{-1}$ is well-defined, then

$$\mathbf{D}^{N_+}_{\sigma,R_\infty}(\mathcal{A}^*) := \left\{ (f, g) \in \mathbf{H}^{N_+}_{\sigma,R_\infty}, \langle f, \mathcal{A}(\cdot)\rangle_{\mathbf{H}^{N_+}_{\sigma,R_\infty}} : \operatorname{ran} \mathcal{R}(s_0) \to \mathbb{C} \text{ is bounded} \right\}.$$

By the Riesz representation theorem, we have that for all $(f, g) \in \mathbf{D}^{N_+}_{\sigma,R_\infty}(\mathcal{A}^*)$, there exists a unique $(\Phi, \Phi') \in \operatorname{ran} \mathcal{R}(s_0)$, such that $\langle (\Phi, \Phi'), (\Psi, \Psi')\rangle_{\mathbf{H}^{N_+}_{\sigma,R_\infty}} = \langle (f, g), \mathcal{A}(\Psi, \Psi')\rangle_{\mathbf{H}^{N_+}_{\sigma,R_\infty}}$ for all $(\Psi, \Psi') \in (C^\infty(\hat{\Sigma}))^2 \cap \mathcal{R}(s_0)$. Then we define the Hermitian adjoint of $\mathcal{A}$ by $\mathcal{A}^* : \mathbf{D}^{N_+}_{\sigma,R_\infty}(\mathcal{A}^*) \to \mathbf{H}^{N_+}_{\sigma,R_\infty}$, with $\mathcal{A}^*(f, g) = (\Phi, \Phi')$, with (f, g) and (Φ, Φ') as above.

Taking the formal adjoint of the right-hand side of (9.1), we obtain

$$\mathcal{A}^\dagger - \overline{s}\mathbf{1} := \begin{pmatrix} \mathbf{1} & -\overline{s}\mathbf{1} \\ 0 & \mathbf{1} \end{pmatrix} \begin{pmatrix} \frac{1}{2M^2h(0)-Mxb_{\tau\tau}} L_s^* & 0 \\ 0 & \mathbf{1} \end{pmatrix} \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & P^* - \overline{s}\mathbf{1} \end{pmatrix}.$$

From the above expression, it follows that $\mathcal{A}^\dagger(f,g)$ is well-defined and agrees with $\mathcal{A}^*(f,g)$ if $g \in D^{N_+}_{\sigma,R_\infty}(L_s^*) \cap D^{N_+}_{\sigma,R_\infty}(P^*)$ and $f \in H^{N_+}_{\sigma,R_\infty}$, with $D^{N_+}_{\sigma,R_\infty}(P^*)$ the domain of the Hermitian adjoint P^*, with respect to the inner product on $H^{N_+}_{\sigma,R_\infty}$.

A direct corollary of the standard orthogonality property between the kernel of a linear operator and the range of its Hermitian adjoint results in the following:

Corollary 9.3. *Assume additionally that there exists a $s_0 \in \mathbb{C}$ such that $\ker L_{s_0} = \{0\}$. We have that*

$$\begin{aligned} \operatorname{ran}(\mathcal{A}^* - \overline{s}\mathbf{1}) &\perp \ker(\mathcal{A} - s\mathbf{1}), \\ \ker(\mathcal{A}^* - \overline{\tilde{s}}\mathbf{1}) &\perp \ker(\mathcal{A} - s\mathbf{1}) \quad \text{if } s \neq \tilde{s}. \end{aligned}$$

Let $\widehat{\psi}_s \in \ker L_s$ and denote $(\Psi_s, \Psi_s') = (\widehat{\psi}_s, s\widehat{\psi}_s) \in \ker(\mathcal{A} - s\mathbf{1})$. Let $g_{\tilde{s}} \in \ker L_{\tilde{s}}^*$ and denote $(\tilde{\Psi}_{\tilde{s}}, \tilde{\Psi}_{\tilde{s}}') = (-(P^* - \overline{\tilde{s}})g_{\tilde{s}}, g_{\tilde{s}})$. Then $(\tilde{\Psi}_{\tilde{s}}, \tilde{\Psi}_{\tilde{s}}') \in \ker(\mathcal{A}^\dagger - \overline{\tilde{s}}\mathbf{1})$ and

$$\langle (\Psi_s, \Psi_s'), (\tilde{\Psi}_{\tilde{s}}, \tilde{\Psi}_{\tilde{s}}') \rangle_{\mathbf{H}^{N_+}_{\sigma,R_\infty}} = 0$$

if $s \neq \tilde{s}$.

10. Precise statements of the main theorems

In this section, we give precise statements of the main theorems of the paper, making use of the preliminaries introduced in §§6–9, and we give precise definitions of the notions of *regularity quasinormal modes* and *regularity quasinormal frequencies*.

10.1. Regularity quasinormal frequencies.

The following theorem provides a precise formulation of Theorem 1.1(i)–(ii) and includes also an additional statement on orthogonality properties of regularity quasinormal modes.

Theorem 10.1. *Let $\sigma \in \mathbb{R}$ and $N_+ \in \mathbb{N}_0$. Let*

$$\begin{aligned} \Omega_\sigma := \Bigg\{ x + iy \in \mathbb{C} \,\Big|\, & -\frac{4-b}{2b}\sqrt{4-b^2}\frac{1}{2}|\sigma| < x < 0, \\ & y \notin \left[-\left(-1 + 4b^{-2} + \frac{b}{2}\right)\frac{1}{2}|\sigma|, \left(-1 + 4b^{-2} + \frac{b}{2}\right)\frac{1}{2}|\sigma| \right], \ b \in (0,2) \Bigg\} \\ & \cup \left\{ x + iy \in \mathbb{C} \,|\, x \in [0,\infty),\ x^2 + y^2 > \frac{1}{4}\sigma^2 \right\} \end{aligned}$$

and

$$\Omega_{\sigma,N_+} := \Omega_\sigma \cap \left\{ \operatorname{Re} z > -\left(\frac{1}{2} + N_+\right)\kappa_+ \right\}.$$

(i) *Then there exists a suitably large $R_\infty > r_+$ such that the following restrictions of the infinitesimal generator of time translations corresponding to $\mathcal{S}(\tau)$ are closed linear operators:*

$$\mathcal{A} : \mathbf{H}^{N_+}_{\sigma,R_\infty} \supset D^{N_+}_{\sigma,R_\infty}(L_s) \oplus H^{N_+}_{\sigma,R_\infty}(P) \to \mathbf{H}^{N_+}_{\sigma,R_\infty}$$

and

$$\operatorname{Spect}(\mathcal{A}) \cap \Omega_{\sigma,N_+} = \operatorname{Spect}_{\text{point}}(\mathcal{A}) \cap \Omega_{\sigma,N_+}$$

consists of isolated eigenvalues, which are independent of the precise choice of foliation-defining function $\mathbb{h}$. *We define* ***regularity quasinormal frequencies*** *to be the union of eigenvalues:*

$$\mathcal{Q}_{\mathrm{reg}} := \bigcup_{N_+\in\mathbb{N}_0,\sigma\in\mathbb{R}} \mathrm{Spect}_{\mathrm{point}}\left(\mathcal{A}\right)\cap\Omega_{\sigma,N_+}$$

and ***regularity quasinormal modes*** *to be the corresponding eigenvectors.*

(ii) *The union* $\Omega := \bigcup_{N_+\in\mathbb{N}_0,\sigma\in\mathbb{R}}\Omega_{\sigma,N_+}$ *satisfies*

$$\Omega = \{z\in\mathbb{C}\setminus\{0\}\,,\,|\arg z|<\varphi_0\}\,,$$

with $\varphi_0 = \pi - \arcsin\left(\sqrt{\frac{1}{\gamma_{GR}}}\right)\approx 0.712\pi$, *with* $\gamma_{GR} := \frac{1}{2}(1+\sqrt{5})$ *the golden ratio. The set of quasinormal frequencies* $\mathcal{Q}_{\mathrm{reg}}$ *has no accumulation points in* Ω *and is independent of the choice of* R_∞.

(iii) *The eigenspaces of* $\mathcal{A}:\mathbf{H}^{N_+}_{\sigma,R_\infty}\supset D^{N_+}_{\sigma,R_\infty}(L_s)\oplus H^{N_+}_{\sigma,R_\infty}(P)\to\mathbf{H}^{N_+}_{\sigma,R_\infty}$ *are finite dimensional.*

(iv) *We define* ***regularity quasinormal co-modes*** *to be eigenvectors of* $\mathcal{A}^\dagger$ *in* $((D^{N_+}_{\sigma,R_\infty})(L_s^*)\cap D^{N_+}_{\sigma,R_\infty}(P^*))\oplus H^{N_+}_{\sigma,R_\infty}$. *The following orthogonality relations hold with respect to the inner product on* $\mathbf{H}^{N_+}_{\sigma,R_\infty}$:

$$\mathrm{ran}(\mathcal{A}^\dagger-\overline{s}\mathbf{1})\perp\ker(\mathcal{A}-s\mathbf{1}),$$
$$\ker(\mathcal{A}^\dagger-\overline{\tilde{s}}\mathbf{1})\perp\ker(\mathcal{A}-s\mathbf{1})\quad\text{if } s\neq\tilde{s}.$$

Proof. Theorem 10.1 follows directly from Theorem 10.2 below, in combination with Corollaries 9.2 and 9.3. □

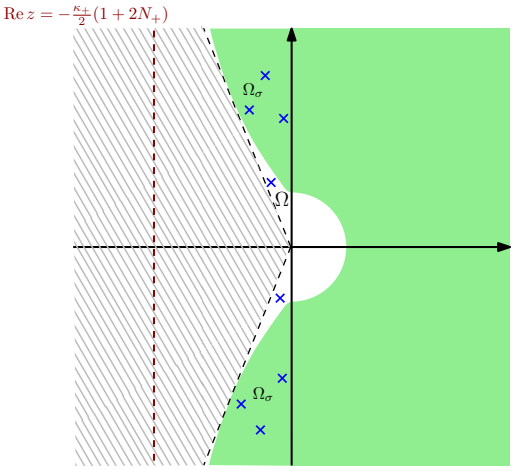


FIGURE 3. A (schematic) pictorial representation of the following subsets of $\mathbb{C}$: Ω (the sector of the complex plane that is not dashed), Ω_σ (the shaded region), and possible regularity quasinormal frequencies represented by crosses.

The proof of Theorem 10.1 uses crucially the meromorphicity of the inverse L_s^{-1} of the Laplace transform of the wave operator in Kerr and $L_{s,\Lambda}^{-1}$ in Kerr–de Sitter (see §18 for a definition of $L_{s,\Lambda}$). Meromorphicity in this context is defined as follows:

Definition 10.1. *Let* $O\subset\mathbb{C}$ *be a connected and open subset,* X, Y *Banach spaces and* $B(X,Y)$ *the space of bounded linear operators from* X *to* Y. *Let* $A: O\to B(X,Y)$ *be a family of bounded operators. Then* A *or* $\{A(s)\}$ *is* meromorphic *if for any* $s_0\in O$ *there exists a small neighbourhood*

U of s_0, such that for all $s \in U \setminus \{s_0\}$, we can write $A(s) = A_0(s) + \sum_{k=1}^{m} (s-s_0)^{-k} A_k$ in U, where $A_0(s)$ is holomorphic at s_0, i.e. $\lim_{s\to s_0} \frac{A_0(s)-A_0(s_0)}{s-s_0}$ is well-defined with respect to the operator topology on $B(X,Y)$, and for all $1 \leq i \leq m$ $A_i \in B(X,Y)$ are finite-rank operators. If at least one of the operators A_i with $1 \leq i \leq m$ is non-zero, we refer to s_0 as a pole *of A.*

The orthogonality properties of regularity quasinormal modes are related to the cokernel of L_s, which is defined as follows:

Definition 10.2. *Let $A : X \to Y$ be a linear operator between Banach spaces X, Y. Then the* cokernel of A *is defined as the quotient space* $\operatorname{coker} A = Y/\operatorname{ran} A$.

The following theorem is a precise version of Theorem 4.1(i)–(ii) and includes an additional statement on the orthogonality relations between the kernel and cokernel of L_s. *This theorem constitutes the core of the present paper.*

Theorem 10.2. *Let $\sigma \in \mathbb{R}$ and $N_+ \in \mathbb{N}_0$.*

(i) *For suitably large $R_\infty > r_+$ and suitably small $M^2\Lambda \geq 0$, the family of operators $\{L_{s,\Lambda}^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_++1}_{\sigma,R_\infty,2}\}$, with $s \in \Omega_{\sigma,N_+}$, is meromorphic. We denote with $\mathscr{Q}_{\sigma,R_\infty,N_+}$ the set of poles of $\{L_{s,\Lambda}^{-1}\}$, which does not depend on the precise choice of foliation-defining function $\mathbb{h}$. Furthermore, there exists an $s_0 \geq 0$ such that the family of operators is holomorphic when restricted to $s \in \{\operatorname{Re} z > s_0\}$.*

(ii) *Let $\widetilde{R}_\infty \geq R_\infty$, $\widetilde{N}_+ \in \mathbb{N}_0$ and $\widetilde{\sigma} \in \mathbb{R}$. Then*

$$\mathscr{Q}_{\sigma,\widetilde{R}_\infty,\widetilde{N}_+} \cap \Omega_{\sigma,N_+} \cap \Omega_{\widetilde{\sigma},\widetilde{N}_+} = \mathscr{Q}_{\sigma,R_\infty,N_+} \cap \Omega_{\sigma,N_+} \cap \Omega_{\widetilde{\sigma},\widetilde{N}_+}$$

and $\bigcup_{N_+\in\mathbb{N}_0,\sigma\in\mathbb{R}} \mathscr{Q}_{\sigma,R_\infty,N_+}$ has no accumulation points in Ω.

(iii) *Let $s \in \Omega_{\sigma,N_+}$. Then the space $\ker L_{s,\Lambda}$ is finite dimensional with $\dim\ker L_{s,\Lambda} = \dim\operatorname{coker} L_{s,\Lambda}$ and $\operatorname{ran} L_{s,\Lambda}^* = (\ker L_{s,\Lambda})^\perp \subset H^{N_++1}_{\sigma,R_\infty,2}$, with $L_{s,\Lambda}^*$ the Hermitian adjoint and $\operatorname{coker} L_{s,\Lambda}^*$ isomorphic to $\ker L_{s,\Lambda}$.*

Proof. By standard, local energy estimates for $\Box_{g_{M,a}}\phi = 0$ and a Grönwall inequality, one can obtain uniform constants $C, s_0 > 0$ such that:

$$\|(\phi, \partial_\tau\phi)\|_{H^1(\Sigma)\times L^2(\Sigma)}(\tau) \leq Ce^{s_0\tau}\|(\phi,\partial_\tau\phi)\|_{H^1(\Sigma)\times L^2(\Sigma)}(0). \tag{10.1}$$

Suppose $\{L_{s,\Lambda}^{-1}\}$ had a pole in $s_0 \in \{\operatorname{Re} z > s_0\}$. Then the solution ϕ corresponding to initial data $(\phi_{s_0}, s\phi_{s_0})$ with $\psi_{s_0} = \varrho\phi_{s_0} \in \ker L_{s_0,\Lambda}$ takes the form $(\phi,\partial_\tau\phi)(\tau) = (e^{s_0\tau}\phi_{s_0}, s\phi_{s_0})$ and hence

$$\|(\phi, \partial_\tau\phi)\|_{H^1(\Sigma)\times L^2(\Sigma)}(\tau) = e^{(\operatorname{Re} s_0)\tau}\|(\phi,\partial_\tau\phi)\|_{H^1(\Sigma)\times L^2(\Sigma)}(0)$$

for all $\tau \geq 0$, which is in contradiction with (12.11). Hence, $\{L_{s,\Lambda}^{-1}\}$ is holomorphic when restricted to $\{\operatorname{Re} z > s_0\}$. In the $\Lambda = 0$ case, the remaining statement in (i) as well as (iii) follow from Proposition 16.2 below. Part (ii) follows from Proposition 16.4 below. This is then generalized to $\Lambda \geq 0$ in Corollary 18.3. □

Remark 10.1. *Uniform boundedness (and decay) of solutions to the wave equation $\Box_{g_{M,a}}\phi = 0$, established by Dafermos–Rodnianski–Shlapentokh-Rothman in* [DRSR16] *implies that in the $\Lambda = 0$ case, the set $\mathscr{Q}_{\mathrm{reg}}$ is in fact entirely contained in the left-half plane $\{\operatorname{Re} z < 0\}$ (s_0 in the proof of Theorem 10.2(i) can be set to 0). The validity of Theorem 10.2, however, is independent of a boundedness statement and applies to settings where such a boundedness statement is false, like the class of wave equations with potentials on Kerr studied in* [Mos17].

Closely related to uniform boundedness for $\Box_{g_{M,a}}\phi = 0$, is the statement of "mode stability" [Whi89, SR15]*, which implies the absence of regularity quasinormal modes of the form (10.3) below. Since regularity quasinormal modes need not satisfy (10.3), mode stability alone is not enough to ensure that $\mathscr{Q}_{\mathrm{reg}}$ is contained in the left-half plane when $\Lambda = 0$; see Remark 10.2 below.*

For $\Lambda > 0$ with $M^2\Lambda \ll 1$, the results of [Hin24] imply that $\mathcal{Q}_{\mathrm{reg}}$ is contained in $\{\mathrm{Re}\, z < 0\}$. Note that in the general $\Lambda > 0$ case however, not even mode stability is known; see [CT22] for partial results in this direction.

Remark 10.2. *Consider the following angular ODE:*

$$-\frac{1}{\sin\theta}\partial_\theta(\sin\theta\partial_\theta S_{m\ell}) + \left[\frac{1}{\sin^2\theta}m^2 + a^2\omega^2\sin^2\theta\right] S_{m\ell} = \Lambda_{m\ell}(a\omega)S_{m\ell}, \tag{10.2}$$

with $S_{m\ell}(\theta; a\omega)$ in $L^2(\sin\theta d\theta)$ the oblate spheroidal harmonics. *When $a = 0$, (10.2) is simply the Laplace equation on $\mathbb{S}^2$ restricted to functions with azimuthal angular dependence $e^{im\varphi}$ and the solutions are spherical harmonics.*

When $\omega \in \mathbb{R}$, (10.2) admits solutions for a discrete set of $\Lambda_{m\ell}(a\omega) \in \mathbb{R}$, labelled by $\ell \in \mathbb{N}_{\geq |m|}$ such that $\Lambda_{m\ell}(a\omega) \geq \Lambda_{m\ell'}(a\omega)$ if $\ell \geq \ell'$. Furthermore, when $\omega \in \mathbb{R}$, the functions $\{S_{m\ell}(\theta; a\omega)\}_{m\in\mathbb{Z}, l\geq|m|}$ form an orthonormal basis of $L^2(\sin\theta d\theta)$, the self-adjointness of the differential operator on the left-hand side of (10.2) and the compactness of its inverse.

The functions $\omega \mapsto \Lambda_{m\ell}(a\omega)$ can be analytically continued to the rest of the complex plane as complex functions, with a finite number of branch points, for which the corresponding (10.2) admits solutions in $L^2(\sin\theta d\theta)$; see for example [MS54][Proposition 5 of Chapter 1.5, Proposition 2 of Chapter 3.22] for a proof in the context of general ODE and [TdC20][Proposition 2.1] for a discussion on this matter in the context of (10.2). However, when $\omega \notin \mathbb{R}$, $\{S_{m\ell}(\theta; a\omega)\}_{m\in\mathbb{Z}, l\geq|m|}$ no longer need to form an orthonormal basis.

In much of the physics literature, see for example [Lea85], quasinormal mode solutions ϕ to the wave equation (1.1) are required to take the separated form

$$\phi(t, r_*, \theta, \varphi) = e^{-i\omega t}\frac{u_{\ell m}(r_*;\omega)}{\sqrt{r^2+a^2}}S_{m\ell}(\theta; a\omega)e^{im\varphi}, \tag{10.3}$$

with $S_{m\ell}(\theta; a\omega)$ a solution to (10.2) and $u_{\ell m}(r;\omega)$ a solution to the ODE:

$$u''_{\ell m} + (\omega^2 - V^{(a\omega)}_{m\ell})u_{\ell m} = 0, \tag{10.4}$$

where $V^{(a\omega)}_{m\ell}(r_*)$ is a potential function that depends on $\Lambda_{m\ell}(a\omega)$.

In the present paper, we do <u>not</u> impose the restriction (10.3) and we instead allow for more general solutions of the form:

$$\phi(t, r, \theta, \varphi) = e^{-i\omega t}\sum_{m\in\mathbb{Z}}\hat{\phi}_m(r,\theta)e^{im\varphi}.$$

In view of the finite dimensionality of the space of regularity quasinormal modes that follows from Theorem 10.1(iii), we can infer that quasinormal modes corresponding to a single quasinormal frequency $s = -i\omega$ must be supported on a finite number of m. If $\{S_{m\ell}(\theta; a\omega)\}_{m\in\mathbb{Z}, l\geq|m|}$ formed an orthonormal basis for $L^2(\mathbb{S}^2)$, then we would also be able to conclude that any regularity quasinormal mode can be written as a finite linear combination of regularity quasinormal modes satisfying (10.3). This is the case when $a = 0$.

As mentioned above, we are not, however, guaranteed an orthonormal basis of functions $S_{m\ell}$ when $a \neq 0$, so we cannot draw such a conclusion. The set of regularity quasinormal frequencies could therefore in principle be <u>strictly larger</u> than the set of frequencies corresponding to quasinormal modes that can be separated as in (10.3).

10.2. **Application 1: stability of the quasinormal spectrum and convergence of Kerr–de Sitter quasinormal frequencies.** We first give a definition of *stability* of the spectrum of a family of linear operators that is suitable for the setting of the present paper.

Definition 10.3 (Stability of the spectrum for a family of restricted operators)**.** *Let $O = \bigcup_{\alpha\in I} O_\alpha \subseteq \mathbb{C}$, with $O_\alpha \subset \mathbb{C}$ open and I an index set. Let $\mathbf{H}$ be a Hilbert space and let $A : \mathrm{dom}(A) \subseteq \mathbf{H} \to \mathbf{H}$*

be a densely defined linear operator. Let $\mathbf{H}'_\alpha \subseteq \mathbf{H}_\alpha \subseteq \mathbf{H}$ *be families of closed subspaces, such that the following restrictions of* A *are densely defined operators:* $A : \mathbf{H}'_\alpha \supseteq \mathrm{dom}(A_\alpha) \to \mathbf{H}_\alpha$. *Denote the corresponding restricted operators* A_α.

Let B *be a linear operator, such that its restrictions to* $\mathbf{H}'_\alpha$ *are operators* $B_\alpha : \mathbf{H}'_\alpha \to \mathbf{H}_\alpha$ *that are bounded with respect to the operator norms* $\|\cdot\|_{\mathbf{H}'_\alpha \to \mathbf{H}_\alpha}$ *for all* $\alpha \in I$.

Then we say that A *has a* stable spectrum *in* O *in the direction* B, *with respect to the families of subspaces* $\{\mathbf{H}_\alpha, \mathbf{H}'_\alpha\}_{\alpha\in I}$, *if the following holds: for any bounded domain* $K \subset O$ *and for all* $\epsilon > 0$, *there exists a* $\delta_0 > 0$ *such that for all* $\alpha \in I$ *and for all* $0 < \delta \leq \delta_0$:

(1) $\mathrm{Spect}(A_\alpha + \delta \cdot B_\alpha) \cap O_\alpha \cap K \subseteq \bigcup_{s\in \mathrm{Spect}(A_\alpha)\cap O_\alpha} D_\epsilon(s)$, *with* $D_\epsilon(s)$ *the open disc of radius* ϵ *around* $s \in \mathbb{C}$.

(2) For all $s \in \mathrm{Spect}(A_\alpha) \cap O_\alpha \cap K$,

$$\mathrm{Spect}(A_\alpha + \delta \cdot B_\alpha) \cap K \cap D_\epsilon(s) \neq \emptyset.$$

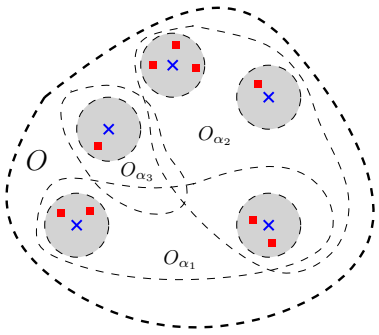


FIGURE 4. A pictorial representation of stability of the spectrum in the case of a spectrum of isolated eigenvalues of A_α represented by crosses. The eigenvalues of $A_\alpha + \delta \cdot B_\alpha$ are represented by squares and the shaded disks represent $D_\epsilon(s)$ with $s \in \mathrm{Spect}(A_\alpha)$.

Using the above definition, the following theorem gives a precise formulation of Theorem 1.2(i):

Theorem 10.3. *Let* $\mathcal{A}_{\delta\cdot Q}$ *denote the infinitesimal generator of time translations corresponding to the following perturbed wave operator:*

$$\Box_{g_{M,a}} + \delta \cdot \varrho^{-3} Q(\varrho\cdot)$$

with Q *the following second-order differential operator:*

$$Q = Q^{\tau\tau}(r,\vartheta,\varphi_*)\partial_\tau^2 + Q^{\tau x}(r,\vartheta,\varphi_*)\partial_\tau\partial_x + Q^{\tau\vartheta}(r,\vartheta,\varphi_*)\partial_\tau\partial_\vartheta + Q^{\tau\varphi}(r,\vartheta,\varphi_*)\partial_\tau\partial_{\varphi_*} + Q^{\tau}(r,\vartheta,\varphi_*)\partial_\tau + \tilde{Q},$$

where $Q^\tau, Q^{\tau\tau}, Q^{\tau x}, Q^{\tau\vartheta}, Q^{\tau\varphi_*}$ *are bounded and* $\tilde{Q}$ *is a differential operator of order* ≤ 2 *acting functions on* Σ, *such that* $\tilde{Q}_s = \tilde{Q} : H^{N_+}_{\sigma,R_\infty,2} \to H^{N_+}_{\sigma,R_\infty}$ *is bounded for all* $|\sigma| < \sigma_0$, *with* $\sigma_0 > 0$ *or* $\sigma_0 = \infty$. *Let* B *be the operator satisfying*

$$\mathcal{A} + \delta \cdot B = \mathcal{A}_{\delta\cdot Q}.$$

Let $s_0 \in \mathbb{C}$ *be such that* $L_{s_0}^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_++1}_{\sigma,R_\infty,2}$ *is well-defined. Then the spectrum of* $\mathcal{A} : \mathrm{dom}(\mathcal{A}) \supset \mathbf{H}^1 \to \mathbf{H}^1$ *is stable in* $\{\mathrm{Re}\, s > -(\frac{1}{2} + N_+)\kappa_+\} \cap D_{\sigma_0^2}(0) \cap \Omega = D_{\sigma_0^2}(0) \cap \bigcup_{\sigma\in\mathbb{R}} \Omega_{\sigma,N_+} \subset \mathbb{C}$ *with respect to the families of subspaces* $\{\mathbf{H}^{N_+}_\sigma, D^{N_+}_{\sigma,R_\infty}(L_{s_0}) \oplus H^{N_++1}_{\sigma,R_\infty,1})\}_{\sigma\in(-\sigma_0,\sigma_0)}$ *in the direction* B.

In particular, if Q *vanishes on* $\Sigma \cap \{r \geq R_\infty\}$ *and has smooth coefficients, then the above stability result applies in the full* Ω.

Proof. By a slight modification of the proof of Proposition 9.1, we obtain the following relation:

$$\mathcal{A}_{\delta\cdot Q} - s\mathbf{1} = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & P_{\delta\cdot Q} - s\mathbf{1} \end{pmatrix} \begin{pmatrix} \frac{1}{2M^2h(0) - Mxb_{\tau\tau} - \delta\cdot Q^{\tau\tau}}(L_s + \delta\cdot\tilde{Q}) & 0 \\ 0 & \mathbf{1} \end{pmatrix} \begin{pmatrix} \mathbf{1} & 0 \\ -s\mathbf{1} & \mathbf{1} \end{pmatrix},$$

with $P_{\delta\cdot Q} = \frac{2M^2h(0) - Mxb_{\tau\tau}}{2M^2h(0) - Mxb_{\tau\tau} - \delta\cdot Q^{\tau\tau}} P + \frac{\delta}{2M^2h(0) - Mxb_{\tau\tau} - \delta\cdot Q^{\tau\tau}} \left(Q^{\tau x}\partial_x + Q^{\tau\vartheta}\partial_\vartheta + Q^{\tau\varphi_*}\partial_{\varphi_*} + Q^{\tau}\mathbf{1}\right)$.

Note finally, that for all $\delta > 0$ $P_{\delta\cdot Q} f \in H^{N_+}_{\sigma,R_\infty}$ if $f \in H^{N_+ + 1}_{\sigma,R_\infty,1}$, so we can restrict the domain of $\mathcal{A}_{\delta\cdot Q}$ to $D^{N_+}_{\sigma,R_\infty}(L_{s_0}) \oplus H^{N_+ + 1}_{\sigma,R_\infty,1}$.

By the above expression for $\mathcal{A}_{\delta\cdot Q} - s\mathbf{1}$ and Corollary 9.2, invertibility of $\mathcal{A}_{\delta\cdot Q} - s\mathbf{1}$ is equivalent to invertibility of $L_s + \delta\cdot\tilde{Q} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+ + 1}_{\sigma,R_\infty,2}$, so we can apply Proposition 17.1 below to obtain the desired stability statement. □

The following theorem is a precise version of Theorem 4.2(ii) (which leads to Theorem 1.2(ii)).

Theorem 10.4. *Let $\{\Lambda_n\}$ be a sequence of cosmological constants satisfying $M^2\Lambda_n \ll 1$ and $\Lambda_n \downarrow 0$. Suppose that there exists an $s_* \in \Omega$ with $\ker L_{s_*}^{-1} \neq \{0\}$ and consider $\sigma \in \mathbb{R}$ and $N_+ \in \mathbb{N}$ such that $s_* \in \Omega_{\sigma,N_+}$.*

Then there exists a sequence $s_n \in \Omega_{\sigma,N_+}$, such that $\ker L^{-1}_{s_n,\Lambda_n} \neq \{0\}$ and $s_n \to s_*$ as $n \to \infty$. Furthermore, for all $\widehat{\psi} \in \ker L^{-1}_{s_*}$, there exists a sequence $\{\widehat{\psi}_n\}$ in $H^{N_+}_{\sigma,R_\infty}$ with $\widehat{\psi}_n \in \ker L^{-1}_{s_n,\Lambda_n}$ and

$$\|\widehat{\psi} - \widehat{\psi}_n\|_{H^{N_+}_{\sigma,R_\infty}} \to 0 \quad (n \to \infty).$$

Conversely, let $\{s_n\}$ be a convergent sequence of elements in $\ker L_{s,\Lambda_n}$ and $\Lambda_n \to 0$ as $n \to \infty$, such that the limit s_* satisfies: $s_* \in \{|\arg z| < \varphi_0\}$. Then $s_* \in \ker L_{s,0}$.

Proof. In the context of the κ-regularized operator $L_{s,\kappa}$, this proposition is proved in Lemma 16.3. This is then generalized to the true Kerr–de Sitter operator $L_{s,\Lambda}$ in §18. □

10.3. **Application 2: a characterization of scattering resonances.** In order to justify the use of the term "regularity quasinormal frequencies", we need to make a connection with other definitions of "quasinormal frequencies" that have previously appeared in the literature.

It is common to define "quasinormal frequencies" as scattering resonances associated to the meromorphic continuation of the cut-off resolvent operator corresponding to a "standard" time function whose level sets are asymptotically flat hypersurfaces. In the case of Kerr, this is typically take to be the Boyer–Lindquist time function t. One first considers the operator

$$A(\omega) = e^{i\omega t}\Box_{g_{M,a}}(e^{-i\omega t}\cdot(\cdot)),$$

which can be inverted on suitable spaces of functions on $\Sigma' = \{t = 0\}$ with so-called outgoing boundary conditions and where $\omega \in \mathbb{C}$ with suitably large $\operatorname{Im}\omega > 0$. The inverse, denoted $R(\omega)$, is also called the *resolvent operator*.

While $R(\omega)$ fails to make sense in a region intersecting both $\{\operatorname{Im} z > 0\}$ and $\{\operatorname{Im} z < 0\} \subset \mathbb{C}$, one can conjugate $R(\omega)$ with a multiplication operator $\chi\mathbf{1}$, where $\chi \in C^\infty_c(\Sigma')$ and try to meromorphically continue the *cut-off resolvent*:

$$\chi R(\omega)\chi$$

as an operator on suitable function spaces. The poles of the meromorphic continuation of $\chi R(\omega)\chi$ are called *scattering resonances*.

The theorem below provides a more precise version of Theorem 1.1(iii). This theorem demonstrates that scattering resonances form a *subset* of the set of regularity quasinormal frequences, thereby justifying the nomenclature.

Theorem 10.5. *Let $N_+ \in \mathbb{N}_0$. The cut-off resolvent operator:*

$$\chi R(\omega)\chi : H^{N_+}_{\nabla\!\!\!/}(\Sigma') \to H^{N_++1}_{\nabla\!\!\!/}(\Sigma')$$

is well-defined for any cut-off function $\chi \in C^\infty_c(\Sigma')$ and for all $\omega \in \mathbb{C}$ with $\operatorname{Im}\omega > \omega_0 > 0$ suitably large and $\omega \mapsto \chi R(\omega)\chi$ can be meromorphically continued to $i\Omega \cap \{\operatorname{Im}\omega > -\left(\frac{1}{2}+N_+\right)\kappa_+\}$. Let $i\mathscr{Q}_{\mathrm{res}}$ denote the union over the set of poles of $\omega \mapsto \chi R(\omega)\chi$ over N_+. Then

$$\mathscr{Q}_{\mathrm{res}} \subseteq \mathscr{Q}_{\mathrm{reg}}.$$

Furthermore, for $\frac{|a|}{M}$ suitably small, or under the restriction to functions with a bounded azimuthal number m,

$$\chi R(\omega)\chi : L^2(\Sigma') \to H^2(\Sigma')$$

is well-defined and can be meromorphically continued to $i\Omega$, with poles in the set $\mathscr{Q}_{\mathrm{res}}$.

Proof. Consider the extension of Σ' to the bifurcation sphere, denoted $\overline{\Sigma'}$. By standard local-in-time energy estimates (see also the proof of Theorem 10.2), it follows that the time evolution map $\mathcal{S}(\tau) : H^1(\overline{\Sigma'}) \times L^2(\overline{\Sigma'}) \to H^1(\overline{\Sigma'}) \times L^2(\overline{\Sigma'})$ corresponding to initial data $(\phi, \partial_t\phi)$ on $\overline{\Sigma'}$ is a bounded linear operator for each τ, with a norm that can be bounded by $e^{s_0\tau}$, for suitably large $s_0 > 0$. By standard results of functional analysis, see for example [Eva98][§7.4, Theorem 2] and [Hil48][Theorem 11.6.1], the corresponding infinitesimal generator $\mathcal{A} : H^1(\overline{\Sigma'}) \times L^2(\overline{\Sigma'}) \supset \operatorname{dom}(\mathcal{A}) \to H^1(\overline{\Sigma'}) \times L^2(\overline{\Sigma'})$ is closed and densely defined with $(\mathcal{A} - s\mathbf{1})^{-1} : H^1(\overline{\Sigma'}) \times L^2(\overline{\Sigma'}) \to \operatorname{dom}(\mathcal{A})$ a well-defined bounded operator for $\operatorname{Re} s > s_0$.

In this case, we can apply (9.1), with $\frac{L_s}{(2Mh(0)-Mxb_{\tau\tau})}$ replaced with $\frac{r^2A(\omega)}{\mathbb{h}(2(r^2+a^2)-\Delta\mathbb{h})-a^2\sin^2\theta}$, where $\mathbb{h} = \frac{r^2+a^2}{\Delta}$ with $\omega = i\cdot s$ and P replaced with an appropriate first order operator to conclude via (9.2) that:

$$R(\omega) : \frac{\Delta}{r^2}L^2(\overline{\Sigma'}) \to H^1(\overline{\Sigma'})$$

is well-defined for suitably large $\operatorname{Im}\omega > 0$ and we can consider $\chi R(\omega)\chi : L^2(\Sigma') \to H^1(\Sigma')$.

In Corollary 19.3 below, we prove the existence of the meromorphic continuation of $\chi R(\omega)\chi$, defined on appropriate subspaces, and its relation to $\mathscr{Q}_{\mathrm{reg}}$. □

Remark 10.3. *The results of Theorem 10.5 in fact apply also when considering cut-off resolvents with respect to time functions whose level sets intersect $\mathcal{H}^+$ to the future of the bifurcation sphere and are asymptotically flat. We denote the level set corresponding to "time zero" in this case by $\widetilde{\Sigma}$. Now, the cut-off functions χ are allowed to be supported near $\mathcal{H}^+$ but the support remains compact. In the domains and codomains of the maps in Theorem 10.5, Σ' is then replaced with $\widetilde{\Sigma}$.*

11. Guide to the rest of the paper

We outline the structure of the rest of the paper and relate each section to the steps sketched in §4, which concern the operator L_s. We first focus on the $\Lambda = 0$ setting.

Theorem 4.1(i):

- In §12, we derive red-shift estimates and elliptic estimates (featuring a finite number of derivatives) and couple them to obtain a single estimate in the bounded region $\varrho \leq \eta^{-1}R_\infty$.
- In §13, we derive Gevrey-type estimates (featuring an infinite number of derivatives) in the region $\varrho \geq 2R_\infty$: estimates with higher-order radial derivatives are derived in §13.1 and estimates with lower-order radial derivatives are derived in §13.2.
- The Gevrey estimates are coupled with the red-shift and elliptic estimates in §14 to obtain a single, global estimate.
- In §15, we establish the invertibility of $\mathfrak{L}_{s,\lambda,\kappa}$ with $\kappa > 0$ and construct $\mathfrak{L}^{-1}_{s,\lambda}$ as a $\kappa \downarrow 0$ limit.

- In §16, we establish the meromorphicity and Fredholm properties of L_s^{-1}.

Theorem 4.1(ii) and Theorem 4.2(ii):

- In §16, we also establish the independence of the regularity quasinormal spectrum with respect to a change in Hilbert spaces, corresponding to different N_+, R_∞ and σ. We also establish Theorem 4.2(ii) for the κ-regularized operator.

We subsequently prove Theorem 4.2(i):

- In §17, we establish stability of the quasinormal spectrum under suitably small perturbations at the level of L_s.

Since the analysis in the $\Lambda > 0$ case actually follows, with minor modifications, from the analysis that is already required for the $\Lambda = 0$ case, we only take $\Lambda > 0$ in §18, where we focus primarily on the differences with the $\Lambda = 0$ case. Theorem 4.1(i) for $\Lambda > 0$ and Theorem 4.2(ii) for the true Kerr–de Sitter operator follow from §18.

Finally, we establish the relation between regularity quasinormal frequencies and scattering resonances:

- Theorem 4.1(iii): In §19, we construct the meromorphic continuation of the cut-off resolvent operator and relate scattering resonances to regularity quasinormal frequencies.

12. Red-shift and elliptic estimates

We will consider different types of estimates in the following two regions:

(A) $r_+ \leq r \leq \frac{1}{2}R_\infty$,
(B) $\frac{1}{4}R_\infty \leq \varrho \leq \eta^{-1}R_\infty$,

with $\mathbb{R} \ni \eta < 2$ and where $R_\infty M^{-1}$ will be chosen suitably large. We will later also need to restrict to suitably small η.

The estimates in (A) are the spacelike analogues of the spacetime *red-shift estimates* of Dafermos–Rodnianski [DR09] and they establish, in particular, control of the non-degenerate H^1 norm of $\hat{\psi} \in C^\infty(\widehat{\Sigma})$ in terms of the L^2 norm of $\mathfrak{L}_{s,\lambda}\hat{\psi}$. These estimates rely crucially on the positivity of the surface gravity κ_+ associated to the event horizon at $r = r_+$. In contrast with the spacetime red-shift estimates, which apply only to regions where r is sufficiently close to r_+, they hold in regions of arbitrarily large r.

Note that for $M^{-1}R_\infty$ sufficiently large, the region (A) covers the ergoregion of Kerr, where the vector field T is not timelike, and therefore "standard" (degenerate) elliptic estimates do not hold.

The estimates in (B) are more similar to standard elliptic estimates, in the sense that they establish control of a (weighted) H^2 norm of $\hat{\psi} \in C^\infty(\widehat{\Sigma})$ in terms of a (weighted) L^2 norm of $\mathfrak{L}_{s,\lambda}\hat{\psi}$ and rely on the fact that the spacetime metric is sufficiently close to the Minkowski metric in appropriate coordinates, by the asymptotic flatness property of the Kerr metric. The main difference with standard elliptic estimates, is the addition of a degenerate weight function in the norms, which will play an important role in the coupling with the Gevrey estimates of §13.

12.1. Red-shift estimate in (A).

We first consider the region with:

$$r_+ \leq \varrho \leq \frac{1}{2}R_\infty.$$

Recall from (7.10) that in this region $\varrho = r$, $\vartheta = \theta$ and we can write:

$$r^{-2}\rho^2\mathfrak{L}_{s,\lambda,\kappa} = r\Delta\partial_r^2(r^{-1}\cdot)+r\frac{d\Delta}{dr}\partial_r(r^{-1}\cdot)+2ar\partial_r\partial_{\varphi_*}(r^{-1}\cdot)+\not{\Delta}_{\mathbb{S}^2}(\cdot)+sr(4Mr-2(1-\mathbb{h})\Delta)\partial_r(r^{-1}\cdot)-\lambda\pi_{\leq l}.$$

Proposition 12.1. *Let $\operatorname{Re} s > -\frac{\kappa_+}{2}$ and $\delta = 1 + 2\kappa_+^{-1}\min\{\operatorname{Re} s, 0\}$. Then there exists a constant $C_s = C_s(\kappa_+, s) > 0$ and a numerical constant $C > 0$ such that*

$$\int_{r_+}^{\frac{1}{3}R_\infty}\int_{\mathbb{S}^2}\frac{1}{2}(r-M)(1+2\kappa_+^{-1}\min\{\mathrm{Re}\, s,0\})r^\delta|\partial_r(r^{-1}\widehat{\psi})|^2$$
$$+\frac{1}{2}(1+2\kappa_+^{-1}\min\{\mathrm{Re}\, s,0\})r^{-3+\delta}\left[|\nabla\!\!\!/_{\mathbb{S}^2}\widehat{\psi}|^2+\lambda|\pi_{\leq l}\widehat{\psi}|^2\right]\,d\sigma dr$$
$$+\int_{\mathbb{S}^2}r^{\delta-2}\left[|\nabla\!\!\!/_{\mathbb{S}^2}\widehat{\psi}|^2+\lambda|\pi_{\leq l}\widehat{\psi}|^2\right]\Big|_{r=r_+}d\sigma$$
$$\leq\boxed{C\int_{\frac{1}{3}R_\infty}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2}r^{-3+\delta}\left[|\nabla\!\!\!/_{\mathbb{S}^2}\widehat{\psi}|^2+\lambda|\pi_{\leq l}\widehat{\psi}|^2\right]\,d\sigma dr}+C_s\int_{r_+}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2}r^{-3+\delta}|\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2\,d\sigma dr. \quad (12.1)$$

Proof. Let $\widehat{\phi}=r^{-1}\widehat{\psi}$. Without loss of generality, we will assume $\widehat{\phi}=\pi_{\leq l}\widehat{\phi}$ in the derivations below. The $\pi_{>l}\widehat{\phi}$ part of general $\widehat{\phi}$ can be estimated in exactly the same way by simply taking $\lambda=0$.

We multiply $r^{-2}\rho^2\mathfrak{L}_{s,\lambda}\widehat{\psi}$ with $\chi_{r\leq\frac{1}{3}R_\infty}r^{\delta-1}\overline{\partial_r\widehat{\phi}}$, where $\delta>0$ will be fixed later, and where $\chi_{r\leq\frac{1}{3}R_\infty}$ is a smooth cut-off function that is equal to 1 for $r\leq\frac{1}{3}R_\infty$ and vanishes when $r\geq\frac{1}{2}R_\infty$. We then apply the Leibniz rule to obtain:

$$\mathrm{Re}(\chi_{r\leq\frac{1}{3}R_\infty}r^{\delta-1}\overline{\partial_r\widehat{\phi}}\cdot r^{-2}\rho^2\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})=\Delta r^\delta\chi_{r\leq\frac{1}{3}R_\infty}\,\mathrm{Re}(\overline{\partial_r\widehat{\phi}}\partial_r^2\widehat{\phi})+\frac{d\Delta}{dr}\chi_{r\leq\frac{1}{3}R_\infty}r^\delta|\partial_r\widehat{\phi}|^2$$
$$+2ar^\delta\chi_{r\leq\frac{1}{3}R_\infty}\,\mathrm{Re}(\overline{\partial_r\widehat{\phi}}\partial_r\partial_{\varphi_*}\widehat{\phi})+r^\delta\,\mathrm{Re}(\overline{\partial_r\widehat{\phi}}\not{\Delta}_{\mathbb{S}^2}\widehat{\phi})+(4Mr-2(\mathbb{h}-1)\Delta))r^\delta\chi_{r\leq\frac{1}{3}R_\infty}\mathrm{Re}\, s|\partial_r\widehat{\phi}|^2-\lambda r^\delta\,\mathrm{Re}(\overline{\partial_r\widehat{\phi}}\widehat{\phi})$$
$$=\frac{1}{2}\partial_r\left(\Delta r^\delta\chi_{r\leq\frac{1}{3}R_\infty}|\partial_r\widehat{\phi}|^2-r^\delta\chi_{r\leq\frac{1}{3}R_\infty}|\nabla\!\!\!/_{\mathbb{S}^2}\widehat{\phi}|^2-\lambda\chi_{r\leq\frac{1}{3}R_\infty}|\widehat{\phi}|^2\right)+\partial_{\varphi_*}(\ldots)+\mathrm{div}_{\mathbb{S}^2}(\ldots)$$
$$+\frac{1}{2}r^\delta\chi_{r\leq\frac{1}{3}R_\infty}\left[\frac{d\Delta}{dr}-\delta r^{-1}\Delta+(8Mr-4(\mathbb{h}-1)\Delta))\mathrm{Re}\, s\right]|\partial_r\widehat{\phi}|^2+\frac{\delta}{2}r^{\delta-1}\chi_{r\leq\frac{1}{3}R_\infty}\left[|\nabla\!\!\!/_{\mathbb{S}^2}\widehat{\phi}|^2+\lambda|\widehat{\phi}|^2\right]$$
$$-\frac{1}{2}\frac{d\chi_{r\leq\frac{1}{3}R_\infty}}{dr}\left(\Delta r^\delta\chi_{r\leq\frac{1}{3}R_\infty}|\partial_r\widehat{\phi}|^2-r^\delta\chi_{r\leq\frac{1}{3}R_\infty}|\nabla\!\!\!/_{\mathbb{S}^2}\widehat{\phi}|^2-\lambda\chi_{r\leq\frac{1}{3}R_\infty}|\widehat{\phi}|^2\right),$$

where we employed the schematic notation $\partial_{\varphi_*}(\ldots)+\mathrm{div}_{\mathbb{S}^2}(\ldots)$ for total derivatives and divergences on $\mathbb{S}^2$, as these terms vanish after integration over $\mathbb{S}^2$.

We first estimate via Young's inequality:

$$\left|\mathrm{Re}\left[r^{\delta-1}\overline{\partial_r\widehat{\phi}}\cdot r^{-2}\rho^2\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}\right]\right|\leq\frac{\delta}{2}r^\delta(r-M)|\partial_r\widehat{\phi}|^2+\delta^{-1}r^{-2+\delta}(r-M)^{-1}|\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2.$$

Furthermore, using that $\mathbb{h}\leq\frac{r^2+a^2}{\Delta}$, we obtain

$$8Mr-4(\mathbb{h}-1)\Delta\geq 8Mr-4(r^2+a^2)+4\Delta=0$$

so for $\mathrm{Re}\, s\geq 0$, we have that $(8Mr-4(\mathbb{h}-1)\Delta))\mathrm{Re}\, s\geq 0$.

For $\mathrm{Re}\, s<0$, we use that $\mathbb{h}\geq 1$ to obtain

$$(8Mr-4(\mathbb{h}-1)\Delta)\,\mathrm{Re}\, s\geq 8Mr\,\mathrm{Re}\, s$$

and we have that

$$\frac{d\Delta}{dr}-\delta r^{-1}\Delta+8Mr\mathrm{Re}\, s-\frac{\delta}{2}(r-M)>2(r-M)\left(1-\frac{3}{4}\delta+2\frac{r_+-M}{r-M}\kappa_+^{-1}\mathrm{Re}\, s\right), \quad (12.2)$$

where we recall the following expression for the surface gravity κ_+:

$$\kappa_+=\frac{r_+-M}{2Mr_+}.$$

Hence, for $\mathrm{Re}\, s>-\frac{\kappa_+}{2}$ and $0<\frac{3}{4}\delta<1+\min\{2\kappa_+^{-1}\mathrm{Re}\, s,0\}$, the right-hand side of (12.2) is non-negative definite.

Since $\frac{d\chi_{r\leq\frac{1}{3}R_\infty}}{dr} > 0$ and there exists a numerical constant $C > 0$ such that $-\frac{d\chi_{r\leq\frac{1}{3}R_\infty}}{dr} \leq CR_\infty^{-1}$, we can estimate:

$$-\frac{1}{2}\frac{d\chi_{r\leq\frac{1}{3}R_\infty}}{dr}\left(\Delta r^\delta \chi_{r\leq\frac{1}{3}R_\infty}|\partial_r\widehat{\phi}|^2 - r^\delta\chi_{r\leq\frac{1}{3}R_\infty}|\nabla\!\!\!/_{\mathbb{S}^2}\widehat{\phi}|^2 - \lambda\chi_{r\leq\frac{1}{3}R_\infty}|\widehat{\phi}|^2\right) \geq -CR_\infty^{-1}r^\delta\left(|\nabla\!\!\!/_{\mathbb{S}^2}\widehat{\phi}|^2 + \lambda|\widehat{\phi}|^2\right).$$

After setting $\delta = 1 + 2\kappa_+^{-1}\min\{\operatorname{Re} s, 0\}$ and integrating over $[r_+, \frac{1}{2}R_\infty]\times\mathbb{S}^2$, we obtain the following estimate: for $\operatorname{Re} s > -\frac{\kappa_+}{2}$, there exists a $C_s = C(\kappa_+, \operatorname{Re} s, \delta) > 0$, such that

$$\begin{aligned}
&\int_{r_+}^{\frac{1}{3}R_\infty}\int_{\mathbb{S}^2}\frac{1}{2}(r-M)(1+2\kappa_+^{-1}\min\{\operatorname{Re} s,0\})r^\delta|\partial_r\widehat{\phi}|^2\,d\sigma dr\\
&\quad+\frac{1}{2}(1+2\kappa_+^{-1}\min\{\operatorname{Re} s,0\})r^{-1+\delta}\left[|\nabla\!\!\!/_{\mathbb{S}^2}\widehat{\phi}|^2+\lambda|\widehat{\phi}|^2\right]+\int_{\mathbb{S}^2}r^\delta\left[|\nabla\!\!\!/_{\mathbb{S}^2}\widehat{\phi}|^2+\lambda|\widehat{\phi}|^2\right]\Big|_{r=r_+}d\sigma\\
&\qquad\leq C\int_{\frac{1}{3}R_\infty}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2}r^{-1+\delta}\left[|\nabla\!\!\!/_{\mathbb{S}^2}\widehat{\phi}|^2+\lambda|\widehat{\phi}|^2\right]d\sigma dr + C_s\int_{r_+}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2}r^{-3+\delta}|\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2\,d\sigma dr.
\end{aligned}$$

□

In order to improve the range of allowed values of $\operatorname{Re} s$, we need to consider higher-order r-derivatives of $\widehat{\psi}$. In addition, we will consider higher angular derivatives.

Lemma 12.2. *Let $k, m \in \mathbb{N}_0$ and $r_+ \leq \varrho \leq \frac{1}{2}R_\infty$. Then, for $\widehat{\phi} = r^{-1}\widehat{\psi}$:*

$$\begin{aligned}
&\partial_r^k\nabla\!\!\!/_{\mathbb{S}^2}^m(r^{-3}\rho^2\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}) = \Delta\partial_r^{2+k}\nabla\!\!\!/_{\mathbb{S}^2}^m\widehat{\phi} + (1+k)\frac{d\Delta}{dr}\partial_r^{k+1}\nabla\!\!\!/_{\mathbb{S}^2}^m\widehat{\phi} + 2a\partial_{\varphi_*}\partial_r^{k+1}\nabla\!\!\!/_{\mathbb{S}^2}^m\widehat{\phi} + \nabla\!\!\!/_{\mathbb{S}^2}^{1+m}\partial_r^k\widehat{\phi}\\
&+4Mrs\partial_r^{k+1}\nabla\!\!\!/_{\mathbb{S}^2}^m\widehat{\phi} - (\lambda\pi_{\leq l} - k(k+1) - 4kMs)\partial_r^k\nabla\!\!\!/_{\mathbb{S}^2}^m\widehat{\phi} - 2\sum_{j=0}^{k}\frac{k!}{j!(k-j)!}\partial_r^{k-j}((\mathbb{h}-1)\Delta)\partial_r^j\nabla\!\!\!/_{\mathbb{S}^2}^m\widehat{\phi}.
\end{aligned}\tag{12.3}$$

Proof. We will first prove (12.3) with $m = 0$ via induction. The $k = 0$ case follows from (7.10). Assume (12.3) holds for $k = K$. Then

$$\begin{aligned}
\partial_r^{K+1}(r^{-3}\rho^2\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}) &= \partial_r\Big(\Delta\partial_r^{2+K}\widehat{\phi} + (1+K)\frac{d\Delta}{dr}\partial_r^{K+1}\widehat{\phi} + 2a\partial_{\varphi_*}\partial_r^{K+1}\widehat{\phi} + \nabla\!\!\!/_{\mathbb{S}^2}\partial_r^K\widehat{\phi} + 4Mrs\partial_r^{K+1}\widehat{\phi}\\
&\quad - (\lambda\pi_{\leq l} - K(K+1) - 4KMs)\partial_r^K\widehat{\phi} - 2\sum_{j=0}^{K}\frac{K!}{j!(K-j)!}\partial_r^{K-j}((\mathbb{h}-1)\Delta)\partial_r^j\nabla\!\!\!/_{\mathbb{S}^2}^m\widehat{\phi}\Big)\\
&= \partial_r^{3+K}\widehat{\phi} + (K+2)\frac{d\Delta}{dr}\partial_r^{K+2}\widehat{\phi} + 2a\partial_{\varphi_*}\partial_r^{K+2}\widehat{\phi} + \nabla\!\!\!/_{\mathbb{S}^2}\partial_r^{K+1}\widehat{\phi} + 4Mrs\partial_r^{K+2}\widehat{\phi}\\
&\quad - \left(\lambda\pi_{\leq l} - K(K+1) - \frac{d^2\Delta}{dr^2}(K+1) - 4M(K+1)s\right)\partial_r^{K+1}\widehat{\phi}\\
&\quad - 2\sum_{j=0}^{K+1}\frac{(K+1)!}{j!(K+1-j)!}\partial_r^{K+1-j}((\mathbb{h}-1)\Delta)\partial_r^j\nabla\!\!\!/_{\mathbb{S}^2}^m\widehat{\phi}.
\end{aligned}$$

Using that $K(K+1) + \frac{d^2\Delta}{dr^2}(K+1) = (K+2)(K+1)$, it therefore follows that (12.3) with $m = 0$ holds for $k = K+1$.

The case $m > 0$ follows immediately from the fact that $\nabla\!\!\!/_{\mathbb{S}^2}$ and $\partial_r^{K+1}(r^{-3}\rho^2\mathfrak{L}_{s,\lambda,\kappa}(\cdot))$ commute. □

We state now a higher-order version of Proposition 12.1.

Corollary 12.3. *Let $N_+, m \in \mathbb{N}_0$, $\operatorname{Re} s > -\kappa_+\left(\frac{1}{2}+N_+\right)$ and $\delta = 1+2N_+ + 2\kappa_+^{-1}\min\{\operatorname{Re} s, 0\}$. Assume that*

$$\lambda - N_+(N_+ + 1) \gg \frac{N_+^2 M^2 |s|^2}{1 + 2N_+ + 2\kappa_+^{-1}\min\{\operatorname{Re} s, 0\}}.$$

Then there exists a constant $C_s = C_s(\kappa_+, s) > 0$ and $C = C(N_+) > 0$ such that:

$$\begin{aligned}
&\sum_{k=0}^{N_+}(1+2N_+ + 2\kappa_+^{-1}\min\{\operatorname{Re} s,0\})R_\infty^{2k}\int_{r_+}^{\frac{1}{3}R_\infty}\int_{\mathbb{S}^2}(r-M)r^\delta|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^{k+1}(r^{-1}\widehat{\psi})|^2 \\
&\qquad + r^{-3+\delta}\left[|\nabla\!\!\!/_{\mathbb{S}^2}^{m+1}\partial_r^k\widehat{\psi}|^2 + (\lambda\pi_{\leq l} - k(k+1))|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^k\widehat{\psi}|^2\right] d\sigma dr \\
&\leq \boxed{C\sum_{k=0}^{N_+}R_\infty^{2k-2}\int_{\frac{1}{3}R_\infty}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2} r^{-1+\delta}\left[|\nabla\!\!\!/_{\mathbb{S}^2}^{m+1}\partial_r^k\widehat{\psi}|^2 + (\lambda\pi_{\leq l} - k(k+1))|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^k\widehat{\psi}|^2\right] d\sigma dr} \\
&\qquad + C_s R_\infty^{2N_+}\int_{r_+}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2} r^{-1+\delta}|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^{N_+}(r^{-3}\rho^2\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})|^2\, d\sigma dr. \qquad (12.4)
\end{aligned}$$

Proof. We repeat the proof of Proposition 12.1, but with (7.10) replaced by the higher-order equation (12.3) and, for $m > 0$, we additionally integrate by parts in the $\mathbb{S}^2$ direction. This gives for $k \in \mathbb{N}_0$:

$$\begin{aligned}
&\int_{r_+}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2}(r-M)(1+2k+2\kappa_+^{-1}\min\{\operatorname{Re} s,0\})r^\delta\chi_{r\leq\frac{1}{3}R_\infty}|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^{k+1}\widehat{\phi}|^2 \\
&+\frac{2}{3}(1+2k+2\kappa_+^{-1}\min\{\operatorname{Re} s,0\})r^{-1+\delta}\chi_{r\leq\frac{1}{3}R_\infty}\left[|\nabla\!\!\!/_{\mathbb{S}^2}^{m+1}\partial_r^k\widehat{\phi}|^2 + (\lambda\pi_{\leq l} - k(k+1))|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^k\widehat{\phi}|^2\right] d\sigma dr \\
&\qquad + \int_{\mathbb{S}^2} r^\delta\left[|\nabla\!\!\!/_{\mathbb{S}^2}^{m+1}\partial_r^k\widehat{\phi}|^2 + (\lambda\pi_{\leq l} - k(k+1))|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^k\widehat{\phi}|^2\right]\Big|_{r=r_+} d\sigma \\
&\qquad \leq C\int_{\frac{1}{3}R_\infty}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2} r^{-1+\delta}\left[|\nabla\!\!\!/_{\mathbb{S}^2}^{m+1}\partial_r^k\widehat{\phi}|^2 + (\lambda\pi_{\leq l} - k(k+1))|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^k\widehat{\phi}|^2\right] d\sigma dr \\
&+Ck^2M^2|s|^2\int_{r_+}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2} r^\delta\chi_{r\leq\frac{1}{3}R_\infty}|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^k\widehat{\phi}|^2\, d\sigma dr + C_k M^2|s|^2\sum_{j=0}^{k-1}\int_{r_+}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2} r^\delta\chi_{r\leq\frac{1}{3}R_\infty}|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^j\widehat{\phi}|^2\, d\sigma dr \\
&\qquad + C_s\int_{r_+}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2} r^{-1+\delta}|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^k(r^{-3}\rho^2\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})|^2\, d\sigma dr. \qquad (12.5)
\end{aligned}$$

Note that we can absorb the term with a factor $Ck^2M^2|s|^2$ into

$$+\frac{2}{3}(1+2k+2\kappa_+^{-1}\min\{\operatorname{Re} s,0\})r^{-1+\delta}\chi_{r\leq\frac{1}{3}R_\infty}\left[|\nabla\!\!\!/_{\mathbb{S}^2}^{m+1}\partial_r^k\widehat{\phi}|^2 + (\lambda\pi_{\leq l} - k(k+1))|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^k\widehat{\phi}|^2\right] d\sigma dr,$$

if we take $\lambda - k(k+1) \gg \frac{k^2|s|^2}{1+2k+2\kappa_+^{-1}\min\{\operatorname{Re} s,0\}}$. This gives the desired control for $k = N_+$ in the sum on the left-hand side of (12.5).

In order to control the $k < N_+$ terms in the sum on the left-hand side of (12.5) as well as bound the $k < N+1$ terms on the right-hand side of (12.5), we will use that we can absorb these lower-order terms into the $k = N_+$ terms by applying a standard Hardy inequality $N_+ - k$ times, see for example [AAG23a][Lemma 2.6, eq. (2.21)], and by taking λ to be appropriately large. For example,

$$\int_{r_+}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2}\chi_{r\leq\frac{1}{3}R_\infty}r^{-1+\delta}|\nabla\!\!\!/_{\mathbb{S}^2}\partial_r^{k-1}\widehat{\phi}|^2\, d\sigma dr \leq C\int_{r_+}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2}\chi_{r\leq\frac{1}{3}R_\infty}r^{1+\delta}|\nabla\!\!\!/_{\mathbb{S}^2}\partial_r^k\widehat{\phi}|^2\, d\sigma dr$$

$$+ C\int_{\frac{1}{3}R_\infty}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2} r^{-1+\delta}|\nabla\!\!\!/_{\mathbb{S}^2}\partial_r^{k-1}\widehat{\phi}|^2\,d\sigma dr$$

$$\leq CR_\infty^2\int_{r_+}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2}\chi_{r\leq\frac{1}{3}R_\infty}r^{-1+\delta}|\nabla\!\!\!/_{\mathbb{S}^2}\partial_r^k\widehat{\phi}|^2\,d\sigma dr + C\int_{\frac{1}{3}R_\infty}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2} r^{-1+\delta}|\nabla\!\!\!/_{\mathbb{S}^2}\partial_r^{k-1}\widehat{\phi}|^2\,d\sigma dr. \quad \square$$

In the sections below, we will consider a coupled system of estimates consisting of (12.4) with $k+l\leq N_+$ and $l\geq 1$, and appropriate elliptic estimates in the region $\{r\geq\frac{1}{4}R_\infty\}$. In this way, we will be able to absorb the integral over the region $\{\frac{1}{3}R_\infty\leq r\leq\frac{1}{2}R_\infty\}$ in the boxed term on the right-hand side of (12.4).

We can in fact improve the estimate in (12.3) in the case of $|a|\ll M$ or under the restriction to fixed azimuthal modes by applying a standard elliptic estimate in $r\leq\frac{1}{2}R_\infty$ in order to include higher-order derivative terms on the right-hand side, with weights that degenerate at $r=r_+$. This will be relevant when considering the cut-off resolvent operator.

Proposition 12.4. *Let $N_+, m\in\mathbb{N}_0$, $\mathrm{Re}\,s>-\kappa_+\left(\frac{1}{2}+N_+\right)$ and $\delta=\frac{2}{3}(1+2N_++2\kappa_+^{-1}\min\{\mathrm{Re}\,s,0\})$. Assume that*

$$\lambda-N_+(N_++1)\gg\frac{N_+^2M^2|s|^2}{1+2N_++2\kappa_+^{-1}\min\{\mathrm{Re}\,s,0\}}.$$

Then for $|a|\ll M$ or $\widehat{\psi}=\sum_{m=-m_0}^{m_0}\widehat{\psi}_m$, with $m_0>0$ and $\Phi\psi_m=im\psi_m$, there exists a constant $C_s=C_s(a,m_0,\kappa_+,s)>0$ and $C=C(N_+,m_0)>0$ such that:

$$\sum_{k=0}^{N_+}(1+2N_++2\kappa_+^{-1}\min\{\mathrm{Re}\,s,0\})R_\infty^{2k}\int_{r_+}^{\frac{1}{3}R_\infty}\int_{\mathbb{S}^2}(r-M)r^\delta|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^{k+1}(r^{-1}\widehat{\psi})|^2+\Delta^2r^{\delta-1}|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^{k+2}(r^{-1}\widehat{\psi})|^2$$
$$+r^{-3+\delta}\left[|\nabla\!\!\!/_{\mathbb{S}^2}^{m+2}\partial_r^k\widehat{\psi}|^2+\Delta|\nabla\!\!\!/_{\mathbb{S}^2}^{m+1}\partial_r^{k+1}\widehat{\psi}|^2+(\lambda\pi_{\leq l}-k(k+1))^2|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^k\widehat{\psi}|^2\right]d\sigma dr$$
$$\leq\boxed{C\sum_{k=0}^{N_+}R_\infty^{2k}\int_{\frac{1}{3}R_\infty}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2}r^{-3+\delta}\left[|\nabla\!\!\!/_{\mathbb{S}^2}^{m+1}\partial_r^k\widehat{\psi}|^2+(\lambda\pi_{\leq l}-k(k+1))|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^k\widehat{\psi}|^2\right]d\sigma dr}$$
$$+C_sR_\infty^{2N_+}\int_{r_+}^{\frac{1}{2}R_\infty}\int_{\mathbb{S}^2}r^{-1+\delta}|\nabla\!\!\!/_{\mathbb{S}^2}^m\partial_r^{N_+}(r^{-3}\rho^2\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})|^2\,d\sigma dr. \quad (12.6)$$

Proof. We complement the estimates in the proof of Proposition 12.3 with a standard elliptic estimate. In the $N_+=0$ case, we integrate

$$\chi|r^{-2}\rho^2\mathfrak{L}_{s,\lambda,\kappa}\widehat{\phi}+\lambda\pi_{\leq l}\widehat{\phi}-sr(4Mr-2(1-\mathbb{h})\Delta)\partial_r\widehat{\phi}-r\frac{d\Delta}{dr}\partial_r\widehat{\phi}-2ar\partial_r\partial_{\varphi_*}\widehat{\phi}|^2=\chi|r\Delta\partial_r^2\widehat{\phi}+\Delta\!\!\!/_{\mathbb{S}^2}\widehat{\phi}|^2$$

and integrate by parts the mixed term arising from the square norm on the right-hand side.

We moreover use that the integral of $\chi r^2a^2|\partial_r\partial_{\varphi_*}\widehat{\phi}|^2$ can be absorbed into the remaining terms of $|a|$ sufficiently small or for $\widehat{\phi}$ supported on bounded set of azimuthal modes. $\square$

12.2. **Weighted elliptic estimates in (B).** Here we will use that

$$R_\infty\gg M,$$

so in the region $\varrho\geq\frac{1}{4}R_\infty$, the operator $\mathfrak{L}_{s,\lambda,\kappa}$ is suitably close to the analogous operator in Minkowski (i.e. the $M=0$ case).

To make the elliptic estimates compatible with the Gevrey estimates in §13.2, we will need to add a sufficiently degenerate weight to the elliptic estimates: $(\mathfrak{l}+1)^{-w_p}$, where $w_p:[0,r_+^{-1}]\to\mathbb{R}$

was defined in (8.1). Recall,

$$w_p(x) = \begin{cases} p & x \leq \frac{1}{4R_\infty}, \\ 0 & x \geq \frac{1}{3R_\infty}. \end{cases}$$

We make the following observations regarding w:

- We have that $w \geq 0$ and $w \equiv 0$ in the region $\varrho \leq 3R_\infty$ where $\mathfrak{L}_{s,\lambda,\kappa}$ does not agree with the corresponding operator in Minkowski.
- By employing standard cut-off functions, we can define w such that

$$\left|\left| \frac{dw_p}{dx} \right|\right|_{L^\infty[0,r_+^{-1}]} \leq CpR_\infty, \tag{12.7}$$

  for some numerical constant $C > 0$.

From (7.12), it follows that we can split

$$\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi} = \partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi}) + (\not{\Delta}_{\mathbb{S}^2,\vartheta} - \lambda\pi_{\leq l} - \chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l})\widehat{\psi} + \mathfrak{R}_{s,\lambda}, \text{ with}$$
$$\mathfrak{R}_{s,\lambda}\widehat{\psi} := 2s\partial_x\widehat{\psi} + \chi_{\varrho\leq R_\infty}(\mathcal{K}_s^\infty\widehat{\psi} + Mx\mathcal{B}_s^\infty\widehat{\psi}) - \chi_{\varrho\leq\frac{1}{2}R_\infty}\mathcal{K}_s^+\widehat{\psi}.$$

We will treat $\mathfrak{R}_{s,\lambda}\widehat{\psi}$ as an inhomogeneous term that can be absorbed, because it will come with additional factors of x compared to the terms on the left-hand side of the elliptic estimates and x will be small in the regions under consideration; see Lemma 12.5 below. Note that $\mathfrak{R}_{s,\lambda}\widehat{\psi} - 2s\partial_x$ contains all deviations from corresponding operator in Minkowski.

Lemma 12.5. *Let $k \in \mathbb{N}_0$. Then there exists a numerical constant $C = C(k) > 0$, such that*

$$\int_{\mathbb{S}^2} |\partial_x^k\mathfrak{R}_{s,\lambda}\widehat{\psi}|^2\, d\sigma \leq C\int_{\mathbb{S}^2}\Big[|s|^2|\partial_x^{k+1}\widehat{\psi}|^2 + M^2\chi_{\varrho\leq R_\infty}\Big(x^6|\partial_x^{k+2}\widehat{\psi}|^2 + x^4|\nabla\!\!\!/_{\mathbb{S}^2}\partial_x^{k+1}\widehat{\psi}|^2 + x^2|\not{\Delta}_{\mathbb{S}^2}\partial_x^k\widehat{\psi}|^2$$
$$+ x^2|\partial_x^{k+1}\widehat{\psi}|^2 + |s|^2|\nabla\!\!\!/_{\mathbb{S}^2}\partial_x^k\widehat{\psi}|^2 + (|s|^2+x^2)|\partial_x^k\widehat{\psi}|^2\Big)\Big]\, d\sigma.$$

Proof. When $\varrho > R_\infty$, we simply have that

$$\mathfrak{R}_{s,\lambda}\widehat{\psi} = 2s\partial_x\widehat{\psi},$$

so the estimate is immediate. When $\varrho \leq R_\infty$, we observe that the top-order derivative terms in $x\mathcal{B}_s^\infty$ and $\mathcal{K}_s^+$ have additional powers of x compared to the top-order derivative terms in the remaining terms of $\mathfrak{L}_{s,\lambda,\kappa}$. □

Proposition 12.6. *Let $0 < \delta \leq 1$ and assume the following relations:*

$$0 < \eta < \frac{1}{4},$$
$$\kappa \ll \eta R_\infty^{-1},$$
$$M^{-1}R_\infty \gg 1,$$
$$\lambda \gg 1 + |s|^2\eta^{-2}R_\infty^2 + p.$$

Then, there exists a numerical constant $C > 0$ such that:

$$\sum_{\ell\in\mathbb{N}_0}\int_{\eta R_\infty^{-1}}^{3R_\infty^{-1}} x^{1-\delta}(\mathfrak{l}+1)^{-w_p}\Bigg(|\partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi}_\ell)|^2 + |(\not{\Delta}_{\mathbb{S}^2} - \lambda\pi_{\leq l} - \chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l})\widehat{\psi}_\ell|^2$$
$$+ 2x(x+\kappa\chi_{\varrho\geq R_\infty})\left[|\nabla\!\!\!/_{\mathbb{S}^2}\partial_x\widehat{\psi}_\ell|^2 + \lambda\pi_{\leq l}|\partial_x\widehat{\psi}_\ell|^2 + \chi_{\varrho\geq R_\infty}|\sqrt{S_\lambda}\pi_{\leq l}\partial_x\widehat{\psi}_\ell|^2\right]\Bigg)\, d\sigma dx$$

$$\leq \boxed{C\int_{\frac{1}{4}R_\infty}^{\frac{1}{3}R_\infty}\int_{\mathbb{S}^2} r^{-2+\delta}\left|(\not\Delta_{\mathbb{S}^2}-\lambda\pi_{\leq l})\widehat{\psi}\right|\cdot\left|\partial_r\widehat{\psi}\right|\,d\sigma dr}$$

$$+\boxed{C(\mathfrak{l}+1)^{-p}\sum_{\ell\in\mathbb{N}_0}\int_{\mathbb{S}^2} x^{2-\delta}(x+\kappa)\mathfrak{l}(\mathfrak{l}+1)|\widehat{\psi}_\ell|\cdot|\partial_x\widehat{\psi}_\ell|\,d\sigma\Big|_{x=\eta R_\infty^{-1}}}$$

$$+C\sum_{\ell\in\mathbb{N}_0}\int_{\eta R_\infty^{-1}}^{4R_\infty^{-1}} x^{1-\delta}(\mathfrak{l}+1)^{-w_p}|(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2\,d\sigma dx. \quad (12.8)$$

Proof. We split:

$$\partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi})+(\not\Delta_{\mathbb{S}^2}-\lambda\pi_{\leq l}-\chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l})\widehat{\psi}=\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}-\mathfrak{R}_{s,\lambda}\widehat{\psi}$$

Then we consider the ℓ-th spherical harmonics $\widehat{\psi}_\ell$, take the square norm of both sides, multiply both sides with the weight factor $x^q(\mathfrak{l}+1)^{-w_p}\chi_{\varrho\geq\frac{1}{3}R_\infty}$, with $q\in\mathbb{R}$ and $\chi_{\varrho\geq\frac{1}{3}R_\infty}$ a smooth cut-off function which equals 1 when $\varrho\geq\frac{1}{3}R_\infty$ and vanishes for $\varrho\leq\frac{1}{4}R_\infty$, and we integrate over $\mathbb{S}^2$ to obtain:

$$\overbrace{\int_{\mathbb{S}^2} x^q(\mathfrak{l}+1)^{-w_p}\chi_{\varrho\geq\frac{1}{3}R_\infty}\left(|\partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi}_\ell)|^2+|(\not\Delta_{\mathbb{S}^2}-\lambda\pi_{\leq l}-\chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l})\widehat{\psi}_\ell|^2\right)d\sigma}^{=:J_0}$$

$$+\int_{\mathbb{S}^2} 2x^q(\mathfrak{l}+1)^{-w_p}\chi_{\varrho\geq\frac{1}{3}R_\infty}\,\mathrm{Re}\left((\not\Delta_{\mathbb{S}^2}-\lambda\pi_{\leq l}-\chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l})\overline{\widehat{\psi}}_\ell\cdot\partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi}_\ell)\right)d\sigma$$

$$=\int_{\mathbb{S}^2} x^q(\mathfrak{l}+1)^{-w_p}\chi_{\varrho\geq\frac{1}{3}R_\infty}|(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell-(\mathfrak{R}_{s,\lambda}\widehat{\psi})_\ell|^2\,d\sigma.$$

For the sake of convenience, we will restrict to $\ell\leq l$. The $\ell>l$ case proceeds entirely analogously, but without the λ and S_λ terms.

We integrate by parts on $\mathbb{S}^2$ to obtain:

$$\int_{\mathbb{S}^2} 2x^q(\mathfrak{l}+1)^{-w_p}\chi_{\varrho\geq\frac{1}{3}R_\infty}\,\mathrm{Re}\left((\not\Delta_{\mathbb{S}^2}-\lambda-\chi_{\varrho\geq R_\infty}S_\lambda)\overline{\widehat{\psi}}_\ell\cdot\partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi}_\ell\right)d\sigma$$

$$=\overbrace{2\int_{\mathbb{S}^2}\partial_x\left[x^q(\mathfrak{l}+1)^{-w_p}\chi_{\varrho\geq\frac{1}{3}R_\infty}\,\mathrm{Re}\left((\not\Delta_{\mathbb{S}^2}-\lambda-\chi_{\varrho\geq R_\infty}S_\lambda)\overline{\widehat{\psi}}_\ell\cdot x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi}_\ell\right)\right]d\sigma}^{=:J_1}$$

$$+\overbrace{2\int_{\mathbb{S}^2} x^q(\mathfrak{l}+1)^{-w_p}x(x+\kappa\chi_{\varrho\geq R_\infty})\chi_{\varrho\geq\frac{1}{3}R_\infty}\left[|\not\nabla_{\mathbb{S}^2}\partial_x\widehat{\psi}_\ell|^2+\lambda|\partial_x\widehat{\psi}_\ell|^2+\chi_{\varrho\geq R_\infty}|\sqrt{S_\lambda}\partial_x\widehat{\psi}_\ell|^2\right]d\sigma}^{=:J_2}$$

$$\overbrace{-2\int_{\mathbb{S}^2}\partial_x(x^q(\mathfrak{l}+1)^{-w_p})x(x+\kappa\chi_{\varrho\geq R_\infty})\chi_{\varrho\geq\frac{1}{3}R_\infty}\,\mathrm{Re}\left((\not\Delta_{\mathbb{S}^2}-\lambda-\chi_{\varrho\geq R_\infty}S_\lambda)\overline{\widehat{\psi}}_\ell\cdot\partial_x\widehat{\psi}_\ell\right)d\sigma}^{=:J_3}$$

$$+\overbrace{2\int_{\mathbb{S}^2} x^q(\mathfrak{l}+1)^{-w_p}x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x(\chi_{\varrho\geq R_\infty})\chi_{\varrho\geq\frac{1}{3}R_\infty}\,\mathrm{Re}(S_\lambda\overline{\widehat{\psi}}_\ell\cdot\partial_x\widehat{\psi}_\ell)\,d\sigma}^{=:J_4}$$

$$\overbrace{-2\int_{\mathbb{S}^2} x^q(\mathfrak{l}+1)^{-w_p}x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x(\chi_{\varrho\geq\frac{1}{3}R_\infty})\,\mathrm{Re}\left((\not\Delta_{\mathbb{S}^2}-\lambda-\chi_{\varrho\geq R_\infty}S_\lambda)\overline{\widehat{\psi}}_\ell\cdot\partial_x\widehat{\psi}_\ell\right)d\sigma}^{=:J_5}.$$

Note first that J_2 is non-negative definite. We will show that we can absorb J_3 and J_4 into J_0 and J_2 by taking λ suitably large.

We apply Young's inequality together with (12.7) to obtain for $x \in [\eta R_\infty^{-1}, 4R_\infty^{-1}]$:

$$
\begin{aligned}
&|J_3| \leq \\
&\int_{\mathbb{S}^2} \left(|q| x^{-1} + \left| \frac{dw_p}{dx} \right| (\mathfrak{l}+1)^{-1} \right) x^{q+1} (x + \kappa \chi_{\varrho \geq R_\infty}) (\mathfrak{l}+1)^{-w_p} (\ell(\ell+1) + \lambda + 2l\ell \chi_{\varrho \geq R_\infty}) |\widehat{\psi}_\ell| |\partial_x \widehat{\psi}_\ell| \, d\sigma \\
&\leq \left(\frac{|q|}{\mathfrak{l} x} + CpR_\infty (\mathfrak{l}+1)^{-1} \mathfrak{l}^{-1} \right) \mathfrak{l} \int_{\mathbb{S}^2} x^{q+1} (x + \kappa \chi_{\varrho \geq R_\infty}) (\mathfrak{l}+1)^{-w_p} (\ell(\ell+1) + \lambda + 2l\ell \chi_{\varrho \geq R_\infty}) |\widehat{\psi}_\ell| |\partial_x \widehat{\psi}_\ell| \, d\sigma \\
&\qquad \leq \frac{1}{4} \int_{\mathbb{S}^2} x^q (\mathfrak{l}+1)^{-w_p} |(\not{\Delta}_{\mathbb{S}^2} - \lambda - \chi_{\varrho \geq R_\infty} S_\lambda) \widehat{\psi}_\ell|^2 \, d\sigma \\
&\quad + C \left(\frac{|q|}{\mathfrak{l} x} + CpR_\infty (\mathfrak{l}+1)^{-1} \mathfrak{l}^{-1} \right)^2 x (x + \kappa \chi_{\varrho \geq R_\infty}) \int_{\mathbb{S}^2} x^q (\mathfrak{l}+1)^{-w_p} x (x + \kappa \chi_{\varrho \geq R_\infty}) \mathfrak{l}^2 |\partial_x \widehat{\psi}_\ell|^2 \, d\sigma.
\end{aligned}
$$

The first term on the very right-hand side above can be absorbed into J_0. The second term on the right-hand side can be absorbed into J_2 by taking $\kappa \ll \eta R_\infty^{-1}$ and $\lambda \gg q^2 + p$ to obtain:

$$
C \left(\frac{|q|}{\mathfrak{l} x} + CpR_\infty \mathfrak{l}^{-2} \right)^2 x (x + \kappa \chi_{\varrho \geq R_\infty}) = C \left(|q| \mathfrak{l}^{-1} + Cpx R_\infty \mathfrak{l}^{-2} \right)^2 (1 + \kappa x^{-1} \chi_{\varrho \geq R_\infty}) \leq \frac{1}{4}.
$$

and noting that

$$
\mathfrak{l}(\mathfrak{l}+1) = \ell(\ell+1) + l(l+1) + 2l\ell \leq 2(\ell(\ell+1) + 2l(l+1)).
$$

We conclude that $|J_3| \leq \frac{1}{4} J_0 + \frac{1}{4} J_2$.

To estimate $|J_4|$, we first note that there exists a constant $C > 0$ such that $\left| \frac{d\chi_{\varrho \geq R_\infty}}{d\varrho} \right| \leq C R_\infty^{-1}$, so that

$$
\left| \frac{d\chi_{\varrho \geq R_\infty}}{dx} \right| \leq C R_\infty.
$$

By repeating the estimates for $|J_3|$ and taking $\lambda \gg M^{-1} R_\infty$, we conclude that $|J_4| \leq \frac{1}{4} J_0 + \frac{1}{4} J_2$. We conclude that

$$
\sum_{j=0}^{4} J_j \geq \frac{1}{2} (J_0 + J_2).
$$

We can estimate J_5 by observing that $\partial_x \chi_{\varrho \geq \frac{1}{3} R_\infty}$ is supported in $\frac{1}{4} R_\infty \leq \varrho \leq \frac{1}{3} R_\infty$, where $\varrho = r$, $w(x) = 0$ and there exists a numerical constant $C > 0$ such that $|\partial_x \chi_{\varrho \geq \frac{1}{3} R_\infty}| \leq C R_\infty$. We obtain:

$$
|J_5| \leq \int_{\mathbb{S}^2} r^{-q+1} |(\not{\Delta}_{\mathbb{S}^2} - \lambda) \widehat{\psi}_\ell| \cdot |\partial_r \widehat{\psi}_\ell| \, d\sigma
$$

and $\operatorname{supp} J_5 \subset \{ \frac{1}{4} R_\infty \leq r \leq \frac{1}{3} R_\infty \}$.

We apply Young's inequality to estimate:

$$
\begin{aligned}
&\int_{\mathbb{S}^2} x^q (\mathfrak{l}+1)^{-w_p} |(\mathfrak{L}_{s,\lambda,\kappa} \widehat{\psi})_\ell - (\mathfrak{R}_{s,\lambda} \widehat{\psi})_\ell|^2 \, d\sigma \\
&\qquad \leq 2 \int_{\mathbb{S}^2} x^q (\mathfrak{l}+1)^{-w_p} |(\mathfrak{L}_{s,\lambda,\kappa} \widehat{\psi})_\ell|^2 \, d\sigma + 2 \int_{\mathbb{S}^2} x^q (\mathfrak{l}+1)^{-w_p} |(\mathfrak{R}_{s,\lambda} \widehat{\psi})_\ell|^2 \, d\sigma.
\end{aligned}
$$

Note that $(\mathfrak{R}_{s,\lambda} \widehat{\psi})_\ell$ does not need to agree with $(\mathfrak{R}_{s,\lambda}(\widehat{\psi}_\ell))$. However, since $w \equiv 0$ in $\{\varrho \leq 2R_\infty\}$ and $\mathfrak{R}_{s,\lambda} \widehat{\psi} = 2s \partial_x \widehat{\psi}$ in $r \geq R_\infty$, we can conclude that:

$$
\sum_{\ell \in \mathbb{N}_0} \int_{\mathbb{S}^2} (\mathfrak{l}+1)^{-w_p} |(\mathfrak{R}_{s,\lambda} \widehat{\psi})_\ell|^2 \, d\sigma = \begin{cases} \int_{\mathbb{S}^2} |\mathfrak{R}_{s,\lambda} \widehat{\psi}|^2 \, d\sigma & \text{if} \quad x \geq \frac{1}{2R_\infty}, \\ \sum_{\ell \in \mathbb{N}_0} \int_{\mathbb{S}^2} (\mathfrak{l}+1)^{-w_p} |\mathfrak{R}_{s,\lambda}(\widehat{\psi}_\ell)|^2 \, d\sigma & \text{if} \quad x < \frac{1}{2R_\infty}. \end{cases}
$$

By applying Lemma 12.5, we can take $MR_\infty^{-1} \ll 1$ and $\lambda \gg |s|^2\eta^{-2}R_\infty^2$ to obtain:

$$2\sum_{\ell\in\mathbb{N}_0}\int_{\mathbb{S}^2} x^q(\mathfrak{l}+1)^{-w_p}|(\mathfrak{R}_{s,\lambda}\widehat{\psi})_\ell|^2\,d\sigma \leq \frac{1}{4}\sum_{\ell\in\mathbb{N}_0}(J_0+J_2).$$

We now integrate over x to obtain:

$$\begin{aligned}\frac{1}{4}\sum_{\ell\in\mathbb{N}_0}\int_{x(\eta^{-1})}^{x(\frac{1}{4}R_\infty)} x^q(\mathfrak{l}+1)^{-w_p}\Big(&|\partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi}_\ell)|^2+|(\not{\Delta}_{\mathbb{S}^2}-\lambda-\chi_{\varrho\geq R_\infty}S_\lambda)\widehat{\psi}_\ell|^2\\ &+2x(x+\kappa\chi_{\varrho\geq R_\infty})\left[|\not{\nabla}_{\mathbb{S}^2}\partial_x\widehat{\psi}_\ell|^2+\lambda|\partial_x\widehat{\psi}_\ell|^2+\chi_{\varrho\geq R_\infty}|\sqrt{S_\lambda}\partial_x\widehat{\psi}_\ell|^2\right]\Big)\,d\sigma dx\\ &\leq C\int_{\frac{1}{4}R_\infty}^{\frac{1}{3}R_\infty}\int_{\mathbb{S}^2} r^{-q-1}\left|(\not{\Delta}_{\mathbb{S}^2}-\lambda)\widehat{\psi}\right|\cdot\left|\partial_r\widehat{\psi}\right|\,d\sigma dr\\ &+2(\mathfrak{l}+1)^{-p}\sum_{\ell\in\mathbb{N}_0}\int_{\mathbb{S}^2} x^{q+1}(x+\kappa)\mathfrak{l}(\mathfrak{l}+1)|\widehat{\psi}_\ell|\cdot|\partial_x\widehat{\psi}_\ell|\,d\sigma\Big|_{x=\eta R_\infty^{-1}}\\ &+2\sum_{\ell\in\mathbb{N}_0}\int_{x(\eta^{-1})}^{x(\frac{1}{4}R_\infty)} x^q(\mathfrak{l}+1)^{-w_p}|(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2\,d\sigma dx,\end{aligned}$$

and we conclude that (12.8) holds after taking $q = 1-\delta$. □

We will also consider higher-order analogues of (12.8). For this it is useful to define:

$$\widehat{\psi}^{(n)} := \partial_x^n\widehat{\psi}.$$

Then it can be easily verified that:

$$\begin{aligned}\partial_x^n(\partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi})+(\not{\Delta}_{\mathbb{S}^2}-\lambda\pi_{\leq l}-\chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l})\widehat{\psi}) &= \partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi}^{(n)}\\ &+(\not{\Delta}_{\mathbb{S}^2}-\lambda\pi_{\leq l}-\chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l})\widehat{\psi}^{(n)}+2(k+1)\left(x+\frac{1}{2}\kappa\chi_{\varrho\geq R_\infty}\right)\partial_x\widehat{\psi}^{(n)}\\ &+\kappa\sum_{k=0}^{n+1}\binom{n+1}{k}\partial_x^{n+1-k}(x\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi}^{(k)}-\sum_{k=0}^{n}\binom{n}{k}\partial_x^{n-k}(\chi_{\varrho\geq R_\infty})S_\lambda\widehat{\psi}^{(k)}.\end{aligned}$$

Define

$$\begin{aligned}\mathfrak{R}_{s,\lambda}^{(n)} := \partial_x^n\mathfrak{R}_{s,\lambda}-2(k+1)\left(x+\frac{1}{2}\kappa\chi_{\varrho\geq R_\infty}\right)\partial_x\widehat{\psi}^{(n)}-\kappa\sum_{k=0}^{n+1}\binom{n+1}{k}\partial_x^{n+1-k}(x\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi}^{(k)}\\ +\sum_{k=0}^{n}\binom{n}{k}\partial_x^{n-k}(\chi_{\varrho\geq R_\infty})\pi_{\leq l}S_\lambda\widehat{\psi}^{(k)}.\end{aligned}$$

Then

$$\partial_x^n(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}) = \partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x\widehat{\psi}^{(n)})+(\not{\Delta}_{\mathbb{S}^2}-\lambda\pi_{\leq l}-\chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l})\widehat{\psi}^{(n)}+\mathfrak{R}_{s,\lambda}^{(n)}.$$

We obtain the following analogue of Proposition 12.8, including additional commutation with vector fields $(x^2\partial_x)^k$.

Corollary 12.7. *Let $N\in\mathbb{N}_0$, $0<\delta<1$ and $p\geq 0$ and assume the following smallness relations:*

$$0<\eta<\frac{1}{4},$$
$$\kappa \ll \eta R_\infty^{-1},$$

$$M^{-1}R_\infty \gg 1,$$
$$\lambda - N(N+1) \gg 1 + |s|^2\eta^{-2}R_\infty^2 + p.$$

Then, there exists a numerical constant $C = C(N) > 0$ such that:

$$\sum_{k=0}^{N}\sum_{\ell\in\mathbb{N}_0}\int_{x(\eta^{-1})}^{3R_\infty^{-1}} x^{1-\delta}(\mathfrak{l}+1)^{-w_p}\Bigg(|\partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x(x^2\partial_x)^k\widehat{\psi}_\ell)|^2+|(\not\Delta_{\mathbb{S}^2}-\lambda\pi_{\leq l}-\chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l})(x^2\partial_x)^k\widehat{\psi}_\ell|^2$$
$$+2x(x+\kappa\chi_{\varrho\geq R_\infty})\left[|\not\nabla_{\mathbb{S}^2}\partial_x(x^2\partial_x)^k\widehat{\psi}_\ell|^2+\lambda\pi_{\leq l}|\partial_x(x^2\partial_x)^k\widehat{\psi}_\ell|^2+\chi_{\varrho\geq R_\infty}|\sqrt{S_\lambda\pi_{\leq l}}\partial_x(x^2\partial_x)^k\widehat{\psi}_\ell|^2\right]\Bigg)\,d\sigma dx$$
$$\leq \sum_{k=0}^{N}\boxed{C\int_{\frac{1}{4}R_\infty}^{\frac{1}{3}R_\infty}\int_{\mathbb{S}^2} r^{-2+\delta}\left|(\not\Delta_{\mathbb{S}^2}-\lambda\pi_{\leq l})\partial_r^k\widehat{\psi}\right|\cdot\left|\partial_r^{k+1}\widehat{\psi}\right|\,d\sigma dr}$$
$$+\boxed{C(\mathfrak{l}+1)^{-p}\sum_{\ell\in\mathbb{N}_0}\int_{\mathbb{S}^2} x^{2+4k-\delta}(x+\kappa)\mathfrak{l}(\mathfrak{l}+1)|\widehat{\psi}_\ell^{(k)}|\cdot|\partial_x\widehat{\psi}_\ell^{(k)}|\,d\sigma\Big|_{x=\eta R_\infty^{-1}}}$$
$$+C\sum_{\ell\in\mathbb{N}_0}\int_{x(\eta^{-1})}^{4R_\infty^{-1}} x^{1-\delta}(\mathfrak{l}+1)^{-w_p}|(x^2\partial_x)^k(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2\,d\sigma dx. \quad (12.9)$$

Proof. We repeat the proof of Proposition 12.6 with $\widehat{\psi}^{(k)}$ replacing $\widehat{\psi}$ and with $q = 1-\delta+4k$, sum over k, and use that all the terms coming from $\mathfrak{R}_{s,\lambda}^{(n)}$ can be absorbed in the terms J_0 and J_2. □

12.3. **Combining the estimates in (A) and (B).** In this section we combine the estimates from Corollary 12.3 and Corollary 12.7 by taking λ suitably large.

Corollary 12.8. *Let $N_+ \in \mathbb{N}_0$, $0 < \delta < 1$, $p \geq 0$, and $\operatorname{Re} s > -\kappa_+\left(\frac{1}{2}+N_+\right)$, and assume the following smallness relations:*

$$0 < \eta < \frac{1}{4},$$
$$\kappa \ll \eta R_\infty^{-1},$$
$$M^{-1}R_\infty \gg 1,$$
$$\lambda - N_+(N_++1) \gg p + N_+|s|^2(1+2N_++2\kappa_+^{-1}\min\{\operatorname{Re} s,0\})^{-1} + (1+2N_++2\kappa_+^{-1}\min\{\operatorname{Re} s,0\})^{-2}$$
$$+|s|^2\eta^{-2}R_\infty^2 + (M^{-1}R_\infty)^{2(1+N_+)}.$$

Then, there exists a constant $C = C(N_+) > 0$ such that:

$$\sum_{k=0}^{N_+}\int_{r_+}^{\frac{1}{2}R_\infty}|\partial_r^{1+k}\not\nabla_{\mathbb{S}^2}\widehat{\psi}|^2+\lambda|\partial_r^{1+k}\widehat{\psi}|^2+|(\not\Delta_{\mathbb{S}^2}-\lambda\pi_{\leq l})\widehat{\psi}|^2+\chi_{r\geq\frac{1}{3}R_\infty}|\partial_r^{2+k}\psi|^2\,d\sigma dr$$
$$+\sum_{k=0}^{N_+}\sum_{\ell\in\mathbb{N}_0}\int_{\eta R_\infty^{-1}}^{x(\frac{1}{2}R_\infty)} x^{1-\delta}(\mathfrak{l}+1)^{-w_p}\Bigg(|\partial_x(x(x+\kappa\chi_{\varrho\geq R_\infty})\partial_x(x^2\partial_x)^k\widehat{\psi}_\ell)|^2+|(\not\Delta_{\mathbb{S}^2}-\lambda\pi_{\leq l}-\chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l})(x^2\partial_x)^k\widehat{\psi}_\ell|^2$$
$$+2x(x+\kappa\chi_{\varrho\geq R_\infty})\left[|\not\nabla_{\mathbb{S}^2}\partial_x(x^2\partial_x)^k\widehat{\psi}_\ell|^2+\lambda\pi_{\leq l}|\partial_x(x^2\partial_x)^k\widehat{\psi}_\ell|^2+\chi_{\varrho\geq R_\infty}|\sqrt{S_\lambda\pi_{\leq l}}\partial_x(x^2\partial_x)^k\widehat{\psi}_\ell|^2\right]\Bigg)\,d\sigma dx$$
$$\leq \boxed{C(\mathfrak{l}+1)^{-p}\sum_{k=0}^{N_+}\sum_{\ell\in\mathbb{N}_0}\int_{\mathbb{S}^2} x^{2+4k-\delta}(x+\kappa)\mathfrak{l}(\mathfrak{l}+1)|\widehat{\psi}_\ell^{(k)}|\cdot|\partial_x\widehat{\psi}_\ell^{(k)}|\,d\sigma\Big|_{x=\eta R_\infty^{-1}}}$$

$$+ C_s \sum_{k=0}^{N_+} \int_{r_+}^{\frac{1}{2}R_\infty} |\partial_r^k \mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2 + \chi_{r\leq \frac{1}{3}R_\infty}(|\not\nabla_{\mathbb{S}^2}\partial_r^k \mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2 + \lambda\pi_{\leq l}|\partial_r^k \mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2)\, d\sigma dr$$
$$+ C_s \sum_{k=0}^{N_+}\sum_{\ell\in\mathbb{N}_0} \int_{\eta R_\infty^{-1}}^{x(\frac{1}{2}R_\infty)} x^{1-\delta}(\mathfrak{l}+1)^{-w_p}|(x^2\partial_x)^k(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2\, d\sigma dx. \quad (12.10)$$

Furthermore, for suitably small $\frac{|a|}{M}$ *or for* $\widehat{\phi} = \sum_{m=-m_0}^{m_0}\widehat{\phi}_m$*, with* $m_0 > 0$*, we can omit the* $|\not\nabla_{\mathbb{S}^2}\partial_r^k \mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2$ *term on the right-hand side.*

Proof. We apply Corollary 12.3 with arbitrary $N_+ \in \mathbb{N}_0$ and $m \in \{0,1\}$ and we also apply Corollary 12.7 with the same value of N_+. We estimate the first term on the right-hand side of (12.9) as follows: let $\epsilon > 0$ be arbitrarily small, then we apply Young's inequality and use that $\widehat{\psi} = r\phi$ in the region under consideration to obtain:

$$C\sum_{k=0}^{N_+}\int_{\mathbb{S}^2} r^{-2+\delta}\left|(\not\Delta_{\mathbb{S}^2} - \lambda\pi_{\leq l})\partial_r^k\widehat{\psi}\right|\cdot\left|\partial_r^{k+1}\widehat{\psi}\right| d\sigma \leq \sum_{k=0}^{N_+}\int_{\mathbb{S}^2}\epsilon r^{-1+\delta}\left|(\not\Delta_{\mathbb{S}^2} - \lambda\pi_{\leq l})\partial_r^k\widehat{\phi}\right|^2 + \frac{C^2}{4\epsilon\lambda}\lambda r^{1+\delta}\left|\partial_r^{k+1}\widehat{\phi}\right|^2 d\sigma.$$

We take λ suitably large so that $\frac{C^2}{4\epsilon\lambda} < \epsilon$ and $\epsilon \ll (1 + 2N_+ + 2\kappa_+^{-1}\mathrm{Re}\, s)$. Then we can absorb the right-hand side above into the left-hand side of (12.4) with $m = 0$, and $m = 1$, after multiplying the $m = 0$ estimates by a factor λ.

By taking ϵ suitably small compared to the constant $CR_\infty^{2N_+}$ appearing on the right-hand side of (12.4), we can also absorb the first integral on the right-hand side of (12.4) into the left-hand side of (12.9). □

Remark 12.1. *Note that the estimate* (12.10) *still has* $\widehat{\psi}$ *and its derivatives appearing in the boxed boundary term at* $x = \eta R_\infty^{-1}$*. In the next section we will show that we can absorb this term into the left-hand side of appropriate Gevrey-type estimates.*

When $\mathrm{Re}\, s > 0$ is suitably large, we can also derive estimates for $(\mathfrak{L}_{s,\kappa} + K_{s,\lambda})\widehat{\psi}$, rather than $\mathfrak{L}_{s,\kappa}\widehat{\psi}$. This may be thought of a a frequency-space manifestation of the Grönwall inequality.

Proposition 12.9. *Let* $s_0 > 0$ *and* $\mathrm{Re}\, s \geq s_0$*. For suitably large* s_0*, there exists a constant* $C_{s_0} > 0$*, such that:*

$$||(1 - \not\Delta_{\mathbb{S}^2})^{\frac{-w_p}{4}}\widehat{\psi}||_{H^{N+1}(\Sigma)} \leq C_{s_0}||(1 - \not\Delta_{\mathbb{S}^2})^{\frac{-w_p}{4}}(\mathfrak{L}_{s,\kappa} + K_{s,\lambda})\widehat{\psi}||_{H^N(\Sigma)}. \quad (12.11)$$

Proof. Note that $(\mathfrak{L}_{s,\kappa} + K_{s,\Lambda})(1 - \not\Delta_{\mathbb{S}^2})^{\frac{w_p}{4}}\widehat{\psi} = (1 - \not\Delta_{\mathbb{S}^2})^{\frac{w_p}{4}}(\mathfrak{L}_{s,\kappa} + K_{s,\lambda})$, using that $\mathfrak{L}_{s,\kappa}$ is a spherically symmetric operator in the support of w_p. Now, consider the multiplier $\overline{s\widehat{\psi} + f(r)\partial_\varrho\widehat{\psi}}$, such that $f \equiv 0$ in $\varrho \geq \frac{1}{3}R_\infty$ and $N = T + f\partial_\varrho$ is timelike with respect to $g_{M,a}$. Then local energy estimates applied to $e^{s\tau}(1 - \not\Delta_{\mathbb{S}^2})^{\frac{w_p}{4}}\widehat{\psi}$ for $\mathrm{Re}\, s \geq s_0$ immediately give (12.11). □

13. Gevrey estimates

In this section, we consider estimates in the region $x \leq x_0$, where we define:

$$x_0 := \frac{1}{2R_\infty}.$$

By (7.11), we can express in the region $x \leq x_0$:

$$(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell = \partial_x(x(\kappa + x)\partial_x\widehat{\psi}_\ell) + 2s\partial_x\widehat{\psi}_\ell - \mathfrak{l}(\mathfrak{l}+1)\widehat{\psi}_\ell.$$

Hence, **all the estimates in this section are estimates for the rescaled Laplace-transformed wave operator in Minkowski**, restricted to fixed spherical harmonic modes with angular frequencies $\geq \mathfrak{l}$. We will moreover require these estimates to be uniform in $\mathfrak{l}$.

Denote $\widehat{\psi}^{(n)} = \partial_x^n \widehat{\psi}$. Then it can be easily verified that:

$$(\partial_x^n \mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell = x(\kappa+x)\partial_x \widehat{\psi}_\ell^{(n+1)} + 2(n+1)\left(x+\frac{\kappa}{2}\right)\partial_x\widehat{\psi}^{(n)} + 2s\partial_x\widehat{\psi}_\ell^{(n)} + (n(n+1)-\mathfrak{l}(\mathfrak{l}+1))\widehat{\psi}_\ell^{(n)}. \tag{13.1}$$

We will estimate higher-order energy norms of $\widehat{\psi}$ in terms of similar energy norms of $\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}$ in two steps.

Step 1: We consider $\widehat{\psi}_\ell^{(n)} = \partial_x^n\widehat{\psi}_\ell$ and derive $\dot{H}^1$ estimates for $\widehat{\psi}_\ell^{(n)}$ with $n \geq \mathfrak{l}$. We then sum the estimates for different values of n to obtain an L^2-based Gevrey-type estimate. *Here it is important that we will not see any terms $\widehat{\psi}_\ell^{(n)}$ with $n < \mathfrak{l}$.*

Step 2: We estimate weighted $\dot{H}^1$ norms of $\widehat{\psi}_\ell^{(n)}$ with $n < \mathfrak{l}$ in terms of a weighted $\dot{H}^1$ norm of $\widehat{\psi}_\ell^{(n)}$. To be able to sum these lower-order estimates, we apply elliptic-type estimates in the region $x \leq x_0$.

13.1. **Step 1: higher-order derivatives.** We will first derive weighted L^2 estimates for $\widehat{\psi}^{(n)}$ with $n \geq \mathfrak{l}$. We obtain these estimates by simply rearranging the terms in (13.1) to obtain:

$$x(\kappa+x)\partial_x\widehat{\psi}_\ell^{(n+1)} + 2(n+1)\left(x+\frac{\kappa}{2}\right)\partial_x\widehat{\psi}_\ell^{(n)} + 2s\partial_x\widehat{\psi}_\ell^{(n)} = (\mathfrak{l}(\mathfrak{l}+1)-n(n+1))\widehat{\psi}_\ell^{(n)} + (\partial_x^n\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell \tag{13.2}$$

and by taking the square norm on both sides, treating the right-hand side as an inhomogeneous term. By considering appropriate weighted sums in n of these L^2 estimates, we will then be able to absorb the terms arising from $(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))\widehat{\psi}_\ell^{(n)}$ on the right-hand side of (13.2).

Proposition 13.1. *Let $\kappa \geq 0$, $M = 1$, $\ell, l \in \mathbb{N}_0$, $N_\infty \in \mathbb{N}_0$, with $N_\infty > \mathfrak{l}$, and let $\alpha, \beta, \nu, \tilde{\sigma}, \epsilon > 0$. Then, for suitably large $\mathfrak{l}$, there exists a constant $C > 0$ such that:*

$$\begin{aligned}
&\sum_{n=\mathfrak{l}}^{N_\infty} \frac{|2s|^{2n}\tilde{\sigma}^{2n}}{n!^2(n+1)!^2}\int_0^{x_0}\left(1-\frac{|\operatorname{Im} s|}{|s|}\mu\nu^{-1}\right)x^2(x+\kappa)^2|\hat{\psi}_\ell^{(n+2)}|^2\,dx \\
&\qquad + \sum_{n=\mathfrak{l}}^{N_\infty}\frac{|2s|^{2n}\tilde{\sigma}^{2n}}{n!^2(n+1)!^2}\int_0^{x_0}\left[4-\beta-\frac{|\operatorname{Im} s|}{|s|}(1-\mu)(\alpha+\tilde{\sigma}^{-2}\alpha^{-1})-6(n+1)^{-1}\right] \\
&\qquad\qquad \times (n+1)^2\left(x+\frac{1}{2}\kappa\right)^2|\hat{\psi}_\ell^{(n+1)}|^2\,dx \\
&+\sum_{n=\mathfrak{l}}^{N_\infty}\frac{|2s|^{2n}\tilde{\sigma}^{2n}}{n!^2(n+1)!^2}\int_0^{x_0}\left(1-\frac{|\operatorname{Im} s|}{|s|}\mu\nu-4\beta^{-1}\left(\frac{\max\{-\operatorname{Re} s,0\}}{|s|}\right)^2-(1+\epsilon)\tilde{\sigma}^2\left(1+2(n-1)^{-1}\right)^2\right) \\
&\qquad\qquad \times |2s|^2|\hat{\psi}_\ell^{(n+1)}|^2\,dx \\
&\boxed{-\frac{|2s|^{2(N_\infty+1)}\tilde{\sigma}^{2(N_\infty+1)}}{(N_\infty+1)!^2(N_\infty+2)!^2}\int_0^{x_0}\frac{|\operatorname{Im} s|}{|s|}(1-\mu)\alpha^{-1}\tilde{\sigma}^{-2}(N_\infty+2)^2\left(x+\frac{1}{x^{-1}+2\kappa^{-1}}\right)^2|\hat{\psi}_\ell^{(N_\infty+2)}|^2\,dx} \\
&\qquad + \sum_{n=\mathfrak{l}}^{N_\infty}\frac{|2s|^{2n}\tilde{\sigma}^{2n}}{n!^2(n+1)!^2}x_0^3(n+1)|\widehat{\psi}^{(n+1)}|^2\big|_{x=x_0} \\
&\qquad\qquad \leq C(1+\epsilon^{-1})\sum_{n=\mathfrak{l}}^{N_\infty}\frac{|2s|^{2n}\tilde{\sigma}^{2n}}{n!^2(n+1)!^2}\int_0^{x_0}|\partial_x^n(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2\,dx.
\end{aligned} \tag{13.3}$$

When x_0 is suitably small, depending on s, the estimate (13.3) holds also for with $\mathfrak{L}_{s,\lambda,\kappa}$ replaced by $\mathfrak{L}_{s,\lambda,\kappa}+K_{s,\lambda}$ and for all $\ell \in \mathbb{N}_0$, provided we replace $\mathfrak{l}$ with ℓ and we add the additional term $C|s|^4h^2(0)x_0\frac{|2s|^{2\ell}\tilde{\sigma}^{2\ell}}{\ell!^2(\ell+1)!^2}|\widehat{\psi}_\ell^{(\ell)}|^2|_{x=x_0}$ on the right-hand side.

Proof. We will omit the subscript ℓ in $\widehat{\psi}_\ell$ for the sake of convenience. Rearranging (13.1) and taking the square norm on both sides, we obtain:

$$x^2(x+\kappa)^2|\hat{\psi}^{(n+2)}|^2 + 4(n+1)^2\left(x+\frac{1}{2}\kappa\right)^2|\hat{\psi}^{(n+1)}|^2 + 4|s|^2|\hat{\psi}^{(n+1)}|^2 + T_1 + T_2 + T_3 = \left|[\mathfrak{l}(\mathfrak{l}+1) - n(n+1)]\hat{\psi}^{(n)} + \partial_x^n(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell\right|^2,$$

with

$$\begin{aligned} T_1 &:= 2(n+1)x\left(x+\frac{1}{2}\kappa\right)(x+\kappa)\partial_x(|\hat{\psi}^{(n+1)}|^2),\\ T_2 &:= 2\,\mathrm{Re}(2s)(n+1)\,(2x+\kappa)\,|\hat{\psi}^{(n+1)}|^2,\\ T_3 &:= 2x(x+\kappa)\,\mathrm{Re}\left(\hat{\psi}^{(n+2)}2\bar{s}\overline{\hat{\psi}^{(n+1)}}\right). \end{aligned}$$

We integrate T_1 by parts to obtain:

$$\begin{aligned} \int_0^{x_0} T_1\,dx &\geq 2(n+1)x_0\left(x_0+\frac{1}{2}\kappa\right)(x_0+\kappa)|\hat{\psi}^{(n+1)}|^2|_{x=x_0}\\ &\quad - \int_0^{x_0}(n+1)(6x^2+6\kappa x+\kappa^2)|\hat{\psi}^{(n+1)}|^2\,dx\\ &\geq 2(n+1)x_0^2(x_0+\kappa)|\hat{\psi}^{(n+1)}|^2|_{x=x_0} - \int_0^{x_0} 6(n+1)\left(x+\frac{1}{2}\kappa\right)^2|\hat{\psi}^{(n+1)}|^2\,dx. \end{aligned}$$

We split T_3 as follows:

$$\begin{aligned} T_3 &= 2x(x+\kappa)\,\mathrm{Re}\left(\hat{\psi}^{(n+2)}\,\mathrm{Re}(2s)\overline{\hat{\psi}^{(n+1)}}\right) - 2x(x+\kappa)\,\mathrm{Re}\left(\hat{\psi}^{(n+2)}i\,\mathrm{Im}(2s)\overline{\hat{\psi}^{(n+1)}}\right)\\ &= x(x+\kappa)\,\mathrm{Re}(2s)\partial_x(|\overline{\hat{\psi}^{(n+1)}}|^2) - 2x(x+\kappa)\,\mathrm{Re}\left(\hat{\psi}^{(n+2)}i\,\mathrm{Im}(2s)\overline{\hat{\psi}^{(n+1)}}\right)\\ &= \partial_x(x(x+\kappa)\,\mathrm{Re}(2s)|\hat{\psi}^{(n+1)}|^2) - (2x+\kappa)\,\mathrm{Re}(2s)|\hat{\psi}^{(n+1)}|^2 - 2x(x+\kappa)\,\mathrm{Re}\left(\hat{\psi}^{(n+2)}i\,\mathrm{Im}(2s)\overline{\hat{\psi}^{(n+1)}}\right). \end{aligned}$$

Note that since $\mathrm{Re}(s)\geq 0$, the total derivative produces only non-negative boundary terms after integrating over $[0,x_0]$. When $\mathrm{Re}(s)<0$, we can instead absorb the corresponding boundary term at $x=x_0$ into the boundary term generated by integrating by parts T_1 above, provided n is sufficiently large compared to $x_0^{-1}|\,\mathrm{Re}\,s|$. We will group the term $-(2x+\kappa)\,\mathrm{Re}(2s)|\hat{\psi}^{(n+1)}|^2$ with T_2.

If $\mathrm{Re}\,s\geq 0$, then $T_2 - (2x+\kappa)\,\mathrm{Re}(2s)|\hat{\psi}^{(n+1)}|^2 \geq 0$.

If $\mathrm{Re}\,s<0$, then we introduce a constant $\beta>0$ and apply Young's inequality to split:

$$|T_2 - (2x+\kappa)\,\mathrm{Re}(2s)|\hat{\psi}^{(n+1)}|^2| \leq \beta\left(n+\frac{1}{2}\right)^2\left(x+\frac{1}{2}\kappa\right)^2|\hat{\psi}^{(n+1)}|^2 + 4\beta^{-1}\left(\frac{|\,\mathrm{Re}\,s|}{|s|}\right)^2|2s|^2|\hat{\psi}^{(n+1)}|^2.$$

We introduce additionally $0\leq\mu\leq 1$ and split

$$\begin{aligned} |2x(x+\kappa)\,\mathrm{Re}\left(\hat{\psi}^{(n+2)}i\,\mathrm{Im}(2s)\overline{\hat{\psi}^{(n+1)}}\right)| &\leq 2x(x+\kappa)\frac{|\,\mathrm{Im}\,s|}{|s|}|2s||\hat{\psi}^{(n+1)}||\hat{\psi}^{(n+2)}|\\ &= \mu\frac{|\,\mathrm{Im}\,s|}{|s|}\cdot 2x(x+\kappa)|2s||\hat{\psi}^{(n+1)}||\hat{\psi}^{(n+2)}| + (1-\mu)\frac{|\,\mathrm{Im}\,s|}{|s|}\cdot 2x(x+\kappa)|2s||\hat{\psi}^{(n+1)}||\hat{\psi}^{(n+2)}|. \end{aligned}$$

We further estimate via Young's inequality with an additional weight $\nu>0$:

$$\mu\frac{|\,\mathrm{Im}\,s|}{|s|}\cdot 2x(x+\kappa)|2s||\hat{\psi}^{(n+1)}||\hat{\psi}^{(n+2)}| \leq \mu\frac{|\,\mathrm{Im}\,s|}{|s|}\nu|2s|^2|\hat{\psi}^{(n+1)}|^2 + \mu\frac{|\,\mathrm{Im}\,s|}{|s|}\nu^{-1}x^2(x+\kappa)^2|\hat{\psi}^{(n+2)}|^2,$$

and we estimate the remaining term with a slightly different Young's inequality, involving the weight α and an additional factor $\frac{|2s|^2\tilde{\sigma}^2}{(n+1)^2(n+2)^2}$ that appears in the summation below:

$$\frac{|\operatorname{Im} s|}{|s|}(1-\mu)\cdot 2x(x+\kappa)|2s\|\hat{\psi}^{(n+1)}\|\hat{\psi}^{(n+2)}| \leq \alpha(1-\mu)\frac{|\operatorname{Im} s|}{|s|}(n+1)^2\left(x+\frac{1}{2}\kappa\right)^2|\hat{\psi}^{(n+1)}|^2 + (1-\mu)\frac{|\operatorname{Im} s|}{|s|}\alpha^{-1}\tilde{\sigma}^{-2}\cdot\frac{|2s|^2\tilde{\sigma}^2}{(n+1)^2(n+2)^2}(n+1+1)^2\frac{x^2(x+\kappa)^2}{\left(x+\frac{1}{2}\kappa\right)^2}|\hat{\psi}^{(n+1+1)}|^2.$$

We moreover have that $\frac{x^2(x+\kappa)^2}{\left(x+\frac{1}{2}\kappa\right)^2} = (x+\frac{1}{x^{-1}+2\kappa^{-1}})^2 \leq (x+\frac{1}{2}\kappa)^2$.

Finally, we estimate

$$|[\mathfrak{l}(\mathfrak{l}+1)-n(n+1)]\hat{\psi}^{(n)}+\partial_x^n((\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell)|^2 \leq (1+\epsilon)(n(n+1)-\mathfrak{l}(\mathfrak{l}+1))^2|\hat{\psi}^{(n+1-1)}|^2 + (1+\epsilon^{-1})|(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2.$$

Now we integrate the above estimates over the interval $[0,x_0]$ to obtain for $\operatorname{Re} s<0$:

$$\int_0^{x_0}(1-\frac{|\operatorname{Im} s|}{|s|}\mu\nu^{-1})x^2(x+\kappa)^2|\hat{\psi}^{(n+2)}|^2\,dx + 2(n+1-|\operatorname{Re} s|x_0^{-1})x_0^2(x_0+\kappa)|\hat{\psi}^{(n+1)}|^2|_{x=x_0} + \int_0^{x_0}\left[4-\beta-\alpha\frac{|\operatorname{Im} s|}{|s|}(1-\mu)-6(n+1)^{-1}\right](n+1)^2\left(x+\frac{1}{2}\kappa\right)^2|\hat{\psi}^{(n+1)}|^2\,dx$$

$$\boxed{-\int_0^{x_0}\frac{|\operatorname{Im} s|}{|s|}(1-\mu)\alpha^{-1}\tilde{\sigma}^{-2}\cdot\frac{|2s|^2\tilde{\sigma}^2}{(n+1)^2(n+2)^2}(n+1+1)^2\left(x+\frac{1}{x^{-1}+2\kappa^{-1}}\right)^2|\hat{\psi}^{(n+1+1)}|^2\,dx}$$

$$+\int_0^{x_0}\left(1-\frac{|\operatorname{Im} s|}{|s|}\mu\nu-4\beta^{-1}\left(\frac{|\operatorname{Re} s|}{|s|}\right)^2\right)|2s|^2|\hat{\psi}^{(n+1)}|^2\,dx$$

$$\boxed{-\int_0^{x_0}(1+\epsilon)|2s|^2\tilde{\sigma}^2\left(\frac{n(n+1)-\mathfrak{l}(\mathfrak{l}+1)}{n(n-1)}\right)^2\frac{n^2(n-1)^2}{|2s|^2\tilde{\sigma}^2}|\hat{\psi}^{(n+1-1)}|^2\,dx}$$

$$\leq (1+\epsilon^{-1})\int_0^{x_0}|\partial_x^n(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2\,dx.$$

For $\operatorname{Re} s\geq 0$, we conclude:

$$\int_0^{x_0}(1-\frac{|\operatorname{Im} s|}{|s|}\mu\nu^{-1})x^2(x+\kappa)^2|\hat{\psi}^{(n+2)}|^2\,dx + 2(n+1)x_0^2(x_0+\kappa)|\hat{\psi}^{(n+1)}|^2|_{x=x_0} + \int_0^{x_0}\left[4-\alpha\frac{|\operatorname{Im} s|}{|s|}(1-\mu)-6(n+1)^{-1}\right](n+1)^2\left(x+\frac{1}{2}\kappa\right)^2|\hat{\psi}^{(n+1)}|^2\,dx$$

$$\boxed{-\int_0^{x_0}\frac{|\operatorname{Im} s|}{|s|}(1-\mu)\alpha^{-1}\tilde{\sigma}^{-2}\cdot\frac{|2s|^2\tilde{\sigma}^2}{(n+1)^2(n+2)^2}(n+1+1)^2\left(x+\frac{1}{x^{-1}+2\kappa^{-1}}\right)^2|\hat{\psi}^{(n+1+1)}|^2\,dx}$$

$$+\int_0^{x_0}\left(1-\frac{|\operatorname{Im} s|}{|s|}\mu\nu\right)|2s|^2|\hat{\psi}^{(n+1)}|^2\,dx$$

$$\boxed{-\int_0^{x_0}(1+\epsilon)|2s|^2\tilde{\sigma}^2\left(\frac{n(n+1)-\mathfrak{l}(\mathfrak{l}+1)}{n(n-1)}\right)^2\frac{n^2(n-1)^2}{|2s|^2\tilde{\sigma}^2}|\hat{\psi}^{(n+1-1)}|^2\,dx}$$

$$\leq (1+\epsilon^{-1})\int_0^{x_0}|\partial_x^n(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2\,dx.$$

We can sum the above estimate over n with the following weights:

$$\sum_{n=\mathfrak{l}}^{N_\infty} \frac{|2s|^{2n}\tilde{\sigma}^{2n}}{n!^2(n+1)!^2}\,[\cdot]\,,$$

and absorbing all but the top order term with a bad sign into the terms with good signs, we obtain (13.3).

Note that we can also express:

$$x^2(x+\kappa)^2|\hat{\psi}^{(n+2)}|^2 + 4(n+1)^2\left(x+\frac{1}{2}\kappa\right)^2|\hat{\psi}^{(n+1)}|^2 + 4|s|^2|\hat{\psi}^{(n+1)}|^2$$
$$+ T_1 + T_2 + T_3 = \left|[\ell(\ell+1) - 2M^2h(0)s^2 - n(n+1)]\hat{\psi}^{(n)} + \partial_x^n((\mathfrak{L}_{s,\lambda,\kappa} + K_{s,\lambda} - K_s^\infty)\widehat{\psi})_\ell\right|^2.$$

Then we apply a standard Hardy inequality to estimate:

$$M^2h^2(0)|s|^4\int_0^{x_0}|\hat{\psi}^{(n)}|^2\,dx \leq 4M^2h^2(0)|s|^4\int_0^{x_0}x^2|\hat{\psi}^{(n+1)}|^2\,dx + 2M^2h^2(0)x_0|s|^4|\hat{\psi}^{(n)}|^2(x_0).$$

We can absorb the integral of $x^2|\hat{\psi}^{(n+1)}|^2$ into $4|s|^2|\hat{\psi}^{(n+1)}|^2$, provided x_0 is chosen so that $M^2h^2(0)x_0^2|s|^2$ is suitably small. □

In order for the unboxed terms on the left-hand side of (13.3) to be non-negative definite, the coefficients $\tilde{\sigma}, \alpha, \beta, \mu, \nu$ need to satisfy the following compatibility conditions:

$$1 - \frac{|\operatorname{Im} s|}{|s|}\mu\nu^{-1} > 0, \tag{13.4}$$

$$4 - \beta - \frac{|\operatorname{Im} s|}{|s|}(1-\mu)(\alpha + \tilde{\sigma}^{-2}\alpha^{-1}) > 0 \quad (\operatorname{Re} s < 0), \tag{13.5}$$

$$4 - \frac{|\operatorname{Im} s|}{|s|}(1-\mu)(\alpha + \tilde{\sigma}^{-2}\alpha^{-1}) > 0 \quad (\operatorname{Re} s \geq 0), \tag{13.6}$$

$$1 - \frac{|\operatorname{Im} s|}{|s|}\mu\nu - 4\beta^{-1}\left(\frac{|\operatorname{Re} s|}{|s|}\right)^2 - \tilde{\sigma}^2 > 0 \quad (\operatorname{Re} s < 0), \tag{13.7}$$

$$1 - \frac{|\operatorname{Im} s|}{|s|}\mu\nu - \tilde{\sigma}^2 > 0 \quad (\operatorname{Re} s \geq 0) \tag{13.8}$$

and we need to assume that $\mathfrak{l}$ is suitably large.

Lemma 13.2. *Let $\delta \in (0,1)$ be arbitrarily small.*

(i) *There exist $\tilde{\sigma}$, α, β, μ, ν satisfying (13.4)–(13.8) if: $\frac{|\operatorname{Im} s|^2}{|s|^2} > \frac{2}{1+\sqrt{5}}$,*

(ii) *Fix $2|s|\tilde{\sigma} = \sigma$, with $\sigma \in \mathbb{R}$. Then there exist α, β, μ, ν satisfying (13.4)–(13.8) if $s \in \Omega_\sigma$, where*

$$\Omega_\sigma := \left\{x+iy \in \mathbb{C}\,\middle|\, -\frac{4-b}{2b}\sqrt{4-b^2}\frac{1}{2}|\sigma| < x < 0,\right.$$
$$\left. y \notin \left[-\left(-1+4b^{-2}+\frac{b}{2}\right)\frac{1}{2}|\sigma|, \left(-1+4b^{-2}+\frac{b}{2}\right)\frac{1}{2}|\sigma|\right],\ b \in (0,2)\right\}$$
$$\cup \left\{x+iy \in \mathbb{C}\,|\, x \in [0,\infty),\ x^2+y^2 > \frac{1}{4}\sigma^2\right\}.$$

Note that

$$\Omega := \bigcup_{\sigma\in\mathbb{R}} \Omega_\sigma = \left\{ z\in\mathbb{C}\,,\ \frac{|\operatorname{Im} z|^2}{|z|^2} > \frac{2}{1+\sqrt{5}} \right\} = \left\{ z\in\mathbb{C}\,,\ |\arg z| < \pi - \arcsin\left(\sqrt{\frac{2}{1+\sqrt{5}}}\right) \right\}.$$

Proof. Appendix A. □

See Figure 3 for a schematic picture of Ω_σ.

By combining Proposition 13.1 with Lemma 13.2, we immediately obtain:

Corollary 13.3. *Let $\kappa \geq 0$, $M = 1$, $\ell, l \in \mathbb{N}_0$, $N_\infty \in \mathbb{N}_0$, with $N_\infty > l$. Let $\sigma \in \mathbb{R}$ and let $s \in \Omega_\sigma$. Then, for suitably large ℓ, there exists a constant $C_{s,\sigma} = C(s,\sigma) > 0$, such that:*

$$\begin{aligned}\sum_{n=l}^{N_\infty} \frac{\sigma^{2n}}{n!^2(n+1)!^2} \int_0^{x_0} x^2(x+\kappa)^2 |\hat{\psi}_\ell^{(n+2)}|^2\,dx + \sum_{n=l}^{N_\infty} \frac{\sigma^{2n}}{n!^2(n+1)!^2}(n+1)x_0^3 |\widehat{\psi}^{(n+1)}|^2\big|_{x=x_0} \\ + \sum_{n=l}^{N_\infty} \frac{\sigma^{2n}}{n!^2(n+1)!^2} \int_0^{x_0} \left[|2s|^2 + (n+1)^2 \left(x+\frac{1}{2}\kappa\right)^2 \right] |\hat{\psi}_\ell^{(n+1)}|^2\,dx \\ \boxed{-C_{s,\sigma} \frac{|2s|^{2(N_\infty+1)}\tilde{\sigma}^{2(N_\infty+1)}}{(N_\infty+1)!^2(N_\infty+2)!^2} \int_0^{x_0} (N_\infty+2)^2 \left(x + \frac{1}{x^{-1}+2\kappa^{-1}}\right)^2 |\hat{\psi}_\ell^{(N_\infty+2)}|^2\,dx} \\ \leq C_{s,\sigma} \sum_{n=l}^{N_\infty} \frac{\sigma^{2n}}{n!^2(n+1)!^2} \int_0^{x_0} |\partial_x^n (\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2\,dx. \quad (13.9)\end{aligned}$$

When x_0 is suitably small, depending on s, the estimate (13.9) holds also for with $\mathfrak{L}_{s,\lambda,\kappa}$ replaced by $\mathfrak{L}_{s,\lambda,\kappa} + K_{s,\lambda}$ and for all $\ell \in \mathbb{N}_0$, provided we replace l with ℓ and we add the additional term $C|s|^4 h^2(0) x_0 \frac{|2s|^{2\ell}\tilde{\sigma}^{2\ell}}{\ell!^2(\ell+1)!^2} |\widehat{\psi}_\ell^{(\ell)}|^2|_{x=x_0}$ on the right-hand side.

We now restrict to the case $\kappa > 0$ to absorb the boxed boundary term in (13.10) and arrive at a closed, uniform estimate in κ:

Corollary 13.4. *Let $\kappa > 0$, $M = 1$, $\ell, l \in \mathbb{N}_0$, $N_\infty \in \mathbb{N}_0$, with $N_\infty > l$. Let $\sigma \in \mathbb{R}$ and let $s \in \Omega_\sigma$. Then there exists a constant $C_{s,\sigma} = C(s,\sigma) > 0$ and a $N_\kappa > l$, such that for all $N_\infty \geq N_\kappa$:*

$$\begin{aligned}\sum_{n=l}^{N_\infty} \frac{\sigma^{2n}}{n!^2(n+1)!^2} \int_0^{x_0} x^2(x+\kappa)^2 |\hat{\psi}_\ell^{(n+2)}|^2\,dx + \sum_{n=l}^{N_\infty} \frac{\sigma^{2n}}{n!^2(n+1)!^2}(n+1)x_0^3 |\widehat{\psi}^{(n+1)}|^2\big|_{x=x_0} \\ + \sum_{n=l}^{N_\infty} \frac{\sigma^{2n}}{n!^2(n+1)!^2} \int_0^{x_0} \left[|2s|^2 + (n+1)^2 \left(x+\frac{1}{2}\kappa\right)^2 \right] |\hat{\psi}_\ell^{(n+1)}|^2\,dx \\ \leq C_{s,\sigma} \sum_{n=l}^{N_\infty} \frac{\sigma^{2n}}{n!^2(n+1)!^2} \int_0^{x_0} |\partial_x^n (\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2\,dx. \quad (13.10)\end{aligned}$$

When x_0 is suitably small, depending on s, the estimate (13.10) holds also for with $\mathfrak{L}_{s,\lambda,\kappa}$ replaced by $\mathfrak{L}_{s,\lambda,\kappa} + K_{s,\lambda}$ and for all $\ell \in \mathbb{N}_0$, provided we replace l with ℓ and we add the additional term $C|s|^4 h^2(0) x_0 \frac{|2s|^{2\ell}\tilde{\sigma}^{2\ell}}{\ell!^2(\ell+1)!^2} |\widehat{\psi}_\ell^{(\ell)}|^2|_{x=x_0}$ on the right-hand side.

Proof. Note that

$$C_{s,\sigma} \frac{|2s|^2\tilde{\sigma}^2}{(N_\infty+1)^2(N_\infty+2)^2}(N_\infty+2)^2 \left(x + \frac{1}{x^{-1}+2\kappa^{-1}}\right)^2 < 4\frac{C_{s,\sigma}|2s|^2\tilde{\sigma}^2\kappa^{-2}}{(N_\infty+1)^2}\kappa^2 x^2,$$

so, for $N_\infty + 1 > 4\sqrt{C_{s,\sigma}}|2s||\tilde{\sigma}|\kappa^{-1} =: N_\kappa + 1$, we can absorb the boxed term in (13.9) into the first term on the right-hand side of (13.9). □

Remark 13.1. *Note that the estimate* (13.10) *is* closed, *i.e. there are only terms involving* $(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell$ *on the right-hand side. This means that we can control weighted* L^2*-norms of* $\widehat{\psi}_\ell^{(n)}$ *with* $n \geq \mathfrak{l}+1$ *in* $x \leq x_0$ *independently from* $\widehat{\psi}_\ell^{(n)}$ *with* $n \leq \mathfrak{l}$.

13.2. **Step 2: lower-order derivatives.** In this section, we will establish estimates for appropriately weighted $\dot{H}^1$ estimates in the region $\{0 < x \leq x_0\}$ for $\widehat{\psi}^{(n)}$ with $n \leq \mathfrak{l}$ in terms of a $\dot{H}^1$ norm for $\widehat{\psi}^{(\mathfrak{l}+1)}$ and lower-order boundary terms.

The estimates in this section rely on a different rearranging of (13.1), compared to Step 1:

$$x(\kappa+x)\partial_x\widehat{\psi}_\ell^{(n+1)} + 2(n+1)\left(x+\frac{\kappa}{2}\right)\partial_x\widehat{\psi}_\ell^{(n)} - (\mathfrak{l}(\mathfrak{l}+1)-n(n+1))\widehat{\psi}_\ell^{(n)} = -2s\partial_x\widehat{\psi}_\ell^{(n)} + (\partial_x^n\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell.$$

The main estimates are obtained by considering the square norm of both sides of the equation and summing over n with a appropriately weighted sum.

Proposition 13.5. *Let* $\kappa \leq x_0$ *and* $\gamma > 1$. *Then there exists a constant* $C = C(\gamma) > 0$ *such that*

$$\begin{aligned}\sum_{n=0}^{\mathfrak{l}-1}\frac{\gamma^{2n}|2s|^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2}\int_0^{x_0}\int_{\mathbb{S}^2}(x^4+\kappa^2x^2)|\hat{\psi}^{(n+2)}|^2+(x^2(\mathfrak{l}+1)^2+\kappa^2n(n+1)+|2s|^2)|\hat{\psi}^{(n+1)}|^2\\ +(\mathfrak{l}-n)^2(\mathfrak{l}+n+1)^2|\hat{\psi}^{(n)}|^2\,d\sigma dx\\ \leq C\frac{|2s|^{2(\mathfrak{l}-1)}\gamma^{2(\mathfrak{l}-1)}}{(2\mathfrak{l}!)^2}\int_0^{x_0}\int_{\mathbb{S}^2}|\hat{\psi}^{(\mathfrak{l}+1)}|^2\,dx + C\frac{|2s|^{2(\mathfrak{l}-1)}\gamma^{2(\mathfrak{l}-1)}}{(2\mathfrak{l}!)^2}\int_{\mathbb{S}^2}|\hat{\psi}^{(\mathfrak{l})}|^2|_{x=x_0}\,d\sigma\\ +C\sum_{n=0}^{\mathfrak{l}-1}\frac{\gamma^{2n}|2s|^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2}(\mathfrak{l}-n)(\mathfrak{l}+n+1)(\mathfrak{l}+1)^2x_0^{-1}\int_{\mathbb{S}^2}|\widehat{\psi}^{(n)}|^2|_{x=x_0}d\sigma\\ +C\sum_{n=0}^{\mathfrak{l}-1}\frac{\gamma^{2n}|2s|^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2}\int_0^{x_0}\int_{\mathbb{S}^2}|\partial_x^n\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2\,d\sigma dx. \quad (13.11)\end{aligned}$$

The estimates (13.11) *holds also with* $\mathfrak{L}_{s,\lambda,\kappa}$ *replaced by* $\mathfrak{L}_{s,\lambda,\kappa}+K_{s,\lambda}$, *provided we replace* $\mathfrak{l}$ *with* ℓ.

Furthermore, there exists a constant $C = C(\gamma,|s|) > 0$ *such that*

$$\begin{aligned}(\mathfrak{l}+1)^{-(4N_++1)}\sum_{n=0}^{N_++1}\int_{\mathbb{S}^2}x|\hat{\psi}^{(n)}|^2|_{x=\eta x_0}d\sigma \leq C\frac{|2s|^{2(\mathfrak{l}-1)}\gamma^{2(\mathfrak{l}-1)}}{(2\mathfrak{l}!)^2}\int_0^{x_0}\int_{\mathbb{S}^2}|\hat{\psi}^{(\mathfrak{l}+1)}|^2\,dx\\ +C\frac{|2s|^{2(\mathfrak{l}-1)}\gamma^{2(\mathfrak{l}-1)}}{(2\mathfrak{l}!)^2}\int_{\mathbb{S}^2}|\hat{\psi}^{(\mathfrak{l})}|^2|_{x=x_0}\,d\sigma\\ +C\sum_{n=0}^{\mathfrak{l}-1}\frac{\gamma^{2n}|2s|^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2}(\mathfrak{l}-n)(\mathfrak{l}+n+1)(\mathfrak{l}+1)^2x_0^{-1}\int_{\mathbb{S}^2}|\widehat{\psi}^{(n)}|^2|_{x=x_0}d\sigma\\ +C\sum_{n=0}^{\mathfrak{l}-1}\frac{\gamma^{2n}|2s|^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2}\int_0^{x_0}\int_{\mathbb{S}^2}|\partial_x^n\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2\,d\sigma dx. \quad (13.12)\end{aligned}$$

Proof. Consider (13.1), suppress the ℓ subscript and rearrange the terms to obtain:

$$\begin{aligned}|(\partial_x^n\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})-2s\widehat{\psi}^{(n+1)}|^2 &= |x(\kappa+x)\partial_x^2\widehat{\psi}^{(n)}+2(n+1)\left(x+\frac{\kappa}{2}\right)\partial_x\widehat{\psi}^{(n)}-(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))\widehat{\psi}^{(n)}|^2\\ &= x^2(x+\kappa)^2|\hat{\psi}^{(n+2)}|^2+4(n+1)^2\left(x+\frac{1}{2}\kappa\right)^2|\hat{\psi}^{(n+1)}|^2+(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))^2|\hat{\psi}^{(n)}|^2\\ &\quad +T_1+T_2+T_3, \quad (13.13)\end{aligned}$$

with

$$T_1 := 2(n+1)x\left(x+\frac{1}{2}\kappa\right)(x+\kappa)\partial_x(|\hat{\psi}^{(n+1)}|^2),$$
$$T_2 := -2(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))x(\kappa+x)\,\mathrm{Re}(\partial_x^2\widehat{\psi}^{(n)}\cdot\overline{\widehat{\psi}^{(n)}}),$$
$$T_3 := -2(n+1)(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))\left(x+\frac{\kappa}{2}\right)\partial_x(|\widehat{\psi}^{(n)}|^2).$$

We now integrate both sides of (13.13) over $[0,x_0]\times\mathbb{S}^2$, but we suppress the integration over $\mathbb{S}^2$ in the notation.

We first estimate the integral of T_1 by integrating by parts:

$$\begin{aligned}\int_0^{x_0} T_1\,dx &= -\int_0^{x_0}(n+1)(6x^2+6\kappa x+\kappa^2)|\widehat{\psi}^{(n+1)}|^2\,dx\\ &\qquad +2(n+1)x_0\left(x_0+\frac{1}{2}\kappa\right)(x_0+\kappa)|\hat{\psi}^{(n+1)}|^2|_{x=x_0}\\ &\geq -\int_0^{x_0}6(n+1)(x(x+\kappa)+\frac{1}{6}\kappa^2)|\hat{\psi}^{(n+1)}|^2\,dx+2(n+1)x_0\left(x_0+\frac{1}{2}\kappa\right)(x_0+\kappa)|\hat{\psi}^{(n+1)}|^2|_{x=x_0}\end{aligned}$$

To estimate T_2, we integrate by parts twice:

$$\begin{aligned}\int_0^{x_0} T_2\,dx &= \int_0^{x_0}2(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))x(\kappa+x)|\widehat{\psi}^{(n+1)}|^2\,dx-\int_0^{x_0}(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))(\kappa+2x)\partial_x(|\widehat{\psi}^{(n)}|^2)\,dx\\ &\qquad -2(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))x_0(\kappa+x_0)\,\mathrm{Re}(\widehat{\psi}^{(n+1)}\cdot\overline{\widehat{\psi}^{(n)}})|_{x=x_0}\\ &\geq \int_0^{x_0}2(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))x(\kappa+x)|\widehat{\psi}^{(n+1)}|^2\,dx+2\int_0^{x_0}(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))|\widehat{\psi}^{(n)}|^2\,dx\\ &-2(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))x_0(\kappa+x_0)\,\mathrm{Re}(\widehat{\psi}^{(n+1)}\cdot\overline{\widehat{\psi}^{(n)}})|_{x=x_0}-(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))(\kappa+2x_0)|\widehat{\psi}^{(n)}|^2|_{x=x_0}\end{aligned}$$

We also integrate T_3 by parts to obtain:

$$\int_0^{x_0}T_3\,dx\geq\int_0^{x_0}2(n+1)(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))|\widehat{\psi}^{(n)}|^2\,dx-2(n+1)(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))\left(x_0+\frac{\kappa}{2}\right)|\widehat{\psi}^{(n)}|^2|_{x=x_0}.$$

Finally, we apply Young's inequality to estimate for $\epsilon>0$:

$$|(\partial_x^n\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})-2s\widehat{\psi}^{(n+1)}|^2\leq(1+\epsilon)|2s|^2|\widehat{\psi}^{(n+1)}|^2+(1+\epsilon^{-1})|\partial_x^n\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2.$$

We combine the above estimates to obtain for $\gamma>1$:

$$\begin{aligned}&\int_0^{x_0}x^2(x+\kappa)^2|\hat{\psi}^{(n+2)}|^2\,dx\\ &\qquad+\int_0^{x_0}\left[(2(n+1)(n-1)+2\mathfrak{l}(\mathfrak{l}+1))x(\kappa+x)+(n+1)^2\kappa^2\right]|\hat{\psi}^{(n+1)}|^2\,dx\\ &\qquad+\int_0^{x_0}(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))(\mathfrak{l}(\mathfrak{l}+1)-n(n+1)+2(n+2))|\hat{\psi}^{(n)}|^2\,dx\\ &\boxed{-\int_0^{x_0}\frac{|2s|^2\gamma^2}{(\mathfrak{l}(\mathfrak{l}+1)-(n+1)(n+2))^2}\gamma^{-2}(1+\epsilon)(\mathfrak{l}(\mathfrak{l}+1)-(n+1)(n+2))^2|\hat{\psi}^{(n+1)}|^2\,dx}\\ &\qquad\leq C(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))(\mathfrak{l}+1)^2x_0^{-1}|\widehat{\psi}^{(n)}|^2|_{x=x_0}+(1+\epsilon^{-1})\int_0^{x_0}|\partial_x^n\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2\,dx.\end{aligned}$$

In order to absorb the boxed term above, we will sum over n with the following weights:

$$\sum_{n=0}^{\mathfrak{l}-1} \frac{\gamma^{2n}|2s|^{2n}}{(\mathfrak{l}(\mathfrak{l}+1))^2(\mathfrak{l}(\mathfrak{l}+1)-1(1+1))^2\dots(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))^2}.$$

Using that $\mathfrak{l}(\mathfrak{l}+1)-n(n+1)=(\mathfrak{l}-n)(\mathfrak{l}+n+1)$. We can simplify for $n\leq \mathfrak{l}-1$:

$$(\mathfrak{l}(\mathfrak{l}+1))^2(\mathfrak{l}(\mathfrak{l}+1)-1(1+1))^2\dots(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))^2=\frac{(\mathfrak{l}+n+1)!^2}{(\mathfrak{l}-(n+1))!^2},$$

so that the summation becomes

$$\sum_{n=0}^{\mathfrak{l}-1}\frac{\gamma^{2n}|2s|^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2}.$$

We obtain:

$$\begin{aligned}
&\sum_{n=0}^{\mathfrak{l}-1}\frac{\gamma^{2n}|2s|^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2}\int_0^{x_0}x^4|\hat{\psi}^{(n+2)}|^2\,dx\\
&\qquad+\int_0^{x_0}\left[((n+1)^2+\mathfrak{l}(\mathfrak{l}+1))x^2+|2s|^2\right]|\hat{\psi}^{(n+1)}|^2\,dx\\
&\qquad+\int_0^{x_0}(1-(1+\epsilon)\gamma^{-2})(\mathfrak{l}(\mathfrak{l}+1)-n(n+1))^2|\hat{\psi}^{(n)}|^2\,dx\\
&\qquad\leq C\frac{|2s|^{2(\mathfrak{l}-1)}\gamma^{2(\mathfrak{l}-1)}}{(2\mathfrak{l}!)^2}\int_0^{x_0}|\hat{\psi}^{(\mathfrak{l})}|^2\,dx\\
&+C\sum_{n=0}^{\mathfrak{l}-1}\frac{\gamma^{2n}|2s|^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2}(\mathfrak{l}-n)(\mathfrak{l}+n+1)(\mathfrak{l}+1)^2x_0^{-1}|\widehat{\psi}^{(n)}|^2|_{x=x_0}\\
&\qquad+C(1+\epsilon^{-1})\sum_{n=0}^{\mathfrak{l}-1}\int_0^{x_0}\frac{\gamma^{2n}|2s|^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2}|\partial_x^n\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2\,dx.
\end{aligned}$$

By applying a Hardy inequality we can further estimate:

$$\frac{|2s|^{2(\mathfrak{l}-1)}\gamma^{2(\mathfrak{l}-1)}}{(2\mathfrak{l}!)^2}\int_0^{x_0}|\hat{\psi}^{(\mathfrak{l})}|^2\,dx\leq 4x_0^2\frac{|2s|^{2(\mathfrak{l}-1)}\gamma^{2(\mathfrak{l}-1)}}{(2\mathfrak{l}!)^2}\int_0^{x_0}|\hat{\psi}^{(\mathfrak{l}+1)}|^2\,dx+2\frac{|2s|^{2(\mathfrak{l}-1)}\gamma^{2(\mathfrak{l}-1)}}{(2\mathfrak{l}!)^2}x_0|\widehat{\psi}^{(\mathfrak{l})}|^2|_{x=x_0}.$$

We conclude (13.11).

By integrating by parts T_3 in the region $[\eta x_0,x_0]\times\mathbb{S}^2$ in the argument above and using the integral estimates above, we can moreover estimate boundary terms at $x=\eta x_0$. We obtain in particular also:

$$\begin{aligned}
&(\mathfrak{l}+1)^{-4(N_++1)}\sum_{n=0}^{N_++1}x|\hat{\psi}^{(n)}|^2|_{x=\eta x_0}\leq C\frac{|2s|^{2(\mathfrak{l}-1)}\gamma^{2(\mathfrak{l}-1)}}{(2\mathfrak{l}!)^2}\int_0^{x_0}|\hat{\psi}^{(\mathfrak{l})}|^2\,dx\\
&+C\sum_{n=0}^{\mathfrak{l}-1}\frac{\gamma^{2n}|2s|^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2}(\mathfrak{l}-n)(\mathfrak{l}+n+1)(\mathfrak{l}+1)^2x_0^{-1}|\widehat{\psi}^{(n)}|^2|_{x=x_0}\\
&\qquad+C(1+\epsilon^{-1})\sum_{n=0}^{\mathfrak{l}-1}\int_0^{x_0}\frac{\gamma^{2n}|2s|^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2}|\partial_x^n\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2\,dx.
\end{aligned}$$

□

We now estimate the boundary terms at $x=x_0$ that appear on the right-hand sides of the estimates in Proposition 13.5.

Lemma 13.6. *Let $\kappa \leq x_0$ and $\mathfrak{l}^2 \gg |s|^2 x_0^{-2}$. Then there exists constants $C > 0$ suitably large and $d_0 > 0$ suitably small, with d_0 independent of $|s|$, $\mathfrak{l}$ and x_0, such that for all $n \in \mathbb{N}_0$:*

$$\sum_{n=0}^{\infty} \frac{(d_0 x_0)^{2n}}{(\max\{\mathfrak{l}+1, n+1\})^{2n}} |\hat{\psi}_\ell^{(n+2)}|^2\big|_{x=x_0} \leq C x_0^{-4} \left[(\mathfrak{l}+1)^4 |\hat{\psi}_\ell|^2 + (x_0^2 + |2s|^2) |\hat{\psi}_\ell^{(1)}|^2 \right] \big|_{x=x_0} + C x_0^{-4} \sum_{n=0}^{\infty} \frac{(d_0 x_0)^{2n}}{(\max\{\mathfrak{l}+1, n+1\})^{2n}} |\partial_x^n (\mathfrak{L}_{s,\lambda,\kappa} \widehat{\psi})_\ell|^2\big|_{x=x_0}. \quad (13.14)$$

Furthermore, for any $\gamma > 0$, with $\mathfrak{l} \gg d_0^{-1} x_0^{-1} \gamma |s|$, we can estimate

$$\sum_{n=0}^{\mathfrak{l}-1} \frac{\gamma^{2n} |2s|^{2n} (\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2} |\hat{\psi}_\ell^{(n+2)}|^2\big|_{x=x_0} \leq C x_0^{-4} \left[|\hat{\psi}_\ell|^2 + (x_0^2 + |2s|^2)(\mathfrak{l}+1)^{-4} |\hat{\psi}_\ell^{(1)}|^2 \right] \big|_{x=x_0} + C(\mathfrak{l}+1)^{-4} x_0^{-4} \sum_{n=0}^{\mathfrak{l}-1} \frac{(d_0 x_0)^{2n}}{(\max\{\mathfrak{l}+1, n+1\})^{2n}} |\partial_x^n (\mathfrak{L}_{s,\lambda,\kappa} \widehat{\psi})_\ell|^2\big|_{x=x_0}. \quad (13.15)$$

The estimates (13.14) and (13.15) hold also with $\mathfrak{L}_{s,\lambda,\kappa}$ replaced by $\mathfrak{L}_{s,\lambda,\kappa} + K_{s,\lambda}$, provided we replace $\mathfrak{l}$ with ℓ.

Proof. We will first establish the following estimate inductively: for $n \in \mathbb{N}_0$

$$\frac{(d_0' x_0)^{2n}}{(\max\{\mathfrak{l}+1, n+1\})^{2n}} |\hat{\psi}_\ell^{(n+2)}|^2\big|_{x=x_0} \leq C x_0^{-4} \left[(\mathfrak{l}+1)^4 |\hat{\psi}_\ell|^2 + (x_0^2 + |2s|^2) |\hat{\psi}_\ell^{(1)}|^2 \right] + C x_0^{-4} \sum_{k=0}^{n} \frac{(d_0' x_0)^{2n}}{(\max\{\mathfrak{l}+1, k+1\})^{2k}} |\partial_x^k (\mathfrak{L}_{s,\lambda,\kappa} \widehat{\psi})_\ell|^2\big|_{x=x_0}. \quad (13.16)$$

with $d_0 < d_0'$. Note that the $n = 0$ case follows immediately from (13.1) with $n = 0$. Now suppose (13.16) holds for $n \leq N$. By applying (13.1) with $n = N+1$, we obtain

$$\begin{aligned}
\frac{(d_0' x_0)^{2(N+1)}}{(\max\{\mathfrak{l}+1, N+2\})^{2(N+1)}} |\hat{\psi}^{(N+3)}|^2\big|_{x=x_0} &\leq C \frac{(d_0' x_0)^{2(N+1)}}{(\max\{\mathfrak{l}+1, N+1\})^{2N}} \\
&\times \Big[((N+2)^2 x_0^{-2} + |2s|^2) |\hat{\psi}^{(N+2)}|^2\big|_{x=x_0} + x_0^{-4} (\mathfrak{l}(\mathfrak{l}+1) - (N+1)(N+2))^2 |\hat{\psi}^{(N+1)}|^2\big|_{x=x_0} \\
&\qquad + x_0^{-4} |\partial_x^{N+1} (\mathfrak{L}_{s,\lambda,\kappa} \widehat{\psi})_\ell|^2\big|_{x=x_0} \Big] \\
&\leq C \frac{(d_0' x_0)^{2N}}{(\max\{\mathfrak{l}+1, N+1\})^{2N}} \frac{d_0'^2 ((N+2)^2 + |2s|^2 x_0^{-2})}{(\max\{\mathfrak{l}+1, N+1\})^2} |\hat{\psi}^{(N+2)}|^2\big|_{x=x_0} \\
&+ C \frac{(d_0' x_0)^{2(N-1)}}{(\max\{\mathfrak{l}+1, N\})^{2(N-1)}} \frac{d_0'^4 (\mathfrak{l}(\mathfrak{l}+1) - (N+1)(N+2))^2}{(\max\{\mathfrak{l}+1, N+1\})^2 (\max\{\mathfrak{l}+1, N+2\})^2} |\hat{\psi}^{(N+1)}|^2\big|_{x=x_0} \\
&+ C \frac{(d_0' x_0)^{2(N+1)}}{(\max\{\mathfrak{l}+1, N+2\})^{2(N+1)}} x_0^{-4} |\partial_x^{N+1} (\mathfrak{L}_{s,\lambda,\kappa} \widehat{\psi})_\ell|^2 \\
&\leq \frac{C}{2} \frac{(d_0' x_0)^{2N}}{(\max\{\mathfrak{l}+1, N+1\})^{2N}} |\hat{\psi}^{(N+2)}|^2\big|_{x=x_0} + \frac{C}{2} \frac{(d_0' x_0)^{2(N-1)}}{(\max\{\mathfrak{l}+1, N\})^{2(N-1)}} |\hat{\psi}^{(N+1)}|^2\big|_{x=x_0} \\
&\qquad + C \frac{(d_0' x_0)^{2(N+1)}}{(\max\{\mathfrak{l}+1, N+2\})^{2(N+1)}} x_0^{-4} |\partial_x^{N+1} (\mathfrak{L}_{s,\lambda,\kappa} \widehat{\psi})_\ell|^2,
\end{aligned}$$

where we arrived at the final inequality by taking d_0' suitably small, $|2s|^2 x_0^{-2} \leq \mathfrak{l}^2$. We now conclude (13.16) with $n = N+1$ by applying (13.16) with $n = N$ and $n = N-1$.

To obtain (13.14), we let $0 < d_0 < d_0'$ and we sum both sides of (13.16) to obtain:

$$\begin{aligned}\sum_{n=0}^{\infty} \frac{(d_0 x_0)^{2n}}{(\max\{\mathfrak{l}+1, n+1\})^{2n}} |\hat{\psi}_\ell^{(n+2)}|^2\big|_{x=x_0} &\leq C \sum_{n=0}^{\infty} \left(\frac{d_0}{d_0'}\right)^n x_0^{-4} \left[|\hat{\psi}_\ell|^2 + (x_0^2 + |2s|^2)(\mathfrak{l}(\mathfrak{l}+1))^{-2} |\hat{\psi}_\ell^{(1)}|^2\right] \\ &+ C x_0^{-4} \sum_{n=0}^{\infty} \left(\frac{d_0}{d_0'}\right)^{2(n-k)} \sum_{k=0}^{\infty} \frac{(d_0 x_0)^{2k}}{(\max\{\mathfrak{l}+1, k+1\})^{2k}} |\partial_x^k (\mathfrak{L}_{s,\lambda,\kappa} \widehat{\psi})_\ell|^2\big|_{x=x_0} \\ &\leq C x_0^{-4} \left[|\hat{\psi}_\ell|^2 + (x_0^2 + |2s|^2)(\mathfrak{l}(\mathfrak{l}+1))^{-2} |\hat{\psi}_\ell^{(1)}|^2\right] \\ &+ C x_0^{-4} \sum_{n=0}^{\infty} \frac{(d_0 x_0)^{2n}}{(\max\{\mathfrak{l}+1, n+1\})^{2n}} |\partial_x^n (\mathfrak{L}_{s,\lambda,\kappa} \widehat{\psi})_\ell|^2,\end{aligned}$$

where we used that $\sum_{j=0}^{\infty} \left(\frac{d_0}{d_0'}\right)^{2j} < \infty$. We now conclude (13.14).

By applying Stirling's approximation in the following form: for any $k \in \mathbb{N}$

$$k! \sim k^{k+\frac{1}{2}} e^{-k},$$

we can estimate for $n \leq \mathfrak{l} - 1$:

$$\begin{aligned}\frac{\gamma^{2n}|2s|^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+(n+1))!^2} &\leq C\gamma^{2n}|2s|^{2n} \frac{(\mathfrak{l}-(n+1))^{2\mathfrak{l}-2(n+1)+1}}{(\mathfrak{l}+(n+1))^{2\mathfrak{l}+2(n+1)+1}} e^{4(n+1)} \\ &= C\gamma^{2n}|2s|^{2n} e^{4(n+1)} \left(1 - 2\frac{n+1}{\mathfrak{l}+(n+1)}\right)^{4(n+1)} (\mathfrak{l}+(n+1))^{-4(n+1)} \\ &\leq C \left(e^{-2} d_0 x_0 |2s|^{-1} \gamma^{-1} (\mathfrak{l}+1)\right)^{-2(n+1)} (d_0 x_0)^{2n} (\mathfrak{l}+1)^{-2n-4} \leq (\mathfrak{l}+1)^{-4} (d_0 x_0)^{2n} (\mathfrak{l}+1)^{-2n}\end{aligned}$$

for $\mathfrak{l} \gg d_0^{-1} x_0^{-1} \gamma |2s|$. Hence, (13.15) follows. □

13.3. **Combining all derivatives.** We combine Corollary 13.3 with Proposition 13.5 and Lemma 13.6 to obtain an estimate in the region $\{x \leq x_0\}$, with on the right-hand side only the terms $\partial_x^n (\mathfrak{L}_{s,\lambda,\kappa} \widehat{\psi})_\ell$ and boundary terms at $x = x_0$ involving only the lowest two derivatives $\widehat{\psi}$ and $\widehat{\psi}^{(1)}$:

Corollary 13.7. *Let $M = 1$, $N_+ \in \mathbb{N}_0$, $\mathfrak{l} > |s| x_0^{-1}$ and $0 < \kappa < x_0$ and let d_0 be the constant from Lemma 13.6. Let $s \in \Omega_\sigma$. Then, for λ suitably large, there exists a constant $C_{s,\sigma} = C(s,\sigma) > 0$, an numerical constant $C > 0$, such that for all $\mathbb{N}_0 \ni N_\infty \geq N_\kappa$:*

$$\begin{aligned}&\sum_{n=0}^{\mathfrak{l}-1} \frac{(4\sigma)^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2} \int_0^{x_0} \int_{\mathbb{S}^2} (x^4 + \kappa^2 x^2)|\hat{\psi}^{(n+2)}|^2 + (x^2(\mathfrak{l}+1)^2 + (n+1)^2\kappa^2 + |2s|^2)|\hat{\psi}^{(n+1)}|^2 \\ &\qquad + (\mathfrak{l}-n)^2(\mathfrak{l}+n+1)^2 |\hat{\psi}^{(n)}|^2 \, d\sigma dx \\ &+ (\mathfrak{l}+1)^3 \sum_{n=\mathfrak{l}}^{N_\infty} \frac{\sigma^{2n}}{n!^2(n+1)!^2} \int_0^{x_0} \int_{\mathbb{S}^2} x^4 |s|^{-2} |\widehat{\psi}^{(n+2)}|^2 + \left[1 + (n+1)^2 x^2 |s|^{-2}\right] |\hat{\psi}_\ell^{(n+1)}|^2 \, d\sigma dx \\ &+ \sum_{n=0}^{\infty} \frac{(d_0 x_0)^{2n}}{(\max\{\mathfrak{l}+1, n+1\})^{2n}} \int_{\mathbb{S}^2} |\widehat{\psi}^{(n+2)}|^2\big|_{x=x_0} d\sigma + (\mathfrak{l}+1)^{-4(N_++1)} \sum_{n=0}^{N_++1} \int_{\mathbb{S}^2} x |\hat{\psi}^{(n)}|^2\big|_{x=\eta x_0} d\sigma \\ &\leq \boxed{C \int_{\mathbb{S}^2} |\hat{\psi}_\ell|^2 + (\mathfrak{l}+1)^{-4} |\hat{\psi}_\ell^{(1)}|^2\big|_{x=x_0} d\sigma}\end{aligned}$$

$$+C\sum_{n=0}^{\mathfrak{l}-1}\frac{(4\sigma)^{2n}(\mathfrak{l}-(n+1))!^2}{(\mathfrak{l}+n+1)!^2}\int_0^{x_0}\int_{\mathbb{S}^2}|\partial_x^n(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2\,d\sigma dx$$
$$+C(\mathfrak{l}+1)^3\sum_{n=\mathfrak{l}}^{N_\infty}\frac{\sigma^{2n}}{n!^2(n+1)!^2}\int_0^{x_0}\int_{\mathbb{S}^2}|\partial_x^n(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2\,d\sigma dx$$
$$+C\sum_{n=0}^{\infty}\frac{(d_0x_0)^{2n}}{(\max\{\mathfrak{l}+1,n+1\})^{2n}}\int_{\mathbb{S}^2}|\partial_x^n(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})_\ell|^2|_{x=x_0}d\sigma. \quad (13.17)$$

When x_0 is suitably small, depending on s, the estimate (13.17) holds also with $\mathfrak{L}_{s,\lambda,\kappa}$ replaced by $\mathfrak{L}_{s,\lambda,\kappa}+K_{s,\lambda}$, provided we replace $\mathfrak{l}$ with ℓ.

Proof. Note first that the boundary terms at $x=x_0$ on the right-hand sides of (13.11) and (13.12) can be estimated directly via (13.15).

Furthermore, the x-integral of $\widehat{\psi}^{(\mathfrak{l}+1)}$ on the right-hand sides of (13.11) and (13.12) can be controlled by the x-integral of $\widehat{\psi}^{(\mathfrak{l}+1)}$ on the left-hand side of (13.10), provided we multiply both sides of (13.10) with the factor

$$|s|^{-2}\frac{\gamma^{2\mathfrak{l}}(\mathfrak{l}+1)!^2\mathfrak{l}!^2}{\tilde{\sigma}^{2\mathfrak{l}}(2\mathfrak{l})!^2}=|s|^{-2}(\mathfrak{l}+1)^2\frac{\gamma^{2\mathfrak{l}}}{\tilde{\sigma}^{2\mathfrak{l}}}\frac{\mathfrak{l}!^4}{(2\mathfrak{l})!^2}$$

By applying Stirling's approximation, we have that

$$\frac{\mathfrak{l}!^4}{(2\mathfrak{l})!^2}\sim\frac{\mathfrak{l}^{4\mathfrak{l}+2}e^{-4\mathfrak{l}}}{(2\mathfrak{l})^{4\mathfrak{l}+1}e^{-4\mathfrak{l}}}=2^{-4\mathfrak{l}}\mathfrak{l}.$$

By taking $\gamma=4\tilde{\sigma}$, we therefore obtain:

$$|s|^2\frac{\gamma^{2\mathfrak{l}}(\mathfrak{l}+1)!^2\mathfrak{l}!^2}{\tilde{\sigma}^{2\mathfrak{l}}(2\mathfrak{l})!^2}\sim|s|^{-2}(\mathfrak{l}+1)^3.$$

□

14. A global estimate for $\mathfrak{L}_{s,\lambda,\kappa}$

We combine Corollaries 12.8 and 13.7, to obtain a global estimate for an $\widehat{\psi}$ in terms of $\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}$. Recall the Hilbert spaces $H^{N_+}_{\sigma,\delta}$ and $H^{N_++1}_{\sigma,R_\infty,2}$ introduced in §8.

Proposition 14.1. *Let $M=1$, $N_+\in\mathbb{N}_0$, $0<\kappa\ll\eta x_0$. Take $s\in\Omega_\sigma\cap\{\mathrm{Re}\,s>-(\frac{1}{2}+N_+)\kappa_+\}$ and $p=4(N_++1)+2$. Then, for $M^{-1}R_\infty$ and λ suitably large, there exists a constant $C_{s,\lambda,\delta,R_\infty}>0$ and a constant $N_\kappa>\mathfrak{l}$, such that for $\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}\in H^{N_+}_{\sigma,\delta}$:*

$$\|\widehat{\psi}\|^2_{H^{N_++1}_{\sigma,R_\infty,2}}+\kappa^2(\|x\partial_x^2\widehat{\psi}\|^2_{G_{\sigma,R_\infty}}+\|\not{\Delta}_{\mathbb{S}^2}\widehat{\psi}\|^2_{G_{\sigma,R_\infty}})\leq C_{s,\sigma}\|\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}\|^2_{H^{N_+}_{\sigma,R_\infty}}. \quad (14.1)$$

When $\mathrm{Re}\,s\geq s_0$ with $s_0>0$ suitably large, we moreover have:

$$\|\widehat{\psi}\|_{H^{N_+}_{\sigma,R_\infty}}\leq C_{s,\sigma}\|(\mathfrak{L}_{s,\lambda,\kappa}+K_{s,\lambda})\widehat{\psi}\|_{H^{N_+}_{\sigma,R_\infty}}. \quad (14.2)$$

Proof. We need to show that:

$$\sum_{k=0}^{1}\|\nabla\!\!\!/^k_{\mathbb{S}^2}\widehat{\psi}\|^2_{H^{N_++1}(\Sigma\cap\{\varrho\leq\frac{1}{3}R_\infty\})}+\|\widehat{\psi}\|^2_{H^{N_++2}(\Sigma\cap\{\frac{1}{3}R_\infty\leq\varrho\leq3R_\infty\})}$$
$$+\left\|\left|(-\not{\Delta}_{\mathbb{S}^2}+\mathbf{1})^{-\frac{w_p}{4}}\widehat{\psi}\right|\right\|^2_{H^{N_++2}(\Sigma\cap\{3R_\infty\leq\varrho\leq4R_\infty\})}+\|\widehat{\psi}\|^2_{B_{R_\infty,2}}+\|\widehat{\psi}\|^2_{G_{\sigma,R_\infty,2}}+\kappa^2\|x\partial_x^2\widehat{\psi}\|^2_{G_{\sigma,R_\infty}}$$
$$\leq C_{s,\lambda,R_\infty}\Big[\|\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}\|^2_{H^{N_+}(\Sigma\cap\{\varrho\leq3R_\infty\})}+\|\nabla\!\!\!/_{\mathbb{S}^2}\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}\|^2_{H^{N_+}(\Sigma\cap\{\varrho\leq\frac{1}{3}R_\infty\})}$$

$$+\|(-\not\Delta_{\mathbb{S}^2}+\mathbf{1})^{-\frac{w_p}{4}}(\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi})\|^2_{H^{N_+}(\Sigma\cap\{3R_\infty\leq\varrho\leq 4R_\infty\})}+\|\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}\|^2_{B_{R_\infty}}+\|\mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}\|^2_{G_{\sigma,R_\infty}}\Big]. \quad (14.3)$$

We consider (12.10) and we commute additionally with $\not\Delta_{\mathbb{S}^2}$ to control additional angular derivatives on the left-hand side and obtain control of the desired H^{N_+}, H^{N_++1} and H^{N_++2} norms.

By using that $M^{-1}R_\infty \ll \sqrt{\lambda}$, we can absorb the boxed term on the right-hand side of (13.17) into the left-hand side of (12.10). In order to subsequently absorb the boxed terms that appear on the right-hand side of (12.10) into the left-hand side of (13.17), we take $p = 4(N_+ + 1) + 2$. Finally, we note that

$$\sum_{\ell=\in\mathbb{N}_0}\int_{\mathbb{S}^2}(\mathfrak{l}+1)^{-w_p}|f_\ell|^2\,d\sigma \leq \sum_{\ell\in\mathbb{N}_0}\int_{\mathbb{S}^2}2^{\frac{w_p}{2}}(\ell(\ell+1)+\lambda)^{-\frac{p}{2}}|f_\ell|^2\,d\sigma \leq 2^{\frac{w_p}{2}}(1+\lambda^{-\frac{w_p}{2}})\int_{\mathbb{S}^2}|(-\not\Delta_{\mathbb{S}^2}+\mathbf{1})^{-\frac{w_p}{4}}f|^2\,d\sigma.$$

We also have that

$$\begin{aligned}\sum_{\ell=\in\mathbb{N}_0}\int_{\mathbb{S}^2}(\mathfrak{l}+1)^{-w_p}|f_\ell|^2\,d\sigma &= \sum_{\ell=0}^{l}\int_{\mathbb{S}^2}(\ell+l+1)^{-w_p}|f_\ell|^2\,d\sigma+\sum_{\ell=l+1}^{\infty}(\ell+1)^{-w_p}|f_\ell|^2\,d\sigma\\ &\geq ((2l+1)^{w_p}+1)\sum_{\ell=0}^{\infty}(1+\ell(\ell+1))^{\frac{-w_p}{2}}|f_\ell|^2\,d\sigma \geq ((2l+1)^{w_p}+1)\int_{\mathbb{S}^2}|(-\not\Delta_{\mathbb{S}^2}+\mathbf{1})^{-\frac{w_p}{4}}f|^2\,d\sigma.\end{aligned}$$

Finally, we observe that in the case $\ell > l$, we have that $\mathfrak{l} = \ell$, whereas in the case $\ell \leq l$, we use boundedness of ℓ to ensure that the $\mathfrak{l}$-dependent norms in (12.10) are comparable to the norms appearing in (14.3).

When considering $\mathfrak{L}_{s,\lambda,\kappa} + K_{s,\lambda}$ and $\operatorname{Re} s \geq s_0$, simply apply (12.11) instead of (12.10). □

15. Inverting $\mathfrak{L}_{s,\lambda,\kappa}$

In this section, we will construct $\mathfrak{L}^{-1}_{s,\lambda,\kappa}$ for $\kappa \geq 0$ with an appropriate Hilbert space domain and with a suitably large λ.

We first make use of the regularizing effect of $\kappa > 0$ to invert $\mathfrak{L}_{s,\lambda,\kappa}$ as an operator on suitable Sobolev spaces. We then appeal to the uniform-in-κ estimate from Proposition 14.1 to construct $\mathfrak{L}^{-1}_{s,\lambda,\kappa=0}$ as a limit of a sequence of operators $\mathfrak{L}^{-1}_{s,\lambda,\kappa_n}$, with $\kappa_n > 0$, $\kappa_n \to 0$ as $n \to \infty$.

Define $H^k(\mathfrak{L}_{s,\lambda,\kappa})$ as the closure of $C^\infty([0, r_+^{-1}]_x \times \mathbb{S}^2)$ with respect to the graph norm:

$$\|\cdot\|_{H^k([0,r_+^{-1}]_x\times\mathbb{S}^2)} + \|\mathfrak{L}_{s,\lambda,\kappa}(\cdot)\|_{H^k([0,r_+^{-1}]_x\times\mathbb{S}^2)}.$$

Then the map:

$$\mathfrak{L}_{s,\lambda,\kappa} : H^k(\mathfrak{L}_{s,\lambda,\kappa}) \to H^k([0, r_+^{-1}]_x \times \mathbb{S}^2)$$

is a well-defined bounded linear operator for all $k \in \mathbb{N}_0$.

Theorem 15.1. *Let $N \geq N_+ \in \mathbb{N}_0$ and $\operatorname{Re} s > -(N + \frac{1}{2})\min\{\kappa_+, \kappa\}$. Then, there exists a $\lambda > 0$ suitably large, such that the inverse map:*

$$\begin{aligned}\mathfrak{L}^{-1}_{s,\lambda,\kappa} : H^N([0, r_+^{-1}]_x \times \mathbb{S}^2) \cap \big\{f \in L^2([0, r_+^{-1}]_x \times \mathbb{S}^2)\,|\,\exists m \in \mathbb{Z} : \partial_\varphi f = imf\big\}&\\ \to H^N(\mathfrak{L}_{s,\lambda,\kappa}) \subseteq H^{N+1}([0, r_+^{-1}]_x \times \mathbb{S}^2)&\end{aligned}$$

is a bounded linear operator.

For $\operatorname{Re} s \geq s_0$ with $s_0 > 0$ suitably large, we can replace $\mathfrak{L}_{s,\lambda,\kappa}$ with $\mathfrak{L}_{s,\lambda,\kappa} + K_{s,\lambda}$ in the above inequality.

Proof. Since we are considering a domain restricted to finite azimuthal modes (with φ-dependence $e^{im\varphi}$), we can apply a straightforward modification of [War15][Theorem 5.7] to obtain the existence of the desired inverse. □

Remark 15.1. *The ability to restrict to fixed azimuthal modes on Kerr is not crucial. One can in principle derive an analogue of Theorem 15.1 without making use of this assumption and instead make use of red-shift estimates depending on κ in the large-r-region, instead of Gevrey-type estimates. See also the related results in* [Vas13].

By applying Proposition 14.1, which is uniform in κ and m, we can obtain the existence of $\mathfrak{L}^{-1}_{s,\lambda,\kappa}$ with domain $H^{N_+}_{\sigma,R_\infty}$.

Corollary 15.2. *Let $\sigma \in \mathbb{R}$ and take $s \in \Omega_\sigma \cap \{\mathrm{Re}\, s > (\frac{1}{2} + N_+)\kappa_+\}$. Then, there exists a suitably small $\kappa > 0$ and a suitably large R_∞ and $\lambda \in \mathbb{N}_0$, such that:*

$$\mathfrak{L}^{-1}_{s,\lambda,\kappa} : H^{N_+}_{\sigma,R_\infty} \to H^{N_++1}_{\sigma,R_\infty,2}$$

is a bounded linear operator and there exists a constant $C_{s,\sigma,R_\infty} > 0$ such that:

$$\|\mathfrak{L}^{-1}_{s,\lambda,\kappa}\|_{H^{N_+}_{\sigma,R_\infty} \to H^{N_++1}_{\sigma,R_\infty,2}} \leq C_{s,\sigma,R_\infty}. \tag{15.1}$$

For all $f \in H^{N_+}_{\sigma,R_\infty}$, we also have:

$$\kappa \|x\partial_x^2 \mathfrak{L}^{-1}_{s,\lambda,\kappa}(f)\|_{G_{\sigma,R_\infty}} + \kappa^2 \|\not{\Delta}_{\mathbb{S}^2} \mathfrak{L}^{-1}_{s,\lambda,\kappa}(f)\|_{G_{\sigma,R_\infty}} \leq C_{s,\sigma} \|f\|_{H^{N_+}_{\sigma,R_\infty}}. \tag{15.2}$$

Furthermore, for $\mathrm{Re}\, s \geq s_0$ with $s_0 > 0$ suitably large:

$$\|(\mathfrak{L}_{s,\lambda,\kappa} + K_{s,\lambda})^{-1}\|_{H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}} \leq C_{s,\sigma,R_\infty}. \tag{15.3}$$

Proof. Let $\kappa_+ > \kappa > 0$ and $s \in \Omega_\sigma \cap \{\mathrm{Re}\, z > (\frac{1}{2} + N_+)\kappa_+\}$. Then there exists a $N_\kappa > N_+$ such that $\mathrm{Re}\, s > -(N_\kappa + \frac{1}{2})\kappa$. Let $f_m \in C^\infty([0, r_+^{-1}]_x \times \mathbb{S}^2)) \cap H^{N_+}_{\sigma,R_\infty}$, such that $\partial_\varphi f_m = imf_m$ for some $m \in \mathbb{Z}$. Then we can apply Theorem 15.1 to conclude that $\mathfrak{L}^{-1}_{s,\lambda,\kappa}(f_m) \in C^\infty([0, r_+^{-1}]_x \times \mathbb{S}^2)$ is well-defined for suitably large λ.

By Proposition 14.1, it then follows that for λ suitably large and κ suitably small, we can estimate

$$\|\mathfrak{L}^{-1}_{s,\lambda,\kappa}(f_m)\|^2_{H^{N_++1}_{\sigma,R_\infty,2}} \leq C_{s,\sigma,\lambda} \|f_m\|^2_{H^{N_+}_{\sigma,R_\infty}}.$$

Let $f \in C^\infty([0, r_+^{-1}]_x \times \mathbb{S}^2)) \cap H^{N_+}_{\sigma,R_\infty}$. Then we can decompose $f = \sum_{m\in\mathbb{Z}} f_m$, with $\partial_\varphi f_m = imf_m$. Since the constant in the above estimate does not depend on m, we can sum over m to conclude that for general $f \in C^\infty([0, r_+^{-1}]_x \times \mathbb{S}^2)) \cap H^{N_+}_{\sigma,R_\infty}$, $\mathfrak{L}^{-1}_{s,\lambda,\kappa}(f) = \sum_{m\in\mathbb{Z}} \mathfrak{L}^{-1}_{s,\lambda,\kappa}(f_m)$ is well defined.

Hence, $\mathfrak{L}^{-1}_{s,\lambda,\kappa}(C^\infty([0, r_+^{-1}]_x \times \mathbb{S}^2) \cap H^{N_+}_{\sigma,R_\infty}) \subseteq H^{N_++1}_{\sigma,R_\infty,2}$. Since $C^\infty([0, r_+^{-1}]_x \times \mathbb{S}^2) \cap H^{N_+}_{\sigma,R_\infty}$ is dense in $H^{N_+}_{\sigma,R_\infty}$, we can extend $\mathfrak{L}^{-1}_{s,\lambda,\kappa}$ to act on $H^{N_+}_{\sigma,R_\infty}$ and we must have that $\mathfrak{L}^{-1}_{s,\lambda,\kappa}(H^{N_+}_{\sigma,R_\infty}) \subseteq H^{N_++1}_{\sigma,R_\infty,2}$.

Finally, we conclude (15.3) by applying (14.2). □

Proposition 15.3. *Let $\sigma \in \mathbb{R}$ and take $s \in \Omega_\sigma \cap \{\mathrm{Re}\, z > -(\frac{1}{2} + N_+)\kappa_+\}$. Then, there exists a suitably large $\lambda \in \mathbb{N}_0$, such that:*

$$\mathfrak{L}^{-1}_{s,\lambda} = \mathfrak{L}^{-1}_{s,\lambda,0} : H^{N_+}_{\sigma,R_\infty} \to H^{N_++1}_{\sigma,R_\infty,2} \tag{15.4}$$

is a bounded linear operator and there exists a constant $C_{s,\sigma} > 0$ such that:

$$\|\mathfrak{L}^{-1}_{s,\lambda}\|_{H^{N_+}_{\sigma,R_\infty} \to H^{N_++1}_{\sigma,R_\infty,2}} \leq C_{s,\sigma}.$$

Furthermore, for any sequence $\{\kappa_n\}$, with $\kappa_n \to 0$ as $n \to \infty$

$$\mathfrak{L}^{-1}_{s,\lambda,\kappa_n} \to \mathfrak{L}^{-1}_{s,\lambda}$$

as $n \to \infty$, with respect to the operator norm on $H^{N_+}_{\sigma,R_\infty}$.

Furthermore, for $\mathrm{Re}\, s \geq s_0$ with $s_0 > 0$ suitably large, $(\mathfrak{L}_{s,\lambda} + K_{s,\lambda})^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ is a well-defined bounded linear operator.

Proof. Note that there exist $\sigma_2^2 > \sigma_1^2 > \sigma^2$, such that $s \in \Omega_{\sigma_1} \cap \Omega_{\sigma_2}$. Let $f \in C^\infty([0, r_+^{-1}]_x \times \mathbb{S}^2) \cap H^{N_++1}_{\sigma_2, R_\infty}$.

Let κ_n be a monotonically non-increasing sequence with $\kappa_n > 0$ and $\kappa_n \to 0$ as $n \to \infty$. By Corollary 15.2, $\widehat{\psi}_n := \mathfrak{L}^{-1}_{s,\lambda,\kappa_n}(f)$ is a well-defined, uniformly bounded sequence in $H^{N_++1}_{\sigma_2,R_\infty,2}$. By (B.2), $H^{N_++1}_{\sigma_2,R_\infty,2} \subset\subset H^{N_++1}_{\sigma_1,R_\infty,2}$, so $\widehat{\psi}_n$ admits a convergent subsequence $\{\widehat{\psi}_{n_j}\}$ with respect to $H^{N_++1}_{\sigma_1,R_\infty,2}$, with limit $\widehat{\psi} \in H^{N_++1}_{\sigma_1,R_\infty,2}$. In fact, since $\{\widehat{\psi}_{n_j}\}$ is a bounded sequence in $H^{N_++1}_{\sigma_2,R_\infty,2}$, it admits a further subsequence with a weak limit in $H^{N_++1}_{\sigma_2,R_\infty,2}$. By the uniqueness of limits, this limit must also be $\widehat{\psi}$, so in fact $\widehat{\psi} \in H^{N_++1}_{\sigma_2,R_\infty,2}$.

We will now show that $\mathfrak{L}_{s,\lambda,0}(\widehat{\psi}) = f$. We have that

$$\mathfrak{L}_{s,\lambda,0}(\widehat{\psi}) = f - (\mathfrak{L}_{s,\lambda,\kappa_{n_j}} - \mathfrak{L}_{s,\lambda,0})(\widehat{\psi}) + \mathfrak{L}_{s,\lambda,\kappa_n}(\widehat{\psi} - \widehat{\psi}_{n_j}). \tag{15.5}$$

Note that:

$$(\mathfrak{L}_{s,\lambda,\kappa_{n_j}} - \mathfrak{L}_{s,\lambda,0})(\widehat{\psi}) = \kappa_{n_j}\chi_{\varrho\geq R_\infty}(x\partial_x^2\widehat{\psi} + \partial_x\widehat{\psi}).$$

Hence, by Proposition C.1

$$\begin{aligned}\|(\mathfrak{L}_{s,\lambda,\kappa_{n_j}} - \mathfrak{L}_{s,\lambda,0})(\widehat{\psi})\|_{H^{N_+}_{\sigma,R_\infty}} &\leq 2\kappa_{n_j}(\|\chi_{\varrho\geq R_\infty}\widehat{\psi}^{(1)}\|_{H^{N_+}_{\sigma,R_\infty}} + \|x\chi_{\varrho\geq R_\infty}\widehat{\psi}^{(2)}\|_{H^{N_+}_{\sigma,R_\infty}}) \\ &\leq C\kappa_{n_j}(\|\widehat{\psi}\|_{H^{N_++2}_{\sigma,R_\infty,2}} + \|x\widehat{\psi}^{(2)}\|_{G_{\sigma,R_\infty}}).\end{aligned}$$

To bound $\|x\widehat{\psi}^{(2)}\|_{G_{\sigma,R_\infty}}$, we first apply Lemma C.3, from which it follows that

$$\|x\widehat{\psi}^{(2)}\|_{G_{\sigma,R_\infty}} \leq C\|\widehat{\psi}^{(2)}\|_{G_{\sigma,R_\infty}} \leq C\|\widehat{\psi}^{(1)}\|_{G_{\sigma,R_\infty,2}}.$$

Then we observe that we can use that $\sigma_1 > \sigma$ to further estimate:

$$\|\widehat{\psi}^{(1)}\|_{G_{\sigma,R_\infty,2}} \leq C\|\widehat{\psi}\|_{G_{\sigma_1,R_\infty,2}}.$$

We can similarly bound:

$$\|\mathfrak{L}_{s,\lambda,\kappa_{n_j}}(\widehat{\psi} - \widehat{\psi}_{n_j})\|_{H^{N_+}_{\sigma,R_\infty}} \leq C\|\widehat{\psi} - \widehat{\psi}_{n_j}\|_{H^{N_++2}_{\sigma_1,R_\infty,2}} \to 0 \quad \text{as } j \to \infty.$$

Taking the limit $j \to \infty$ in (15.5) therefore gives $\mathfrak{L}_{s,\lambda,0}(\widehat{\psi}) = f$. Hence, $\mathfrak{L}^{-1}_{s,\lambda,0} : C^\infty([0, r_+^{-1}]_x \times \mathbb{S}^2)) \cap H^{N_+}_{\sigma_2,R_\infty} \to H^{N_++1}_{\sigma_2,R_\infty,2}$ is well-defined. Using density of $C^\infty([0, r_+^{-1}]_x \times \mathbb{S}^2)) \cap H^{N_+}_{\sigma_2,\delta_2}$ in $H^{N_+}_{\sigma_2,\delta_2}$, we conclude (15.4).

To conclude uniform operator convergence, we first write:

$$\mathfrak{L}^{-1}_{s,\lambda,\kappa_m} - \mathfrak{L}^{-1}_{s,\lambda,\kappa_n} = \mathfrak{L}^{-1}_{s,\lambda,\kappa_m} \circ (\mathfrak{L}_{s,\lambda,\kappa_n} - \mathfrak{L}_{s,\lambda,\kappa_m}) \circ \mathfrak{L}^{-1}_{s,\lambda,\kappa_n}$$

Let $f \in H^{N_+}_{\sigma,R_\infty}$, then $\widehat{\psi} := \mathfrak{L}^{-1}_{s,\lambda,\kappa_n}(f) \in H^{N_++1}_{\sigma,R_\infty,2}$. Furthermore,

$$(\mathfrak{L}_{s,\lambda,\kappa_n} - \mathfrak{L}_{s,\lambda,\kappa_m})(\widehat{\psi}) = (\kappa_n - \kappa_m)\partial_x(\chi_{\varrho\geq R_\infty}x\partial_x\widehat{\psi}),$$

so $(\mathfrak{L}_{s,\lambda,\kappa_n} - \mathfrak{L}_{s,\lambda,\kappa_m})(\widehat{\psi})$ vanishes in $\varrho \leq \frac{1}{2}R_\infty$. Define

$$\widehat{\psi}' := \chi_{\varrho\geq R_\infty}\partial_x\widehat{\psi} \in H^{N_+}_{\sigma,R_\infty},$$

then $\|\mathfrak{L}^{-1}_{s,\lambda,\kappa_m}(\widehat{\psi}')\|_{H^{N_++1}_{\sigma,R_\infty,2}} \leq C_{s,\sigma}\|f\|_{H^{N_+}_{\sigma,R_\infty}}$ by (15.1). We now express:

$$\begin{aligned}(\mathfrak{L}_{s,\lambda,\kappa_n} - \mathfrak{L}_{s,\lambda,\kappa_m})(\widehat{\psi}) &= (\kappa_n - \kappa_m)\partial_x(x\widehat{\psi}') = (\kappa_n - \kappa_m)\partial_x(x\mathfrak{L}_{s,\lambda,\kappa_m}\mathfrak{L}^{-1}_{s,\lambda,\kappa_m}(\widehat{\psi}')) \\ &= (\kappa_n - \kappa_m)\mathfrak{L}_{s,\lambda,\kappa_m}(\partial_x \circ x\cdot)\mathfrak{L}^{-1}_{s,\lambda,\kappa_m}(\widehat{\psi}') + (\kappa_n - \kappa_m)[\partial_x \circ x, \mathfrak{L}_{s,\lambda,\kappa_m}\cdot]\mathfrak{L}^{-1}_{s,\lambda,\kappa_m}(\widehat{\psi}')\end{aligned} \tag{15.6}$$

and hence

$$(\mathfrak{L}^{-1}_{s,\lambda,\kappa_m} - \mathfrak{L}^{-1}_{s,\lambda,\kappa_n})(f) = (\kappa_n - \kappa_m)\left(\partial_x(x\mathfrak{L}^{-1}_{s,\lambda,\kappa_m}(\widehat{\psi}')) + \mathfrak{L}^{-1}_{s,\lambda,\kappa_m}[\partial_x \circ x\cdot, \mathfrak{L}_{s,\lambda,\kappa_m}]\mathfrak{L}^{-1}_{s,\lambda,\kappa_m}(\widehat{\psi}')\right).$$

For $x \leq x_0$, we have that

$$[\partial_x \circ x\cdot, \mathfrak{L}_{s,\lambda,\kappa_m}] = -\kappa_m x\partial_x^2 - (2s+\kappa_m)\partial_x.$$

Using that $\|\mathfrak{L}^{-1}_{s,\lambda,\kappa_m}(\widehat{\psi}')\|_{H^{N_++1}_{\sigma,R_\infty,2}} \leq C_{s,\sigma}\|f\|_{H^{N_+}_{\sigma,R_\infty}}$ and additionally that $\kappa_m\|x\partial_x^2\mathfrak{L}^{-1}_{s,\lambda,\kappa_m}(\widehat{\psi}')\|_{G_{\sigma,\delta}} \leq C_{s,\sigma}\|f\|_{H^{N_+}_{\sigma,R_\infty}}$ by (15.2), we conclude that

$$\|(\mathfrak{L}^{-1}_{s,\lambda,\kappa_m} - \mathfrak{L}^{-1}_{s,\lambda,\kappa_n})(f)\|_{H^{N_+}_{\sigma,R_\infty}} \leq C\cdot C_{s,\sigma}|\kappa_n - \kappa_m|\|f\|_{H^{N_+}_{\sigma,R_\infty}}.$$

Since $\{\kappa_n\}$ is converges to 0, it is Cauchy, and by the above estimate, $\{\mathfrak{L}^{-1}_{s,\lambda,\kappa_n}\}$ must be Cauchy in the Banach space of bounded linear operators $B(H^{N_+}_{\sigma,R_\infty}, H^{N_+}_{\sigma,R_\infty})$ with respect to the operator norm, so it converges. By uniqueness of limits, the limit must be equal to the operator $\mathfrak{L}^{-1}_{s,\lambda,0}$ defined above.

To conclude that $(\mathfrak{L}_{s,\lambda} + K_{s,\lambda})^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ is well-defined for $\operatorname{Re} s \geq s_0$, we instead apply (15.3). □

16. Inverting L_s: boundedness, compactness and Fredholm theory

In this section, we will establish the meromorphicity of the maps

$$s \mapsto L^{-1}_{s,\kappa}$$

with $\kappa \geq 0$.

By (7.8) and (7.13), we have the following relation between $L_{s,\kappa}$ and $\mathfrak{L}_{s,\lambda,\kappa}$.

$$L_{s,\kappa} = \mathfrak{L}_{s,\lambda,\kappa} + \epsilon_0 \cdot B_s + K_{s,\lambda},$$

where $\epsilon_0 > 0$ can be taken arbitrarily small.

Proposition 16.1. *Let $\lambda \in \mathbb{R}$. Assume that $\mathbb{h} \equiv 1$ in $\{\varrho \leq \frac{1}{2}R_\infty\}$. Then the linear operators*

$$B_s : H^{N_++1}_{\sigma,R_\infty,2} \to H^{N_+}_{\sigma,R_\infty},$$
$$K_{s,\lambda} : H^{N_++1}_{\sigma,R_\infty,2} \to H^{N_++1}_{\sigma,R_\infty,2}$$

are bounded independently of ϵ_0. Let $\mathcal{R} : H^{N_+}_{\sigma,R_\infty} \to H^{N_++1}_{\sigma,R_\infty,2}$ be bounded. Then $K_{s,\lambda}\circ\mathcal{R} : H^{N_+}_{\sigma,R_\infty} \to H^{N_++1}_{\sigma,R_\infty,2} \subset H^{N_+}_{\sigma,R_\infty}$ is compact.

Proof. Note first that the compactness of $K_{s,\lambda}\circ\mathcal{R}$ follows directly from the boundedness of $K_{s,\lambda}$ by the fact that $H^{N_++1}_{\sigma,R_\infty,2} \subset\subset H^{N_+}_{\sigma,R_\infty}$ by (B.3).

Recall that

$$B_s = M\epsilon_0^{-1}x\chi_{\varrho\geq R_\infty}B_s^\infty,$$
$$K_{s,\lambda} = \lambda\pi_{\leq l} + \chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l} + K_s^\infty + \chi_{\varrho\leq\frac{1}{2}R_\infty}K_s^+.$$

We first consider B_s. Take $M^{-1}R_\infty$ to be suitably large, so that $\epsilon_0^{-1}x \leq 1$ in $\operatorname{supp}\chi_{\varrho\geq R_\infty}$. We need to establish boundedness of $\chi_{\varrho\geq R_\infty}B_s^\infty$.

Recall moreover that

$$B_s^\infty f = b_{xx}\cdot x^2\partial_x^2 f + sb_{x\tau}\cdot\partial_x f + s^2 b_{\tau\tau}f + a^2(xb_{x\vartheta}\cdot x^2\partial_x + sxb_{\tau\vartheta})\sin\vartheta\partial_\vartheta f$$
$$+ a^4x^4\cdot b_{\vartheta\vartheta}(\sin\vartheta\partial_\vartheta)^2 f + b_x\cdot x\partial_x f + a^2x^2b_\vartheta\cdot\sin\vartheta\partial_\vartheta f + sb_\tau f + b\cdot f.$$

We apply Proposition C.1 to estimate $\mathcal{B}_s^\infty f$:

$$\begin{aligned}\|B_s^\infty f\|_{H^{N_+}_{\sigma,R_\infty}} &\leq C_s \left[\sum_{\star\in\{x,\tau,\vartheta\}} \|b_{\star\star}\|_{H^{N_+}_{\sigma,R_\infty;\mathrm{coeff}}} + \|b_\star\|_{H^{N_+}_{\sigma,R_\infty;\mathrm{coeff}}} + \|b\|_{H^{N_+}_{\sigma,R_\infty;\mathrm{coeff}}}\right] \\ &\quad \times \Big(\|f\|_{H^{N_+}_{\sigma,R_\infty}} + \|\partial_x f\|_{H^{N_+}_{\sigma,R_\infty}} + a^2\|x^2\sin\vartheta\partial_\vartheta f\|_{H^{N_+}_{\sigma,R_\infty}} + \|\chi_{\varrho\geq R_\infty}x^2\partial_x^2 f\|_{H^{N_+}_{\sigma,R_\infty}} \\ &\qquad + a^2\|\chi_{\varrho\geq R_\infty}x^4(\sin\vartheta\partial_\vartheta)^2 f\|_{H^{N_+}_{\sigma,R_\infty}}\Big) \\ &\leq C_s \left[\sum_{\star\in\{x,\tau,\vartheta\}} \|b_{\star\star}\|_{H^{N_+}_{\sigma,R_\infty;\mathrm{coeff}}} + \|b_\star\|_{H^{N_+}_{\sigma,R_\infty;\mathrm{coeff}}} + \|b\|_{H^{N_+}_{\sigma,R_\infty;\mathrm{coeff}}}\right] \|f\|_{H^{N_++1}_{\sigma,R_\infty,2}} \\ &\leq C_{\mathrm{coeff},s}\cdot\|f\|_{H^{N_++1}_{\sigma,R_\infty,2}},\end{aligned}$$

where the final inequality follows from the analyticity properties of $b, b_\star$ and $b_{\star\star}$ that were established in Lemma 7.1. Note moreover that we used here that the $H^{N_++1}_{\sigma,R_\infty,2}$ norm controls two more derivatives (with degenerate factors) than the $H^{N_+}_{\sigma,R_\infty}$ norm in the region $\varrho \geq \frac{1}{2}R_\infty$ and one more derivative globally. Boundedness of $\|B_s\|$ follows by combining the above estimates.

We now turn to $K_{s,\lambda}$. Recall that

$$\begin{aligned}K_s^\infty f &= -2M^2h(0)s^2 f, \\ r^{-2}\rho^2 K_s^+ f &= \left[2r - \frac{d}{dr}(\Delta\mathbb{h})\right] s(r^{-1}f) + [\mathbb{h}^2\Delta - 2\mathbb{h}(r^2+a^2) + a^2\sin^2\theta]s^2 f \\ &\quad - 2a(\mathbb{h}-1)s\partial_{\varphi_*} f.\end{aligned}$$

In order for all terms in K_s to be zeroth-order terms, we use the assumption $\mathbb{h}\equiv 1$ in $\varrho \leq R_\infty$.

Hence, applying Proposition C.1, it is straightforward to estimate

$$\|\chi_{\varrho\leq\frac{1}{2}R_\infty}K_s^+ f\|_{H^{N_++1}_{\sigma,R_\infty,2}} \leq C_{\mathrm{coeff},s}\cdot\|f\|_{H^{N_++1}_{\sigma,R_\infty,2}}.$$

Boundedness of $K_s^\infty, \chi_{\varrho\leq\frac{1}{2}R_\infty}K_s^+ : H^{N_++1}_{\sigma,R_\infty,2} \to H^{N_++1}_{\sigma,R_\infty,2}$ is therefore immediate. Consider $\lambda\pi_{\leq l}$ and $\chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l}$. The operator $\lambda\pi_{\leq l} : H^{N_++1}_{\sigma,R_\infty,2} \to H^{N_++1}_{\sigma,R_\infty,2}$ can easily be seen to be bounded. Finally, since $\int_{\mathbb{S}^2}|S_\lambda\pi_{\leq l}f|^2\,d\sigma \leq \lambda^2\int_{\mathbb{S}^2}|\nabla\!\!\!/_{\mathbb{S}^2}f_{\leq l}|^2 \leq \lambda^4\int_{\mathbb{S}^2}|f_{\leq l}|^2\,d\sigma$, we have that

$$\chi_{\varrho\geq R_\infty}S_\lambda\pi_{\leq l} : H^{N_++1}_{\sigma,R_\infty,2} \to H^{N_++1}_{\sigma,R_\infty,2}.$$

is bounded. □

Recall the definitions of a meromorphic family of operators in Definition 10.1 and the definition of the cokernel of an operator in Definition 10.2. In the proposition below, we establish meromorphicity of the inverses $L_{s,\kappa}^{-1}$ with $\kappa \geq 0$, and we relate the poles to the cokernel of $L_{s,\kappa}$.

Proposition 16.2. *Let $\sigma\in\mathbb{R}$ and $N_+\in\mathbb{N}_0$. For $\kappa\geq 0$ sufficiently small and $M^{-1}R_\infty$ sufficiently large:*

(i) *The family of operators $\{L_{s,\kappa}^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_++1}_{\sigma,R_\infty,2}\}$ with $s\in\Omega_\sigma\cap\{\mathrm{Re}\,s > -(\frac{1}{2}+N_+)\kappa_+\}$ is meromorphic and holomorphic if moreover $\mathrm{Re}\,s \geq s_0$ with $s_0 > 0$ suitably large. We denote with $\mathcal{Q}_{\sigma,R_\infty,N_+,\kappa}$ the set of poles of $\{L_{s,\kappa}^{-1}\}$. The set $\mathcal{Q}_{\sigma,R_\infty,N_+,\kappa}$ is independent of the precise choice of $\mathbb{h}$.*

(ii) *Let $s\in\Omega_\sigma\cap\{\mathrm{Re}\,s > -(\frac{1}{2}+N_+)\kappa_+\}$. Then the space $\ker L_s$ is finite dimensional with $\dim\ker L_{s,\kappa} = \dim\mathrm{coker}\,L_{s,\kappa}$ and $\mathrm{ran}\,L_{s,\kappa}^* = (\ker L_{s,\kappa})^\perp \subset H^{N_++1}_{\sigma,R_\infty,2}$, with $L_{s,\kappa}^*$ the Hermitian adjoint and $\mathrm{coker}\,L_{s,\kappa}^*$ is isomorphic to $\ker L_{s,\kappa}$.*

Proof. Without loss of generality, we consider the $\kappa = 0$ case. The argument proceeds in exactly the same way for $\kappa > 0$. We first suppose that $\mathbb{h} = 1$ in $\{\varrho \leq \frac{1}{2}R_\infty\}$. By Proposition 15.3, $\mathfrak{L}_{s,\lambda}^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+ +1}_{\sigma,R_\infty,2}$ is well-defined for suitably large λ. We can express:

$$\mathfrak{L}_{s,\lambda} + \epsilon_0 B_s = (\mathbf{1} + \epsilon_0 B_s \mathfrak{L}_{s,\lambda}^{-1})\mathfrak{L}_{s,\lambda}.$$

By Proposition 16.1, $B_s : H^{N_+ +1}_{\sigma,R_\infty,2} \to H^{N_+}_{\sigma,R_\infty}$ is bounded, so if we take $\epsilon_0 > \frac{1}{\|B_s\| \cdot \|\mathfrak{L}_{s,\lambda}^{-1}\|}$, we have that $\epsilon_0 B_s \mathfrak{L}_{s,\lambda}^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ is bounded with operator norm less than 1. Hence, $(\mathbf{1} + \epsilon_0 B_s \mathfrak{L}_{s,\lambda}^{-1})^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ is a bounded operator that we can express as follows in terms of a Neumann series:

$$(\mathbf{1} + \epsilon_0 B_s \mathfrak{L}_{s,\lambda}^{-1})^{-1} = \sum_{k=0}^{\infty} (-1)^k (\epsilon_0 B_s \mathfrak{L}_{s,\lambda}^{-1})^k.$$

We now conclude that

$$-T_s := (\mathfrak{L}_{s,\lambda} + \epsilon_0 B_s)^{-1} = \mathfrak{L}_{s,\lambda}^{-1}(\mathbf{1} + \epsilon_0 B_s \mathfrak{L}_{s,\lambda}^{-1})^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+ +1}_{\sigma,R_\infty,2}$$

is a well-defined bounded linear operator and we can express:

$$L_s = -(\mathbf{1} - K_{s,\lambda} T_s) T_s^{-1}$$

on $H^{N_+ +1}_{\sigma,R_\infty,2}$.

Formally, $L_s^{-1} = -T_s(\mathbf{1} - K_{s,\lambda}T_s)^{-1}$. By the analytic Fredholm theorem (see for example [RR06][Theorem 8.92]), $(\mathbf{1} - K_{s,\lambda}T_s)^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ defines a meromorphic family of bounded operators if $\{K_{s,\lambda}T_s\}$ is holomorphic and compact as an operator from $H^{N_+}_{\sigma,R_\infty}$ to $H^{N_+}_{\sigma,R_\infty}$. Indeed, we use here that $(\mathfrak{L}_{s,\lambda} + K_{s,\lambda})^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ is bounded for $\operatorname{Re} s \geq s_0$ by Proposition 15.3, so for ϵ_0 suitably small, $L_s^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ is also bounded for $\operatorname{Re} s \geq s_0$. Compactness follows directly from Proposition 16.1. Consider the following difference quotient: let

$$\begin{aligned}
\frac{K_{s,\lambda}T_s - K_{s_0,\lambda}T_{s_0}}{s - s_0}(f) &= \frac{K_{s,\lambda} - K_{s_0,\lambda}}{s - s_0} T_s + K_{s_0,\lambda} \frac{T_s - T_{s_0}}{s - s_0} \\
&= \frac{K_{s,\lambda} - K_{s_0,\lambda}}{s - s_0} T_s - K_{s_0,\lambda} T_s \frac{\mathfrak{L}_{s,\lambda} - \mathfrak{L}_{s_0,\lambda} + \epsilon_0(B_s - B_{s_0})}{s - s_0} T_{s_0} \\
&= \frac{K_{s,\lambda} - K_{s_0,\lambda}}{s - s_0} T_s - K_{s_0,\lambda} T_s \frac{L_s - L_{s_0} - (K_{s,\lambda} - K_{s_0,\lambda})}{s - s_0} T_{s_0}
\end{aligned}$$

Note that

$$\begin{aligned}
\frac{L_s - L_{s_0}}{s - s_0}(f) &= 2\partial_x f + Mx\left[b_{x\tau}\partial_x f + (s + s_0) b_{\tau\tau} f + x b_{\tau\vartheta})\sin\vartheta\partial_\vartheta f + b_\tau f\right] - 2M^2(s + s_0)h(0)f, \\
\frac{K_{s,\lambda} - K_{s_0,\lambda}}{s - s_0} &= -2M^2(s + s_0)h(0)f
\end{aligned}$$

By the above expressions, it is straightforward to see that $\lim_{s \to s_0} \frac{K_{s,\lambda}T_s - K_{s_0,\lambda}T_{s_0}}{s - s_0}$ is a well-defined operator from $H^{N_+}_{\sigma,R_\infty}$ to $H^{N_+}_{\sigma,R_\infty}$.

We conclude that $L_s^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+ +2}_{\sigma,\delta,2}$ constitute a meromorphic family of operators, so (i) holds when $\mathbb{h} = 1$ in $\{\varrho \leq \frac{1}{2}R_\infty\}$.

Since $K_{s,\lambda}T_s$ is compact, we have by the Fredholm alternative, that

$$\dim\ker(T_s L_s)^* = \dim\ker(\mathbf{1} - (K_{s,\lambda}T_s)^*) = \dim\ker(\mathbf{1} - (K_{s,\lambda}T_s)^*) = \dim\ker(T_s L_s) < \infty$$

and

$$\operatorname{ran}((T_s L_s)^*) = \operatorname{ran}(\mathbf{1} - (K_{s,\lambda}T_s)^*) = [\ker(\mathbf{1} - K_{s,\lambda}T_s)]^\perp = (\ker(T_s L_s))^\perp,$$

with T_s^* and L_s^* the Hilbert space adjoints of T_s and L_s, respectively. By the boundedness of T_s and T_sL_s, we have that $(T_sL_s)^* = L_s^*T_s^* : H^{N_+ +1}_{\sigma,R_\infty,2} \to H^{N_+ +1}_{\sigma,R_\infty,2}$.

Suppose $\widehat{\psi} \in \ker L_s$. Then $L_s\widehat{\psi} = 0$, so $T_sL_s\widehat{\psi} = 0$, so $\widehat{\psi} \in \ker(T_sL_s)$. Suppose now that $\widehat{\psi} \in \ker(T_sL_s)$, then $T_sL_s\widehat{\psi} = 0$. Since $\ker T_s = \{0\}$, we must have that $L_s\widehat{\psi} = 0$, so $\ker L_s = \ker(T_sL_s)$. Similarly, since $\ker T_s = \{0\}$, we have that $\operatorname{ran} T_s^* = H^{N_+}_{\sigma,R_\infty}$, so $\operatorname{ran} L_s^* = \operatorname{ran}(L_s^*T_s^*)$.

Finally, observe that

$$\operatorname{coker}(L_s^*) = H^{N_+}_{\sigma,R_\infty}/\operatorname{ran}(L_s^*) \cong (\operatorname{ran}(L_s^*))^\perp = (\operatorname{ran}((T_sL_s)^*))^\perp = \ker(T_sL_s) = \ker L_s$$

and we conclude (ii) when $\mathbb{h} = 1$ in $\{\varrho \leq \frac{1}{2}R_\infty\}$.

Now, consider the case of more general $\mathbb{h}$, which do not need to satisfy $\mathbb{h} \equiv 1$ in $\{\varrho \leq \frac{R_\infty}{2}\}$. Denote with $\Sigma^{\mathbb{h}}$ the hypersurface Σ corresponding to the more general function $\mathbb{h}$. Similarly, denote with $L^{\mathbb{h}}_{s,\kappa}$ and $\tau^{\mathbb{h}}$ the operator L_s and the time function τ with respect to $\mathbb{h}$, respectively. Let $\mathbb{h}_0$ satisfy $\mathbb{h}_0 = 1$ in $\{\varrho \leq \frac{1}{2}R_\infty\}$.

By assumption on the hyperboloidal time functions, we must have that $\mathbb{h}_0 - \mathbb{h}$ is integrable in r, so

$$\tau^{\mathbb{h}} - \tau^{\mathbb{h}_0} = \int_\varrho^\infty (\mathbb{h}_0 - \mathbb{h})(\varrho')\, d\varrho'$$

is well-defined.

We will show that invertibility of $L^{\mathbb{h}_0}_{s,\kappa}$ is equivalent to invertibility of $L^{\mathbb{h}}_{s,\kappa}$.

Let $Q_s : L^2(\Sigma^{\mathbb{h}}) \to L^2(\Sigma^{\mathbb{h}_0})$ be defined as follows: let $\overline{f} : D^+(\Sigma^{\mathbb{h}}) \to \mathbb{C}$ denote the trivial extension of $f \in L^2(\Sigma^{\mathbb{h}})$ to $D^+(\Sigma^{\mathbb{h}})$, i.e. $\overline{f}(\tau^{\mathbb{h}}, \varrho, \vartheta, \varphi_*) = \overline{f}(\varrho, \vartheta, \varphi_*)$. Then we define

$$Q_s(f)(\varrho, \vartheta, \varphi_*) = (e^{s\tau^{\mathbb{h}}}\overline{f})|_{\Sigma^{\tau^{\mathbb{h}_0}}}(\varrho, \vartheta, \varphi_*) = e^{s\int_\varrho^\infty (\mathbb{h}_0 - \mathbb{h})(\varrho')\, d\varrho'} f(\varrho, \vartheta, \varphi_*)$$

to obtain

$$L_s^{\mathbb{h}} = Q_s^{-1} \circ L_s^{\mathbb{h}_0} \circ Q_s,$$

with $Q_s^{-1}(f) = e^{-s\int_\varrho^{R_\infty} (\mathbb{h}_0 - \mathbb{h})(\varrho')\, d\varrho'} f$. Since the factor $e^{s\int_\varrho^{R_\infty} (\mathbb{h}_0 - \mathbb{h})(\varrho')\, d\varrho'}$ is analytic in $x = \frac{1}{\varrho}$ near $x = 0$ by assumption on $\mathbb{h}$, we can apply Proposition C.1 to conclude that there exists a constant $B > 0$ such that

$$\begin{aligned} \|Q_s(f)\|_{H^{N_+}_{\sigma,R_\infty}} &\leq B\|f\|_{H^{N_+}_{\sigma,R_\infty}} \quad \text{and} \quad \|Q_s(f)\|_{H^{N_+ +1}_{\sigma,R_\infty,2}} \leq B\|f\|_{H^{N_+ +1}_{\sigma,R_\infty,2}}, \\ \|Q_s^{-1}(f)\|_{H^{N_+}_{\sigma,R_\infty}} &\leq B\|f\|_{H^{N_+}_{\sigma,R_\infty}} \quad \text{and} \quad \|Q_s^{-1}(f)\|_{H^{N_+ +1}_{\sigma,R_\infty,2}} \leq B\|f\|_{H^{N_+ +1}_{\sigma,R_\infty,2}}. \end{aligned}$$

The restrictions $Q_s, Q_s^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ and $Q_s, Q_s^{-1} : H^{N_+ +1}_{\sigma,R_\infty,2} \to H^{N_+ +1}_{\sigma,R_\infty,2}$ are therefore bounded operators, so

$$(L_s^{\mathbb{h}})^{-1} = Q_s^{-1} \circ (L_s^{\mathbb{h}_0})^{-1} \circ Q_s : H^{N_+}_{\sigma,R_\infty} \to H^{N_+ +1}_{\sigma,R_\infty,2}$$

is well-defined and bounded if and only if $(L_s^{\mathbb{h}_0})^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+ +1}_{\sigma,R_\infty,2}$ is well-defined and bounded. This moreover implies that the poles of $(L_s^{\mathbb{h}})^{-1}$are independent of the choice of $\mathbb{h}$, provided that $\mathbb{h}$ satisfies the assumptions stated in §6.2. □

We denote $\mathscr{Q}_{\sigma,R_\infty,N_+} := \mathscr{Q}_{\sigma,R_\infty,N_+,0}$. In the next proposition, we will show that sets $\mathscr{Q}_{\sigma,R_\infty,N_+}$ do not depend on the choice of R_∞, N_+ and σ, provided R_∞ is taken to be suitably large and σ is restricted to an appropriate interval. We will also show that the union of $\mathscr{Q}_{\mathrm{reg}} := \bigcup_{\sigma\in\mathbb{R}, N_+\in\mathbb{N}_0} \mathscr{Q}_{\sigma,R_\infty,N_+}$ has no accumulation points in $\Omega = \{z \in \mathbb{C} \setminus \{0\}, |\arg(z)| < \varphi_0\}$.

We first need to establish suitably convergence properties of the elements in $\mathscr{Q}_{\sigma,R_\infty,N_+,\kappa}$ as $\kappa \downarrow 0$ and with respect to small perturbations $L_{s,\kappa} + \delta\widetilde{Q}$:

Lemma 16.3. *Let $\{\kappa_n\}$ and $\{\delta_n\}$ be a sequences in $[0,\infty)$ such that $\kappa_n \to 0$ and $\delta_n \to 0$ as $n \to \infty$. Define $T_{s,n} := -(\mathfrak{L}_{s,\lambda,\kappa_n} + \delta_n \cdot \widetilde{Q} + \epsilon_0 B_s)^{-1}$, with $\widetilde{Q} \in B(H^{N_++1}_{\sigma,R_\infty,2}, H^{N_+}_{\sigma,R_\infty})$ and let $T_s := -(\mathfrak{L}_{s,\lambda} + \epsilon_0 B_s)^{-1}$.*

(i) *Suppose that there exists a $s_* \in \Omega_\sigma \cap \{\mathrm{Re}(z) > -(\frac{1}{2}+N_+)\kappa_+\}$ such that $\ker(\mathbf{1} - K_{s_*,\lambda}T_{s_*}) \neq \{0\}$. Then there exist a sequence $\{s_n\}$ in $\Omega_\sigma \cap \{\mathrm{Re}(z) > -(\frac{1}{2}+N_+)\kappa_+\}$ such that $s_n \to s_*$ as $n \to \infty$ and*

$$\ker(\mathbf{1} - K_{s_n,\lambda}T_{s_n,n}) \neq \{0\}.$$

Furthermore, for all $\widehat{\psi}_{s_*} \in \ker L_{s_*}$, there exists a sequence $\{\widehat{\psi}_{s_n}\}$ in $\ker(\mathbf{1} - K_{s_n,\lambda}T_{s_n,n})$, such that

$$\|\widehat{\psi}_{s_n} - \widehat{\psi}_{s_*}\|_{H^{N_+}_{\sigma,R_\infty}} \to 0$$

as $n \to \infty$.

(ii) *Conversely, suppose that there exists a sequence $\{s_n\}$ in $\Omega_\sigma \cap \{\mathrm{Re}(z) > -(\frac{1}{2}+N_+)\kappa_+\}$, $\widehat{\psi}_{s_n}$ with $||\widehat{\psi}_{s_n}||_{H^{N_+}_{\sigma,R_\infty}} = 1$ and $\{\kappa_n\}$, such that $s_n \to s_* \in \Omega_\sigma \cap \{\mathrm{Re}(z) > -(\frac{1}{2}+N_+)\kappa_+\}$, $\kappa_n \downarrow 0$ and $\widehat{\psi}_n \in \ker(\mathbf{1} - K_{s_n,\lambda}T_{s_n,n})$. Then there exists a $0 \neq \widehat{\psi} \in \ker L_{s_*}$.*

Proof. The key ingredients of the proof of (i) are the convergence property $\mathfrak{L}^{-1}_{s,\lambda,\kappa_n} \to \mathfrak{L}^{-1}_{s,\lambda}$ with respect to the operator norm, which we established in (15.3), together with the fact that $(\mathfrak{L}_{s,\lambda,\kappa} + \delta_n \cdot \widetilde{Q})^{-1}$ is a well-defined bounded operator for $\delta_n < \frac{1}{\|\widetilde{Q}\| \cdot \|\mathfrak{L}^{-1}_{s,\lambda,\kappa}\|}$, which follows from the proof of Proposition 16.2, and $(\mathfrak{L}_{s,\lambda,\kappa} + \delta_n \cdot \widetilde{Q})^{-1}$ converges to $\mathfrak{L}^{-1}_{s,\lambda,\kappa}$ with respect to the operator norm. We also use that $\ker L_s = \ker(\mathbf{1} - K_{s_*,\lambda}T_{s_*})$. Then Hurwitz's theorem is applied. We refer the reader to [GW21][Proposition 11.6 and 11.7] for the details, where $K_{s_n,\lambda}T_{s_n,n}$ replaces the operator $A_{\kappa_n,s}$ from [GW21].

We prove (ii) by first proceeding as in the proof of Proposition 15.3. Let $\sigma' < \sigma$. Then $\widehat{\psi}_{s_n} \in H^{N_++1}_{\sigma,R_\infty,2} \subset\subset H^{N_++2}_{\sigma',R_\infty,2}$, so $\{\widehat{\psi}_n\}$ admits a convergent subsequence with non-trivial limit $\widehat{\psi}_{s_*} \in H^{N_++1}_{\sigma',R_\infty,2}$. Since $\{\widehat{\psi}_n\}$ is also weakly convergent in $H^{N_++1}_{\sigma,R_\infty,2}$ and there is uniqueness of limits, we must have that $\widehat{\psi}_{s_*} \in H^{N_++1}_{\sigma,R_\infty,2}$.

We will now show that $\widehat{\psi}_{s_*} \in \ker L_{s_*}$. By splitting

$$\mathbf{1} - K_{s_*,\lambda}T_{s_*} = (\mathbf{1} - K_{s_n,\lambda}T_{s_n,n}) + (K_{s_n,\lambda}T_{s_n,n} - K_{s_*,\lambda}T_{s_n}),$$

we obtain

$$(\mathbf{1} - K_{s_*,\lambda}T_{s_*})\widehat{\psi}_{s_*} = (\mathbf{1} - K_{s_*,\lambda}T_{s_*})(\widehat{\psi}_{s_*} - \widehat{\psi}_{s_n}) + (K_{s_n,\lambda}T_{s_n,n} - K_{s_*,\lambda}T_{s_n})\widehat{\psi}_{s_*}.$$

The terms on the right-hand side go to zero in $H^{N_++1}_{\sigma',R_\infty,2}$ as $n \to \infty$, so we conclude that $\widehat{\psi}_{s_*} \in \ker L_{s_*}$. □

Proposition 16.4. *Let $N+ \in \mathbb{N}_0$ and let $R_\infty > 0$ be suitably large, such that Proposition 16.2 applies.*

(i) *Let $\tilde{N}_+ \geq N_+$. Then $\mathscr{Q}_{\sigma,R_\infty,\tilde{N}_+} = \mathscr{Q}_{\sigma,R_\infty,N_+}$, $\ker L_s \subset C^\infty(\hat{\Sigma})$ and $\ker(\mathfrak{L}_{s,\lambda,\kappa} + \epsilon_0 B_s + K_{s,\lambda}) \subset C^\infty(\hat{\Sigma})$ for $s \in \Omega_\sigma \cap \{\mathrm{Re}(z) > -(\frac{1}{2}+N_+)\kappa_+\}$ with $\sigma \in \mathbb{R}$.*

(ii) *Let $\widetilde{R}_\infty \geq R_\infty$. Then $\mathscr{Q}_{\sigma,\widetilde{R}_\infty,N_+} = \mathscr{Q}_{\sigma,R_\infty,N_+}$.*

(iii) *Let $|\sigma'| > |\sigma|$ and $\Omega_{\sigma'} \cap \Omega_\sigma \neq \emptyset$. Then*

$$\mathscr{Q}_{\sigma,R_\infty,N_+} \cap (\Omega_{\tilde{\sigma}} \cap \Omega_\sigma) = \mathscr{Q}_{\sigma',R_\infty,N_+} \cap (\Omega_{\sigma'} \cap \Omega_\sigma).$$

In particular, $\mathscr{Q}_{\mathrm{reg}} = \bigcup_{\sigma\in\mathbb{R},N_+\in\mathbb{N}_0} \mathscr{Q}_{\sigma,R_\infty,N_+}$ has no accumulation points in $\Omega = \{z \in \mathbb{C} \setminus \{0\}\,,\ |\arg(z)| < \varphi_0\}$.

Proof. We will first prove (i). Let $\widehat{\psi}_s \in \ker L_s \subset H^{N_+ +1}_{\sigma,R_\infty,2}$. Then

$$(\mathfrak{L}_{s,\lambda} + \epsilon_0 B_s)\widehat{\psi}_s = -K_{s,\lambda}\widehat{\psi}_s.$$

In the proof of Proposition 16.2, we showed that for ϵ_0 suitably small, $(\mathfrak{L}_{s,\lambda} + \epsilon_0 B_s)^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+ +1}_{\sigma,R_\infty,2}$ is a well-defined bounded operator and hence, there exists a constant $C > 0$, such that

$$\|\widehat{\psi}_s\|_{H^{N_+ +1}_{\sigma,R_\infty,2}} = \|(\mathfrak{L}_{s,\lambda} + \epsilon_0 B_s)^{-1} K_{s,\lambda}\widehat{\psi}_s\|_{H^{N_+ +1}_{\sigma,R_\infty,2}} \leq C\|K_{s,\lambda}\widehat{\psi}_s\|_{H^{N_+}_{\sigma,R_\infty}}. \tag{16.1}$$

By Proposition 16.1, we have in fact that $\|K_{s,\lambda}\widehat{\psi}_s\|_{H^{N_+ +1}_{\sigma,R_\infty,2}} < \infty$.

Now let $\tilde{N}_+ = N_+ + 1$, then (16.1) applies with $\tilde{N}_+$ replacing N_+:

$$\|\widehat{\psi}_s\|_{H^{\tilde{N}_+ +1}_{\sigma,R_\infty,2}} \leq C\|K_{s,\lambda}\psi_s\|_{H^{\tilde{N}_+}_{\sigma,R_\infty}} \leq C\|K_{s,\lambda}\widehat{\psi}_s\|_{H^{N_+ +1}_{\sigma,R_\infty,2}} < \infty.$$

Therefore $\widehat{\psi}_s \in H^{N_+ +2}_{\sigma,R_\infty,2}$.

Conversely, suppose that $\widehat{\psi} \in \ker L_s \subset H^{\tilde{N}_+ +1}_{\sigma,R_\infty,2}$ with $\tilde{N}_+ = N_+ + 1$. By (16.1), we have that

$$\|\widehat{\psi}_s\|_{H^{N_+ +1}_{\sigma,R_\infty,2}} \leq \|K_{s,\lambda}\widehat{\psi}_s\|_{H^{N_+}_{\sigma,R_\infty}} \leq \|K_{s,\lambda}\widehat{\psi}_s\|_{H^{\tilde{N}_+ +1}_{\sigma,R_\infty,2}} < \infty,$$

where we used in particular that for $w_{\tilde{p}}$ defined with $\tilde{p} = 4(\tilde{N}_+ + 1) + 2 = 4(N_+ + 1) + 2 + 4$ and w_p defined with $p = 4(N_+ + 1) + 2$:

$$\|(\mathbf{1} - \not{\Delta}_{\mathbb{S}^2})^{-\frac{w_p}{4}} K_{s,\lambda}\widehat{\psi}_s\|_{H^{N_+}_{\sigma,R_\infty}} \leq \|\not{\nabla}_{\mathbb{S}^2}(\mathbf{1} - \not{\Delta}_{\mathbb{S}^2})^{-\frac{w_{\tilde{p}}}{4}} K_{s,\lambda}\widehat{\psi}_s\|_{H^{\tilde{N}_+}_{\sigma,R_\infty}} \leq \|(\mathbf{1} - \not{\Delta}_{\mathbb{S}^2})^{-\frac{w_{\tilde{p}}}{4}}\widehat{\psi}_s\|_{H^{\tilde{N}_+ +1}_{\sigma,R_\infty,2}}.$$

Hence, $\widehat{\psi} \in \ker L_s \subset H^{N_+ +1}_{\sigma,R_\infty,2}$.

We can repeat this procedure inductively to conclude that the kernel of L_s with respect to $H^{N_+ +1}_{\sigma,R_\infty,2}$ agrees with the kernel of L_s with respect to $H^{\tilde{N}_+ +1}_{\sigma,R_\infty,2}$ for arbitrary $\tilde{N}_+ \geq N_+$. Therefore, we must have that $\mathscr{Q}_{\sigma,R_\infty,\tilde{N}_+} = \mathscr{Q}_{\sigma,R_\infty,N_+}$.

Note that $(\mathfrak{L}_{s,\lambda} + \epsilon_0 B_s)(\not{\nabla}_{\mathbb{S}^2}\widehat{\psi}_s) = -\not{\nabla}_{\mathbb{S}^2} K_{s,\lambda}\widehat{\psi}_s + [\not{\nabla}_{\mathbb{S}^2}, \mathfrak{L}_{s,\lambda} + \epsilon_0 B_s]\widehat{\psi}_s$.

It is straightforward to show that

$$\|[\not{\nabla}_{\mathbb{S}^2}, \mathfrak{L}_{s,\lambda} + \epsilon_0 B_s]\widehat{\psi}_s\|_{H^{N_+}_{\sigma,R_\infty}} \leq C\|\widehat{\psi}_s\|_{H^{N_+ +1}_{\sigma,R_\infty,2}}.$$

Hence, we can proceed as above to conclude that

$$\|\not{\nabla}_{\mathbb{S}^2}\widehat{\psi}_s\|_{H^{N_+ +1}_{\sigma,R_\infty,2}} \leq \|\not{\nabla}_{\mathbb{S}^2} K_{s,\lambda}\psi_s\|_{H^{N_+}_{\sigma,R_\infty}} + \|[\not{\nabla}_{\mathbb{S}^2}, \mathfrak{L}_{s,\lambda} + \epsilon_0 B_s]\psi_s\|_{H^{N_+}_{\sigma,R_\infty}} \leq C\|\psi_s\|_{H^{N_+ +1}_{\sigma,R_\infty,2}} < \infty.$$

Hence, $\not{\nabla}_{\mathbb{S}^2}\widehat{\psi}_s \in H^{N_+ +1}_{\sigma,R_\infty,2}$. By repeating this procedure for $\not{\nabla}^k_{\mathbb{S}^2}\widehat{\psi}_s$, we conclude in particular that

$$\|\widehat{\psi}_s\|_{H^{N_+ +1}(\hat{\Sigma})} + \|\psi_s\|_{H^{N_+ +1}_{\sigma,R_\infty,2}} \leq C\|\psi_s\|_{H^{N_+ +1}_{\sigma,R_\infty,2}} \tag{16.2}$$

so $\widehat{\psi}_s \in H^{N_+ +1}_{\sigma,R_\infty,2} \cap H^{N_+ +1}(\hat{\Sigma})$, and in fact $\widehat{\psi}_s \in H^{\tilde{N}_+ +1}_{\sigma,R_\infty,2} \cap H^{\tilde{N}_+ +1}(\hat{\Sigma})$ for all $\tilde{N}_+ \geq N_+$, so $\widehat{\psi}_s \in C^\infty(\hat{\Sigma}) \cap H^{N_+ +1}_{\sigma,R_\infty,2}$. To conclude that $\ker(\mathfrak{L}_{s,\lambda,\kappa} + \epsilon_0 B_s + K_{s,\lambda}) \subset C^\infty(\hat{\Sigma})$, we simply repeat the above arguments with $\mathfrak{L}_{s,\lambda,\kappa}$ replacing $\mathfrak{L}_{s,\lambda}$.

We now turn to (ii). Since $\widehat{\psi}_s \in C^\infty(\hat{\Sigma})$, it follows immediately that for $\widehat{\psi}_s \in \ker L_s \subseteq H^{N_+ +1}_{\sigma,R_\infty,2}$ with $\tilde{R}_\infty \geq R_\infty$, $\widehat{\psi}_s \in H^{N_+ +1}_{\sigma,\tilde{R}_\infty,2}$. It remains to show that $\widehat{\psi}_s \in \ker L_s \subseteq H^{N_+ +1}_{\sigma,\tilde{R}_\infty,2}$ implies that $\widehat{\psi}_s \in H^{N_+ +1}_{\sigma,R_\infty,2}$. For this, we will apply Lemma 16.3 with $Q \equiv 0$, which says that there exists a

sequence $\{\kappa_n\}$ in $(0,\infty)$ and $s_n \in \Omega_\sigma \cap \{\mathrm{Re}(z) > -(\frac{1}{2}+N_+)\kappa_+\}$ with $\kappa_n \downarrow 0$ as $n \to \infty$, and there exists a sequence $\psi_{s_n} \in \ker(\mathfrak{L}_{s,\lambda,\kappa_n} + \epsilon B_s + K_{s,\lambda})$ such that

$$\|\widehat{\psi}_{s_n} - \widehat{\psi}_s\|_{H^{N_+}_{\sigma,\tilde{R}_\infty}} \to 0$$

as $n \to \infty$.

Observe that

$$(\mathfrak{L}_{s,\lambda,\kappa_n} + \epsilon_0 B_s)\widehat{\psi}_{s_n} = -K_{s,\lambda}\widehat{\psi}_{s_n}.$$

Since any estimates involving $\mathfrak{L}_{s,\lambda,\kappa_n}$ apply with $H^{N_++1}_{\sigma,R_\infty}$ and $H^{N_++1}_{\sigma,R_\infty,2}$ norms where the infinite sums $\sum_{n=\dots}^{\infty}$ are replaced with sums $\sum_{n=\dots}^{N_\infty}$ with $N_\infty < \infty$ suitably large depending on κ_n, we can first apply the estimates with finite N_∞, using that $\widehat{\psi}_{s_n}, \widehat{\psi}_{s_m} \in C^\infty(\hat{\Sigma})$, and subsequently take the limit $N_\infty \to \infty$ to obtain: there exists an $N > 0$ (independent of n) such that

$$\|\widehat{\psi}_{s_n}\|_{H^{N_++1}_{\sigma,R_\infty,2}} \leq C\|\widehat{\psi}_{s_n}\|_{H^N(\hat{\Sigma})},$$

where we moreover used that all sufficiently higher-order derivatives can appearing on the right-hand side of the $N_\infty < \infty$ estimates can be absorbed to the left-hand side, leaving only lower-order derivatives on the right-hand side. For N_+ suitably large depending on N, we obtain from (16.2) applied to the $\kappa > 0$ setting with $\tilde{R}_\infty$ replacing R_∞:

$$\|\widehat{\psi}_{s_n}\|_{H^{N_++1}_{\sigma,R_\infty,2}} + \|\widehat{\psi}_{s_n}\|_{H^{N_++1}(\hat{\Sigma})} \leq C\|\widehat{\psi}_{s_n}\|_{H^{N_++1}_{\sigma,\tilde{R}_\infty,2}}.$$

In particular, $\{\widehat{\psi}_{s_n}\}$ is a bounded sequence in $H^{N_++1}_{\sigma,R_\infty,2} \cap H^{N_++1}(\hat{\Sigma})$, with respect to the norm $\|\cdot\|_{H^{N_++1}_{\sigma,R_\infty,2}} + \|\cdot\|_{H^{N_++1}(\hat{\Sigma})}$ so it admits a weakly convergent subsequence. Since $H^{N_++1}_{\sigma,R_\infty,2} \cap H^{N_++1}(\hat{\Sigma}) \subset H^{N_++1}_{\sigma,\tilde{R}_\infty,2}$ and $\widehat{\psi}_{s_n}$ converges strongly to ψ_s with respect to $\|\cdot\|_{H^{N_++1}_{\sigma,\tilde{R}_\infty,2}}$, we can conclude by uniqueness of limits that $\widehat{\psi}_s \in H^{N_++1}_{\sigma,\tilde{R}_\infty,2}$.

We therefore conclude that $\mathscr{Q}_{\sigma,\tilde{R}_\infty,N_+} = \mathscr{Q}_{\sigma,R_\infty,N_+}$.

Finally, consider (iii). First of all, we have that $H^{N_++1}_{\tilde{\sigma},R_\infty,2} \subset H^{N_++1}_{\sigma,R_\infty,2}$, so if $\psi_s \in \ker L_s$ with respect to $H^{N_+}_{\tilde{\sigma},R_\infty}$, then $\psi_s \in \ker L_s$ with respect to $H^{N_+}_{\sigma,R_\infty}$.

Conversely, suppose that $\psi_s \in \ker L_s$ with respect to $H^{N_+}_{\sigma,R_\infty}$. Suppose also that $s \in \Omega_{\sigma'}$. As above, we can find a sequence $\{\kappa_n\}$ in $(0,\infty)$, $s_n \in \Omega_\sigma \cap \Omega_{\sigma'} \cap \{\mathrm{Re}(z) > -(\frac{1}{2}+N_+)\kappa_+\}$ with $\kappa_n \downarrow 0$ as $n \to \infty$ and $\psi_{s_n} \in \ker(\mathfrak{L}_{s,\lambda,\kappa_n} + \epsilon B_s + K_{s,\lambda})$ with

$$\|\widehat{\psi}_{s_n} - \widehat{\psi}_s\|_{H^{N_+}_{\sigma,R_\infty}} \to 0.$$

We use the same strategy as before: we apply the $N_\infty < \infty$ estimates with respect to $H^{N_++1}_{\sigma',R_\infty}$ to $(\mathfrak{L}_{s,\lambda,\kappa_n} + \epsilon_0 B_s)\widehat{\psi}_{s_n}$ and use that any higher-order derivatives can be absorbed into the left-hand side, leaving

$$\|\widehat{\psi}_{s_n}\|_{H^{N_++1}_{\sigma,R_\infty,2}} \leq C\|\widehat{\psi}_{s_n}\|_{H^N(\hat{\Sigma}\cap\{x\leq x_0\})} \leq \|\widehat{\psi}_{s_n}\|_{H^{N_++1}_{\sigma,R_\infty,2}}.$$

Hence, $\{\widehat{\psi}_{s_n}\}$ admits a weakly convergent subsequence with limit in $H^{N_++1}_{\sigma',R_\infty,2}$. Since $H^{N_++1}_{\sigma',R_\infty,2} \subset H^{N_++1}_{\sigma,R_\infty,2}$ and uniqueness of limits holds, we conclude that $\psi_s \in H^{N_++1}_{\sigma',R_\infty,2}$. Hence, s is an element of $\ker L_s$ with respect to $H^{N_++1}_{\sigma',R_\infty,2}$, from which (iii) follows. □

17. Stability of the quasinormal spectrum

Recall the definition of the stability of the spectrum of a family of operators in Definition 10.3. The following proposition forms the main ingredient for the stability results in Theorem 10.3.

Proposition 17.1. *Consider the operator $L_s + \delta \cdot \widetilde{Q} : D^{N_+}_{\sigma,R_\infty}(L_s) \to H^{N_+}_{\sigma,R_\infty}$, with $\widetilde{Q} : H^{N_+}_{\sigma,R_\infty,2} \to H^{N_+}_{\sigma,R_\infty}$ bounded for all $|\sigma| < \sigma_0$, with $\sigma_0 > 0$ or $\sigma_0 = \infty$. Let $\epsilon > 0$ and $D \subset \Omega_\sigma$. Then for all $\epsilon > 0$ and $K \subset \Omega_\sigma$ compact, there exists a $\delta_0 > 0$ such that for all $0 < \delta \leq \delta_0$:*

(i) *For $|\sigma| < \sigma_0$, the family $s \mapsto (L_s + \delta \cdot \widetilde{Q})^{-1} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty,2}$ is meromorphic on K and holomorphic on $K \setminus \bigcup_{z \in \mathscr{Q}_{\mathrm{reg}}} D_\delta(z) \cap K$.*

(ii) *For all $z \in \mathscr{Q}_{\mathrm{reg}} \cap D_{\sigma_0^2}(0)$, there exists an $s \in D_\delta(z)$, such that s is a pole of $s \mapsto (L_s + \delta \cdot \widetilde{Q})^{-1}$.*

Proof. Part (i) follows from the fact that for $s \in K \setminus (\bigcup_{z \in \mathscr{Q}_{\mathrm{reg}}} B_\delta(z) \cap K)$, we can write.

$$L_s + \delta \cdot \widetilde{Q} = (1 + \delta \cdot \widetilde{Q} L_s^{-1}) L_s$$

and for suitably small δ, $\|\widetilde{Q} L_s^{-1}\|_{H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}} < 1$ for all $s \in K \setminus (\bigcup_{z \in \mathscr{Q}_{\mathrm{reg}}} D_\delta(z) \cap K)$, provided $|\sigma| < \sigma_0$, so

$$(L_s + \delta \cdot \widetilde{Q})^{-1} = L_s^{-1}(1 + \delta \cdot Q L_s^{-1})^{-1}$$

is well-defined.

Part (ii) follows directly from Lemma 16.3 with $\kappa_n = 0$ and the observation that for $|\sigma| < \sigma_0$ we have that

$$\bigcup_{\sigma \in (-\sigma_0, \sigma_0)} \Omega_\sigma = \Omega \cap D_{\sigma_0^2}(0). \qquad \square$$

18. Kerr–de Sitter quasinormal modes

Let $(\mathcal{M}_{M,a,\Lambda}, g_{M,a,\Lambda})$ be Kerr de Sitter spacetimes with mass M, angular momentum/mass a and cosmological constant Λ, such that κ_+ remains uniformly bounded away from zero as $\Lambda \downarrow 0$. We denote by r_+ the radius of the event horizon, by r_c the radius of the cosmological horizon and by κ_c the surface gravity of the cosmological horizon. Then it can easily be verified that:

$$\kappa_c \sim \sqrt{\frac{\Lambda}{3}} \sim r_c^{-1}.$$

The wave operator $\Box_{g_{M,a,\Lambda}}$ takes the following form with respect to Boyer–Linquist coordinates (t, r, θ, φ):

$$\rho^2 \Box_{g_{M,a,\Lambda}} \phi = \partial_r(\Delta_r \partial_r \phi) - \Xi^2 \left[\frac{(r^2+a^2)^2}{\Delta_r} - \frac{a^2 \sin^2\theta}{\Delta_\theta} \right] \partial_t^2 \phi - 2a \frac{\Xi^2}{\Delta_r} \left[(r^2+a^2) - \frac{\Delta_r}{\Delta_\theta} \right] \partial_t \partial_\varphi \phi + \frac{1}{\sin\theta} \partial_\theta(\Delta_\theta \sin\theta \partial_\theta \phi) + \Xi^2 \left(\frac{1}{\Delta_\theta \sin^2\theta} - \frac{a^2}{\Delta_r} \right) \partial_\varphi^2 \phi,$$

with

$$\begin{aligned} \rho^2 &= r^2 + a^2 \cos^2\theta, \\ \Delta_r &= (r^2+a^2)\left(1 - \frac{\Lambda}{3} r^2\right) - 2Mr, \\ \Delta_\theta &= 1 + \frac{\Lambda a^2}{3} \cos^2\theta, \\ \Xi &= 1 + \frac{\Lambda a^2}{3}. \end{aligned}$$

Furthermore, $\kappa_c = -\frac{\Xi}{2(r_c^2+a^2)}\frac{d}{dr}\Delta_r|_{r=r_c}$. By using that $\Delta(r_c)=0$, we moreover obtain:

$$0 \leq 1 - \frac{\Lambda}{3r_c^2} = \frac{2M}{1+a^2r_c^{-2}}r_c^{-1} \leq 2Mr_c^{-1}. \tag{18.1}$$

With respect to the vector fields

$$\begin{aligned} T =& \Xi\partial_t, \\ \Phi =& \Xi\partial_\varphi, \\ Y =& \partial_r - \frac{r^2+a^2}{\Delta_r}\Xi\partial_t - \frac{a}{\Delta_r}\Xi\partial_\varphi, \end{aligned}$$

we can express:

$$\begin{aligned} \rho^2\Box_{g_{M,a,\Lambda}}\phi = Y(\Delta_r Y\phi) + 2aY\Phi\phi + \frac{1}{\sin\theta}\partial_\theta(\Delta_\theta\sin\theta\partial_\theta\phi) + \frac{1}{\Delta_\theta\sin^2\theta}\Phi^2\phi \\ + 2(r^2+a^2)YT\phi + 2rT\phi + \frac{1}{\Delta_\theta}a^2\sin^2\theta T^2\phi + 2a\frac{1}{\Delta_\theta}T\Phi\phi. \end{aligned}$$

Define $v = \Xi^{-1}t + r_*$ with $\frac{dr_*}{dr}(r) = \frac{r^2+a^2}{\Delta_r}$ and $\varphi_* = \Xi^{-1}\varphi + \int_{r_0}^r \frac{a}{\Delta_r} \mod 2\pi$. With respect to $(v, r, \theta, \varphi_*)$ coordinates, we have that $T = \partial_v$, $\Phi = \partial_{\varphi_*}$ and $Y = \partial_r$.

In a region where $r - r_c$ is suitably small, we will view $\Box_{g_{M,a,\Lambda}}\phi$ as a perturbation of $\Box_{g_{0,a,\Lambda_c}}\phi$, which is the de Sitter wave operator with cosmological constant $\Lambda_c = \frac{3}{r_c^2}$. We denote $\mathring{\Delta}_\theta = \frac{\Lambda_c a^2}{3}\cos^2\theta$ and $\mathring{\Delta}_r = (r^2+a^2)\left(1 - \frac{\Lambda_c}{3}r^2\right)$, which correspond to the expressions for Δ_r and Δ_θ, respectively, in the case $M=0$ and $\Lambda = \Lambda_c$.

Note that we can estimate

$$\left|\Delta_\theta - \mathring{\Delta}_\theta\right| = a^2\left|\frac{\Lambda}{3} - r_c^{-2}\right|\cos^2\theta \leq Ca^2\kappa_c^2.$$

Then

$$\begin{aligned} \rho^2\Box_{g_{M,a,\Lambda}}\phi = \rho^2\Box_{g_{0,a,\Lambda_c}}\phi + Y((\Delta_r - \mathring{\Delta}_r)Y\phi) + \frac{1}{\sin\theta}\partial_\theta((\Delta_\theta - \mathring{\Delta}_\theta)\sin\theta\partial_\theta\phi) \\ + (\Delta_\theta^{-1} - \mathring{\Delta}_\theta^{-1})\left(\frac{1}{\sin^2\theta}\Phi^2\phi + a^2\sin^2\theta T^2\phi + 2aT\Phi\phi\right). \end{aligned}$$

Consider the following coordinate transformations in the region $\{r \geq \frac{R_\infty}{\sqrt{2}}\}$ with $r_+ < R_\infty < r_c$:

$$\begin{aligned} \varrho(r,\theta) &= \sqrt{\frac{1}{1+a^2r_c^{-2}}(r^2\mathring{\Delta}_\theta + a^2\sin^2\theta)}, \\ \tan\vartheta &= \sqrt{\frac{1+a^2r^{-2}}{1+a^2r_c^{-2}}}\tan\theta, \\ \tilde{\varphi} &= \varphi - \frac{a}{r_c^2}t. \end{aligned}$$

Note that

$$\varrho(r,\theta) = r + \frac{a^2}{r_c^2+a^2}\sin^2\theta(r_c - r) + r^{-1}O_1((r_c-r)^2), \tag{18.2}$$

$$\vartheta(r,\theta) = \theta + \frac{a^2}{r^2}O_1(r_c - r). \tag{18.3}$$

The coordinates $(t, \varrho, \vartheta, \tilde{\varphi})$ correspond to standard static coordinates on de Sitter with cosmological constant Λ_c (see for example [GV17][§V]). That is to say, we can write:

$$g_{0,a,\Lambda_c} = -\left(1 - \Lambda_c \varrho^2\right) dt^2 + \left(1 - \Lambda_c \varrho^2\right)^{-1} d\varrho^2 + \varrho^2 (d\vartheta^2 + \sin^2 \vartheta d\tilde{\varphi}^2).$$

Let $\tilde{u}(t, \varrho) = t - \int_{R_\infty}^{\varrho} \frac{r_c^2}{r_c^2 - \rho'^2}\, d\varrho'$. Let $T = \partial_{\tilde{u}}$ and $X = \partial_\varrho$. Then we can express:

$$\Box_{g_{0,a,\Lambda_c}} \phi = r_c^{-2} \varrho^{-2} X(\varrho^2 (r_c^2 - \varrho^2) X\phi) + \varrho^{-2} \not{\Delta}_{\mathbb{S}^2,\vartheta} \phi - 2\varrho^{-1} XT(\varrho\phi).$$

Let $\psi = \varrho\phi$. Then

$$\varrho^3 \left(\Box_{g_{0,a,\Lambda_c}} \phi - \frac{2}{r_c^2} \phi \right) = \varrho^2 X((1 - r_c^{-2}\varrho^2) X\psi) + \not{\Delta}_{\mathbb{S}^2,\vartheta} \psi - 2\varrho^2 XT\psi.$$

Let $x = \varrho^{-1} - r_c^{-1}$, so that $\partial_x = -\rho^2 X$.

We then obtain:

$$\varrho^3 \left(\Box_{g_{0,a,\Lambda_c}} \phi - \frac{2}{r_c^2} \phi \right) = \partial_x (x(x + 2r_c^{-1}) \partial_x \psi) + \not{\Delta}_{\mathbb{S}^2,\vartheta} \psi + 2\partial_x T\psi.$$

We now extend the functions ϱ, ϑ, $\tilde{\varphi}$ to $\{r \geq r_+\}$, such that $\varrho = r$, $\vartheta = \theta$ and $\tilde{\varphi} = \varphi$ for $r \leq \frac{R_\infty}{2}$. We consider the following time function:

$$\tau(t, \varrho) = \Xi^{-1} t + \int_{r_0}^{r} \frac{(\varrho^2 + a^2)(1 + \mathbb{h}(\varrho'))}{\Delta_r(\varrho')}\, d\varrho'.$$

We consider $\mathbb{h}$ such that τ are smooth, spacelike hypersurfaces that intersect $\mathcal{H}^+$ and $\mathcal{C}^+$ to the future of the bifurcation spheres. We moreover demand that the τ-level sets are analytic in a neighbourhood of $\mathcal{C}^+$ and that $\mathbb{h}$ and its derivatives can be uniformly bounded as $\Lambda \downarrow 0$.

We then consider the coordinate chart $(\tau, x = \varrho^{-1} - r_c^{-1}, \vartheta, \tilde{\varphi})$.

Proposition 18.1. *The conformal wave operator on Kerr–de Sitter can be decomposed as follows:*

$$\varrho^3 \left(\Box_{g_{M,a,\Lambda}} \phi - \frac{2}{3} \Lambda \phi \right) = \partial_x (x(2\kappa_c + x) \partial_x \psi) + \not{\Delta}_{\mathbb{S}^2,\vartheta} \psi - 2\partial_x T\psi + \epsilon_0 \mathcal{B}_\Lambda^\infty + \mathcal{K}_\Lambda^\infty + \kappa_c^2 \mathcal{B}_\Lambda^{\mathcal{C}^+},$$

where the coefficients of the differential operators $\mathcal{B}_\Lambda^\infty$, $\mathcal{K}_\Lambda^\infty$ and $\mathcal{B}_\Lambda^{\mathcal{C}^+}$ stay bounded as $\Lambda \downarrow 0$. Furthermore, $\mathcal{B}_\Lambda^\infty$ and $\mathcal{K}_\Lambda^\infty$ have the same form as the operators $\mathcal{B}^\infty$ and $\mathcal{K}^\infty$ from Lemma 7.1 and $\mathcal{B}_\Lambda^{\mathcal{C}^+}$ can contain up to second-order angular derivatives and τ-derivatives, up to first order x-derivatives and second-order x-derivatives of the form $\kappa_c x \partial_x^2$. The constant $\epsilon_0 > 0$ can be chosen arbitrarily small for $M^{-1} R_\infty$ suitably large.

Proof. The decomposition follows by applying Lemma 7.1 (the $\Lambda = 0$) case, combined with the observation that additional terms that vanish when $\Lambda = 0$ contain a factor of Λ and hence can be written as follows $\kappa_c^2 \mathcal{B}_\Lambda^{\mathcal{C}^+}$, where $\mathcal{B}_\Lambda^{\mathcal{C}^+}$ is a second-order operator with coefficients that stay bounded as $\Lambda \downarrow 0$.

We will not derive the precise form of $\mathcal{B}_\Lambda^\infty$, $\mathcal{K}_\Lambda^\infty$ and $\mathcal{B}_\Lambda^{\mathcal{C}^+}$ in detail, but will will include a schematic version of the key relevant computation.

By applying (18.2) and (18.3), we can write:

$$\begin{aligned}
\varrho^3 \rho^{-2} Y((\Delta_r - \mathring{\Delta}_r) Y\phi) &= \varrho^3 \rho^{-2} Y((\Delta_r - \mathring{\Delta}_r) Y(\varrho^{-1}\psi)) \\
&= \varrho Y(\varrho^{-1}(\Delta_r - \mathring{\Delta}_r) Y\psi) - \varrho Y(\varrho^{-2}(\Delta_r - \mathring{\Delta}_r)\psi) \\
&= \frac{d}{dr}(\Delta_r - \mathring{\Delta}_r) \left(X\psi - \varrho^{-1}\psi \right) + \dots \\
&= -2(\kappa_c - r_c^{-1}) r_c^2 X\psi + \frac{d}{dr} \left(\frac{\Lambda - \Lambda_c}{3} r^2 (r^2 + a^2) + 2Mr \right) \varrho^{-1}\psi + \dots
\end{aligned}$$

$$= +2(\kappa_c - r_c^{-1})\partial_x\psi + \frac{4}{3}(\Lambda - 3r_c^{-2})r_c^2\psi + \dots,$$

where the terms in $\dots$ are part of $\epsilon_0\mathcal{B}^\infty_\Lambda$, $\mathcal{K}^\infty_\Lambda$ and $\kappa_c^2\mathcal{B}^{\mathcal{C}^+}_\Lambda$, for $M^2\Lambda$ suitably small depending on ϵ_0.

Note moreover that by (18.1):

$$\frac{4}{3}\left(\Lambda - 3r_c^{-2}\right) = r_c^{-2}(\Lambda r_c^{-2} - 3) = MO(r_c^{-3}),$$
$$\frac{2}{3}\Lambda_c r_c^2\psi = \frac{2}{3}\Lambda r_c^2\psi + MO(r_c^{-1})\psi.$$

□

Denote $L_{s,\Lambda} := \varrho^3\left(\Box_{g_{M,a,\Lambda}} - \frac{2}{3}\Lambda\right)(\varrho^{-1}\cdot)$, $B^\infty_{s,\Lambda} = e^{-s\tau}\mathcal{B}^\infty_\Lambda(e^{s\tau}(\cdot))$, $K^\infty_{s,\Lambda} = e^{-s\tau}\mathcal{K}^\infty_\Lambda(e^{s\tau}(\cdot))$ and $B^{\mathcal{C}^+}_{s,\Lambda} = e^{-s\tau}\mathcal{B}^{\mathcal{C}^+}_\Lambda(e^{s\tau}(\cdot))$. Then

$$L_{s,\Lambda} = \partial_x(x(x+2\kappa_c)\partial_x\psi) + \not\Delta_{\mathbb{S}^2,\vartheta}\psi - 2s\partial_x\psi + MxB^\infty_{s,\Lambda} + K^\infty_{s,\Lambda} + \kappa_c^2 B^{\mathcal{C}^+}_{s,\Lambda}.$$

Proposition 18.2. *We can decompose:*

$$L_{s,\Lambda} = \mathfrak{L}_{s,\lambda,2\kappa_c} + \epsilon_0\cdot B_{s,\Lambda} + \kappa_c^2 B^{\mathcal{C}^+}_{s,\Lambda} + K_{s,\lambda,\Lambda},$$

where

$$r^{-2}\rho^2\mathfrak{L}_{s,\lambda,\kappa_c} = r\partial_r(\Delta_r\partial_r(r^{-1}\cdot)) + 2ar\partial_r\partial_{\varphi_*}(r^{-1}\cdot) + \not\Delta_{\mathbb{S}^2}(\cdot) \tag{18.4}$$
$$+ sr(4Mr - 2(1-\mathbb{h})\Delta_r)\partial_r(r^{-1}\cdot) - \lambda\pi_{\leq l} \quad \text{for} \quad r = \varrho \leq \frac{R_\infty}{2},$$
$$(\mathfrak{L}_{s,\lambda,2\kappa_c}f)_\ell = \partial_x(x(2\kappa_c + x)\partial_x f_\ell) + 2s\partial_x f_\ell - \mathfrak{l}(\mathfrak{l}+1)f_\ell \quad \text{for} \quad \varrho \geq R_\infty. \tag{18.5}$$

Furthermore:

(i) *There exists a constant $C_{s,\sigma,R_\infty} > 0$ such that:*

$$\|\mathfrak{L}^{-1}_{s,\lambda,2\kappa_c}\|_{H^{N_+}_{\sigma,R_\infty}\to H^{N_++1}_{\sigma,R_\infty,2}} \leq C_{s,\sigma,R_\infty}.$$

For all $f \in H^{N_+}_{\sigma,R_\infty}$ we have additionally that

$$\kappa_c\|x\partial_x^2\mathfrak{L}^{-1}_{s,\lambda,2\kappa_c}(f)\|_{G_{\sigma,R_\infty}} + \kappa_c\|\not\Delta_{\mathbb{S}^2}\mathfrak{L}^{-1}_{s,\lambda,\kappa}(f)\|_{G_{\sigma,R_\infty}} \leq C_{s,\sigma}\|f\|_{H^{N_+}_{\sigma,R_\infty}}.$$

(ii) *For $M^2\Lambda$ suitably small, $B_{s,\Lambda}\circ\mathfrak{L}^{-1}_{s,2\kappa_c,\lambda} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ and $\kappa_c B^{\mathcal{C}^+}_s\circ\mathfrak{L}^{-1}_{s,2\kappa_c,\lambda} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ are bounded operators and $K_{s,\lambda,\Lambda}\circ\mathfrak{L}^{-1}_{s,2\kappa_c,\lambda} : H^{N_+}_{\sigma,R_\infty} \to H^{N_+}_{\sigma,R_\infty}$ is compact.*

Proof. The statement in (i) follows from an analogue of Corollary 15.2. Due to the presence of the factor Δ_r instead of Δ in (18.4), we cannot directly apply Corollary 15.2. Instead we need to prove an analogue of Corollary 12.3 for Δ replaced with Δ_r. Since the surface gravity κ_+ of $\mathcal{H}^+$ is still positive, by assumption, and away from $r = r_+$ the difference $\Delta_r - \Delta$ can be made arbitrarily small in $\{r \leq \frac{1}{2}R_\infty\}$ for $M^2\Lambda$ suitably small, the red-shift estimates in Corollary 12.3 still hold.

The statement in (ii) then follows straightforwardly by repeating the proof of Proposition 16.1. The only difference is the presence of the operator $\kappa_c B^{\mathcal{C}^+}_s$, whose boundedness follows from the estimates in (i) that feature κ_c-dependent terms on the left-hand side. □

We then directly obtain the following corollary:

Corollary 18.3. *The conclusions of Propositions 16.2–16.4 hold also for $L_{s,2\kappa_c}$ replacing $L_{s,\kappa}$.*

19. Scattering resonances

In this section, we will establish meromorphicity of cut-off resolvents associated to the Boyer–Lindquist time coordinate t.

Let $R_\infty > 0$ be arbitrarily large. Let $\chi_\infty \in C^\infty(\Sigma)$, such that $\operatorname{supp}\chi_\infty \subseteq [r_+, r_{\max}]$ for some arbitrary $r_{\max} < \frac{1}{2}R_\infty$.

We first show that we can simplify the domain of L_s^{-1} by considering $\chi_\infty L_s^{-1}\chi_\infty$.

Corollary 19.1. *The cut-off operators $L_s^{-1}\chi_\infty$ and $\chi_\infty L_s^{-1}\chi_\infty$ satisfy the following properties:*

(i) *Let $N_+ \in \mathbb{N}_+$ and $\delta_0 > 0$. Then there exists a $\sigma_0(\delta_0) \in \mathbb{R}$ such that $L_s^{-1}\chi_\infty : H^{N_+}_{\mathbb{V}}(\Sigma) \to H^{N_++1}_{\sigma_0,R_\infty,2}$ defines a meromorphic family of operators in $\{|\arg z| < \varphi_0\} \cap \{\operatorname{Re} z > -(\frac{1}{2} + N_+)\kappa_+\} \cap \{|z| \geq \delta_0\}$, with poles that lie in $\mathcal{Q}_{\mathrm{reg}}$.*

(ii) *$\chi_\infty L_s^{-1}\chi_\infty : H^{N_+}_{\mathbb{V}}(\Sigma) \to H^{N_++1}_{\mathbb{V}}(\Sigma)$ defines a meromorphic family of operators for $s \in \{|\arg z| < \varphi_0\} \cap \{\operatorname{Re} s > -(\frac{1}{2} + N_+)\kappa_+\}$, with poles that lie in $\mathcal{Q}_{\mathrm{reg}}$.*

(iii) *$\chi_\infty L_s^{-1}\chi_\infty : C^\infty(\Sigma) \to C^\infty(\Sigma)$ is well-defined for all $s \in \Omega \setminus \mathcal{Q}_{\mathrm{reg}}$.*

Proof. Let $s \in \Omega \setminus \mathcal{Q}_{\mathrm{reg}}$. By Proposition 16.2, there exists an N_+ such that $\operatorname{Re} s > -(\frac{1}{2} + N_+)\kappa_+$, so $L_s^{-1} : H^N_{\sigma,R_\infty} \to H^{N+1}_{\sigma,R_\infty,2}$ is well-defined for all $N \geq N_+$ and for an appropriate choice of σ. Let $f \in H^{N_+}_{\mathbb{V}}(\Sigma)$. Then $\chi_\infty f \in H^{N_+}_{\sigma,R_\infty}$ for all $\sigma \in \mathbb{R}$ and $L_s^{-1}(\chi_\infty f) \in H^{N+1}_{\sigma,R_\infty,2}$ for all σ such that $s \in \Omega_\sigma$. Since $H^{N+1}_{\sigma,R_\infty,2} \subset H^{N+1}_{\sigma',R_\infty,2}$, for all $\sigma \geq \sigma'$, we obtain (i) by taking σ_0 to be the minimum of σ, such that Ω^2_σ cover the region $\{|\arg z| < \varphi_0\} \cap \{\operatorname{Re} z > -(\frac{1}{2} + N_+)\kappa_+\} \cap \{|z| \geq \delta_0\}$. The above implies moreover that $\chi_\infty L_s^{-1}(\chi_\infty f) \in H^{N_++1}_{\mathbb{V}}(\Sigma)$, so (ii) also follows.

In particular, if $f \in C^\infty(\Sigma)$, then we can take N_+ arbitrarily large in (ii) to conclude (iii). □

Now let $\chi_{\infty,+} \in C^\infty(\Sigma)$, such that $\operatorname{supp}\chi_{\infty,+} \subseteq [r_{\min}, r_{\max}]$, for some arbitrary $r_{\min} > r_+$. Then can moreover obtain further meromorphicity results in the full set Ω, in the case $\frac{|a|}{M} \ll 1$ or for bounded azimuthal frequencies m.

Proposition 19.2. *The operator $\chi_\infty L_s^{-1}\chi_{\infty,+} : \pi_{|m|\leq m_0}(L^2(\Sigma)) \to \pi_{|m|\leq m_0}(H^2(\Sigma))$ is well-defined for all $s \in \Omega \setminus \mathcal{Q}_{\mathrm{reg}}$ and defines a meromorphic family of operators on Ω with poles that lie in $\mathcal{Q}_{\mathrm{reg}}$. Furthermore, for $\frac{|a|}{|M|}$ suitably small, $\chi_\infty L_s^{-1}\chi_{\infty,+} : L^2(\Sigma) \to H^2(\Sigma)$ is well-defined and defines a meromorphic family of operators on Ω with poles that lie in $\mathcal{Q}_{\mathrm{reg}}$. In both cases, there exists a constant $C > 0$, with $C = C(a, r_{\min}, r_{\max}, R_\infty, s) > 0$ or $C = C(m_0, r_{\min}, r_{\max}, R_\infty, s) > 0$ such that*

$$\|\chi_\infty L_s^{-1}(\chi_{\infty,+} f)\|_{H^2(\Sigma)} \leq C_s \|f\|_{L^2(\Sigma)}.$$

Proof. For $\frac{|a|}{|M|}$ suitably small or when restricting to the span of azimuthal modes with $|m| \leq m_0$, we can omit the angular derivatives $|\nabla\!\!\!/_{\mathbb{S}^2}\partial_r^k \mathfrak{L}_{s,\lambda,\kappa}\widehat{\psi}|^2$ on the right-hand side of (12.10) by applying Proposition 12.4 to obtain the following improved version of Corollary 19.1(i): for $s \notin \mathcal{Q}_{\mathrm{reg}}$

$$\|\chi_\infty L_s^{-1}(\chi_{\infty,+} f)\|_{H^{N+1}(\Sigma)} + \left\|\left(1 - \frac{r_+}{r}\right)\chi_\infty L_s^{-1}(\chi_{\infty,+} f)\right\|_{H^{N+2}(\Sigma)} \leq C_s \|f\|_{H^N(\Sigma)}. \tag{19.1}$$

It is straightforward to improve (19.1) in the region $r \leq r_{\min}$ by applying the estimates in the proof of Corollary 12.3, but restricted to $r \in [r_+, r_{\min}]$. It then follows that

$$\|\chi_\infty L_s^{-1}(\chi_{\infty,+} f)\|_{H^{N_++2}(\Sigma)} \leq C\|f\|_{H^N_+(\Sigma)} + C\|\partial_r^{N_++1}(\chi_{\infty,+} f)\|_{L^2(\Sigma\cap\{r\leq r_{\min}\})} \leq C\|f\|_{H^{N_+}(\Sigma)}, \tag{19.2}$$

where the final inequality follows from the fact that $\chi_{\infty,+} f = 0$ in $\{r \leq r_{\min}\}$.

Suppose that $s \in \Omega$ satisfies $\operatorname{Re} s > -\frac{1}{2}\kappa_+$, then the proposition holds. Suppose now instead that $-\frac{3}{2}\kappa_+ < \operatorname{Re} s \leq -\frac{1}{2}\kappa_+$ and let $s' \in \Omega$ be such that $\operatorname{Re} s' > -\frac{1}{2}\kappa_+$.

Let $f \in C^\infty(\Sigma)$ and define $\widehat{\psi} = L_s^{-1}(\chi_{\infty,+} f)$. We can express:

$$L_s^{-1} = L_s^{-1} L_{s'} L_{s'}^{-1} = L_s^{-1}(L_s + (L_{s'} - L_s))L_{s'}^{-1} = L_{s'}^{-1} + L_s^{-1}(L_{s'} - L_s)L_{s'}^{-1}.$$

We can now estimate

$$\begin{aligned}
\|\chi_\infty \widehat{\psi}\|_{H^2(\Sigma)} &= \|\chi_\infty L_s^{-1}(\chi_{\infty,+} f)\|_{H^2(\Sigma)} \\
&\leq \|\chi_\infty L_{s'}^{-1}(\chi_{\infty,+} f)\|_{H^2(\Sigma)} + \|\chi_\infty L_s^{-1}(L_{s'} - L_s)L_{s'}^{-1}(\chi_{\infty,+} f)\|_{H^2(\Sigma)} \\
&\leq C_{s',r_{\min}} \|f\|_{L^2(\Sigma)} + C_s \|\chi_\infty (L_{s'} - L_s) L_{s'}^{-1}(\chi_{\infty,+} f)\|\|_{H^1(\Sigma)} \\
&\leq C_{s',r_{\min}} \|f\|_{L^2(\Sigma)} + C_s |s - s'| \|\chi_\infty L_{s'}^{-1}(\chi_{\infty,+} f)\|_{H^2(\Sigma)} \\
&\leq C_{s',r_{\min}} (1 + C_s |s - s'|) \|f\|_{L^2(\Sigma)},
\end{aligned}$$

where we applied (19.2) with $N = 0$ and Corollary 19.1(i) with $N_+ = 1$ to arrive at the second line, and then we applied (19.2) with $N_+ = 0$ once again to arrive at the last line. By a density argument, we therefore obtain $\|\chi_\infty \widehat{\psi}\|_{H^2(\Sigma)} \leq C \|f\|_{L^2(\Sigma)}$ for all $f \in L^2(\Sigma)$.

We can repeat this procedure to conclude that the desired inequality holds for all $s \in \Omega$. □

We consider now the differential operator

$$A(\omega) := e^{i\omega t} \Box_{g_{M,a}} (e^{-i\omega t}(\cdot)) : C^\infty(\mathbb{R}_r \times \mathbb{S}^2_{(\theta,\varphi)}) \to C^\infty(\mathbb{R}_r \times \mathbb{S}^2_{(\theta,\varphi)}).$$

We will construct the corresponding resolvent operator $R(\omega) = A(\omega)^{-1}$ and the associated cut-off resolvent operator $\chi_{\infty,+} R(\omega) \chi_{\infty,+}$ by relating $A(\omega)$ to L_s. Denote $\Sigma' := \{t = 0\}$.

Let $Q_s : L^2(\Sigma') \to L^2(\Sigma \setminus \{r = r_+\})$ be defined as follows: let $\overline{f} : D^+(\Sigma') \to \mathbb{C}$ denote the trivial extension of $f \in L^2(\Sigma')$ to $D^+(\Sigma')$, i.e. $\overline{f}(t, r, \theta, \varphi_*) = \overline{f}(r, \theta, \varphi_*)$. Then

$$Q_s(f)(r, \theta, \varphi_*) = (e^{st} \overline{f})|_\Sigma (r, \theta, \varphi^*).$$

Furthermore, $Q_s \chi : L^2(\Sigma') \to L^2(\Sigma)$ is well-defined and bounded. We consider also the inverse $Q_s^{-1} : L^2(\Sigma) \to L^2(\Sigma')$, to obtain the bounded operator:

$$\chi_{\infty,+} Q_s^{-1} : H^2(\Sigma) \to H^2(\Sigma').$$

Let $s = -i\omega$, then we can express

$$A(\omega) = Q_s^{-1} \circ (\varrho^{-3} L_s(\varrho \cdot)) \circ Q_s.$$

so that

$$\chi_{\infty,+} R(\omega) \chi_{\infty,+} = (\chi_{\infty,+} Q_s^{-1}) \circ (\varrho^{-1} L_s(\varrho^3 \cdot))^{-1} \circ (Q_s \chi_{\infty,+}).$$

Corollary 19.3. *The cut-off resolvent operator:*

$$\chi_{\infty,+} R(\omega) \chi_{\infty,+} : H^{N_+}_{\mathbb{V}}(\Sigma') \to H^{N_+ + 1}_{\mathbb{V}}(\Sigma')$$

can be meromorphically continued to $\omega \in (i\Omega \setminus i\mathcal{Q}_{\mathrm{reg}}) \cap \{\mathrm{Im}\,\omega > -(\frac{1}{2} + N_+)\kappa_+\}$ *and* $\omega \mapsto \chi_{\infty,+} R(\omega) \chi_{\infty,+}$ *is meromorphic on* $i\Omega \cap \{\mathrm{Im}\,\omega > -(\frac{1}{2} + N_+)\kappa_+\}$. *Let* $i\mathcal{Q}_{\mathrm{res}}$ *denote the set of poles of* $\omega \mapsto \chi_{\infty,+} R(\omega) \chi_{\infty,+}$ *in* $i\Omega$. *Then*

$$\mathcal{Q}_{\mathrm{res}} \subseteq \mathcal{Q}_{\mathrm{reg}}.$$

Furthermore, the above results hold also with the following domain changes:

$$\chi_{\infty,+} R(\omega) \chi_{\infty,+} : \pi_{|m| \leq m_0}(L^2(\Sigma')) \to \pi_{|m| \leq m_0}(H^2(\Sigma')) \quad \text{for any } m_0 \in [0, \infty),$$

$$\chi_{\infty,+} R(\omega) \chi_{\infty,+} : L^2(\Sigma') \to H^2(\Sigma') \quad \text{for suitably small } \frac{|a|}{M}.$$

Proof. By taking $s = -i\omega$ and and using that $r = \varrho$ in $\{r \leq \frac{1}{2}R_\infty\}$, we obtain

$$\chi_{\infty,+}R(\omega)\chi_{\infty,+} = (\chi_{\infty,+}Q_s^{-1}) \cdot r^{-1}L_s^{-1} \cdot (r^3 Q_s \chi_{\infty,+}).$$

We conclude by applying Corollary 19.1 and Proposition 19.2 and using boundedness of $\chi_{\infty,+}Q_s^{-1}$ and $Q_s\chi_{\infty,+}$. □

Appendix A. Proof of Lemma 13.2

By (13.4), we can take $\nu = \frac{|\operatorname{Im} s|}{|s|}\mu$ in the remaining inequalities without loss of generality. First, consider $\operatorname{Re} s \geq 0$. Then (13.6) and (13.8) can be satisfied if:

$$\frac{1}{\frac{4}{\frac{|\operatorname{Im} s|}{|s|}(1-\mu)}\alpha - \alpha^2} < \tilde{\sigma}^2 < 1 - \frac{|\operatorname{Im} s|^2}{|s|^2}\mu^2.$$

We can minimize the term on the very left-hand side by taking $\alpha = \frac{1}{2}\frac{4-\beta}{\frac{|\operatorname{Im} s|}{|s|}(1-\mu)}$ to obtain:

$$\frac{\frac{|\operatorname{Im} s|^2}{|s|^2}(1-\mu)^2}{4} < \tilde{\sigma}^2 < 1 - \frac{|\operatorname{Im} s|^2}{|s|^2}\mu^2. \tag{A.1}$$

There exists a $\tilde{\sigma}$ so that (A.1) is satisfied if and only if:

$$\left[\frac{(1-\mu)^2}{4} + \mu^2\right]\frac{|\operatorname{Im} s|^2}{|s|^2} < 1.$$

Taking the derivative leads to:

$$\frac{d}{d\mu}\left[\frac{(1-\mu)^2}{4} + \mu^2\right] = \frac{1}{2}(5\mu - 1),$$

so the term in square brackets attains a minimum at $\mu = \frac{1}{5}$ and we obtain

$$\frac{|\operatorname{Im} s|^2}{|s|^2} < 5,$$

which is trivially satisfied. This means that for all $s \in \mathbb{C}$ with $\operatorname{Re} s \geq 0$, we can find $\alpha, \mu, \beta, \tilde{\sigma}$ so that the relevant compatibility conditions are satisfied. Now fix $\sigma = |2s|\tilde{\sigma}$. Write $X = \sigma^{-1}|\operatorname{Re}(2s)|$ and $Y = \sigma^{-1}|\operatorname{Im}(2s)|$. Then (A.1) is equivalent to:

$$\frac{1}{2}Y(1-\mu) < 1,$$
$$X^2 + (1-\mu^2)Y^2 > 1.$$

Note that the left-hand side of the second inequality is decreasing in μ, so we would like to consider the smallest allowed μ so that the first inequality is still satisfied. We take $\mu \downarrow \mu_{\min} := \max\{0, 1-\frac{2}{Y}\}$. Then we fill this into the second equation to obtain:

$$1 < X^2 + (1-\mu_{\min}^2)Y^2.$$

Suppose that $\mu_{\min} = 0$, then we need $X^2 + Y^2 > 1$. Suppose that $\mu_{\min} = 1 - \frac{2}{Y}$. Then $Y \geq 2$ and

$$1 < X^2 + (1-\mu_{\min}^2)Y^2 = X^2 - 4 + 4Y.$$

Note that $X^2 - 4 + 4Y \geq X^2 + 5 > 1$, so the required condition is automatically satisfied. We conclude that we can find $\alpha, \mu, \beta, \tilde{\sigma}$ so that the relevant compatibility conditions are satisfied if $X^2 + Y^2 < 1$, or equivalently $|s|^2 > \frac{1}{4}\sigma^2$.

In the case $\operatorname{Re} s < 0$, (13.5) and (13.7) imply that:

$$\frac{1}{\frac{4-\beta}{\frac{|\operatorname{Im} s|}{|s|}(1-\mu)}\alpha - \alpha^2} < \tilde{\sigma}^2 < 1 - \frac{|\operatorname{Im} s|^2}{|s|^2}\mu^2 - 4\beta^{-1}\left(\frac{|\operatorname{Re} s|}{|s|}\right)^2.$$

We can minimize the term on the very left-hand side by taking $\alpha = \frac{1}{2}\frac{4-\beta}{\frac{|\operatorname{Im} s|}{|s|}(1-\mu)}$, so that the above equation becomes:

$$\frac{4\frac{|\operatorname{Im} s|^2}{|s|^2}(1-\mu)^2}{(4-\beta)^2} < \tilde{\sigma}^2 < 1 - 4\beta^{-1} + (4\beta^{-1} - \mu^2)\frac{|\operatorname{Im} s|^2}{|s|^2}. \tag{A.2}$$

Hence, we can find a $\tilde{\sigma}$ satisfying the above inequality if and only if

$$\left[1 + \frac{\beta(1-\mu)}{4-\beta}\left((1+\mu) - \frac{4(1-\mu)}{(4-\beta)^2}\right)\right]\frac{|\operatorname{Im} s|^2}{|s|^2} > 1.$$

We will first pick $\mu \in [0,1]$ so that the factor in square brackets above is maximized. Note the factor is equal to 1 for $\mu = 1$. Furthermore, a maximum is attained in $(0,1)$ when

$$\frac{d}{d\mu}\left(1 + \frac{\beta(1-\mu)}{4-\beta}\left((1+\mu) - \frac{4(1-\mu)}{(4-\beta)^2}\right)\right) = \frac{2\beta[4 - \mu(20 - \beta(8-\beta))]}{(4-\beta)^3}$$

vanishes, which is the case for

$$\mu = \frac{4}{4 + (4-\beta)^2}.$$

Then

$$\left[1 + \frac{\beta(1-\mu)}{4-\beta}\left((1+\mu) - \frac{4(1-\mu)}{(4-\beta)^2}\right)\right] = 1 + \frac{\beta(4-\beta)}{20 - \beta(8-\beta)} > 1.$$

We will now maximize the right-hand side of the above identity in β, noting that the minima are attained at $\beta = 0$ and $\beta = 4$. We have that

$$\frac{d}{d\beta}\left(\frac{\beta(4-\beta)}{20 - \beta(8-\beta)}\right) = \frac{80 - 4(10-\beta)\beta}{(20 - (8-\beta)\beta)^2},$$

which vanishes for $\beta = 5 - \sqrt{5}$. In that case, we conclude:

$$\left[1 + \frac{\beta(1-\mu)}{4-\beta}\left((1+\mu) - \frac{4(1-\mu)}{(4-\beta)^2}\right)\right] = \frac{1}{2}(1+\sqrt{5}),$$

which is the golden ratio.

Hence,

$$\left(\frac{|\operatorname{Im} s|}{|s|}\right)^2 > \frac{2}{1+\sqrt{5}}.$$

Note that this corresponds to the sector $|\arg z| < \pi - \arcsin\left(\frac{2}{1+\sqrt{5}}\right)$.

Now fix σ. Then as in the $\operatorname{Re} s > 0$ case let $X = \sigma^{-1}|\operatorname{Re}(2s)|$ and $Y = \sigma^{-1}|\operatorname{Im}(2s)|$ and write $b = 4 - \beta$. Then (A.2) is equivalent to:

$$\frac{2}{b}Y(1-\mu) < 1,$$
$$-\frac{b}{4-b}X^2 + (1-\mu^2)Y^2 > 1.$$

As above, the left-hand side of the second equation increases as μ decreases, so we need to minimize μ while keeping the first equation valid. Analogously to above, we defined $\mu_{\min} = \max\{0, 1 - \frac{b}{2Y}\}$.

Suppose first that $\mu_{\min} = 1 - \frac{b}{2Y}$. Then $Y \leq 2b^{-1}$ and we need

$$-\frac{b^2}{4} + bY - \frac{b}{4-b}X^2 > 1.$$

Suppose now that $\mu_{\min} = 0$. Then $Y \geq \frac{b}{2}$ and we need:

$$Y^2 - \frac{b}{4-b}X^2 > 1.$$

Since the left-hand side is decreasing as b increases, the left-hand side is maximized when $b = 2Y$ (the smallest choice in this case). But in that case $\mu_{\min} = 1 - \frac{b}{2Y}$, so the previous case applies. We may therefore assume without loss of generality restrict to $\mu_{\min} = 1 - \frac{b}{2Y}$.

To obtain the largest (X, Y)-region, we therefore need to consider:

$$\sup_{b\in(0,4)} bY - \frac{b}{4-b}X^2 - \frac{b^2}{4}.$$

Let $G(b, X, Y) = bY - \frac{b}{4-b}X^2 - \frac{b^2}{4} - 1$. Then the boundary of the relevant (X, Y)-region is the level set $G(b, X, Y) = 0$. The boundary of the largest region is obtained by considering the envelope of these level sets, which corresponds to $G(b, X, Y) = 0$, $\partial_b G(b, X, Y) = 0$. Note that:

$$0 = G(b, X, Y) = bY - \frac{b}{4-q}X^2 - \frac{b^2}{4} - 1,$$
$$0 = \partial_b G(b, X, Y) = Y - \frac{4}{(4-b)^2}X^2 - \frac{1}{2}b.$$

Assuming $0 < b \leq 2$ and solving for (X, Y) then results in:

$$X(b) = \frac{4-b}{2b}\sqrt{4-b^2},\ Y(b) = -1 + 4b^{-2} + \frac{b}{2}.$$

After rescaling we obtain the following region: write $s = x + iy$, then we can find $\alpha, \beta, \tilde{\mu}, \nu$ such that (13.4), (13.5), (13.7) are satisfied if (x, y) lie in the set:

$$\left\{(x,y) \in \mathbb{R}^2 \,\middle|\, -\frac{4-b}{2b}\sqrt{4-b^2}\frac{1}{2}|\sigma| < x < 0,\ y \notin \left[\frac{1}{2}(-1+4b^{-2}+\frac{b}{2})|\sigma|, \frac{1}{2}(-1+4b^{-2}+\frac{b}{2})|\sigma|\right],\ b \in (0,2)\right\}.$$

This finishes the proof of Lemma 13.2.

Appendix B. Compact embeddings

The following Hilbert spaces are compactly embedded:

Proposition B.1. *Let $R_\infty \geq 3r_+$ and $N_+ \in \mathbb{N}_0$. Then*

$$H^{N_++1}_{\sigma_2,R_\infty} \subset\subset H^{N_+}_{\sigma_1,R_\infty} \quad \text{for all } 0 > |\sigma_2| > |\sigma_1|, \tag{B.1}$$
$$H^{N_++2}_{\sigma_2,R_\infty,2} \subset\subset H^{N_++1}_{\sigma_1,R_\infty,2} \quad \text{for all } 0 > |\sigma_2| > |\sigma_1|, \tag{B.2}$$
$$H^{N_++1}_{\sigma,R_\infty,1} \subset\subset H^{N_+}_{\sigma,R_\infty} \quad \text{for all } \sigma \in \mathbb{R}. \tag{B.3}$$

Proof. We first consider (B.3). Let $\{f_k\}$ be a sequence in $H^{N_++1}_{\sigma,R_\infty,1}$ satisfying $\|f_k\|_{H^{N_++1}_{\sigma,R_\infty,1}} = 1$. To conclude (B.3), we need to show that it admits a Cauchy subsequence with respect to the $H^{N_+}_{\sigma,R_\infty}$-norm.

We consider first the B_{R_∞}-part of the norm:

$$\|f_k\|^2_{B_{R_\infty}} = \sum_{\ell\in\mathbb{N}_0}\sum_{n=0}^{\infty} \frac{(d_0x_0)^{2n}}{(\max\{\ell+1, n+1\})^{2n}} \|\|\partial_x^n(f_k)_\ell\|^2_{L^2(\mathbb{S}^2)}(x_0)$$

$$= \sum_{\ell\in\mathbb{N}_0} \|(f_k)_\ell\|^2_{L^2(\mathbb{S}^2)}(x_0) + \sum_{\ell\in\mathbb{N}_0}\sum_{n=0}^{\infty} \frac{1}{(\max\{\ell+1,n+1\})^2}\frac{(d_0x_0)^{2n}}{(\max\{\ell+1,n+1\})^{2n}}\||\partial_x^{n+1}(f_k)_\ell\|^2_{L^2(\mathbb{S}^2)}(x_0)$$

$$= \sum_{\ell\in\mathbb{N}_0}\sum_{|m|\leq\ell} |(f_k)_{\ell m}|^2(x_0)$$

$$+ \sum_{\ell\in\mathbb{N}_0}\sum_{|m|\leq\ell}\sum_{n=0}^{\infty} \frac{1}{(\max\{\ell+1,n+1\})^2}\frac{(d_0x_0)^{2n}}{(\max\{\ell+1,n+1\})^{2n}}|\partial_x^{n+1}(f_k)_{\ell m}|^2(x_0)$$

Since $\|f_k\|^2_{B_{R_\infty}} \leq 1$, we have by Bolzano–Weierstrass that there exists a subsequence f_{k_l} such that for fixed $L > 0$

$$\sum_{\ell=0}^{L}\sum_{|m|\leq\ell} |(f_{k_l})_{\ell m}|^2(x_0) \to a_{L,N},$$

$$\sum_{\ell=0}^{L}\sum_{|m|\leq\ell}\sum_{n=0}^{N} \frac{1}{(\max\{\ell+1,n+1\})^2}\frac{(d_0x_0)^{2n}}{(\max\{\ell+1,n+1\})^{2n}}|\partial_x^{n+1}(f_{k_l})_{\ell m}|^2(x_0) \to b_{L,N}$$

for fixed L, N. We moreover have that the "high-frequency part" of the norm of the difference $f_k - f_r$ can be estimated as follows:

$$\sum_{\ell\geq L} \|(f_k - f_r)_\ell\|^2_{L^2(\mathbb{S}^2)}(x_0) \leq \frac{1}{L(L+1)}\sum_{\ell\geq L}\|(\not\nabla_{\mathbb{S}^2}(f_k - f_r))_\ell\|^2_{L^2(\mathbb{S}^2)}(x_0) \leq \frac{C}{L(L+1)},$$

$$\sum_{\ell\in\mathbb{N}_0, n\in\mathbb{N}_0, \ell\geq L \text{ or } n\geq N} \frac{1}{(\max\{\ell+1,n+1\})^2}\frac{(d_0x_0)^{2n}}{(\max\{\ell+1,n+1\})^{2n}}\||\partial_x^{n+1}(f_k - f_r)_\ell\|^2_{L^2(\mathbb{S}^2)}(x_0)$$

$$\leq \frac{1}{(\min\{N,L\}+1)^2}(\|\not\nabla_{\mathbb{S}^2}(f_k - f_r)\|^2_{B_{R_\infty}} + \|\partial_x(f_k - f_r)\|^2_{B_{R_\infty}}) \leq \frac{C}{(\min\{N,L\}+1)^2}$$

Hence, $\{f_{k_l}\}$ is Cauchy with respect to $\|\cdot\|_{B_{R_\infty}}$.

Consider now the G_{σ,R_∞}-part of the norm:

$$\|f_k\|^2_{G_{\sigma,R_\infty}} = \sum_{\ell\in\mathbb{N}_0}\sum_{n=0}^{\max\{\ell-1,0\}} \frac{(4\sigma)^{2n}(\ell-(n+1))!^2}{(\ell+n+1)!^2}\int_0^{x_0}\|\partial_x^n f_\ell\|_{L^2(\mathbb{S}^2)}(x)\,dx$$

$$+ (\ell+1)^3\sum_{n=\ell}^{\infty}\frac{\sigma^{2n}}{n!^2(n+1)!^2}\int_0^{x_0}\|\partial_x^n f_\ell\|_{L^2(\mathbb{S}^2)}(x)\,dx$$

$$= \sum_{\ell\in\mathbb{N}_0}\sum_{|m|\leq\ell}\sum_{n=0}^{\max\{\ell-1,0\}} \frac{(4\sigma)^{2n}(\ell-(n+1))!^2}{(\ell+n+1)!^2}\int_0^{x_0}|\partial_x^n f_{m\ell}|^2(x)\,dx$$

$$+ (\ell+1)^3\sum_{n=\ell}^{\infty}\frac{\sigma^{2n}}{n!^2(n+1)!^2}\int_0^{x_0}|\partial_x^n f_{m\ell}|^2(x)\,dx.$$

Now we can proceed as above: we first split the sum into a part where $\ell \leq L$ and $n \leq N$ and, rather than applying Bolzano–Weierstrass, we apply Arzelà–Ascoli for functions on $[0, x_0]$ to obtain a convergent subsequence. Here we moreover use that we can consider the $C^N([0, x_0])$ norm rather than the $H^N([0, x_0])$ norm via a Sobolev inequality on $[0, x_0]$, since the norm $\|f_k\|_{G_{\sigma,R_\infty}}$ bounds arbitrarily many x-derivatives.

Subsequently, we bound the remaining high-frequency part part of $\|f_k - f_r\|^2_{G_{\sigma,R_\infty}}$, where $n > N$ or $\ell \geq L$, with a constant $\frac{1}{(\min\{N,L\}+1)^2}$ by using the boundedness of $\|f_k - f_r\|_{G_{\sigma,R_\infty}} + \|\nabla\!\!\!/_{\mathbb{S}^2}(f_k - f_r)\|_{G_{\sigma,R_\infty}} + \|\partial_x(f_k - f_r)\|_{G_{\sigma,R_\infty}}$.

To conclude the proof of (B.3), we consider $\|f\|^2_{H^{N_+}(\Sigma\cap\{\varrho\leq 2R_\infty\})} + \|\nabla\!\!\!/_{\mathbb{S}^2} f\|^2_{H^{N_+}(\Sigma\cap\{\varrho\leq \frac{1}{3}R_\infty\})} + \sum_{\ell\in\mathbb{N}_0} |(\ell(\ell+1)+1)^{-\frac{w_p}{4}} f_\ell\|^2_{H^{N_+}(\Sigma\cap\{\frac{1}{3}R_\infty\leq\varrho\leq 3R_\infty\})}$. We proceed as before, but rather than applying Arzelà–Ascoli to estimate the bounded angular frequencies $\ell \leq L$, we apply Rellich–Kondrachov on $[r_+^{-1}, \frac{1}{R_\infty}]$.

We can prove (B.1) and (B.2) in an analogous manner, where we use obtain an smallness factor in the high-frequency estimates by using that $(4\sigma_1)^{2n} = (\frac{\sigma_2}{\sigma_1})^{-2n}(4\sigma_2)^{2n}$. □

Appendix C. Norms of pointwise products

In this section, we consider norms of products of functions: $\|g \cdot f\|_{H^{N_+}_{\sigma,R_\infty}}$, where g will take the role of a coefficient in a differential operator and f will take the role of a function in the domain of that operator. Since the $H^{N_+}_{\sigma,R_\infty}$-norm features an infinite number of derivatives, one cannot immediately estimate $\|g \cdot f\|_{H^{N_+}_{\sigma,R_\infty}}$ by a product of $\|f\|_{H^{N_+}_{\sigma,R_\infty}}$ with an appropriate norm for g.

In this section, we will prove:

Proposition C.1. *Let $\beta > 2$. Then there exists a constant $C > 0$ such that for all appropriately regular functions $f, g; [0, x_0] \times \mathbb{S}^2 \to \mathbb{C}$, we can estimate:*

$$\|g \cdot f\|_{H^{N_+}_{\sigma,R_\infty}} \leq C\|g\|_{H^{N_+}_{\sigma,R_\infty;\mathrm{coeff}}} \cdot \|f\|_{H^{N_+}_{\sigma,R_\infty}},$$

with

$$\begin{aligned}\|g\|^2_{H^{N_+}_{\sigma,R_\infty;\mathrm{coeff}}} &:= \|g\|^2_{H^{N_++2}(\Sigma\cap\{\varrho\leq 3R_\infty\})} + \|\nabla\!\!\!/_{\mathbb{S}^2} g\|^2_{H^{N_++2}(\Sigma\cap\{\varrho\leq\frac{1}{3}R_\infty\})}\\ &+\sum_{\ell\in\mathbb{N}_0}\|(1+\Delta\!\!\!/_{\mathbb{S}^2})^{\frac{w_p+5}{4}} g_\ell\|^2_{H^{N_++2}(\Sigma\cap\{3R_\infty\leq\varrho\leq 4R_\infty\})}+\sum_{j=0}^{\infty}\sum_{\ell\in\mathbb{N}_0} e^{2\ell}(\ell+1)^5\frac{(d_0x_0)^{2j}(j+1)^2}{j!^2}\|\partial_x^j g_\ell|_{x=x_0}\|^2_{L^2(\mathbb{S}^2)}\\ &+\sum_{j\in\mathbb{N}_0}\sum_{\ell\in\mathbb{N}_0}\frac{e^{5(\ell+j)}}{j!^4}\|\partial_x^j g_\ell\|^2_{L^\infty([0,x_0]\times\mathbb{S}^2)}.\end{aligned}$$

The proof follows from combining Lemmas C.2 and C.3 below, which each deal with a different part of the norm $\|g \cdot f\|_{H^{N_+}_{\sigma,R_\infty}}$.

Lemma C.2. *Let $\beta > 2$. Then there exists a constant $C > 0$ such that for all appropriately regular functions $f, g; [0, x_0] \times \mathbb{S}^2 \to \mathbb{C}$, we can estimate:*

$$\|gf\|^2_{B_{R_\infty}} \leq C\|f\|^2_{B_{R_\infty}}\left[\sum_{j=0}^{\infty}\sum_{\ell\in\mathbb{N}_0} e^{2\ell}(\ell+1)^5\frac{(d_0x_0)^{2j}(j+1)^2}{j!^2}\|\partial_x^j g_\ell|_{x=x_0}\|^2_{L^2(\mathbb{S}^2)}\right]. \tag{C.1}$$

Furthermore,

$$\begin{aligned}&\|gf\|^2_{H^{N_+}(\Sigma\cap\{r\leq 3R_\infty\})} + \|\nabla\!\!\!/_{\mathbb{S}^2}(gf)\|^2_{H^{N_+}(\Sigma\cap\{r\leq\frac{1}{3}R_\infty\})} + \|(-\Delta\!\!\!/_{\mathbb{S}^2}-\mathbf{1})^{-\frac{w_p}{4}}(gf)_\ell\|^2_{H^{N_+}(\Sigma\cap\{3R_\infty\leq r\leq 4R_\infty\})}\\ &\leq \|f\|^2_{H^{N_+}(\Sigma\cap\{r\leq 3R_\infty\})}\|g\|^2_{H^{N_++2}(\Sigma\cap\{r\leq 3R_\infty\})} + \|\nabla\!\!\!/_{\mathbb{S}^2} f\|^2_{H^{N_+}(\Sigma\cap\{r\leq\frac{1}{3}R_\infty\})}\|\nabla\!\!\!/_{\mathbb{S}^2} g\|^2_{H^{N_++2}(\Sigma\cap\{r\leq\frac{1}{3}R_\infty\})}\\ &\quad + \|(\mathbf{1}-\Delta\!\!\!/_{\mathbb{S}^2})^{-\frac{w_p}{4}} f\|^2_{H^{N_+}(\Sigma\cap\{3R_\infty\leq r\leq 4R_\infty\})}\|(\mathbf{1}-\Delta\!\!\!/_{\mathbb{S}^2})^{\frac{w_p+5}{4}} g\|^2_{H^{N_+}(\Sigma\cap\{3R_\infty\leq r\leq 4R_\infty\})}.\end{aligned} \tag{C.2}$$

Proof. Consider the functions $g, f : [r_+, \infty)_\varrho \times \mathbb{S}^2_{\vartheta,\varphi*} \to \mathbb{C}$. We denote with $g^{(n)} = \partial_x^n g$ and $f^{(n)} = \partial_x^n f$. We can expand the product gf in spherical harmonics:

$$(gf)_{\ell m}(x) = \sum_{\tilde\ell\in\mathbb{N}_0} \sum_{\tilde m=-\tilde\ell}^{\tilde\ell} \sum_{\ell'\in\mathbb{N}_0} \sum_{m'=-\ell'}^{m=\ell'} g_{\ell' m'}(x) f_{\tilde\ell\tilde m}(x) \langle Y_{\ell' m'} Y_{\tilde\ell\tilde m}, Y_{\ell m}\rangle_{L^2(\mathbb{S}^2)}.$$

Since $\langle Y_{\ell' m'} Y_{\tilde\ell\tilde m}, Y_{\ell m}\rangle_{L^2(\mathbb{S}^2)} = 0$ if $m \neq m' + \tilde m$, we can write

$$\int_{\mathbb{S}^2} |(gf)_\ell|^2 \, d\sigma = \sum_{\tilde\ell\in\mathbb{N}_0} \sum_{\tilde m=-\tilde\ell}^{\tilde\ell} \sum_{\ell'\in\mathbb{N}_0} \sum_{m'=-\ell'}^{m=\ell'} |g_{\ell' m'}(x)|^2 |f_{\tilde\ell\tilde m}(x)|^2 |\langle Y_{\ell' m'} Y_{\tilde\ell\tilde m}, Y_{\ell(m'+\tilde m)}\rangle_{L^2(\mathbb{S}^2)}|^2.$$

By applying Cauchy–Schwarz and using that spherical harmonics have unit norm on $\mathbb{S}^2$, we can further estimate:

$$\begin{aligned}|\langle Y_{\ell' m'} Y_{\tilde\ell\tilde m}, Y_{\ell(m'+\tilde m)}\rangle_{L^2(\mathbb{S}^2)}|^2 &\leq \|Y_{\ell' m'} Y_{\tilde\ell\tilde m}\|^2_{L^2(\mathbb{S}^2)} \|Y_{\ell(m'+\tilde m)}\|^2_{L^2(\mathbb{S}^2)} \\ &\leq \|Y_{\ell' m'}\|^2_{L^\infty(\mathbb{S}^2)} \|Y_{\tilde\ell\tilde m}\|^2_{L^2(\mathbb{S}^2)} \|Y_{\ell(m'+\tilde m)}\|^2_{L^2(\mathbb{S}^2)} \leq \|Y_{\ell' m'}\|^2_{L^\infty(\mathbb{S}^2)}.\end{aligned}$$

By a standard Sobolev inequality on $\mathbb{S}^2$, we can further estimate:

$$\|Y_{\ell' m'}\|^2_{L^\infty(\mathbb{S}^2)} \leq C \|Y_{\ell' m'}\|^2_{H^2(\mathbb{S}^2)} \leq C(\ell'+1)^4.$$

Furthermore, from the vanishing properties of Clebsch–Gordan coefficients (see for example [Sak94][§3.7]) $\langle Y_{\ell' m'} Y_{\tilde\ell\tilde m}, Y_{\ell(m'+\tilde m)}\rangle_{L^2(\mathbb{S}^2)} = 0$ if $\ell' + \tilde\ell < \ell$ or $|\tilde\ell - \ell'| > \ell$. Finally, observe that by Cauchy–Schwarz and $\sum_{j=1} j^{-2} = \frac{\pi^2}{6}$:

$$\begin{aligned}|(gf)^{(n)}_\ell|^2 = \left|\sum_{k=0}^n \binom{n}{k} (g^{(n-k)} f^{(k)})_\ell\right|^2 &\leq \sum_{k=0}^n (n-k+1)^{-2} \sum_{k=0}^n \binom{n}{k}^2 (n-k+1)^2 |(g^{(n-k)} f^{(k)})_\ell|^2 \\ &\leq \frac{\pi^2}{6} \sum_{k=0}^n \binom{n}{k}^2 (n-k+1)^2 |(g^{(n-k)} f^{(k)})_\ell|^2.\end{aligned}$$

Combining the above observations, we obtain:

$$\begin{aligned}&\sum_{\ell\in\mathbb{N}_0} \sum_{n=0}^\infty \frac{(d_0 x_0)^{2n}}{(\max\{\ell+1, n+1\})^{2n}} \int_{\mathbb{S}^2} |(gf)^{(n)}_\ell|^2 \, d\sigma \\ &\qquad \leq C \sum_{\ell\in\mathbb{N}_0} \sum_{\{\ell',\tilde\ell\in\mathbb{N}_0 \,:\, |\tilde\ell-\ell'|\leq\ell\leq\ell+\ell'\}} \sum_{n=0}^\infty \frac{(d_0 x_0)^{2n}}{(\max\{\ell+1, n+1\})^{2n}} \\ &\qquad\qquad \times \sum_{k=0}^n \binom{n}{k}^2 |f^{(k)}_{\tilde\ell\tilde m}|^2 (\ell'+1)^4 (n-k+1)^2 |g^{(n-k)}_{\ell' m'}|^2.\end{aligned}$$

We introduce the summation index $j = n - k$, we rearrange the summation and separate the cases $\tilde\ell \leq \ell$ and $\tilde\ell > \ell$ to obtain:

$$\begin{aligned}&\sum_{n=0}^\infty \sum_{\ell\in\mathbb{N}_0} \frac{(d_0 x_0)^{2n}}{(\max\{\ell+1, n+1\})^{2n}} \int_{\mathbb{S}^2} |(gf)^{(n)}_\ell|^2 \, d\sigma \\ &\qquad \leq C \sum_{k=0}^\infty \sum_{\tilde\ell\in\mathbb{N}_0} \sum_{\tilde m=-\tilde\ell}^{\tilde\ell} \frac{(d_0 x_0)^{2k}}{(\max\{\tilde\ell+1, k+1\})^{2k}} |f^{(k)}_{\tilde\ell\tilde m}|^2\end{aligned}$$

$$\times \Bigg[\sum_{\ell' \in \mathbb{N}_0} \sum_{m'=-\ell'}^{\ell'} \sum_{\ell=|\tilde{\ell}-\ell'|}^{\tilde{\ell}+\ell'} \sum_{j=0}^{\infty} (d_0 x_0)^{2j} \frac{(\max\{\tilde{\ell}+1, k+1\})^{2k}}{(\max\{\ell+1, k+j+1\})^{2(k+j)}} \binom{k+j}{k}^2 \times (j+1)^2 (\ell'+1)^4 |g^{(j)}_{\ell' m'}|^2 \Bigg]$$

We will now bound the terms inside the square brackets uniformly in k and $\tilde{\ell}$.

Note first of all that

$$(k+j+1)^{-2j} \binom{k+j}{k}^2 = \frac{1}{j!^2} (k+j+1)^{-2j} (k+j)^2 (k+j-1)^2 \dots (k+1)^2 \leq \frac{1}{j!^2}.$$

Now suppose $\tilde{\ell}+1 \leq k+1$. Then

$$\left(\frac{\max\{\tilde{\ell}+1, k+1\}}{\max\{\ell+1, k+j+1\}} \right)^{2k} \leq \left(\frac{k+1}{k+j+1} \right)^{2k} \leq 1.$$

Suppose $\tilde{\ell}+1 > k+1$. Then

$$\left(\frac{\max\{\tilde{\ell}+1, k+1\}}{\max\{\ell+1, k+j+1\}} \right)^{2k} \leq \left(\frac{\tilde{\ell}+1}{\max\{\ell+1, k+1\}} \right)^{2k} = e^{2k \log\left(\frac{\tilde{\ell}+1}{\max\{\ell+1, k+1\}} \right)}.$$

If $\tilde{\ell} \leq \ell$, then the right-hand side can be bounded by 1. Suppose therefore that $\tilde{\ell} > \ell$. We have that $\tilde{\ell} \leq \ell' + \ell$, so we can bound further:

$$e^{2k \log\left(\frac{\tilde{\ell}+1}{\max\{\ell+1, k+1\}} \right)} \leq e^{2k \log\left(1 + \frac{\ell'}{k+1}\right)} \leq e^{2\ell' \frac{k+1}{\ell'} \log\left(1 + \frac{\ell'}{k+1}\right)} \leq e^{2\ell'},$$

where we obtained the final inequality by using that $x^{-1} \log(1+x) \leq 1$ for $x \geq 0$.

We can therefore infer that:

$$\sum_{n=0}^{\infty} \sum_{\ell \in \mathbb{N}_0} \frac{(d_0 x_0)^{2n}}{(\max\{\ell+1, n+1\})^{2n}} \int_{\mathbb{S}^2} |(gf)^{(n)}_{\ell}|^2 \, d\sigma$$

$$\leq C \sum_{k=0}^{\infty} \sum_{\tilde{\ell} \in \mathbb{N}_0} \sum_{\tilde{m}=-\tilde{\ell}}^{\tilde{\ell}} \frac{(d_0 x_0)^{2k}}{(\max\{\tilde{\ell}+1, k+1\})^{2k}} |f^{(k)}_{\tilde{\ell}\tilde{m}}|^2 \times \Bigg[\sum_{\ell' \in \mathbb{N}_0} \sum_{m'=-\ell'}^{m=\ell'} \sum_{\ell=|\tilde{\ell}-\ell'|}^{\tilde{\ell}+\ell'} \sum_{j=0}^{\infty} (\ell'+1)^4 e^{2\ell'} \frac{(j+1)^2 (d_0 x_0)^{2j}}{j!^2} |g^{(j)}_{\ell' m'}|^2 \Bigg]$$

Then (C.1) follows from the observation that $\sum_{\ell=|\tilde{\ell}-\ell'|}^{\tilde{\ell}+\ell'} 1 = \tilde{\ell} + \ell' - |\tilde{\ell} - \ell'| \leq 2\ell'$.

The estimate (C.2) follows in an analogous manner, where the above computations become simplified since the relevant norms depend only a finite number of derivatives, and we use that for $\tilde{\ell} \leq \ell$, $(\ell(\ell+1)+1)^{-w} \leq (\tilde{\ell}(\tilde{\ell}+1)+1)^{-w}$ and for $\tilde{\ell} > \ell$ (in which case $\tilde{\ell} < \ell + \ell'$):

$$(\ell(\ell+1)+1)^{-w} = (\tilde{\ell}(\tilde{\ell}+1)+1)^{-w} \left(\frac{\tilde{\ell}(\tilde{\ell}+1)+1}{\ell(\ell+1)+1} \right)^{w} \leq (\tilde{\ell}(\tilde{\ell}+1)+1)^{-w} \left(\frac{(\ell+\ell')(\ell+\ell'+1)}{\ell(\ell+1)+1} \right)^{w}$$

$$= (\tilde{\ell}(\tilde{\ell}+1)+1)^{-w} \left(1 + \frac{\ell'(\ell'+1)}{\ell(\ell+1)+1} + \frac{2\ell\ell'}{\ell(\ell+1)+1} \right)^{w}$$

$$\leq (\tilde{\ell}(\tilde{\ell}+1)+1)^{-w} \left(1 + \ell'(\ell'+1) \right)^{w}. \quad \text{(C.3)}$$

□

Lemma C.3. *There exists constants $C, \beta > 0$ such that for all appropriately regular functions $f, g; [0, x_0] \times \mathbb{S}^2 \to \mathbb{C}$, we can estimate:*

$$\|gf\|^2_{G_{\sigma,R_\infty}} \leq C\|f\|^2_{G_{\sigma,R_\infty}} \left[\sum_{j\in\mathbb{N}_0}\sum_{\ell'\in\mathbb{N}_0} \frac{e^{5(\ell'+j)}}{j!^4}\|\partial_x^j g_{\ell'}\|^2_{L^\infty([0,x_0]\times\mathbb{S}^2)}\right]. \tag{C.4}$$

Proof. We can express:

$$\|gf\|^2_{G_{\sigma,R_\infty}} = \sum_{\ell\in\mathbb{N}_0}\sum_{n=0}^{\ell+l-1} \frac{(4\sigma)^{2n}(\ell-(n+1))!^2}{(\ell+n+1)!^2}\int_0^{x_0}\int_{\mathbb{S}^2}|\partial_x^n(gf)_\ell|^2\,d\sigma dx$$
$$+ (\mathfrak{l}+1)^3\sum_{n=\ell+l}^{\infty}\frac{\sigma^{2n}}{n!^2(n+1)!^2}\int_0^{x_0}\int_{\mathbb{S}^2}|\partial_x^n(gf)_\ell|^2\,d\sigma dx.$$

We then proceed as in the proof of Lemma C.2 by decomposing g and f into spherical harmonics: $g = \sum_{\ell'\in\mathbb{N}_0}\sum_{m=-\ell'}^{\ell'} g_{\ell'm'}Y_{\ell'm'}$, $f = \sum_{\tilde\ell\in\mathbb{N}_0}\sum_{\tilde m=-\tilde\ell}^{\tilde\ell} f_{\tilde\ell\tilde m}Y_{\tilde\ell\tilde m}$ and using that

$$|(gf)_\ell^{(n)}|^2 \leq \frac{\pi^2}{6}\sum_{k=0}^{n}\binom{n}{k}^2(n-k+1)^2|(g^{(n-k)}f^{(k)})_\ell|^2.$$

We then obtain:

$$\|gf\|^2_{G_{\sigma,R_\infty}} \leq C\sum_{\ell\in\mathbb{N}_0}\sum_{\{\ell',\tilde\ell\in\mathbb{N}_0\,|\,|\tilde\ell-\ell'|\leq\ell\leq\ell+\ell'\}}\sum_{\tilde m=-\tilde\ell}^{\tilde\ell}\sum_{m=-\ell'}^{\ell'}\sum_{n=0}^{\ell-1}\frac{(4\sigma)^{2n}(\ell-(n+1))!^2}{(\ell+n+1)!^2}$$
$$\times\int_0^{x_0}|f^{(k)}_{\tilde\ell\tilde m}|^2\binom{n}{k}^2(n-k+1)^2\|Y_{\ell'm'}\|^2_{L^\infty(\mathbb{S}^2)}|g^{(n-k)}_{\ell'm'}|^2\,d\sigma dx$$
$$+\left(\frac{4\sigma}{d_0x_0}\right)^{-2\ell}(\ell+1)^3\sum_{n=\ell}^{\infty}\frac{\sigma^{2n}}{n!^2(n+1)!^2}\int_0^{x_0}\int_{\mathbb{S}^2}|f^{(k)}_{\tilde\ell\tilde m}|^2\binom{n}{k}^2(n-k+1)^2\|Y_{\ell'm'}\|^2_{L^\infty(\mathbb{S}^2)}|g^{(n-k)}_{\ell'm'}|^2\,d\sigma dx.$$

By introducing the summation index $j = n-k$ and taking $\sup_{x\in[0,x_0]}|g^{(n-k)}_{\ell'm'}|^2(x)$ outside of the x-integral, we can rewrite the right-hand side above to obtain:

$$\|gf\|^2_{G_{\sigma,R_\infty}} \leq C\sum_{\tilde\ell\in\mathbb{N}_0}\sum_{\tilde m=-\tilde\ell}^{\tilde\ell}\sum_{k=0}^{\tilde\ell-1}\frac{(4\sigma)^{2k}(\tilde\ell-(k+1))!^2}{(\tilde\ell+k+1)!^2}$$
$$\times\int_0^{x_0}|f^{(k)}_{\tilde\ell\tilde m}|^2\,dx$$
$$\times\left[\sum_{\ell'\in\mathbb{N}_0}\sum_{\ell=|\tilde\ell-\ell'|}^{\tilde\ell+\ell'}\sum_{m=-\ell'}^{\ell'}\left(\sum_{j=0}^{\ell-1-k}(4\sigma)^{2j}\underbrace{\frac{(\tilde\ell+k+1)!^2(\ell-(k+j+1))!^2}{(\ell+k+j+1)!^2(\tilde\ell-(k+1))!^2}\frac{(k+j)!^2}{k!^2j!^2}}_{=:J_1}(j+1)^2\|g^{(j)}_{\ell'm'}Y_{\ell'm'}\|^2_{L^\infty}\right.\right.$$
$$\left.\left.+\sum_{j=\ell-k}^{\infty}4^{-2k}(4\sigma)^{2j}(\ell+1)^3\underbrace{\frac{(\tilde\ell+k+1)!^2}{(\tilde\ell-(k+1))!^2(k+j+1)!^2(k+j)!^2}\frac{(k+j)!^2}{k!^2j!^2}}_{=:J_2}(j+1)^2\|g^{(j)}_{\ell'm'}Y_{\ell'm'}\|^2_{L^\infty}\right)\right]$$
$$+\sum_{k=\tilde\ell+l}^{\infty}\frac{\sigma^{2k}(\tilde\ell+1)^3}{(k+1)!^2k!^2}\int_0^{x_0}|f^{(k)}_{\tilde\ell\tilde m}|^2\,dx\left[\sum_{\ell'\in\mathbb{N}_0}\sum_{\ell=|\tilde\ell-\ell'|}^{\tilde\ell+\ell'}\sum_{m=-\ell'}^{\ell'}\left(\sum_{j=0}^{\min\{\ell-1-k,0\}}4^{2k}(4\sigma)^{2j}(\tilde\ell+1)^{-3}\right.\right.$$

$$\times \underbrace{\frac{(k+1)!^2 k!^2 (\ell-(k+j+1))!^2}{(\ell+k+j+1)!^2}\frac{(k+j)!^2}{k!^2 j!^2}}_{=:J_3}(j+1)^2\|g^{(j)}_{\ell' m'}Y_{\ell' m'}\|^2_{L^\infty}$$

$$+\sum_{j=\max\{\ell+l-k,0\}}^{\infty}\sigma^j\left(\frac{\ell+1}{\tilde{\ell}+1}\right)^3\underbrace{\frac{(k+1)!^2k!^2}{(k+j+1)!^2(k+j)!^2}\frac{(k+j)!^2}{k!^2j!^2}}_{=:J_4}(j+1)^2\|g^{(j)}_{\ell' m'}Y_{\ell' m'}\|^2_{L^\infty}\Bigg)\Bigg].$$

We will simplify the terms inside square brackets. We will first estimate J_1. We have that:

$$J_1=\frac{(\tilde{\ell}+k+1)^2(\tilde{\ell}+k)^2\dots(\tilde{\ell}-k)^2}{(\ell+k+j+1)^2\dots(\ell+k+j)^2\dots(\ell-k-j)^2}\frac{(k+j)!^2}{k!^2j!^2}$$

Suppose that $\tilde{\ell}\leq\ell$. Then, using that $\ell\geq k+j+1$, we estimate

$$J_1\leq\frac{1}{(\ell+k+j+1)^2\dots(\ell+k+1)^2(\ell-k-1)^2\dots(\ell-k-j)^2}\frac{(k+j)!^2}{k!^2j!^2}$$
$$\leq\frac{(k+j)^2(k+j-1)^2\dots(k+1)^2}{(2k+j+2+j)^2\dots(2k+j+2)^2}\frac{1}{j!^4}\leq\frac{1}{j!^4}.$$

Now suppose that $\tilde{\ell}\geq\ell+1$. Then we estimate using that $k+j\leq\ell-1$

$$J_1\leq\frac{(\tilde{\ell}+k+1)^2\dots(\tilde{\ell}-k)^2}{(\ell+k+1)^2\dots(\ell-k)^2}$$
$$\times\frac{1}{(\ell+k+j+1)^2\dots(\ell+k+2)^2(\ell-k-1)^2\dots(\ell-k-j)^2}\frac{(k+j)!^2}{k!^2j!^2}$$
$$\leq\frac{(\tilde{\ell}+k+1)^2\dots(\ell+k+2)^2}{(\tilde{\ell}-k-1)^2\dots(\ell-k)^2}\frac{1}{j!^4}$$
$$\leq\frac{(2\ell+1+(\tilde{\ell}-\ell-1))^2\dots(2\ell+1)^2}{(\tilde{\ell}-\ell)!^2}\frac{1}{j!^4}$$
$$\leq Ce^{2(\tilde{\ell}-\ell)}(\tilde{\ell}-\ell)^{-2(\tilde{\ell}-\ell-1)-3}(2\ell+l+1+(\tilde{\ell}-\ell-1))^{2(\tilde{\ell}-\ell-1)}\frac{1}{j!^4}$$
$$\leq Ce^{2(\tilde{\ell}-\ell)}(\tilde{\ell}-\ell)^{-3}\left(1+\frac{2\ell+1}{\tilde{\ell}-\ell}\right)^{2(\tilde{\ell}-\ell-1)}\frac{1}{j!^4}$$

Suppose now that $\tilde{\ell}-\ell\geq\frac{1}{2}(2\ell+1)$. Then we can further estimate:

$$J_1\leq Ce^{2(1+\log(\frac{3}{2}))(\tilde{\ell}-\ell)}(\tilde{\ell}-\ell)^{-3}\frac{1}{j!^4}\leq Ce^{2(1+\log(\frac{3}{2}))\ell'}\ell^{-3}\frac{1}{j!^4}.$$

If instead $\tilde{\ell}-\ell<\frac{1}{2}(2\ell+1)$, then $\tilde{\ell}\leq 2\ell+\frac{1}{2}$ and we estimate:

$$J_1\leq\frac{(\tilde{\ell}+k+1)^2\dots(\tilde{\ell}-k)^2}{(\ell+k+1)^2\dots(\ell-k)^2}\frac{1}{j!^4}\leq 4^{2k+1}\frac{1}{j!^4}.$$

We now turn to J_2.

Note first that $k\geq\ell-j\geq\tilde{\ell}-(\ell'+j)$, so $\tilde{\ell}\leq k+(\ell'+j)$. Hence,

$$\frac{(\tilde{\ell}+k+1)!^2}{(\tilde{\ell}-(k+1))!^2(k+1)!^4}\leq\frac{(\ell'+j-1+2(k+1))!^2}{(\ell'+j-1)!^2(k+1)!^4}$$

Via Stirling's approximation, we can further estimate

$$\frac{(\ell'+j-1+2(k+1))!^2}{(\ell'+j-1)!^2(k+1)!^4} \leq C\left(1+\frac{2(k+1)}{\ell'+j-1}\right)^{2(\ell'+j-1)+1}\left(2+\frac{\ell'+j-1}{k+1}\right)^{4(k+1)}$$
$$\leq Ce^{2(k+1)\frac{\ell'+j-1}{2(k+1)}\log\left(1+\frac{2(k+1)}{\ell'+j-1}\right)}e^{4(\ell'+j-1)\frac{k+1}{\ell'+j-1}\log\left(2+\frac{\ell'+j-1}{k+1}\right)}$$
$$\leq Ce^{2(k+1)\frac{\ell'+j-1}{2(k+1)}\log\left(1+\frac{2(k+1)}{\ell'+j-1}\right)}e^{4\log(1+\ell'+j)},$$

where we obtained the final inequality by using that $x^{-1}\log(2+x)$ is strictly decreasing. Suppose that $2(k+1)\geq \ell'+j-1$. Then, by the decreasing property of $x^{-1}\log(1+x)$, we have that

$$e^{2(k+1)\frac{\ell'+j-1}{2(k+1)}\log\left(1+\frac{2(k+1)}{\ell'+j-1}\right)} \leq e^{2\log(2)(k+1)} \leq 2^{2(k+1)}.$$

Suppose now that $2(k+1)<\ell'+j-1$. Then we can estimate instead:

$$e^{2(k+1)\frac{\ell'+j-1}{2(k+1)}\log\left(1+\frac{2(k+1)}{\ell'+j-1}\right)} = \left(1+\frac{2(k+1)}{\ell'+j-1}\right)^{2(\ell'+j-1)+1} \leq 2^{2(\ell'+j-1)+1}.$$

Hence, there exists constants $C,\beta>0$ such that

$$\frac{(\tilde{\ell}+k+1)!^2}{(\tilde{\ell}-(k+1))!^2(k+1)!^4} \leq C2^{2\ell'+2j}2^{2k}.$$

From the above, we can therefore estimate J_2 as follows:

$$4^{-2k}J_2 \leq 2^{-2k}2^{2\ell'+2j}\frac{(k+1)^2(k+1)!^2}{(k+j+1)!^2j!^2} \leq C2^{-2k}k^2\frac{1}{j!^4}$$

Note also that by $\ell\leq k+j$

$$(\ell+1)^3 \leq (k+j+1)^3 \leq (k+1)^3\left(1+\frac{j}{k+1}\right)^3 \leq (k+1)^3(j+1)^3.$$

Consider now J_3. Note that by the Stirling approximation and the fact that $\ell\geq k+j$

$$\frac{(\ell-(k+j+1))!^2(k+j+1)!^4}{(\ell+(k+j+1))!^2} \leq \frac{(k+j+1)!^4}{(2(k+j+1))!^2} \leq C2^{-4(k+j+1)}(k+j+1),$$

so

$$J_3 \leq C4^{-2(k+j)}\frac{(k+1)!^2}{(k+j+1)!^2(k+j+1)}\frac{1}{j!^2} \leq C4^{-2(k+j)}(k+j+1)^{-1}\frac{1}{j!^4}.$$

We consider J_4. Note that

$$J_4 = \frac{(k+1)!^2}{(k+j+1)!^2}\frac{1}{j!^2} = \frac{1}{(k+j)^2\dots(k+2)!^2}\frac{1}{j!^2} \leq \frac{1}{j!^4}.$$

Furthermore,

$$\left(\frac{\ell+1}{\tilde{\ell}+1}\right)^3 = \left(1+\frac{\ell-\tilde{\ell}}{\tilde{\ell}+1}\right)^3 \leq C(1+\ell')^3.$$

The inequality (C.4) follows by combining the above estimates. □

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